The Mathematics of the pmsimstats-ng Manuscript Compendium

A unified account of the notation, data-generating processes, and analysis models across the manuscripts under `analysis/report/` and the whitepapers under `docs/`

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Abstract

The manuscripts of the pmsimstats-ng compendium share one outcome model, one family of trial designs, and one analysis-model convention, but each paper varies a different piece of that shared machinery to ask a different question. This document collects, in one place and in the compendium's own notation, the mathematics that each manuscript actually states, ordered as a pipeline in five parts. Part I fixes the scope and the notation. Part II states the data-generating process: the response trajectory, the trial designs and treatment exposure, carryover on the mean, the encodings by which a biomarker moderates treatment response (mean moderation, covariance moderation, their dual-channel combination, and a latent-variable reading of covariance moderation), the validity of covariance binding, a pharmacodynamic sensitivity model, further features a data-generating process can vary, the decomposition of the treatment effect into its biological, placebo, and natural-history parts, and informative dropout. Part III gives the closed-form theory of the interaction test and of design efficiency. Part IV treats the analysis: the common analysis model and its carryover specifications, baseline coding and a prognostic biomarker, the inferential machinery used to test and calibrate the interaction estimand, and the stress-test construct. Part V gives the treatment main-effect track and the performance measures shared across papers. Where a manuscript's own presentation is thin, inconsistent, or silent on a point the compendium's design nonetheless raises, that gap is flagged explicitly rather than papered over with new derivation. Every equation below is attributed to the manuscript it came from.

The document was revised and extended on 2026-10-08: the original sections were corrected against the current sources, and results drawn from the whitepapers `docs/09a` to `docs/52` and the restated Paper 01 were added and placed with the material they extend. Each result is marked as derived, verified by a check script in `analysis/scripts/compendium-mathematics/`, or argued. The studies the results describe are exploratory.

(Updated 2026-10-09: results at $t_{1/2} > 0$ now use anchored carryover; see `docs/53`. Values changed by at most 0.021 in power (closed form at most 0.017, in CO; the larger Monte Carlo changes of Table 37 are mostly reseeding), 0.53 points in the drug coefficient of Table 48, and 2.7 in the CO path-mean term; conclusions unchanged. The stress-test and covariance-study results still come from runs with the published recursion; their anchored reruns are pending.)

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Part I
Scope and notation

1 Scope and reading order

[2026-10-08: `analysis/report/` now holds fourteen manuscript folders. This scope paragraph describes the eleven covered in 2026-08; Paper 12 is drawn on in Section 15, and Papers 13 and 14 (two-component variants of Paper 06) only for their component-count result on the coupling ceiling (§9.12).]

The compendium's original eleven manuscripts fall into two families. Nine share the biomarker-treatment interaction $\beta_{bm:D}$ as their estimand and the common outcome components of Section 2.2; one (Paper 04) targets the treatment *main* effect β_D under a simpler, biomarker-free model; one (Paper 10) is written in deliberately general, non-N-of-1-specific notation to isolate what drives realized Type I error in *any* Wald test on a mixed-model interaction coefficient. This document orders the material as a pipeline in five parts, so that each section can be read against the ones before it:

Part I: scope and notation.

1. **Notation** (§2) fixes indices, the sign convention, and the rule that N is always a total across randomization paths.

Part II: the data-generating process.

2. **The response trajectory** (§3, Papers 06–07) gives the modified-Gompertz curve that every latent factor in the outcome follows, its landmarks and numerical conditioning, and the alternative curve families used to test whether the compendium's conclusions depend on that choice.

3. **Trial designs and treatment exposure** (§4, Papers 02, 05) gives the four designs and the Hybrid paths, the clocks t_{od}, t_{sd} and t_{pb}, and the binary and continuous treatment-exposure indicators, with the rule that D_{bc} is zero before first exposure.

4. **Carryover on the mean** (§5, Papers 01, 02, 05) gives the carryover decay families, the published compounding recursion and the anchored form that replaces it, and the scale factor of the 2024 revision of the published code.

5. **Two architectures for a biomarker-treatment interaction** (§6, Paper 01) states the two encodings of the interaction: in the outcome's first moment (mean moderation) or in its second moment (covariance moderation), with the step and graded rules that couple the biomarker to the drug response under carryover. As restated in October 2026, Paper 01 shows that, conditional on the biomarker, the covariance encoding is mean moderation with a reduced residual variance, so the two differ in parameterization and in the exposure profile of the interaction rather than in kind. The section closes with the pharmacokinetic argument for graded coupling and the historical formulations of both encodings.

6. **The dual-channel architecture** (§7, Paper 11) activates both channels at once and shows their contributions to the interaction test are additive to first order, with the two channel weights not separately identifiable from a single fitted coefficient.

7. **A latent-variable reading of covariance moderation** (§8, Paper 03) asks what generative mechanism a fitted covariance-moderation model may stand in for, and gives the moment conditions under which a two-class mixture and a single MVN are close in their first two moments.

8. **Validity of covariance binding** (§9, `docs/32`, `docs/34`–`docs/36`) gives the largest coupling for which the covariance matrix is positive definite, under compound symmetry and separable AR(1), its factorization into per-step costs, its dependence on the schedule and on the number of response components, and why a generator must stop rather than repair a matrix.

9. **A pharmacodynamic sensitivity model** (§10, `docs/38`) derives the coupling profile from a turnover model of drug effect.

10. **Further features of a data-generating process** (§11, `docs/49`, `docs/51`) gives the effect on the estimand and its power of biomarker reliability, threshold response, adherence, unblinding, person-level half-lives, regression to the mean and range restriction, and states phase-specific measurement error and the prognostic pathways of a biomarker as options of the data-generating process.

11. **Partitioning the treatment effect** (§12, Paper 06) decomposes the outcome itself into biological, placebo-belief, and natural-history components, and derives closed-form bias identities for what a one-component analysis actually estimates when the true generative process has three.

12. **Informative dropout** (§13, Paper 09) gives the hazard mechanism used to censor participants as a function of their own symptom trajectory, the missingness classification that follows from it, and the MCAR and MAR mechanisms of the stress test.

Part III: closed-form theory of the interaction test.

13. **Coupling and closed-form power** (§14, `docs/37`, Paper 01 Appendix A) derives the moments and power of the paired-difference statistics E9 and E9b in closed form, shows that power depends on the coupling profile only through the coupling gap, collapses to the test size under the step rule and declines monotonically under graded coupling, and extends the result to other designs and to AR(1) response structures, with its validation against simulation.

14. **Design efficiency and time adjustment** (§15, `docs/39`, `docs/42`, Papers 08 and 12) gives contrast variances by design and correlation structure, the design search, the optimal spacing and the most efficient contrast, and the limit of the drug coefficient under each time adjustment.

Part IV: analysis.

15. **The analysis model and its carryover specifications** (§16, Papers 01, 02) gives the analysis model common to the compendium, its open-label branch, the nine carryover specifications E1–E9 and two historical analyses of carryover.

16. **A prognostic biomarker** (§17, `docs/41`, `docs/45`, `docs/48`, `docs/49`) treats the coding of baseline, derives the spurious interaction that a shared biomarker effect produces when the biomarker is prognostic, and what each remedy removes.

17. **Inference: from a fitted coefficient to a calibrated test** (§18, Papers 08, 10, 02) collects the inferential machinery: a closed-form repeated-measures ANOVA derivation, a general four-channel decomposition of realized Type I error, and the cluster-robust sandwich-variance family used to correct it.

18. **The stress-test construct** (§19, `docs/43`, `docs/46`, `docs/47`) states as model formulas the data-generating process and the fourteen analyses of the exploratory stress test of the Hybrid design, with its AIC selection of the analysis half-life, the implied outcome covariance, and the sign symmetry of power.

Part V: the main-effect track and shared measures.

19. **The treatment main-effect track** (§20, Paper 04) is the compendium's odd paper out: no biomarker channel at all, a linear (not Gompertz) DGP, and the classical closed-form power theory of Senn and of Zucker et al. that most of the compendium replaces with simulation.

20. **Shared performance measures** (§21) collects the Monte Carlo standard error, power, and calibration-ratio definitions used identically across every paper, size-adjusted power, and the selection criteria of the stress test (§21.2).

Every symbol is defined at first use and, where the compendium's own `NOTATION.md` governs it, cross-referenced to that governing rule. Sign convention throughout: the three outcome components are non-negative *reductions* in symptom severity, so treatment and interaction coefficients are negative and moderation parameters are positive (`NOTATION.md` Rule 5).

1.1 Verification scripts

The document, first written on 2026-08-20, was revised and extended on 2026-10-08. The revision corrected the original sections against the current sources and added mathematics established after 2026-08-20, mostly in the whitepapers `docs/32` to `docs/51` and in the restated Paper 01, together with the results of an inventory of `docs/09a` to `docs/52`. The added results are placed with the material they extend, are written in the notation of Section 2, and cross-reference the results they refine or correct. Each result is marked derived, verified (with its

check script) or argued; numbers quoted from a source without recomputation are marked as such.

The check scripts are in `analysis/scripts/compendium-mathematics/` and run from the repository root with `Rscript`: `check-coupling-closed-form.R` (Section 14), `check-ceilings.R` and `check-ceilings-graded.R` (Section 9), `check-prognostic.R`, `check-prognostic-car1.R` and `check-design.R` (Sections 17 and 15), and `check-pd-features.R` (Sections 10 and 11). The checks of the revised 2026-08 sections are in `check-existing-sections.R` and `check-paper06-decomposition.R`. The results added by the inventory of `docs/09a` to `docs/52`, which appear throughout, are checked by `check-historical-results.R` (`docs/09a`, 19, 21–23), `check-gompertz-rmanova-cycles.R` (`docs/24`–28), `check-landmarks-hybrid-ceilings.R`, `check-extracted-parameters.R` and `check-crossover-binding-path.R` (`docs/29`–31, 33, 40), `check-stress-test-construct.R` and `check-open-label-ceiling.R` (`docs/43`, 44, 46, 47, 50, 52, `docs/54` and the Architecture C plan, and Section 19), `check-closed-form-extensions.R` (`docs/32`, 34–38), `check-baseline-design-search.R` (`docs/39`, 41, 42, 45, 48, 49, 51), and `check-reconciliations.R` (figures reconciled when the inventory was merged: the drug-expectancy collinearity of §14.9, the week-12 landmark share of Table 3 and the rounding of Table 8). Some read stored outputs under `analysis/data/quick-sim/` for comparison only. The checks confirm the formulas and their stated numbers; except where a section says otherwise, they do not rerun the package's simulation code.

2 Notation and conventions

2.1 Indices and structural counts

Symbol	Meaning
i	participant index
t	timepoint index (weekly measurement occasions throughout)
j, ij	period or occasion index within participant (Papers 04, 06, 08)
N	**total** number of participants, across all randomization paths
P	number of randomization paths in a design; expected allocation N/P per path
n_{reps}, n_{sim}	Monte Carlo replicates per cell
k	cycles per participant in a design sweep; also the Weibull shape parameter

Rule (`NOTATION.md` Rule 4). N is always the total sample size across randomization paths, never a per-path count. Table 1 gives the path count P for each design; the per-path allocation is $\lfloor N/P \rfloor$ with the remainder spread one per path.

Table 1: Trial designs and their randomization-path counts.

Design	P	Paths
OL (open-label)	1	on drug throughout
CO (crossover)	2	drug-then-placebo, placebo-then-drug
OL+BDC (blinded discontinuation)	2	discontinuation at either of two visits
Hybrid	4	two discontinuation points × two crossover orders

This distinction is not cosmetic: `NOTATION.md` documents a 2026-08-08 compendium-wide correction in which several drivers had read `modelparam$N` as a per-path count rather than a total. Re-matching every design to a common total $N = 70$ moved Paper 01's Hybrid-design power from a ceiling of ≈ 0.99 down to 0.84 (tripling its measured carryover loss) and withdrew

Paper 09's second headline conclusion, that its most powerful design was also its most dropout-robust: the earlier near-immunity was a ceiling effect at an unmatched N. An unmatched design comparison therefore does not merely inflate one arm; it can produce an apparent robustness finding from a ceiling effect.

2.2 Outcome and latent components

Symbol	Meaning
Y_{it}	observed symptom score, the modeled outcome (code column `Sx`)
BL_i	participant-specific baseline level
BR_{it}	biological (pharmacological) response component
PB_{it}	placebo-belief (expectancy) response component
TV_{it}	time-variant natural-history component
η_t	design expectancy: 1 open label, 0.5 blinded, 0 at baseline (code `e`, regressor `De`)
ε_{it}, σ^2_ε	observation-level residual and the residual variance of an analysis model
u_i, σ^2_u	participant random intercept and its variance
σ_{TV}, σ_{PB}, σ_{BL}	SDs of the natural course, the placebo response ($\sigma_{PB,t} = \sigma_{PB}\eta_t$) and the baseline level

[2026-10-08: the random intercept was previously listed as "u_i, b_i". b_i is the canonical standardized biomarker, so the random intercept is u_i only, and σ^2_u replaces the σ^2_b that Paper 10 writes for its variance. The expectancy η_t is added as in `docs/52`, which renamed the e_t of earlier documents; Paper 01 was renamed to η_t on 2026-10-09. 2026-10-09: `NOTATION.md` now gives η_t this design meaning (formerly "measured belief, paper 06"; Paper 06's η(phase) is the same design object), and reserves $\tilde{\eta}_{it}$ for a measured placebo-belief covariate, which no section of this document uses. σ^2_ε is the residual variance of an analysis model (not σ^2_e).]

2.3 Treatment exposure and carryover

Symbol	Meaning
D_{it}	binary drug state: 1 on drug, 0 off drug
$D_{bc,it}$	continuous exposure-decayed drug indicator (code `Dbc`)
t_{sd}	time since discontinuation (code `tsd`)
t_{od}	cumulative time on drug (code `tod`)
t_{pb}	time-by-placebo-belief accumulation (code `tpb`)
w_t	calendar time (weeks) of visit t
$t_{1/2}$	carryover half-life of the data-generating process (code `carryover_t1half`)
$t^a_{1/2}$	half-life assumed by the analysis when building D_{bc}
λ	carryover decay rate, $\lambda = \ln 2 / t_{1/2}$ (code `lambda_cor`)
$\phi(\cdot)$, ϕ_t	decay-family weight (exponential, linear, Weibull, power) and its value at visit t
k, λ_w	Weibull shape and rate, $\lambda_w = (\ln 2)^{1/k}/t_{1/2}$
$G(t_{od}; m, d, r)$	modified Gompertz mean, ceiling m, displacement d, rate r (code `maxr`, `disp`, `rate`)
$\mu_{BR,t}$	mean drug response at visit t after carryover

Units. $t_{1/2}$ is quoted in *weeks* in the interaction-focused manuscripts (canonical grid $\{0, 0.5, 1.0\}$) and in the main-effect manuscript (Paper 04, rebuilt under the refined process), and in *days* in the test-procedure manuscripts, on a pharmacokinetic scale with a 3-day baseline (1 week = 7 days). The day scale with the 3-day baseline also belongs to the version 1 study of Paper 04, which §20 describes. [2026-10-09: aligned with `NOTATION.md`, Units (user decision of 2026-10-09).]

2.4 Biomarker, moderation, and estimand

Symbol	Meaning
B_i	participant-level pre-treatment biomarker (code `bm`)
b_i	standardized biomarker, $(B_i - \bar{B})/s_B$ (code `bm_z`)
B_i^c	biomarker centered at the baseline sample mean, not scaled (code `bmc`)
μ_{bm}, σ_{bm}	population mean and SD of the biomarker
c_{bm}	covariance-moderation parameter: on-drug $\mathrm{Cor}(B_i, BR_{it})$
β_{bm}	mean-moderation parameter: multiplier scaling the treatment effect by b_i
$c_{bm,A}$, $c_{bm,B}$	independent mean-channel and covariance-channel weights (dual-channel)
g_t	coupling profile of the covariance encoding: $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} g_t$ (Paper 01)
$\mathcal{G}$	coupling gap, $\bar{g}_{\text{on}} - \bar{g}_{\text{off}}$
$c_{bm,TV}$, $c_{bm,PB}$, $c_{bm,BL}$	prognostic couplings of the biomarker to TV, PB and baseline (not c_{bl})
s_{prog}, f	prognostic slope, and the share of it absorbed by a shared β_B
β_{bm}^{TV}	prognostic biomarker slope on the natural course
$\beta_{bm:D}$	the target estimand, the biomarker-by-treatment interaction (code `bm:Dbc`)
ρ	AR(1) within-component correlation per week
ρ_{cs}	common within-component correlation of compound symmetry (published 0.8)
π	power, $\Pr(p < \alpha)$; $\alpha = 0.05$ throughout

Rule (`NOTATION.md` Rule 2). c_{bm} is a correlation; β_{bm} is a multiplier. They are calibrated to a common numeric scale (the reference value 0.45 denotes matched moderation strength under either architecture, because the mean-channel shift is applied as $\beta_{bm}\, b_i\, \sigma_{BR}$ on the standardized biomarker), but the two symbols are not interchangeable, and a paper using both architectures at a matched value must say so explicitly.

Architecture names. Following `NOTATION.md` (Part 3), the architectures are named by the channel that carries the interaction: the mean-moderation, covariance-moderation and dual-channel architectures, formerly Architectures A, B and C. The single letters survive only in the subscripts of the dual-channel weights $c_{bm,A}$ and $c_{bm,B}$ and in quotations of code comments.

2.5 Correlation structures, contrasts, and inference

Symbol	Meaning
c_1, $c_\times$	cross-component correlation at the same visit and at different visits (code `c.cf1t`, `c.cfct`)
K, A, M	cross-component correlation matrix, temporal correlation matrix, and response block ($M = K \otimes A$ when separable)
n_c	number of response components (3, or 2 in Papers 13 and 14)
ρ_t	step correlation ρ^{d_t} across the gap $d_t = w_t - w_{t-1}$
$\omega_{\min}(\cdot)$	smallest eigenvalue of a correlation matrix
c^*_{bm}	coupling ceiling: largest c_{bm} with a positive-definite joint matrix
β_D, β_B, β_T	treatment main effect, biomarker main effect, and time slope ($\beta_T w_t$) of the analysis model
β_η, $\beta_{bm:\eta}$	expectancy and biomarker-by-expectancy coefficients (code `De`, `bmc:De`)
ρ_a	working CAR(1) correlation per week of the analysis model
τ^2	between-participant variance of the drug effect, treatment heterogeneity (§20)
Δ_i, a_t	paired on-minus-off contrast of E9 and its visit weights
σ^2_Δ, $\sigma^2_{\Delta,p}$	residual variance of the contrast Δ_i given the biomarker, overall and in path p (formerly τ^2, τ^2_p)
ϖ_p	share n_p/N of path p (formerly q_p)
$r_{\Delta B}$	correlation of the contrast with the biomarker
Λ	noncentrality parameter of a t or F test
θ, θ_{true}	generic interaction coefficient and its calibrated true value
κ	mean model-based SE over empirical SD; $\kappa > 1$ is conservative
δ	standardized bias of the test statistic
ν	denominator degrees of freedom
ℓ_g	eigenvalue weights of the CR2 Satterthwaite degrees of freedom
ψ_g	latent-class gating slope (Paper 03)

Section-local symbols. Symbols introduced for one section are defined where they are used and are listed here only when they could be confused with a canonical symbol: φ and Φ, the standard normal density and distribution function (§11); Δ_h, the effect of a threshold moderation; $\mathcal{A}_{it}$, adherence; $e_{B,i}$ and R_B, biomarker measurement error and reliability; $\tilde{\beta}_p$, the deviation of path p's slope from the pooled slope; $\beta_{BL,t}$, the slope of visit t on baseline (§17); $h(t)$, h_0, h_{res}, the drug-response time course of the pharmacodynamic model, and $\mathcal{M}(b)$, the moderation function (§10, §11); $\mathcal{Q}$, $\mathcal{Q}_\xi$, quadratic forms. The RM-ANOVA effects ξ_g, ϕ_j, υ_t of §18.1 and the crossover cell effects $\upsilon_{A,1}$, … of §16.5 are local to those subsections. Letters that carry a second, local meaning that the context separates: ω_i is a participant's drug sensitivity in the pharmacodynamic model (§10) and ω_j, $\omega_{\min}$ are eigenvalues in the ceiling sections; π also denotes the circle constant in closed forms such as $\sqrt{2/\pi}$; $f(t)$ is also the time adjustment of an analysis model (§17, as in `docs/41`) and $f(d)$ a design information function (§15), apart from the canonical absorbed share f; p and q are also occasion counts in the Paper 08 short communication and parameter counts in the stress-test catalog, and $q_{1-\alpha}$ is a quantile. [2026-10-09: notation. The rows for ρ_{cs}, τ^2, σ^2_Δ, ϖ_p, β_η, $\beta_{bm:\eta}$, ρ_a, B^c_i, μ_{bm}, s_{prog}, f, G and $\mu_{BR,t}$ were added to match `NOTATION.md` as merged on 2026-10-09.]

2.6 Glossary

The published process

Hendrickson et al.'s `generateData()` and `buildtrialdesign()` at commit `58b32a9`: com-

pound symmetry within components, constant cross-component correlation, step coupling, the compounding carryover recursion, and time on drug restarting at zero on re-exposure. "The published construct" denotes its covariance construct alone (configuration A).

The refined process
The recommended process of the restated Paper 01: graded coupling, separable AR(1) at $\rho = 0.7$ with $c_1 = 0.2$, anchored carryover, a stop on matrices that are not positive definite, and randomized path allocation.

Configurations A to F
The covariance and coupling constructs compared in `docs/36`: A published; B AR(1) with the `covar` cross block and step coupling; C separable AR(1) with graded coupling; D mean moderation; E all three; F AR(1) with the `covar` cross block and graded coupling. Configuration C is the covariance construct of the refined process, not the refined process itself, and is unrelated to the former "Architecture C".

Step and graded coupling
Off-drug biomarker coupling g_t set to 1 wherever the drug-response mean is nonzero (step) or to the decay weight ϕ_t, with $g_t = 0$ before first exposure (graded).

Anchored and compounding carryover
The off-drug drug-response mean decays from the last on-drug level (anchored) or is carried forward by the published recursion, which compounds across visits.

Mean, sensitivity and covariance binding
The ways a biomarker enters the drug response: as a shift of the mean, as a scaling of a participant's sensitivity ω_i, or through $\mathrm{Cor}(B_i, BR_{it})$.

E1 to E9
The carryover specifications of the analysis model (§16.3); E9 and E9b are the paired-difference statistics.

Coupling ceiling
The largest c_{bm} for which the joint correlation matrix is positive definite.

Part II
The data-generating process

3 The response trajectory: modified-Gompertz progression

Every latent factor in the outcome model (BR, PB, and TV) is generated as a trajectory in t (or in t_{od}, cumulative time on drug) that rises from zero toward an asymptote. The default functional form throughout the package is a *modified Gompertz curve*; Paper 07 exists specifically to ask how much of the compendium's power and bias findings survive if that functional-form choice is wrong.

3.1 The modified Gompertz curve

Paper 07 writes the Gompertz kernel

$$y(t) = m \cdot \exp(-d \cdot \exp(-r \cdot t)) \tag{1}$$

and state that a vertical offset makes the curve pass through the origin, written with the code names `maxr`, `disp` and `rate` of `respparam`. Here, as in `NOTATION.md` (Part 3), the mathematical symbols are m (ceiling, `maxr`), d (displacement, `disp`) and r (rate, `rate`). [2026-10-09: notation.

The Gompertz equations of this document were written with the code names maxr, disp and rate; they now use m, d, r.] The kernel itself does not: $y(0) = m\,e^{-d} > 0$. The curve the package computes (`modgompertz` in `R/utilities.R`, inspected) subtracts that value and rescales so the asymptote is restored. This is the form Paper 05 and Paper 01 state:

$$y(t) = m \cdot \frac{\exp(-d \cdot \exp(-r \cdot t)) - \exp(-d)}{1 - \exp(-d)}, \qquad y(0) = 0, \quad y(\infty) = m. \tag{2}$$

Here m is the asymptotic maximum response (the ceiling) and r the onset rate (per week). The displacement d sets the length of the initial lag: the inflection is at $t^* = \ln(d)/r$ (derived; an affine rescaling does not move it), so a large d gives a long slow start, with the steepest part of the rise later and, at fixed m and r, slightly less steep (the maximum slope decreases in d, Equation (8)), and a small d a curve that rises almost at once. As $d \to 0$ the ratio in Equation (2) tends to $1 - e^{-rt}$ (derived). At the shipped drug-response values $m = 10.99$, $d = 5$, $r = 0.42$ per week, the curve inflects at 3.83 weeks, reaches half its maximum at 4.73 weeks and 90% of it at 9.21 weeks, and stands at 84% of its maximum at week 8 (verified numerically, `check-existing-sections.R`; these match Paper 01, Methods, and `docs/31`; further landmarks in Table 3). [2026-10-08: corrected. This paragraph previously said that a large d gives "a long slow start followed by a steep rise"; only the lag statement holds (§3.3).]

Paper 06 states the offset without the rescaling:

$$C(t) = m_C \exp\big(-d_C \exp(-r_C t)\big) - m_C \exp(-d_C), \tag{3}$$

for a generic component $C \in \{BR, PB, TV\}$. Its asymptote is $m_C(1 - e^{-d_C})$, not m_C, so it differs from Equation (2) by the constant factor $1 - e^{-d_C}$ (0.9933 at $d_C = 5$), which Paper 06 itself notes.

[2026-10-08: corrected. This section previously described Equation (1) as satisfying $y(0) = 0$, called disp "the curvature near $t = 0$", and called Equations (1), (3) and (2) the same curve. Only Equation (2) is the curve the code computes. The meaning of d was stated reversed in `docs/24` and `docs/31` (corrected there on 2026-10-07; see `docs/52`), and Paper 06's Section on the biological response still states it reversed ("high d_{BR} gives an early-onset curve that rises sharply at first").]

3.2 Alternative response-curve families

Paper 07 evaluates three alternatives, each calibrated so its asymptotic maximum and its response level at a single anchor time ($t = 5$ weeks) match the Gompertz reference:

$$\text{Logistic:} \qquad y(t) = \frac{m}{1 + \exp(-r \cdot (t - t_0))} - \text{offset} \tag{4}$$

$$\text{Hyperbolic tangent:} \qquad y(t) = m \cdot \frac{\tanh(r \cdot (t - t_0)) + 1}{2} - \text{offset} \tag{5}$$

$$\text{Piecewise-linear breakpoint:} \qquad y(t) = m \cdot \min\left(1, \frac{t}{t_{\text{bp}}}\right) + \text{post_slope} \cdot \max(0,\, t - t_{\text{bp}}) \tag{6}$$

The logistic and hyperbolic-tangent curves are both symmetric around an inflection at t_0; the piecewise-linear form is the only non-differentiable member of the family, with a breakpoint at t_{bp}. All three are built-in `br_family` options of the package. Paper 07 notes that the hyperbolic-tangent form is formally a rescaled logistic: $(\tanh(x) + 1)/2 = 1/(1 + e^{-2x})$, so Equation (5) at rate r is Equation (4) at rate $2r$ (derived), apart from the offset. [Whether the two calibrated curves of Paper 07 coincide after the matched-anchor calibration was not checked here.]

3.3 Shipped parameters, landmarks, the displacement, and numerical conditioning

Parameter sets. The package dataset `extracted_rp` (`data/extracted_rp.rda`, loaded), the published calibration of Hendrickson et al., gives the drug response $m_{BR} = 10.98604$, $d_{BR} = 5$, $r_{BR} = 0.42$ per week, standard deviation 8, and gives the natural-course and placebo-belief components one common set, $m_{TV} = m_{PB} = 6.50647$, $d = 5$, $r = 0.35$ per week, standard deviation 10 (the placebo-belief mean and standard deviation are then multiplied by η_t). The equality of the TV and PB parameters is Hendrickson et al.'s tabula rasa assumption: the placebo arm of the source trial was fitted as $TV + PB$, and no data separate the two (Source: `docs/40`, Section 3; inspected). The default of the carryover-sensitivity driver (`default_resp_param()` in `analysis/scripts/carryover-sensitivity/simulation-core.R`, inspected) differs: it gives TV the drug-response curve and different standard deviations, so the two sets are not interchangeable (Table 2; Source: `docs/25`, Section 4).

Table 2: Modified-Gompertz parameters by component (rate per week). Verified by loading `data/extracted_rp.rda`; driver values inspected.

	`extracted_rp` (published)				carryover-sensitivity driver			
Component	m	d	r	SD	m	d	r	SD
BR	10.986	5	0.42	8	10.986	5	0.42	8
TV	6.506	5	0.35	10	10.986	5	0.42	6
PB	6.506	5	0.35	10	6.506	5	0.35	2

The inflection fraction. In the rescaled curve of Equation (2), write $G(t; m, d, r)$. The textbook kernel, Equation (1), has $y(t^*) = m/e$, so it inflects at $1/e \approx 0.368$ of its asymptote. Because the rescaling is affine, it does not move the inflection $t^* = \ln(d)/r$, and the share of the ceiling reached there is

$$\frac{G(t^*; m, d, r)}{m} = \frac{e^{-1} - e^{-d}}{1 - e^{-d}}, \tag{7}$$

0.364 at $d = 5$ for both shipped curves (derived; the inflection time and value confirmed by numerical differentiation, `check-gompertz-rmanova-cycles.R`, and by `check-landmarks-hybrid-ceilings.R`). For $d \leq 1$ the inflection time is not positive: the curve is concave from the origin and has no initial lag, as `docs/24` states for small d. A logistic curve inflects at half its asymptote, so at a matched asymptote and a matched inflection time the Gompertz curve has completed less of its rise (about 36% against 50%) before the inflection and more after it (derived; Source: `docs/25`, Section 6 and Section 6b, point 4).

Landmarks. The rate is a pure time scale, $G(t; m, d, kr) = G(kt; m, d, r)$, so multiplying r by k divides every landmark time by k (derived; verified at $k = 2$). Table 3 gives the landmarks of the shipped curves.

Table 3: Landmarks of the shipped modified-Gompertz curves, both with $d = 5$ (Source: `docs/40`, Section 4; `docs/31`, Section 2; `docs/25`, Section 4, for the inflection of TV and PB and the week-12 share of BR, "about 97 percent"; all values verified, `check-landmarks-hybrid-ceilings.R`, block 1, and `check-gompertz-rmanova-cycles.R`).

	Time (weeks) to			Share of m at week			
Curve	inflection	$m/2$	$0.9m$	1	4	8	12
BR ($r = 0.42$)	3.83	4.73	9.21	0.031	0.390	0.839	0.968
TV, PB ($r = 0.35$)	4.60	5.67	11.05	0.023	0.287	0.736	0.927

Half the drug-response ceiling is reached at 33 days (verified). Because time on drug restarts at zero on re-exposure, a four-week on-drug crossover period recovers 0.390 of the ceiling, and

a participant returned to drug for one week shows 0.031 of it. [The week-12 share of TV and PB, 0.927, was computed for this revision from Equation (2) (`check-reconciliations.R`); it is not stated in the sources.]

The displacement delays the rise; it does not steepen it. Differentiating Equation (2) and writing $x = de^{-rt} \in (0, d]$,

$$G'(t) = \frac{m\, r\, x\, e^{-x}}{1 - e^{-d}},$$

and xe^{-x} is maximal at $x = 1$. Hence the steepest slope on $t \geq 0$ is

$$\max_{t \geq 0} G'(t) = \begin{cases} \dfrac{m\, r}{e\,(1 - e^{-d})}, & d \geq 1 \text{ (at } t^* = \ln(d)/r), \\ \dfrac{m\, r\, d}{e^{d} - 1}, & 0 < d < 1 \text{ (at } t = 0), \end{cases} \tag{8}$$

which is continuous at $d = 1$, strictly decreasing in d, and runs from mr as $d \to 0$ to mr/e as $d \to \infty$ (derived; verified against a numerical maximum at d from 0.25 to 20, block 2). At the shipped drug-response values the steepest rise is 1.71 points per week, at week 3.83. Holding m and r fixed, a larger d therefore moves the steepest part of the rise later and makes it slightly less steep, not steeper. [`docs/40`, Section 4, states that a larger displacement gives "a longer lag and a steeper rise once it starts", and `docs/31`, Section 2, says the same; so did the first subsection of Section 3 ("a long slow start followed by a steep rise") before its correction on 2026-10-08. By Equation (8) only the lag statement is correct. Over $d \geq 1$ the maximum slope falls by at most a factor $e/(e-1) = 1.58$, so the qualitative picture of a slow start is unaffected.]

Numerical conditioning of the rescaling. The factor that restores the asymptote in Equation (2) is $m/(m - me^{-d}) = 1/(1 - e^{-d})$, which grows as $1/d$ when $d \to 0$ (100.5 at $d = 0.01$). In exact arithmetic the curve has the finite limit $m(1 - e^{-rt})$ (§3); the difficulty is floating-point cancellation, because the numerator $\exp(-de^{-rt}) - e^{-d}$ subtracts two numbers close to 1. The relative rounding error of the code's evaluation is of order ϵ_{mach}/d: at $r = 0.42$, $t = 5$ it is 6×10^{-9} at $d = 10^{-8}$, 4×10^{-6} at $d = 10^{-12}$ and 3×10^{-4} at $d = 10^{-14}$, and $d = 0$ returns `NaN`, since the code guards only $m = 0$ (verified, `check-gompertz-rmanova-cycles.R`; `R/utilities.R` inspected). The form

$$y(t) = m\, \frac{\operatorname{expm1}(-de^{-rt}) - \operatorname{expm1}(-d)}{-\operatorname{expm1}(-d)}, \qquad \operatorname{expm1}(x) = e^{x} - 1, \tag{9}$$

is algebraically identical and accurate to machine precision down to $d = 10^{-14}$ (verified). At the shipped $d = 5$ the factor is 1.0068 and the issue does not arise. (Source: `docs/25`, Sections 3 and 6.) [`docs/25` describes the rescaling factor itself as "numerically unstable as `disp` shrinks". The factor is benign in exact arithmetic; the instability is the cancellation in the numerator, and it is negligible for $d \gtrsim 10^{-6}$.]

Heterogeneity is level-only. In every current data-generating process the three parameters (m, d, r) of each component are population constants. Participants differ only through the multivariate-normal deviation of each component around its mean curve, with a constant per-visit standard deviation (scaled by η_t for PB) and the serial correlation of §9, plus the architecture's biomarker term. No participant has a different ceiling or onset rate. (`buildSigma()` and `generateData()` in `R/generateData.R`, inspected; Source: `docs/25`, Section 6, and `docs/24`, "The math behind it".)

3.4 The Hill (Emax) family as a comparator

The pharmacodynamic family that `docs/25` sets against the Gompertz curve is

$$y(t) = E_{\max}\, \frac{t^{n_H}}{EC_{50}^{n_H} + t^{n_H}}, \qquad y\big(EC_{50}\, e^{s}\big) = \frac{E_{\max}}{1 + e^{-n_H s}}, \tag{10}$$

so it is a logistic function of $\ln t$, symmetric about EC_{50} on the logarithmic time scale. On the linear scale it has, for $n_H > 1$, a single inflection at

$$t^* = EC_{50}\left(\frac{n_H - 1}{n_H + 1}\right)^{1/n_H}, \qquad \frac{y(t^*)}{E_{\max}} = \frac{n_H - 1}{2n_H}, \tag{11}$$

before EC_{50}, and no inflection for $n_H \leq 1$. Its slope at $t = 0$ is unbounded for $n_H < 1$, equal to $E_{\max}/EC_{50}$ for $n_H = 1$, and zero for $n_H > 1$. (Derived; inflection time, value and slopes verified numerically for $n_H \in \{0.5, 1, 1.5, 2, 3\}$, `check-gompertz-rmanova-cycles.R`. Source: `docs/25`, Sections 2 and 5.) [`docs/25` writes the Hill exponent as h; it is n_H here because h_t is the mean-moderation exposure profile of Equation (26). The fraction $y(t^*)/E_{\max}$ is not stated in the source; it is derived here to set beside Equation (7). The Hill family is not among the package's `br_family` options, and Paper 07 did not evaluate it.]

4 Trial designs and treatment exposure

4.1 Simple versus complex designs

The compendium compares four designs of increasing complexity, defined by their randomization-path structure (Table 1) rather than by a closed-form exposure-sequence equation. No manuscript in the compendium formalizes D_{it} as an explicit function of path or sequence index; the designs are specified operationally, as visit-by-visit exposure schedules, and D_{it} is read off that schedule.

OL (open-label). Unblinded titration; no off-drug contrast, so it supplies no within-subject interaction information.

CO (crossover). Two-period within-subject crossover, blinded throughout.

OL+BDC (blinded discontinuation). Open-label titration followed by a blinded switch to placebo.

Hybrid. The Hendrickson (2020) reference design: open-label titration, blinded discontinuation, then a brief crossover; four randomization paths.

Design *complexity*, in this compendium, is therefore a statement about P and about how many within-subject on/off contrasts a design's path structure supplies, not about a formal information functional of P. Paper 05, whose remit is this comparison, contains no equation relating P or switch count to the power of the interaction estimand $\beta_{bm:D}$; its stated target is the *main* drug effect, not the interaction. The one quantitative link Paper 05 supplies is between serial correlation and the effective information in a *between*-subject contrast, which it uses to explain why the parallel-group RCT loses power under strong serial correlation while the within-subject N-of-1 designs do not:

$$\text{effective-}N \text{ shrinkage factor} = \frac{1}{1 + (k-1)\rho}, \tag{12}$$

where k is the number of repeated observations contributing to a between-subject mean and ρ the serial correlation. Because the N-of-1 designs' primary contrast is within-subject, this shrinkage does not act on their information the way it acts on a parallel-arm comparator's.

[2026-10-08: Equation (12) is the design effect for *exchangeable* (compound-symmetric) correlation ρ. Paper 05 attributes it to AR(1). For AR(1) correlation ρ^h at lag h, the factor is $\left[1+2\sum_{h=1}^{k-1}(1-h/k)\rho^h\right]^{-1}$, which is never smaller than Equation (12) for $0 \leq \rho < 1$ because $\rho^h \leq \rho$ (derived). The exchangeable form therefore overstates the shrinkage under AR(1).]

The Hybrid paths. Table 4 gives the drug state and the time since discontinuation of the four Hybrid paths at their eight post-baseline visits, at weeks 4, 8, 9, 10, 11, 12, 16 and 20; the OL+BDC and CO paths are in Table 31.

Table 4: Hybrid paths: drug state and t_{sd} (weeks) at visits 1–8 (Source: `docs/37`, Section 2).

Path	D_t	t_{sd}	n_{on}	n_{off}
1	1 1 1 1 0 0 1 0	0 0 0 0 1 2 0 4	5	3
2	1 1 1 1 0 0 0 1	0 0 0 0 1 2 6 0	5	3
3	1 1 1 0 0 0 1 0	0 0 0 1 2 3 0 4	4	4
4	1 1 1 0 0 0 0 1	0 0 0 1 2 3 7 0	4	4

4.2 Treatment exposure: binary versus continuous

Rule (`NOTATION.md` Rule 1). D_{it} is binary; $D_{bc,it}$ is continuous. These are different objects and must never be substituted for one another:

$$D_{bc,it} = \begin{cases} 1 & D_{it} = 1 \text{ (on drug)} \\ \exp(-\lambda\, t_{sd,it}) & D_{it} = 0,\ t_{sd,it} > 0 \text{ (off drug after exposure)} \\ 0 & D_{it} = 0 \text{ before any exposure,} \end{cases} \tag{13}$$

with $t_{sd,it}$ the time since discontinuation. The third case matters for placebo-first crossover paths, where $t_{sd} = 0$ before the first exposure and the exponential would otherwise return 1 (`exposure_weight()` in `analysis/scripts/carryover-sensitivity/simulation-core.R`, inspected; Paper 01, Appendix B.4). In the limit of an assumed zero half-life, the continuous predictor collapses onto the binary one, $D_{bc,it} \to D_{it}$ as $\hat{t}_{1/2} \to 0$ (Paper 02); in this sense the "Unadjusted" analysis specification is the boundary member of the continuous-exposure family.

[2026-10-08: the before-exposure case was added; the earlier version of Equation (13) had only two cases. `docs/52` (open items) records that it is not established whether Paper 02's crossover cells were rerun with the corrected exposure coding.]

Pre-exposure visits. As noted above, a decay weight evaluated at $t_{sd} = 0$ equals 1, so a rule $D_{bc} = 1$ on drug and $2^{-t_{sd}/t^a_{1/2}}$ off drug codes off-drug visits before first exposure as fully exposed. The correct coding is Equation (13), with $\lambda = \ln 2/t^a_{1/2}$ at the half-life $t^a_{1/2}$ that the analysis assumes (Source: `docs/45`, Section 8; compare the note after Equation (36)). In CO the placebo-first path is off drug for its first four visits; under the defective coding its indicator is 1 at every visit, $B_i D_{bc,it}$ is collinear with B_i within each such participant, and half the participants contribute no within-participant contrast. The information for $\beta_{bm:D}$ then roughly halves and its standard deviation rises by about $\sqrt{2} = 1.41$ (derived, approximate: those participants still inform the other coefficients); the source reports 1.44 to 1.47 and a power loss of 0.26 to 0.30 (quoted). At the first CO off-drug visit, 2.5 weeks after discontinuation, $D_{bc} = 0.031$ at $t^a_{1/2} = 0.5$ and 0.420 at $t^a_{1/2} = 2$ (derived).

4.3 Time arguments of the components, the shared clock, and the phase contrasts

Clocks. Let visit t follow the previous visit by Δw_t weeks (Δw_1 is the first visit time). The design builder (`buildtrialdesign()` in `R/buildtrialdesign.R`, inspected) defines three running clocks by recursion:

$$\begin{aligned} t_{od,t} &= D_t\,(t_{od,t-1} + \Delta w_t), \\ t_{sd,t} &= (1 - D_t)\,(t_{sd,t-1} + \Delta w_t)\,\mathbf{1}\Big[\textstyle\sum_{s \le t} D_s > 0\Big], \\ t_{pb,t} &= \mathbf{1}[\eta_t > 0]\,(t_{pb,t-1} + \Delta w_t), \end{aligned} \tag{14}$$

with all three zero before the first visit. Thus t_{od} is the time since the current exposure began and restarts at re-exposure; t_{sd} is the time since the last on-drug visit, and zero on drug and before

any exposure (the third case of Equation (13)); and t_{pb} accumulates over consecutive visits with positive expectancy and resets to zero at a visit with $\eta_t = 0$. With G_C the modified-Gompertz curve of Equation (2) with component C's parameters, the component means are

$$\begin{aligned} E[BR_{it}] &= m_{BR}\, G_{BR}(t_{od,t}) \quad \text{on drug,} \\ E[PB_{it}] &= \eta_t\, m_{PB}\, G_{PB}(t_{pb,t}), \\ E[TV_{it}] &= m_{TV}\, G_{TV}(w_t), \end{aligned} \tag{15}$$

with w_t calendar time, $\mathrm{SD}(PB_{it}) = \eta_t \sigma_{PB}$, and the off-drug BR mean given by anchored carryover (Equation (22)). m_{TV} may be negative (natural worsening). (Source: `docs/24`, Components 1 to 3 and "The math behind it"; `docs/25`, Section 4; the code inspected.) [`docs/24` describes t_{pb} as "time accumulated while the expectancy is positive"; the code resets it to zero at every visit with $\eta_t = 0$, which matters only for designs with covert visits, such as the balanced-placebo design of §12.4.]

The shared clock. In every design in which $\eta_t > 0$ at every visit (OL, OL+BDC, CO and Hybrid as coded), $t_{pb,t} = w_t$, so PB and TV are functions of one clock and

$$E[PB_{it} + TV_{it}] = \eta_t\, m_{PB}\, G_{PB}(w_t) + m_{TV}\, G_{TV}(w_t), \tag{16}$$

which under the published parameters, where $G_{PB} = G_{TV}$ (Table 2), is $(\eta_t m_{PB} + m_{TV})\, G(w_t)$. The two components are then distinguished only through the change of η_t (at week 9 in the Hybrid design) and through their assumed curve shapes (derived; Source: `docs/24`, "How the trial design lets us identify the components"). Every off-drug visit of these designs is blinded, so TV is never observed without PB.

Components by phase. Table 5 gives the mean structure of each Hybrid phase. Because η_t and the PB and TV means depend on the visit and not on the path, the on-drug minus off-drug difference between paths at a common blinded visit is $E[BR]$ on drug minus its carryover residual, with PB and TV canceling exactly (derived). The open-label to blinded change within drug, from week 8 to week 9, is $m_{BR}[G_{BR}(9) - G_{BR}(8)] + m_{PB}[0.5\, G_{PB}(9) - G_{PB}(8)] + m_{TV}[G_{TV}(9) - G_{TV}(8)]$, which approximates $-0.5\, E[PB]$ only to the extent that the curves change little over that week (derived; `docs/24` states the qualification, `docs/28`, Section 3.2, omits it).

Table 5: Mean components present by Hybrid phase (Source: `docs/24`, `docs/28`, Section 3.1; expectancy vector $(1, 1, 0.5, \ldots, 0.5)$ inspected).

Phase (weeks)	Drug	η_t	Mean reduction $E[BR + PB + TV]$
Open label (4, 8)	on	1	$m_{BR}G_{BR}(t_{od}) + m_{PB}G_{PB}(w_t) + m_{TV}G_{TV}(w_t)$
Blinded discontinuation (9–12)	by path	0.5	$E[BR_{it}] + 0.5\, m_{PB}G_{PB}(w_t) + m_{TV}G_{TV}(w_t)$
Blinded crossover (16, 20)	by path	0.5	$E[BR_{it}] + 0.5\, m_{PB}G_{PB}(w_t) + m_{TV}G_{TV}(w_t)$

[`docs/28` (superseded by `docs/31`) states that the lumped slope is unbiased "if and only if" the biomarker is orthogonal to PB and TV. By Equation (98) the condition is $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV} = 0$, which orthogonality implies but which a cancellation could also satisfy; the statement in §12 is the correct one.]

5 Carryover on the mean

This section collects the carryover decay families and the rules by which carryover acts on the off-drug mean of the drug response. Its action on the biomarker-response correlation is treated with the covariance encoding (§6.3).

5.1 Carryover decay families

Four decay forms are implemented (`carryover_decay()` in `implementations/nof1power/R/carryover_models.R`, inspected); the first three satisfy $\phi(t_{1/2}) = 0.5$:

$$\text{Exponential:} \quad \phi_{\text{exp}}(t_{sd}) = \exp(-\lambda\, t_{sd}), \qquad \lambda = \frac{\ln 2}{t_{1/2}} \tag{17}$$

$$\text{Linear:} \quad \phi_{\text{lin}}(t_{sd}) = \max\left(0,\, 1 - \frac{t_{sd}}{2\, t_{1/2}}\right) \tag{18}$$

$$\text{Weibull:} \quad \phi_{\text{weib}}(t_{sd}) = \exp\big(-(\lambda_w t_{sd})^k\big), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}} \tag{19}$$

$$\text{Power:} \quad \phi_{\text{pow}}(t_{sd}) = \max\left(0,\, \left(1 - \frac{t_{sd}}{3\, t_{1/2}}\right)^p\right) \tag{20}$$

[2026-10-08: corrected. The section previously stated that all four forms satisfy $\phi(t_{1/2}) = 0.5$, as the function's documentation also does. The power form gives $\phi(t_{1/2}) = (2/3)^p$, which equals 0.5 only at $p = \ln 2 / \ln 1.5 \approx 1.71$; at the default $p = 1.5$ it is 0.544 (derived), so its nominal half-life is not the time at which half the effect remains.]

The Weibull family recovers the exponential exactly at shape $k = 1$; a value $k < 1$ produces a heavier elimination tail than exponential, and $k > 1$ a lighter one. Paper 02, whose remit is this decay family, tests $k \in \{0.25, 0.5, 2.0, 4.0\}$ against the exponential default. Its ranking of the specifications is largely unchanged, but the advantage of the exposure-weighted specification over the unadjusted one, where it leads (Hybrid and OL+BDC), widens under the light-tailed shapes and narrows to statistical insignificance under the two heavy-tailed shapes $k \in \{0.25, 0.5\}$ (Paper 02, Abstract).

[2026-10-08: corrected. This paragraph previously said Paper 02 finds the conclusions "materially unchanged" across Weibull shapes; Paper 02 reports the narrowing just described.]

5.2 Anchored versus compounding carryover on the mean

Paper 05 states a carryover *recursion* for the biological-response mean during off-drug periods,

$$\mu[t] = \text{base}[t] + \mu[t-1] \cdot \phi(t_{sd}), \tag{21}$$

and the published process of Hendrickson et al. uses the same recursion, $\mu_{BR,t} = \mu_{BR,t-1}\, 2^{-t_{sd,t}/t_{1/2}}$ (Paper 01, Methods). Because t_{sd} is the *cumulative* time since discontinuation, the recursion counts elapsed time twice: at the j-th consecutive off-drug visit it applies $\prod_{l \le j} \phi(t_{sd,l})$ instead of $\phi(t_{sd,j})$, and so over-decays every off-drug visit after the first in a run (derived; Paper 01, Appendix B.3). The package now uses *anchored* carryover, decaying from the last on-drug value:

$$\mu_{BR,t} = \text{base}_t + \mu_{BR,\text{last}}\, \phi(t_{sd,t}), \qquad \phi(t_{sd}) = 2^{-t_{sd}/t_{1/2}}\ \text{(exponential)}, \tag{22}$$

with $\mu_{BR,\text{last}}$ the drug-response mean at the most recent on-drug visit (`R/generateData.R`, inspected; Paper 01, Methods, refinement step 3).

[2026-10-08: superseded in part. Equation (21) was previously presented as the carryover mechanism without qualification. It is retained as Paper 05's statement; the current mechanism is Equation (22) (Paper 01 as restated, `docs/51`). Whether the stored results of Paper 05 were generated under the recursion or under anchored carryover was not checked.]

With the exponential decay of the published recursion, the k-th consecutive off-drug visit after a last on-drug mean μ_{last} has

$$\mu_{BR}^{\text{comp}} = \mu_{\text{last}}\, 2^{-\sum_{j=1}^{k} t_{sd,j}/t_{1/2}} \quad \text{against the anchored} \quad \mu_{BR}^{\text{anch}} = \mu_{\text{last}}\, 2^{-t_{sd,k}/t_{1/2}}: \tag{23}$$

the decay compounds from the second off-drug visit and runs faster than the stated half-life (Paper 01, Appendix B.3; `docs/40`, Section 3; derived). On path 1 at $t_{1/2} = 1$ with $\mu_{\text{last}} = 10.19$, both give 5.09 at $t_{sd} = 1$; at $t_{sd} = 2$ the recursion gives 1.27 and the anchored form 2.55 (verified). Paper 01's refinement 3 adopts the anchored form.

[Equation (21) as Paper 05 writes it leaves unstated whether ϕ takes the cumulative t_{sd} or the one-visit increment; with the increment it would not compound. The published code takes the cumulative value, so it compounds as stated above (inspected: `R/buildtrialdesign.R` and `R/generateData.R`, line 88, at commit 58b32a9; `docs/53`, Sections 4 and 5). A paired simulation in `docs/53` (Section 8.3; Hybrid, $N = 70$, $c_{bm} = 0.25$, 500 replicates) found that replacing the recursion by the anchored form changes the interaction estimate by at most 0.005 and at most 4 of 500 test decisions at $t_{1/2} \leq 2$ weeks. With the restart of the Gompertz clock at re-exposure it produces a downward step in the mean at the first re-exposure visit when the residual exceeds the restarted curve (Paper 01, Appendix B.3).]

5.3 The 2024 revision of the published code: a scale factor on the carryover exponent

The vendored copy of the Hendrickson et al. code that served as Paper 01's earlier comparison arm is upstream HEAD `06dac83` (2024-11-27), not the published commit `58b32a9` (`docs/54`, review note and Section 2). Its recursion multiplies the carryover exponent by a scale factor s_f (code `scalefactor`, default 2):

$$\mu_{BR,t} = \mu^{\text{raw}}_{BR,t} + \mu_{BR,t-1}\, 2^{-s_f\, t_{sd,t}/t_{1/2}} \qquad \text{off drug after exposure,} \tag{24}$$

with $\mu^{\text{raw}}_{BR,t} = 0$ off drug. At the k-th consecutive off-drug visit after a last on-drug mean μ_{last}, with cumulative times $t_{sd,1} < \cdots < t_{sd,k}$,

$$\mu^{(s_f)}_{BR} = \mu_{\text{last}}\, 2^{-s_f \sum_{j\leq k} t_{sd,j}/t_{1/2}}, \qquad \frac{\mu^{(s_f)}_{BR}}{\mu^{\text{anch}}_{BR}} = 2^{-\left(s_f \sum_{j\leq k} t_{sd,j} - t_{sd,k}\right)/t_{1/2}}, \tag{25}$$

which is Equation (23) at $s_f = 1$ (derived). With $s_f = 1$ the first off-drug visit of a run is exact and later ones compound; with $s_f = 2$ even the first is understated, by the factor $2^{-t_{sd,1}/t_{1/2}}$. Table 6 gives the values on Hybrid path 1 (Source: `docs/54`, Section 3.5, as corrected 2026-10-07; every entry reproduced, `check-stress-test-construct.R`, Section 3).

Table 6: Drug-response mean at the off-drug visits of Hybrid path 1, $t_{1/2} = 1$ week, under the recursion with $s_f = 1$ and $s_f = 2$ and under anchored decay (Source: `docs/54`, Section 3.5; reproduced).

Visit	t_{sd}	$s_f = 1$	$s_f = 2$	Anchored	Ratio, $s_f = 1$	Ratio, $s_f = 2$
BD3 (week 11)	1	5.094	2.547	5.094	1.000	0.500
BD4 (week 12)	2	1.273	0.159	2.547	0.500	0.0625
COp (week 20)	4	0.268	0.017	0.268	1.000	0.0625

The last on-drug means are 10.187 (week 10) for BD3 and BD4, and 4.281 (week 16, after the Gompertz clock restarts) for COp. The scale factor acts on the mean channel and does not enter the correlation matrix; its counterpart in the correlation matrix is the ratio-graded coupling of §6.6. [The whitepaper's general statement that "at $s_f = 2$ even the first occasion is wrong (0.500x)" holds at $t_{sd} = t_{1/2}$; in general the first-visit ratio is $2^{-(s_f-1)t_{sd,1}/t_{1/2}}$, 0.0625 at COp, which is the first off-drug visit of its run.]

6 Two architectures for a biomarker-treatment interaction

Paper 01 states the two ways in which the target estimand, $\beta_{bm:D}$, is encoded into the data-generating process of the compendium. The version of Paper 01 this document first drew on

(the pre-restatement version, now the file `report-2026-10-07-pre-restatement.Rmd` in the `archive/` folder of the Paper 01 directory) presented them as structurally different encodings whose power diverges under carryover. The restated Paper 01 (October 2026) shows that, conditional on the biomarker, the covariance encoding is mean moderation with a coupling-shaped exposure profile and a reduced residual variance (Equation (30) below), so that the size of the carryover loss is set by the exposure profile of the interaction, not by the channel that carries it.

6.1 Mean moderation

In the mean-moderation architecture (formerly Architecture A) the biomarker shifts the drug response in the conditional mean. The package (`dgp_architecture = 'mean_moderation'`, default `moderation_scaling = 'constant'`; `R/generateData.R`, inspected) and the restated Paper 01 (Methods) apply the shift

$$BR_{it} \;\leftarrow\; BR_{it} + \beta_{bm}\, b_i\, \sigma_{BR}\, h_t, \qquad h_t = \begin{cases} 1 & D_{it} = 1 \\ 0 \text{ (plain) or } 2^{-t_{sd,t}/t_{1/2}} \text{ (decayed)} & D_{it} = 0, \end{cases} \tag{26}$$

after the multivariate-normal draw, with b_i the standardized biomarker. The package implements the plain profile; the decayed profile is Paper 01's comparison arm. Because $Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$, the implied on-drug interaction coefficient on the raw biomarker is

$$\beta_{bm:D} = -\beta_{bm}\, \frac{\sigma_{BR}}{\sigma_{bm}} \tag{27}$$

(Paper 01, Estimand), the coefficient of $D_{it}B_i$ in the standard additive effect-modification form $Y_{it} = \beta_0 + \beta_D D_{it} + \beta_B B_i + \beta_{bm:D} D_{it} B_i + \ldots$ of Kent et al. The contrast between two participants at a common exposure state is

$$E[Y_{it} \mid b_1] - E[Y_{it} \mid b_2] = -\beta_{bm}\, \sigma_{BR}\, h_t\, (b_1 - b_2) + (\text{terms not involving } h_t), \tag{28}$$

so carryover enters the moderation contrast only through the profile h_t. Under the plain profile the moderation stops when the drug stops, and power is flat in $t_{1/2}$ (0.681 at every half-life for the paired-difference statistic E9, Hybrid, $N = 70$, $\beta_{bm} = 0.25$, compound symmetry; Paper 01, Results). Paper 01 calls this flatness a coding choice rather than a finding, because the drug effect itself carries over.

[2026-10-08: superseded. This subsection previously stated mean moderation as $Y_{it} = \mu_0 + \beta_1 D_{it}(1 + \beta_{bm} B_i) + \varepsilon_{it}$ with interaction $\beta_3 = \beta_1 \beta_{bm}$, the multiplicative form of the pre-restatement Paper 01. That form survives only as the trajectory-scaled option of Paper 07 (Equation (37)); it is not the package default, and it used the raw biomarker B_i where the implemented shift uses b_i.]

6.2 Covariance moderation

In the covariance-moderation architecture (formerly Architecture B) the biomarker and the response components, over all visits, are drawn jointly from one multivariate normal whose biomarker-response correlation, not whose mean, depends on exposure state:

$$\begin{pmatrix} B_i \\ BR_{i\cdot} \end{pmatrix} \sim \mathrm{MVN}\left(\begin{pmatrix} \mu_{bm} \\ \mu_{BR,\cdot} \end{pmatrix}, \Sigma \right), \qquad \mathrm{Cor}(B_i, BR_{it}) = c_{bm}\, g_t, \tag{29}$$

with the graded coupling g_t of Equation (31): 1 on drug, $e^{-\lambda t_{sd}} = 2^{-t_{sd}/t_{1/2}}$ off drug after exposure, 0 before any exposure. Without carryover the estimand is $\beta_{bm:D} = -c_{bm}\, \sigma_{BR}/\sigma_{bm}$ (Paper 01, Estimand). Conditioning on the biomarker (Paper 01, Discussion "Covariance-based and mean-based binding"), with $b_i = (B_i - \mu_{bm})/\sigma_{bm}$ standardized by the population moments,

$$E(BR_{it} \mid B_i) = \mu_{BR,t} + c_{bm}\, g_t\, \sigma_{BR}\, b_i, \qquad \mathrm{Cov}(BR_{i\cdot} \mid B_i) = \Sigma_{BR} - c_{bm}^2 \sigma_{BR}^2\, g g^\top, \tag{30}$$

so the covariance encoding is mean moderation (Equation (26)) with exposure profile $h_t = g_t$ and strength $\beta_{bm} = c_{bm}$, and a drug-response covariance reduced by a rank-one term. Here Σ_{BR} is the drug-response covariance block, g the vector of the g_t ($g = (g_1, \ldots, g_8)$ in the Hybrid design), and $\mathrm{Var}(BR_{it}) = \sigma^2_{BR}$ at each visit (derived: re-checked here from the conditional-normal formulas, since $\mathrm{Cov}(BR_{it}, B_i) = c_{bm} g_t \sigma_{BR} \sigma_{bm}$; verified to machine precision on Hybrid path 1 under graded coupling at $t_{1/2} = 1$ week, `check-coupling-closed-form.R`). The TV and PB blocks and their covariances with BR are unchanged, since neither covaries with B. The consequences for the slope, the residual variance and the power of the interaction test, and for the positive-definiteness ceiling, are derived in §14.6 (Paper 01, Discussion).

Under the published step rule ($g_t = 1$ wherever the drug-response mean is nonzero), the coupling gap $1 - \bar{g}_{\text{off}}$ is zero at every positive half-life, and closed-form E9 power falls from 0.721 to 0.050, the test size (§6.3 and §14.3; Paper 01, Results).

[2026-10-08: superseded. This subsection previously gave the off-drug correlation as $c_{bm} e^{-\lambda t_{sd}}$ at every off-drug visit (no zero before exposure, no distinction from the published step rule) and reported, as Paper 01's headline, Hybrid carryover losses of "1.4–6.9 percentage points under mean moderation but 5.1–15.4 points under covariance moderation" with $z = 2.80$ (Hybrid) and $z = 7.64$ (OL+BDC). Two corrections apply. First, those figures were misquoted: `NOTATION.md` (Part 2) reports that matching every design to $N = 70$ moved the Hybrid relative losses from 1.4% to 6.9% (mean moderation) and from 5.1% to 15.4% (covariance moderation); they are before-and-after values of relative percent loss, not ranges of percentage points. Second, the comparison itself is superseded: the restated Paper 01 shows that a larger loss under covariance binding than under plain mean binding compares a fading interaction with a non-fading one. Which coupling rule and carryover recursion the pre-restatement runs used was not checked here. The z statistics are kept only as a record of the earlier version.]

6.3 Covariance moderation with carryover

Under the covariance-moderation architecture (Equation (29)), carryover acts on the biomarker-response correlation itself, not only on the mean, through the coupling profile g_t:

$$\mathrm{Cor}(B_i, BR_{it}) = c_{bm}\, g_t, \qquad g_t = \begin{cases} 1 & D_{it} = 1 \\ \phi(t_{sd,it}) & D_{it} = 0 \text{ after first exposure} \\ 0 & \text{before first exposure,} \end{cases} \tag{31}$$

with ϕ any of Equations (17)–(20) (Paper 02). With exponential decay the off-drug coupling is $2^{-t_{sd}/t_{1/2}} = e^{-\lambda t_{sd}}$. This is the *graded* coupling of Paper 01 (as restated) and `docs/37`, and the rule the package implements (`buildSigma()` in `R/generateData.R`, inspected).

The process published by Hendrickson et al. used a *step* coupling instead: $g_t = 1$ wherever the drug-response mean is nonzero and 0 elsewhere (Paper 01, Methods). Under carryover every off-drug visit that follows an exposure has a positive drug-response mean, so it keeps the full on-drug correlation c_{bm}. The biomarker then moderates the on-drug and off-drug responses equally, and the on-minus-off interaction vanishes at *any* positive half-life, however small (Paper 01, Results; `docs/37`). Graded coupling restores a decline that is smooth and, for the paired-difference statistic, monotone in $t_{1/2}$, and that vanishes faster than any power of $t_{1/2}$ as $t_{1/2} \to 0$ (Paper 01, Appendix A.3).

The conditional-mean reading of this construction, which connects it to the mean-moderation architecture, is the conditional mean of Equation (30), which in the raw biomarker reads $E[BR_{it} \mid B_i] = \mu_{BR}(t) + c_{bm}\, g_t\, (\sigma_{BR}/\sigma_{bm})\, (B_i - \mu_{bm})$ (Paper 01, Discussion), with b_i computed from the population mean and SD of the biomarker.

[2026-10-08: corrected. The section previously gave only the decayed correlation $c_{bm}\phi(t_{sd})$ for every off-drug visit, without the zero before first exposure, and did not distinguish the published step rule from graded coupling. The conditional mean was previously given for $D_{it} = 1$ only, with the off-drug case described in words.]

6.4 The estimand and the coupling rules

The biomarker is correlated with the drug response only, $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} g_t$ with $0 \le g_t \le 1$ and $g_0 = 0$ at baseline; g_t is the *coupling* (`docs/36` writes $r_t = c_{bm} g_t$). The estimand is the on-drug moderation of the drug response, which without carryover is

$$\beta_{bm:D} = -c_{bm}\,\sigma_{BR}/\sigma_{bm} \text{ (covariance binding)}, \qquad \beta_{bm:D} = -\beta_{bm}\,\sigma_{BR}/\sigma_{bm} \text{ (mean binding)}, \tag{32}$$

both -0.1302 at the matched value 0.25 with $\sigma_{BR} = 8$ and $\sigma_{bm} = 15.36$, the setting of §14 (Paper 01, Methods "Estimand"; derived). An analysis built on the binary D_{it} estimates this only when no moderation remains off drug; under carryover it estimates an attenuated on-minus-off contrast, made precise by Equation (107). With $\phi_t = 2^{-t_{sd,t}/t_{1/2}} = \phi_{\exp}(t_{sd,t})$ (Equation (17)) at an off-drug visit after exposure, and $\phi_t = 0$ before any exposure or when $t_{1/2} = 0$, four rules set the off-drug interaction (Source: `docs/37`, Section 2; Paper 01, Table 1):

$$\text{Step (published):} \quad g_t = \mathbf{1}[\mu_{BR,t} \neq 0]; \tag{33}$$

$$\text{Graded:} \quad g_t = 1 \text{ on drug, } \phi_t \text{ off drug}; \tag{34}$$

$$\text{Mean moderation:} \quad g_t = 0;\ BR_{it} \text{ shifted by } \beta_{bm}\sigma_{BR} b_i h_t,\ h_t = D_{it}; \tag{35}$$

$$\text{Decayed mean moderation:} \quad g_t = 0;\ h_t = 1 \text{ on drug, } \phi_t \text{ off drug}, \tag{36}$$

with $\mu_{BR,t}$ the mean drug response and b_i the standardized biomarker. Off-drug visits before first exposure carry coupling and moderation 0 under every rule.

[The pre-exposure zero is a condition, not a detail: a rule "coupling equals the decay weight" with $e^{-\lambda t_{sd}} = 1$ at $t_{sd} = 0$ gives full coupling at baseline and at the placebo-first crossover visits. Equations (13) and (31) state the zero; `docs/37`, Appendix A, notes its absence from Paper 02's indicator.]

6.5 How trajectory shape interacts with carryover and the two architectures

Under the *mean-moderation* architecture Paper 07 uses the trajectory-scaled form, which it writes schematically as

$$Y_{it} = \mu_0 + \beta_D D_{it}\,(1 + \beta_{bm} B_i) + \varepsilon_{it}, \tag{37}$$

so that the biomarker-dependent component of the treatment effect rides the trajectory. The package implements it (`moderation_scaling = 'trajectory'` in `R/generateData.R`, inspected) as an on-drug shift of BR_{it} by

$$\beta_{bm}\, b_i\, \sigma_{BR}\, \frac{\mu_{BR}(t)}{\bar{\mu}_{BR}^{\,\mathrm{on}}}, \tag{38}$$

where $\mu_{BR}(t)$ is the drug-response mean curve and $\bar{\mu}_{BR}^{\,\mathrm{on}}$ its mean over on-drug visits, so that the mean on-drug magnitude of the shift per unit b_i equals σ_{BR}. A family with a slower early rise therefore places less of the interaction signal on the earliest on-drug visits, while the marginal interaction effect size is held fixed. Equation (37) uses the raw biomarker B_i; the implemented shift uses the standardized b_i and is additive in BR, not a multiple of a scalar treatment effect β_D. [2026-10-09: notation. Paper 07 writes the treatment effect as β_1; it is the canonical treatment main effect β_D here.]

Under the *covariance-moderation* architecture, by contrast, the interaction lives in the biomarker-response correlation $\mathrm{Cor}(B_i, BR_{it}) = c_{bm}\, g_t$ of Equation (31), whose coupling profile g_t is set by exposure state and, off drug, by t_{sd} and $t_{1/2}$, not by the mean curve BR follows. The biomarker-response block of the correlation matrix is constructed directly, so trajectory-shape misspecification cannot move the covariance-channel interaction test; Paper 07 confirms this with a matched-seed check. This is the structural reason Paper 07's 16-cell factorial (four families × two architectures × two biomarker-effect levels, $N = 70$, OL+BDC design, $t_{1/2} = 0$

to isolate the trajectory effect from carryover) finds power differences of only a few percentage points across families under both architectures: the mean channel is mildly shape-sensitive, and the covariance channel is not shape-sensitive by construction. Paper 07 is explicit that the grid fixes $t_{1/2} = 0$; at $t_{1/2} > 0$ the off-drug decay would interact with each family's own profile, a combination that paper does not evaluate.

[2026-10-08: the correlation $c_{bm} g_t$ replaces a derivative $\partial \,\mathrm{Cor}(BM, BR)/\partial D_{bc}$ that had no defined meaning. The independence from the trajectory holds for the graded coupling now in the package. Under the published step rule (§6.3) the coupling was switched on wherever the drug-response mean was nonzero, so at $t_{1/2} > 0$ it depended on the mean curve through its support; at $t_{1/2} = 0$, the case Paper 07 evaluates, the two rules agree.]

6.6 The 2024 revision of the published code: ratio-graded coupling and an uncoupled first visit

Ratio-graded coupling. In the 2024 code (§5.3) an off-drug visit after exposure receives the coupling $c_{bm}\, \mu_{BR,t}/\mu_{BR,t-1}$, the ratio of consecutive adjusted means of Equation (24). The history cancels, so

$$g_t = \begin{cases} 1, & D_{it} = 1,\ t \geq 2, \\ 2^{-s_f\, t_{sd,t}/t_{1/2}}, & D_{it} = 0 \text{ after exposure}, \\ 0, & \text{before first exposure, or } t = 1, \end{cases} \tag{39}$$

an exponential decay at half-life $t_{1/2}/s_f$ that does not compound across consecutive off-drug visits, although the mean does (derived; `docs/54`, Section 3.3, as corrected). The first visit is never assigned ($g_1 = 0$), because the loop that forms the ratio starts at the second visit (Section 3.4 of the source; inspected in the vendored file `vendored-hendrickson-generateData.R`). On Hybrid path 1 at $t_{1/2} = 1$ week and $c_{bm} = 0.45$, the off-drug couplings are $0.45 \cdot 2^{-2} = 0.1125$, $0.45 \cdot 2^{-4} = 0.0281$ and $0.45 \cdot 2^{-8} = 0.0018$ at $t_{sd} = 1, 2, 4$, against the graded $0.45e^{-\ln 2} = 0.225$ at $t_{sd} = 1$ (verified, `check-stress-test-construct.R`, Section 3). At $t_{1/2} = 0$ the off-drug mean is zero, so Equation (39), the step rule and graded coupling coincide off drug.

6.7 The pharmacokinetic argument for graded coupling, and the historical coupling rules

The spread argument. On drug, joint normality gives $E(BR_{it} \mid b_i = \pm 1) = \mu_{BR,t} \pm c_{bm}\sigma_{BR}$, a spread $\Delta_{\text{on}} = 2c_{bm}\sigma_{BR}$ between participants one standard deviation above and below the biomarker mean. If carryover is pharmacokinetic, every participant's drug effect decays by the same factor $\phi_t = 2^{-t_{sd,t}/t_{1/2}}$, so the off-drug conditional means are $\phi_t(\mu_{BR,t_0} \pm c_{bm}\sigma_{BR})$, with t_0 the last on-drug visit, and the spread is $\Delta_{\text{off}} = \phi_t \Delta_{\text{on}}$. Matching it to the conditional-normal spread $2\, c_{bm} g_t\, \sigma_{BR}$, which holds when $\mathrm{Var}(BR_{it}) = \sigma^2_{BR}$ at every visit, gives

$$g_t = \phi_t = 2^{-t_{sd,t}/t_{1/2}} = e^{-\lambda t_{sd,t}}, \qquad \lambda = \frac{\ln 2}{t_{1/2}}, \tag{40}$$

the graded coupling of Equation (31), with the coupling decay rate equal to the elimination rate. (Derived. Source: `docs/23`, Sections 5.2–5.4, from `docs/08`.) The premise, a common decay of each participant's conditional mean, is decayed mean moderation, so the argument is the conditional-mean identity of Equation (30) read in reverse: it fixes the conditional mean and says nothing about the conditional variance. The construct keeps $\mathrm{Var}(BR_{it} \mid B_i) = \sigma^2_{BR}(1 - c^2_{bm} g^2_t)$, nearly σ^2_{BR} off drug, whereas a decay of the whole drug-dependent deviation would shrink it by ϕ^2_t (`docs/23`, Section 5.8; argued). [`docs/23` writes the coupled correlation $r_t = c_{bm} g_t$ and calls ϕ_t the decay factor, as `docs/37` does.]

Table 7 evaluates the rules at the first off-drug visit of Hybrid path 1 at $t_{1/2} = 1$ week ($t_{sd} = 1$, with $\mu_{BR} = 10.19$ at the last on-drug visit and 5.09 off drug, the values of §5.2), $c_{bm} = 0.5$, $\sigma_{BR} = 8$.

Table 7: Conditional mean of BR at $b = \pm 1$ under alternative off-drug coupling rules, Hybrid path 1, $t_{1/2} = 1$, $c_{bm} = 0.5$ (Source: `docs/23`, Sections 5.5–5.6; verified, `check-historical-results.R`, block G).

Visit, rule	$c_{bm} g_t$	$b = +1$	$b = -1$	Spread / mean
On drug (week 10)	0.50	14.19	6.19	0.785
Off drug, graded, $\lambda = \ln 2 / t_{1/2}$	0.25	7.09	3.09	0.785
Off drug, zero coupling	0	5.09	5.09	0
Off drug, slow decay, $\lambda = 1$	0.184	6.56	3.62	0.578
Off drug, full coupling (published step rule)	0.50	9.09	1.09	1.57

Only graded coupling keeps the ratio of the biomarker spread to the mean response at its on-drug value (verified). At the published $t_{1/2} = 0.1$ week the drug-response mean one week after discontinuation is $10.19 \cdot 2^{-10} = 0.010$, about 0.1 percent of its last on-drug value; the step rule keeps $c_{bm} = 0.6$ there, so the conditional means at $b = \pm 1$ (biomarker 139.7 and 109.0) are $0.010 \pm 0.6 \cdot 8 = +4.81$ and -4.79, a biomarker-specific spread around a near-zero mean, against 0.010 ± 0.005 under graded coupling (derived; verified, block G; Source: `docs/23`, Section 5.7). [`docs/23` quotes ±4.8 with mean +0.01. Its account of the consequence for power (added noise inflating the standard error) is superseded by the exact one of §14.3: the coupling gap is zero and the slope vanishes. Its quoted publication figure "0.95 to 0.18" was not re-verified; the figure verified in `docs/36` is 0.74 to 0.12 (Hybrid, $N = 70$, $c_{bm} = 0.3$).]

The proportional rule. An intermediate version of the published code (commit `8609f12`) set the off-drug coupling to the ratio of successive drug-response means, $c_{bm} g_t = c_{bm}\, \mu_{BR,t} / \mu_{BR,t-1}$ (`docs/23`, Section 5.1), and introduced a factor 2 in the decay exponent, $2^{-2t_{sd}/t_{1/2}}$, which halves the effective half-life (`docs/23`, Section 2.4). The same rule, with the factor written as a scale factor $s_f = 2$, is in the 2024 upstream code, where it is derived in §6.6 (Equation (39)): with no drug-response base off drug the ratio is $2^{-s_f t_{sd,t}/t_{1/2}}$ at every off-drug visit of a run, so the rule is graded coupling at half-life $t_{1/2}/s_f$ (derived; also checked on a three-visit run, `check-historical-results.R`, block L). [Derived from the formulas as `docs/23` states them; commit `8609f12` was not inspected, and whether its off-drug mean had a zero base was not checked. The ratio is undefined where $\mu_{BR,t-1} = 0$, at the first visit, which is consistent with the `if (p > 1)` guard that `docs/23`, Section 2.5, describes (argued) and with the uncoupled first visit of the 2024 code (§6.6).]

6.8 Historical mean-moderation formulations and the calibration check

`docs/19` records three formulations of mean moderation that preceded Equation (26). Each is a post-draw shift of BR_{it}, and each has a conditional-mean reading in terms of the profile h_t of Equation (26).

(i) **Multiplicative scaling,** $BR_{it} \leftarrow BR_{it}(1 + \beta_{bm} b_i)$. The shift is $\beta_{bm} b_i BR_{it}$, with $E(\beta_{bm} b_i BR_{it} \mid b_i) = \beta_{bm} \sigma_{BR} b_i\, (\mu_{BR,t}/\sigma_{BR})$ when BR is drawn independently of b_i: mean moderation with profile $h_t = \mu_{BR,t}/\sigma_{BR}$, which tracks the drug-response mean and so decays off drug with the carryover, plus a biomarker-by-noise term $\beta_{bm} b_i (BR_{it} - \mu_{BR,t})$ (derived). It is a trajectory-scaled shift (Equation (38)) without the normalization. A profile that decays off drug loses power with the half-life, as decayed mean moderation does (§14.6, Table 25; argued for this profile, which was not evaluated in closed form), which is consistent with the source's report of a loss of about 60 percent under carryover (50 replicates; quoted).

(ii) **Additive with exposure decay,** $BR_{it} \leftarrow BR_{it} + \beta_{bm} b_i \sigma_{BR} h_t$ with $h_t = 2^{-t_{sd}/t_{1/2}}$ off drug: decayed mean moderation, Equation (36). On the crossover schedule (2.5-week spacing, $t_{1/2} = 1$) the off-drug profile is 0.177, 0.031, 0.0055 and 0.0010, and the per-SD shift at $\beta_{bm} = 0.45$, $\sigma_{BR} = 8$ is 0.64 falling to 0.0035 (verified, block C).

(iii) **Raw biomarker times time on drug,** $BR_{it} \leftarrow BR_{it} + \beta_{bm} B_i t_{od,it}$. With $B_i = \mu_{bm} + \sigma_{bm} b_i$ the shift splits into a biomarker-free drift and a moderation term,

$$\beta_{bm} B_i t_{od,it} = \beta_{bm} \mu_{bm} t_{od,it} + \beta_{bm} \sigma_{bm} t_{od,it}\, b_i \tag{41}$$

(derived). At $\beta_{bm} = 0.45$, $\mu_{bm} = 124.33$, $\sigma_{bm} = 15.36$ and $t_{od} = 10$ the drift is 559.5 and the moderation is 69.1 per standard deviation of the biomarker, 8.6 times σ_{BR} and 19.2 times the calibrated 3.6 (verified, block C).

(Source: `docs/19`, `.md` and `.tex`, Section 3; `docs/23`, Section 8.1.) [Correction. `docs/19` and `docs/23` compare the total shift $124 \times 10 \times 0.45 = 558$ with the noise SD 8 ("about 70 to 1"). Almost all of that shift is the biomarker-free drift, which the drug and time terms absorb and which carries no interaction information; the part that the interaction term sees is the per-SD moderation of 69.1, about 8.6 standard deviations of BR rather than 70, and it grows with t_{od}. The conclusion, a grossly overpowered signal, stands. The source's shift of 0.65 in (ii) uses the rounded profile 0.18.]

The calibration check. The adopted shift $\beta_{bm} \sigma_{BR} b_i$ on drug matches the covariance-moderation conditional mean $c_{bm} \sigma_{BR} b_i$ (Equation (30)) when $\beta_{bm} = c_{bm}$, giving $0.45 \times 8 = 3.6$ per standard deviation (§2, Rule 2). `docs/19` (`.tex`, Section 4.1) checks this on 500 crossover participants by regressing the mean on-drug BR on the biomarker, and reports slopes 3.36 and 3.48 and residual SDs 5.23 and 3.89 (mean and covariance moderation). For the mean $\overline{BR}_{i,\text{on}}$ over the $n_{\text{on}} = 4$ on-drug visits of a crossover path at 2.5-week spacing, with A_{on} their AR(1) correlation block ($\rho = 0.7$ per week, `buildSigma()`, inspected),

$$\operatorname{Var}\big(\overline{BR}_{i,\text{on}}\big) = \frac{\sigma_{BR}^2\, \mathbf{1}^\top A_{\text{on}} \mathbf{1}}{n_{\text{on}}^2}, \qquad \sigma_{\text{res}}^2 = \begin{cases} \operatorname{Var}\big(\overline{BR}_{i,\text{on}}\big) & \text{mean moderation,} \\ \operatorname{Var}\big(\overline{BR}_{i,\text{on}}\big) - c_{bm}^2 \sigma_{BR}^2\, \bar{g}_{\text{on}}^2 & \text{covariance moderation,} \end{cases} \tag{42}$$

with slope $c_{bm} \sigma_{BR} = \beta_{bm} \sigma_{BR}$ in both and $\bar{g}_{\text{on}} = 1$ (derived: the mean shift adds no residual variance; the covariance coupling removes the explained part, as in Equation (108)). The values are $\sigma_{\text{res}} = 5.39$ and 4.02. A 2,000-replicate simulation of the 500-participant regression gives sampling SDs of 0.25 and 0.18 for the slopes and 0.17 and 0.13 for the residual SDs, so all four reported values lie within one sampling SD of Equation (42) (verified, block D). [Correction to `docs/23`, Section 8.2, which states that the residual SDs could not be checked and that a regression of a single BR_{it} would give about 8 and 7.1. The `.tex` version of `docs/19` specifies the regression as one of the mean on-drug BR, and under that reading the reported values are reproduced. That the source used this exact schedule and ρ is inferred from the match and from the package defaults, not from the source run, which was not located.]

7 The dual-channel architecture

Paper 11 activates the mean and covariance channels of §6 simultaneously, with independent weights $c_{bm,A}$ (mean channel) and $c_{bm,B}$ (covariance channel).

7.1 Construction

The covariance channel is built into the MVN correlation matrix exactly as in Equation (29), with c_{bm} replaced by $c_{bm,B}$. The mean channel is then applied as a post-draw additive shift at on-drug timepoints, Equation (26) with the plain profile and β_{bm} replaced by $c_{bm,A}$:

$$\Delta BR_{it} = c_{bm,A}\, b_i\, \sigma_{BR}\, \mathbf{1}[D_{it} = 1], \qquad b_i = \frac{B_i - \mu_{bm}}{\sigma_{bm}} \tag{43}$$

(`dgp_architecture = 'combined'` in `R/generateData.R`, inspected). Paper 11 writes the standardized biomarker as $z_{B,i}$; the canonical symbol is b_i. Note the asymmetry this construction builds in: the covariance channel carries the graded off-drug coupling, while the mean channel stops when the drug stops.

7.2 The channels add on the covariance scale

Each channel's on-drug contribution to $\mathrm{Cov}(B, BR)$ can be computed separately:

$$\text{Covariance channel alone:} \qquad \mathrm{Cov}(B, BR \mid \text{on-drug}) = c_{bm,B}\,\sigma_{bm}\sigma_{BR} \tag{44}$$

$$\text{Mean channel alone:} \qquad \mathrm{Cov}(B, \Delta BR) = c_{bm,A}\,\sigma_{bm}\sigma_{BR} \tag{45}$$

and, because the combined-architecture on-drug response is the sum of the correlated draw and the mean shift, the two contributions add:

$$\mathrm{Cov}(B, BR_C \mid \text{on-drug}) = (c_{bm,A} + c_{bm,B})\,\sigma_{bm}\sigma_{BR}. \tag{46}$$

To first order the residual variance is unchanged (the mean channel adds $\sigma^2_{BR}(c^2_{bm,A} + 2c_{bm,A}c_{bm,B})$ to the on-drug variance, a second-order term; derived), so the interaction test's noncentrality is additive across channels, $\Lambda_C \approx \Lambda_A + \Lambda_B$. Paper 11 then argues that power, as a function $\pi(\Lambda)$ of noncentrality, is convex over the sub-ceiling regime, which gives *super*-additivity of power relative to the probability-scale benchmark:

$$\pi(\Lambda_A + \Lambda_B) - \pi(0) \;>\; \big[\pi(\Lambda_A) - \pi(0)\big] + \big[\pi(\Lambda_B) - \pi(0)\big]. \tag{47}$$

At $(c_{bm,A}, c_{bm,B}) = (0.22, 0.22)$ in the Hybrid design, at the matched total $N = 70$, observed power is 0.577 against an additive benchmark $\hat{\pi}(0.22, 0) + \hat{\pi}(0, 0.22) - \hat{\pi}(0, 0) = 0.214 + 0.138 - 0.040 = 0.312$, an excess of 0.265, consistent with Equation (47) (Paper 11, Section 4.3; simulated).

[2026-10-08: notation. Paper 11 writes the noncentrality as η, which `NOTATION.md` reserves for expectancy; it is written Λ here, as in `docs/26`. Convexity of $\pi(\Lambda)$ holds only below the inflection of the power curve; the statement is argued in Paper 11, not derived for a specific test.]

7.3 Boundary reductions and an identifiability consequence, now stated

Setting either weight to zero recovers the corresponding pure architecture exactly, by direct substitution into Equations (43) and (29): $c_{bm,B} = 0$ recovers mean moderation; $c_{bm,A} = 0$ recovers covariance moderation. Paper 11 treats these boundaries as internal validation gates: a correctly implemented combined driver must reproduce the companion manuscripts' pure-architecture power values there, run under matched design, sample-size, and calibration settings. Equation (46) shows that the combined on-drug covariance, and hence the fitted $bm{:}D_{bc}$ coefficient to first order, depends on $c_{bm,A}$ and $c_{bm,B}$ only through their *sum*, so a single fitted interaction coefficient cannot separately recover the two channel weights. Paper 11 now states this non-identifiability explicitly in its §3, after the additive-covariance derivation. Off drug the two channels are not interchangeable, since only the covariance channel carries graded coupling (Equation (43)); at $t_{1/2} > 0$ the sum is therefore not the only quantity that matters (argued; not examined in Paper 11).

[A second gap, investigated rather than left flagged.] Paper 11's driver (`analysis/scripts/dgp-combined/01-run-combined-factorial.R`) already allocates N as a total, via the same `generate_data_multi_path()` helper the companion paper uses, and its stored output already includes the matched $N = 70$ cells, so the sample-size-convention explanation an earlier version of §4.2 gave for the boundary mismatch does not hold. Reading the $N = 70$, E2-specification boundary cells directly from that stored output and comparing them to Paper

01's $N = 70$ baselines shows the two still do *not* agree: at $t_{1/2} = 0$ the mean-moderation boundary reads 0.595/0.738/0.612 (CO/Hybrid/ OL+BDC) against Paper 01's 0.739/0.842/0.751, a gap of 10–19 percentage points, far beyond the $\approx$ 0.013–0.016 MCSE at these cell sizes. The manuscript's §4.2 has been rewritten to report this finding rather than the superseded N-convention explanation, and now names the likely locus of the mismatch (a calibration difference in `resp_param`, `baseline_param`, or the nuisance correlations `c.cf1t/c.cfct/rho` between the two drivers) as an open item for a follow-up revision. Paper 11's own 3×3 channel-weight power table has likewise been recomputed on the $N = 70$ data in that same output file, in place of the earlier $N = 35$ table; Paper 11 states that it should be treated as not yet validated against the companion baselines.

[2026-10-08: the Paper 01 baselines quoted here (0.739, 0.842, 0.751 and 0.745, 0.840, 0.777 at $c_{bm} = 0.45$) are those of the pre-restatement version of Paper 01. The restated Paper 01 does not report them; it works at $c_{bm} = 0.25$ with the paired-difference statistic and Hendrickson et al.'s binary-indicator model, so the boundary check of Paper 11 now has no current reference values in Paper 01.]

7.4 The dual channel conditional on the biomarker, and a matched-total parameterization

Let the covariance channel carry a coupling profile g_t (Equation (31)) with weight $c_{bm,B}$, and let the mean channel add the on-drug shift of Equation (43) with weight $c_{bm,A}$, so that $BR^C_{it} = BR_{it} + c_{bm,A}\, b_i\, \sigma_{BR}\, D_{it}$ with $(B_i, BR_{i\cdot})$ jointly normal as in Equation (29). Because the shift is a fixed linear function of b_i, conditioning on the biomarker gives

$$\begin{aligned} E\big(BR^C_{it} \mid B_i\big) &= \mu_{BR,t} + \big(c_{bm,A}\, D_{it} + c_{bm,B}\, g_t\big)\, \sigma_{BR}\, b_i, \\ \mathrm{Cov}\big(BR^C_{i\cdot} \mid B_i\big) &= \Sigma_{BR} - c^2_{bm,B}\, \sigma^2_{BR}\, g g^\top, \end{aligned} \tag{48}$$

the dual-channel counterpart of Equation (30) (derived; verified to 10^{-14} against the conditional-normal formulas on Hybrid path 1 at $t_{1/2} = 1$ week, $\rho = 0.7$, and consistent within Monte Carlo error with a 4×10^5-draw simulation, `check-stress-test-construct.R`, Section 7). Three consequences follow.

(i) On drug the conditional slope is exactly $(c_{bm,A} + c_{bm,B})\sigma_{BR}$, so the additivity of Equation (46) holds exactly in the conditional mean, not only to first order. The conditional on-drug variance is $\sigma^2_{BR}(1 - c^2_{bm,B})$ and does not involve $c_{bm,A}$: only the covariance channel reduces the residual variance, while the marginal variance $\sigma^2_{BR}(1 + c^2_{bm,A} + 2c_{bm,A}c_{bm,B})$ of §7 rises with the mean channel.

(ii) Off drug after exposure the conditional slope is $c_{bm,B}\, g_t\, \sigma_{BR}$: only the covariance channel carries carryover, which makes exact the statement of §7 that the sum $c_{bm,A} + c_{bm,B}$ is not the only quantity that matters at $t_{1/2} > 0$.

(iii) Positive definiteness constrains $c_{bm,B}$ alone, since the mean shift is applied after the draw: a cell $(c_{bm,A}, c_{bm,B})$ is feasible exactly when $(0, c_{bm,B})$ is, that is when $c_{bm,B} < c^*_{bm}$ of Equation (55).

Matched total. The original plan for the dual-channel architecture proposes, besides the factorial grid, holding the total at 0.45 with a mixing weight (Source: `docs/plan-architecture-c.md`, Parameterization). Writing that weight $w_A \in [0, 1]$,

$$c_{bm,A} = w_A\, c_{\mathrm{tot}}, \qquad c_{bm,B} = (1 - w_A)\, c_{\mathrm{tot}}, \qquad c_{\mathrm{tot}} = 0.45, \tag{49}$$

Equation (48) gives an on-drug conditional slope of $c_{\mathrm{tot}}\sigma_{BR}$ for every w_A, an off-drug slope $(1 - w_A)c_{\mathrm{tot}}\, g_t\, \sigma_{BR}$, and a rank-one variance reduction $(1 - w_A)^2 c^2_{\mathrm{tot}} \sigma^2_{BR} g g^\top$. Without carryover ($g_t = D_{it}$) the conditional mean is the same for every w_A, so the matched-total family fixes the

on-drug estimand of Equation (32) and varies only the residual covariance, which is smallest at $w_A = 0$; it is feasible for $(1 - w_A)c_{\text{tot}} < c^*_{bm}$ (derived). [The plan writes the mixing weight as α, the canonical significance level; it is w_A here. The plan's statement that the $(0.45, 0.45)$ cell carries "approximately double" the interaction of either pure architecture is exact on the conditional-mean scale, by consequence (i). The matched-total family was not run in Paper 11.]

8 A latent-variable reading of covariance moderation

Paper 03 asks what mechanism a fitted covariance-moderation model could be a moment-matched summary of, if the true underlying process is a discrete responder/non-responder dichotomy rather than a continuously varying correlation.

8.1 The faithful latent-class data-generating process

If a candidate predictive biomarker indexes an unobserved responder/non-responder class,

$$Z_i \mid B_i \sim \text{Bernoulli}\big(\pi(B_i)\big), \qquad BR_{it} \mid Z_i = z \sim f_z(t, D_{bc,it}), \tag{50}$$

with Z_i the unobserved class label and $\pi(B_i) = \Pr(Z_i = 1 \mid B_i)$ the class-membership probability, now given an explicit closed form in Paper 03's Notation section,

$$\pi(B_i) = \text{expit}(\psi_0 + \psi_g\, b_i), \qquad b_i = \frac{B_i - \bar{B}}{s_B}, \tag{51}$$

with ψ_g the **gating slope** (called `gs` in the code, and now given the symbol ψ_g in the manuscript rather than left unnamed) and ψ_0 fixed so $\pi(0) = 0.5$ throughout. f_0, f_1 are class-conditional response distributions differing in drug-response parameters. Within-class temporal structure follows the same modified-Gompertz form of §3, with class- specific parameters substituted.

[Notation caveat, resolved.] Paper 03 previously referred to gating-function steepness only informally, as `gs` in prose and code, with no manuscript symbol attached and no closed form for $\pi(B_i)$ ever written out. This compendium originally flagged that gap because a symbol γ *does* appear elsewhere in the compendium (Paper 06, §12) as the unrelated $BR \times PB$ subadditivity coefficient, inviting confusion between the two. Paper 03 now introduces ψ_g and Equation (51) explicitly, and its Notation section states in so many words that ψ_g is not to be confused with Paper 06's γ. `NOTATION.md` has been updated to match: γ is now documented compendium-wide as Paper 06's coefficient only, and ψ_g is added as its own entry for Paper 03's gating slope. The class-membership probability $\pi(B_i)$, and the marginal class proportion π below, are local to this section and are not the power π of §21; `NOTATION.md` uses the same $\pi(B_i)$.

8.2 The MVN approximation to mixture-induced moderation

Under within-class independence between B_i and BR_{it}, a two- component mixture with class-dependent means and common within-class covariance produces a marginal joint distribution of (B_i, BR_{it}) whose first two moments are reproduced, to leading order, by a single MVN with $\text{Cor}(B, BR) = c_{bm}$, where

$$c^2_{bm} = \underbrace{\frac{\pi(1-\pi)(\mu^1_{bm} - \mu^0_{bm})^2}{\text{Var}(B)}}_{f_B\text{: between-class fraction in } B} \cdot \underbrace{\frac{\pi(1-\pi)(\mu^1_{BR} - \mu^0_{BR})^2}{\text{Var}(BR)}}_{f_{BR}\text{: between-class fraction in } BR}. \tag{52}$$

That is, $c^2_{bm} = f_B f_{BR}$ is the *product* of the two between-class variance fractions, so under within-class independence $|c_{bm}|$ equals their geometric mean $\sqrt{f_B f_{BR}}$ (derived: the covariance is carried by class membership alone, $\text{Cov}(B, BR) = \pi(1-\pi)(\mu^1_{bm} - \mu^0_{bm})(\mu^1_{BR} - \mu^0_{BR})$). If either

class separation is small, the achievable c_{bm} is small. Paper 03 phrases this as a bound by "the geometric mean of the two intra-class correlations"; under its stated assumption the relation is an equality. Under this reading the covariance-moderation architecture is the second-moment signature of a latent-class mixture whose class-conditional means separate on drug and whose separation is eroded by carryover off drug, not necessarily a generative covariance mechanism in its own right. The restated Paper 01 (Discussion) adopts the same identity, $c_{bm}^2 = f_B f_{BR}$, and adds that the match is limited to second moments: the mixture's conditional mean is nonlinear in the biomarker, and its higher moments differ. The approximation breaks down for strongly bimodal responses, a step-function (rather than smooth) $\pi(B_i)$, or class-varying (rather than common) within-class covariance.

8.3 Estimation and its two competing test statistics

The class-*unaware* analysis is the compendium's common model, Equation (136). The class-*aware* analysis fits a two-class growth-mixture model (`lcmm::hlme`) and extracts two candidate signals: a Wald test on the biomarker's coefficient in the class-gating equation, and a within-class Wald test on $bm{:}D_{bc}$. A model-comparison gate is applied before either is trusted:

$$\text{prefer } ng = 2 \text{ over } ng = 1 \iff \Delta\text{BIC} > 0 \;\; \text{(or, more conservatively, } > 6), \tag{53}$$

following Raftery (1995), with the gating test interpreted only conditional on that selection. In Paper 03's 240-replicate pilot at $N = 70$, the unconditional class-gating Wald test is markedly anti-conservative under the null (Type I error 0.197, 95% CI $[0.145, 0.250]$), while the BIC-conditional version is severely under-powered (Type I error 0.004; power 0.025 at gating slope $\psi_g = 1.0$ and 0.004 at $\psi_g = 1.5$, against 0.325 and 0.542 for the class-unaware $bm{:}D_{bc}$ Wald test on the same cells). In this pilot the BIC step selects $ng = 1$ in nearly every replicate, so the conditional procedure approximates a never-reject rule; the class-aware tests are dominated by the class-unaware one (Paper 03, Pilot results). These are pilot estimates (MCSE of power about 0.030); Paper 03's production program has not been run.

[2026-10-08: corrected. The power 0.025 was previously attributed to gating slope 1.5; Paper 03 reports it at 1.0, with 0.004 at 1.5. The previous summary, that the class-aware procedure "buys correct calibration at a substantial power cost", was replaced by Paper 03's own reading.]

9 Validity of covariance binding: the coupling ceiling and the separable response covariance

The covariance-moderation architecture (Equation (29)) writes the interaction into a correlation, and a correlation matrix must be positive definite. This bounds the moderation parameter c_{bm} from above, by a ceiling that depends on the response covariance, the visit schedule, and the coupling pattern. This section states that ceiling exactly, evaluates it for the published compound-symmetry construct of Hendrickson et al. and for the separable AR(1) construct that Paper 01 now adopts as its reference structure, and records why a generator must stop, rather than repair, when a requested c_{bm} exceeds it. Mean moderation (Equation (26)) writes nothing into the correlation matrix and has no ceiling. Throughout, c_1 and $c_\times$ denote the cross-component correlations at the same visit and at different visits (code `c.cf1t`, `c.cfct`), and every numerical statement is made at a single value of ρ. The compound-symmetry correlation is ρ_{cs} and the AR(1) correlation per week is ρ (`NOTATION.md`); where a table or a sentence compares the two constructs at a common value, as in Table 8, the ρ it states is ρ_{cs} for the compound-symmetry rows and the AR(1) ρ for the others. [2026-10-09: notation. The compound-symmetry correlation was written ρ; it is ρ_{cs} throughout this document, as in `NOTATION.md`.]

9.1 The joint correlation matrix and the exact ceiling

For one participant on one randomization path with n visits at weeks $w_1 < \cdots < w_n$, the data-generating process draws $(B_i, \mathrm{BL}_i, TV_{1:n}, PB_{1:n}, BR_{1:n})$ jointly from a multivariate normal distribution. On the correlation scale,

$$R = \begin{pmatrix} 1 & 0 & c_{bm}\tilde{g}^\top \\ 0 & 1 & 0^\top \\ c_{bm}\tilde{g} & 0 & M \end{pmatrix}, \qquad \tilde{g} = e_3 \otimes g, \tag{54}$$

where M is the $3n \times 3n$ response block in component-major order (TV, PB, BR), $e_3 = (0, 0, 1)^\top$ selects BR, and $g \in \mathbb{R}^n$ is the coupling pattern. Under graded coupling, Equation (31) with exponential decay, $g_t = 1$ on drug, $2^{-t_{sd,t}/t_{1/2}} = e^{-\lambda t_{sd,t}}$ off drug after first exposure and 0 before first exposure; the published step rule replaces the middle case by $\mathbf{1}[g_t > 0]$, which equals 1 at every post-exposure visit whenever $t_{1/2} > 0$. Variances rescale R to the covariance and do not affect positive definiteness. By the Schur complement on the biomarker row, $R \succ 0$ if and only if $M \succ 0$ and $1 - c_{bm}^2\, \tilde{g}^\top M^{-1} \tilde{g} > 0$, so

$$c_{bm}^* = \left(\tilde{g}^\top M^{-1} \tilde{g}\right)^{-1/2}. \tag{55}$$

The quantity $c_{bm}^2\, \tilde{g}^\top M^{-1} \tilde{g}$ is the squared multiple correlation of the biomarker on the whole response trajectory, so the condition says only that the biomarker cannot be more than perfectly predictable. The bound is attained: at $c_{bm} = c_{bm}^*$ the smallest eigenvalue of R is zero. A design's ceiling is the minimum over its paths, since all paths share one c_{bm}. (Derived. Source: `docs/34`, Section 3.2; `docs/36`, Section 4.2; Paper 01, Methods and Appendix A.4.) [The sources write the padded pattern as u (Paper 01, `docs/36`) or v (`docs/34`, `docs/32`); $\tilde{g}$ is used here because u_i is the canonical random intercept.]

A matrix can therefore fail in two distinct ways: the response block M is itself indefinite, which no choice of c_{bm} repairs, or M is valid and the coupling exceeds what it supports. On every grid of `docs/36` (Section 6.5) only the second occurs.

9.2 The published compound-symmetry construct

Hendrickson et al.'s construct (commit `58b32a9`) sets every within-component correlation to ρ_{cs} and the cross-component correlations to c_1 and $c_\times$. It decomposes exactly as

$$M_{\mathrm{CS}} = K_o \otimes I_n + K_p \otimes J_n, \qquad K_p = \rho_{cs} I_3 + c_\times (J_3 - I_3), \qquad K_o = (1 - \rho_{cs}) I_3 + (c_1 - c_\times)(J_3 - I_3), \tag{56}$$

a person-level term plus an occasion-level term. Since J_n has eigenvalues n and 0, $M_{\mathrm{CS}} \succ 0$ if and only if $K_o \succ 0$ and $K_o + nK_p \succ 0$, which with $K_p \succeq 0$ reduces to $(1 - \rho_{cs}) - (c_1 - c_\times) > 0$, independent of n and of the schedule. At the published values $\rho_{cs} = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$ this smallest eigenvalue is 0.100 for $n = 4$ to 32. (Derived; verified, `check-ceilings.R`. Source: `docs/35`, Section 4.5; Paper 01, Appendix A.5.)

Writing $P_1 = n^{-1} J_n$, $M_{\mathrm{CS}} = K_o \otimes (I_n - P_1) + (K_o + nK_p) \otimes P_1$, and the two projections invert separately. With $a = [K_o^{-1}]_{33}$, $a_n = [(K_o + nK_p)^{-1}]_{33}$ and $\bar{g}$ the mean of g,

$$\frac{1}{c_{bm}^{*\,2}} = a \sum_{t=1}^{n} (g_t - \bar{g})^2 + a_n\, n\, \bar{g}^2 \overset{g \in \{0,1\}^n}{=} a\, \frac{n_{\mathrm{on}}(n - n_{\mathrm{on}})}{n} + a_n\, \frac{n_{\mathrm{on}}^2}{n}, \tag{57}$$

where n_{on} counts visits with $g_t = 1$. (Derived here; the binary form is `docs/34`, Section 5.1, and agrees with the full matrix to machine precision, verified.) Three consequences follow.

1. **Schedule features are irrelevant; visit counts are not.** For a binary pattern only n, n_{on}, ρ_{cs}, c_1 and $c_\times$ enter: spacing, the gap at a switch, the number of switches and their

order do not. The ceiling falls with the number of visits at a fixed on-drug share, roughly as $n^{-1/2}$. With half the visits on drug, at $\rho_{cs} = 0.8$ it is 0.358, 0.256, 0.182 and 0.129 for $n = 4$, 8, 16 and 32 (verified, `check-ceilings.R`). At $\rho_{cs} = 0.7$ it is 0.480, 0.346, 0.247 and 0.176 (`docs/32`, Section 8; verified). The loss comes through the coupling vector, not through the response block, whose smallest eigenvalue does not depend on n.

2. **The published values are infeasible without carryover.** At $\rho_{cs} = 0.8$ and $t_{1/2} = 0$ the ceiling is 0.256 in Hybrid (paths with $n_{\text{on}} = 4$ of 8) and CO, and 0.262 in OL+BDC (verified). Both published effect sizes, $c_{bm} = 0.3$ and 0.6, exceed it; `docs/36` (Section 4.1) reports that 40 of the 162 published design-path-cell matrices fail the code's own check, all through the biomarker row.

3. **The high ceilings under carryover carry no signal.** With step coupling and any $t_{1/2} > 0$, every Hybrid and OL+BDC path, all of which begin on drug, has $g = \mathbf{1}$, so $c^*_{bm} = (n\, a_n)^{-1/2} = 0.893$ at $\rho_{cs} = 0.8$, $n = 8$ (verified). A constant pattern has no on/off contrast, so the cells that become feasible are exactly the cells in which the interaction has no population counterpart. CO stays at 0.256, because its placebo-first path keeps $g_t = 0$ before exposure. (Source: `docs/36`, Sections 4.2 and 5.3; `docs/32`, Section 3.3.)

Under graded coupling (Equation (31)) and compound symmetry at $\rho_{cs} = 0.8$, the Hybrid ceiling rises only slowly with the half-life: 0.256, 0.257, 0.307 and 0.369 at $t_{1/2} = 0.1$, 0.2, 1 and 2 weeks (verified, `check-ceilings-graded.R`; Paper 01, Appendix A.5; `docs/36`, Section 6.5). [Paper 01, Appendix A.5, places this compound-symmetry result under its separable-structure heading and attributes it to the lower on-to-off step cost, which is an AR(1) quantity (Equation (61)). Under compound symmetry the operative mechanism is Equation (57): graded coupling reduces the dispersion $\sum_t (g_t - \bar{g})^2$ that carries the larger weight a.]

9.3 The separable response covariance

Let $A_{ts} = \rho^{|w_t - w_s|}$ (AR(1) in calendar weeks) and $K = (1 - c_1)I_3 + c_1 J_3$, and set

$$M = K \otimes A, \qquad \text{Cor}(X_{c,t}, X_{c',s}) = K_{cc'}\, A_{ts}, \tag{58}$$

for components $c, c' \in \{TV, PB, BR\}$. Lagged cross-component correlation is then $c_1 \rho^{|w_t - w_s|}$ and $c_\times$ is retired. The eigenvalues of $K \otimes A$ are the products of those of K and A. K has eigenvalues $1 + 2c_1$ (once) and $1 - c_1$ (twice), so $K \succ 0$ for $-\frac{1}{2} < c_1 < 1$; A is the correlation matrix of a stationary Ornstein–Uhlenbeck process at the visit times and is positive definite for any distinct times. Hence $M \succ 0$ for every schedule. Generatively, $\mathbf{X} = L_A \mathbf{Z} L_K^\top$ with $L_A L_A^\top = A$, $L_K L_K^\top = K$ and $\mathbf{Z}$ an $n \times 3$ standard normal matrix: the components are fixed mixtures of three independent AR(1) paths on one clock. (Derived. Source: `docs/35`, Section 4.2; Paper 01, Appendix A.5.)

The non-separable AR(1) construct in the package (`covar`, `buildSigma()`) decomposes as $M_{\text{covar}} = K_a \otimes A + K_b \otimes I_n$ with $K_a = (1 - c_\times)I_3 + c_\times J_3$ and $K_b = (c_1 - c_\times)(J_3 - I_3)$. K_b has zero diagonal and is indefinite: a same-visit cross-component excess attached to no variance. Diagonalizing A gives

$$M_{\text{covar}} \succ 0 \iff \omega_{\min}(A) > \frac{c_1 - c_\times}{1 - c_\times}, \tag{59}$$

with $\omega_{\min}(A)$ the smallest eigenvalue of A, which falls as visits become closer or more numerous; at $\rho = 0.7$ and $(c_1, c_\times) = (0.2, 0.1)$ the condition fails for half-week spacing. (Derived, and verified to 10^{-15} in `docs/35`, Section 4.4.) [Paper 01, Appendix A.5, writes the eigenvalues of A as $\lambda_j(A)$; ω is used here because λ is the canonical carryover decay rate.] The published construct (Equation (56)) is a single separable term, $K \otimes A_{\text{CS}}$, only when $c_\times = c_1 \rho_{cs}$ (`docs/35`, Section 4.5).

9.4 The factorized ceiling and the per-step costs

Since $M^{-1} = K^{-1} \otimes A^{-1}$, the quadratic form in Equation (55) factorizes, and the equicorrelated K has a closed-form inverse:

$$\tilde{g}^\top M^{-1} \tilde{g} = [K^{-1}]_{33}\, g^\top A^{-1} g, \qquad K^{-1} = (1-c_1)^{-1}\Big(I_3 - \frac{c_1}{1+2c_1} J_3\Big),$$

so that

$$\begin{aligned} c_{bm}^* &= \sqrt{1-R^2}\,\big(g^\top A^{-1} g\big)^{-1/2}, \\ [K^{-1}]_{33} &= \frac{1+c_1}{(1-c_1)(1+2c_1)} = \frac{1}{1-R^2}, \qquad R^2 = \frac{2c_1^2}{1+c_1}, \end{aligned} \tag{60}$$

where R^2 is the squared multiple correlation of BR on TV and PB at one visit: 0.0182, 0.0667 and 0.1385 at $c_1 = 0.1$, 0.2, 0.3 (derived; verified, `check-ceilings.R`). The second identity follows from $(1-c_1)(1+2c_1) = (1+c_1)(1-R^2)$.

The sampled Ornstein–Uhlenbeck process is Markov, $X_t = \rho_t X_{t-1} + \sqrt{1-\rho_t^2}\,\xi_t$ with step correlation $\rho_t = \rho^{d_t}$, gap $d_t = w_t - w_{t-1}$ and independent standard normal innovations ξ_t. Hence $A^{-1} = L^\top L$ with L lower bidiagonal ($L_{11} = 1$, $L_{tt} = (1-\rho_t^2)^{-1/2}$, $L_{t,t-1} = -\rho_t(1-\rho_t^2)^{-1/2}$), and

$$g^\top A^{-1} g = g_1^2 + \sum_{t=2}^{n} \frac{(g_t - \rho_t\, g_{t-1})^2}{1-\rho_t^2}, \tag{61}$$

a sum of standardized one-step prediction errors of the coupling pattern. For a binary pattern the step from visit $t-1$ to t costs $(1-\rho_t)/(1+\rho_t)$ on to on, $\rho_t^2/(1-\rho_t^2)$ on to off, $1/(1-\rho_t^2)$ off to on, and 0 off to off; all four were checked against a direct inverse (verified). Abrupt switches between closely spaced visits are expensive, which is why designs with many on/off switches carry only small effects. For the constant pattern, $g^\top A^{-1} g = 1 + \sum_t (1-\rho_t)/(1+\rho_t)$, an effective number of independent visits, and the ceiling is $\sqrt{(1-R^2)/n_{\text{eff}}}$. Under graded coupling the first off-drug step costs $(g_t - \rho_t)^2/(1-\rho_t^2)$, smaller than the binary on-to-off cost when g_t is near ρ_t, so the ceiling rises with the half-life. (Derived. Source: `docs/35`, Sections 4.3 and 6; Paper 01, Appendix A.5.) [`docs/35` writes the step correlation as ϕ_t; ρ_t is used here because ϕ denotes the carryover decay family (Equation (31)). `docs/35` also records that `docs/34`, Section 5.5, assigned every switch the off-to-on cost; an on-to-off switch costs exactly one unit less, as Equation (61) gives.]

Table 8 collects the values. At $\rho = 0.8$ the compound-symmetry ceiling without carryover is 0.256 and the separable AR(1) ceiling is 0.440 (Hybrid path A). At $\rho = 0.7$ the separable AR(1) ceiling is 0.474 (OL+BDC path A), and 0.480 in the Hybrid design. At $\rho = 0.8$, $c_{bm} = 0.3$ is feasible in every design under the separable form, and $c_{bm} = 0.6$ exceeds every graded-coupling ceiling in Hybrid and OL+BDC at every half-life examined (at most 0.544 and 0.578 at $t_{1/2} = 1$). At $\rho = 0.7$, $c_{bm} = 0.45$ is feasible in every design under the separable form. (Source: `docs/34`, review note; `docs/35`, Section 6; `docs/36`, Section 6.5; Paper 01, "Room for larger effect sizes".) A higher ceiling is not more power: at the same c_{bm}, AR(1) data carry less interaction information than compound-symmetric data, and `docs/35` and Paper 01 report closed-form paired-difference power in Hybrid at $c_{bm} = 0.25$ of 0.721 under compound symmetry against 0.335 under the separable form. [The power pair is from `docs/37`, Section 5.8, via `docs/35` and Paper 01, neither of which states the ρ at which it holds. It is at $\rho = 0.8$, the published value, with $c_1 = 0.2$ and no measurement error: in `analysis/data/quick-sim/carryover-closed-form/design-power.csv` the Hybrid rows at $t_{1/2} = 0$ give power 0.7209 (compound symmetry) and 0.3347 (separable arm, whose ceiling 0.4400 is the $\rho = 0.8$ value of Table 8) (inspected), and the pair is recomputed in Table 35 (verified, `check-closed-form-extensions.R`).]

9.5 Ceilings by serial correlation and by cross-component form

Table 8 was extended in this revision to $\rho = 0.6$ and 0.5 and to the package's non-separable AR(1) form (configuration F of `docs/36`, the `covar` construct: AR(1) within components, cross-

Table 8: Ceiling c^*_{bm} without carryover ($c_1 = 0.2$; for compound symmetry and `covar`, $c_\times = 0.1$), minimum over each design's paths. Each row is at a single value of ρ. CS: compound symmetry (the published construct); F: the package's non-separable AR(1) `covar` construct, Equation (59); C: separable AR(1), Equation (58). Closed forms (57) and (60) agree with the full matrix to machine precision (verified, `check-ceilings.R`). The F rows and the rows at $\rho = 0.6$ and 0.5 were added from `docs/44`, Step 6 (verified, `check-stress-test-construct.R`, Section 1, and `ceilings-graded-by-rho.csv`); §9.5 discusses them, and Table 11 adds $\rho = 0.3$ and 0.9.

ρ	Structure	CO	Hybrid	OL+BDC	Design minimum
0.8 (published)	CS	0.256	0.256	0.262	0.256
	F	0.627	0.291	0.303	0.291
	C	0.637	0.440	0.468	0.440
0.7 (package)	CS	0.346	0.346	0.352	0.346
	F	0.619	0.456	0.452	0.452
	C	0.616	0.480	0.474	0.474
0.6	CS	0.410	0.410	0.414	0.410
	F	0.584	0.485	0.457	0.457
	C	0.580	0.491	0.462	0.462
0.5	CS	0.458	0.458	0.458	0.458
	F	0.549	0.489	0.448	0.448
	C	0.546	0.490	0.448	0.448

component correlation c_1 at the same visit and $c_\times \rho^{|w_t - w_s|}$ otherwise, Equation (59)). Each entry is Equation (55) with graded coupling minimized over the design's paths. (Source: `docs/44`, Step 6; recomputed independently, and all 144 cells of `ceilings-graded-by-rho.csv`, which also cover $t_{1/2} = 0.5$, 1 and 2 weeks, agree to 6×10^{-16}, `check-stress-test-construct.R`, Section 1; the Hybrid F entry at $\rho = 0.5$, 0.48855 unrounded, is given as 0.489, as in Table 9's source `docs/34`, Section 4.4, and checked in `check-reconciliations.R`.) Four statements follow (verified, `check-stress-test-construct.R`, Section 1). (a) For every structure and ρ in the table, the design minimum over $t_{1/2} \in \{0, 0.5, 1, 2\}$ weeks is attained at $t_{1/2} = 0$, so the last column of Table 8 is also the minimum over those half-lives. Carryover never lowers a ceiling below its no-carryover value except in CO at $\rho = 0.5$ and $t_{1/2} = 2$ weeks, by 0.0029 (F) and 0.0036 (C). (b) Separability raises the ceiling most where serial correlation is strongest (0.291 to 0.440 at $\rho = 0.8$) and hardly at all at $\rho = 0.5$ (0.448 either way); under F at $\rho = 0.8$ the response block is nearly singular in Hybrid and OL+BDC (Table 10). (c) Configuration C at $\rho = 0.7$ admits $c_{bm} = 0.45$ in every design and half-life with a margin of 0.024. (d) Compound symmetry has the lowest ceiling at $\rho \geq 0.6$ and the highest at $\rho = 0.5$ (0.458 against 0.448); the order of the structures depends on ρ.

A constant pattern. For an open-label path coupled at every visit ($g = \mathbf{1}$), on eight visits 2.5 weeks apart, at $\rho = 0.8$ the ceiling is $(na_n)^{-1/2} = 0.893$ under compound symmetry, 0.576 under F and 0.567 under C (the OL rows of Table 14), and 0.535 under AR(1) with the published constant cross-component block (verified, `check-open-label-ceiling.R`). [`docs/44`, Step 4, attributes the fall from 0.893 to 0.576 to AR(1) "applied with the published cross-component form". The value 0.576, like the Hybrid value 0.291, is that of `docs/36`'s configuration B, whose cross-component block is the decaying form $c_\times \rho^{|w_t - w_s|}$ (`docs/36`, Appendix A.6). With the published constant block the values are 0.535 (open label) and 0.363 (Hybrid, $t_{1/2} = 0$). The schedule of the open-label design was not stated in the sources; the equally spaced 2.5-week schedule reproduces all three `docs/36` values, which is the evidence for it.]

9.6 Ceilings at the package value $\rho = 0.7$: the crossed structures

`docs/34` crosses the two blocks that "AR(1)" touches and the coupling rule: within-factor block compound symmetry or AR(1) in calendar weeks; cross-factor entries at different visits constant ($c_\times$) or decaying ($c_\times \rho^{|w_t - w_s|}$); graded or step coupling. At the paper 01 production parameters ($\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$) the design-level ceilings are in Table 9; AR(1) with decaying cross-factor entries and graded coupling is the package's `covar`, and compound symmetry with constant entries and step coupling is the published specification (Source: `docs/34`, Section 4.1; verified, every entry).

Table 9: Design-level ceilings at $\rho = 0.7$, reported as $t_{1/2} = 0$ / 0.5 / 1 weeks (Source: `docs/34`, Section 4.1; verified).

Within	Cross, $t \neq s$	Coupling	CO	Hybrid	OL+BDC
AR(1)	decaying	graded (`covar`)	.619 / .619 / .619	.456 / .506 / .544	.452 / .500 / .533
AR(1)	decaying	step	.619 / .494 / .494	.456 / .519 / .519	.452 / .519 / .519
AR(1)	constant	graded	.591 / .591 / .591	.452 / .490 / .514	.438 / .472 / .491
AR(1)	constant	step	.591 / .430 / .430	.452 / .462 / .462	.438 / .463 / .463
CS	constant	graded	.346 / .346 / .346	.346 / .370 / .411	.352 / .386 / .456
CS	constant	step (published)	.346 / .346 / .346	.346 / .841 / .841	.352 / .841 / .841
CS	decaying	graded	.164 / .164 / .164	.247 / .265 / .299	.151 / .170 / .214
CS	decaying	step	.164 / .164 / .164	.247 / .845 / .845	.151 / .832 / .832

Under `covar` the binding path is path A (path 1) in Hybrid and OL+BDC without carryover, switching to path B in Hybrid at $t_{1/2} = 1$; in CO it is path A without carryover and the placebo-first path B at positive half-lives (`docs/34`, Section 4.1; quoted). The reference value $c_{bm} = 0.45$ of paper 01 lies within 0.002 to 0.006 of the `covar` ceiling in the no-carryover Hybrid and OL+BDC cells. Four readings follow (derived from the table).

(i) With the cross-factor block and graded coupling held fixed and $t_{1/2} = 0$, replacing compound symmetry by AR(1) raises the ceiling at $\rho = 0.7$ in every design: by 0.09 to 0.25 with constant entries and by 0.21 to 0.45 with decaying ones.

(ii) Compound symmetry with decaying cross-factor entries gives the lowest ceilings of all (0.151 in OL+BDC). It asserts that a factor's correlation with *another* factor decays with lag while its correlation with itself does not, and its response block is then nearly singular (Table 10). Alongside AR(1) the same cross-factor decay costs nothing: it raises the ceiling by 0.004 to 0.042 across the graded cells of the table (CO: 0.591 to 0.619). [`docs/32`, Section 5.2, summarizes this as "roughly +0.02 to +0.03"; the table gives +0.004 (Hybrid) to +0.028 (CO) without carryover and up to +0.042 (OL+BDC) at $t_{1/2} = 1$.]

(iii) The step-rule reversal (low ceiling without carryover, high with any) occurs in Hybrid and OL+BDC, whose paths all begin on drug, and not in CO, whose placebo-first path keeps pre-exposure zeros at every half-life. With an AR(1) within-factor block the step ceiling at positive half-lives is 0.519, not 0.841: the large jump requires both the step and compound symmetry.

(iv) Under `covar` the ceiling rises with the half-life in Hybrid and OL+BDC and is flat in CO, so a grid feasible at positive half-lives may be infeasible at $t_{1/2} = 0$; the no-carryover cell binds.

The compound-symmetry value is $(1 - \rho_{cs}) - (c_1 - c_\times)$ exactly (Section 9.2). The AR(1) values depend on the design because the one-week gaps of Hybrid and OL+BDC make adjacent visits strongly correlated. On every grid of `docs/34` and `docs/36` the response block itself is valid, so every failure is a ceiling failure. [At $\rho_{cs} = 0.8$ compound symmetry with decaying

Table 10: Smallest eigenvalue of the response block M, minimum over paths (Source: `docs/34`, Section 4.2, at $\rho = 0.7$; `docs/36`, Section 6.5, at $\rho = 0.8$, where OL has the CO schedule and the CO value; verified).

ρ	Response block	CO	Hybrid	OL+BDC
0.7	AR(1), decaying (`covar`)	0.288	0.074	0.074
0.7	AR(1), constant	0.331	0.093	0.093
0.7	CS, constant (published)	0.200	0.200	0.200
0.7	CS, decaying	0.034	0.039	0.015
0.8	CS, constant (published)	0.100	0.100	0.100
0.8	AR(1), decaying (`covar`)	0.153	0.010	0.010
0.8	Separable AR(1)	0.225	0.097	0.098

cross-factor entries is not positive definite in any design (smallest eigenvalues -0.082, -0.056, -0.095; computed here, not in either source).]

Dependence on ρ. Table 8 gives the no-carryover ceilings of the compound-symmetry, `covar` and separable structures at $\rho = 0.5$ to 0.8; Table 11 adds the end points $\rho = 0.3$ and 0.9. Under compound symmetry the ceiling falls steadily with ρ_{cs} and M reaches singularity at $\rho_{cs} = 0.9$, where $(1 - \rho_{cs}) - (c_1 - c_\times) = 0$. Under `covar` the ceiling rises with ρ through 0.8 in CO, and on the grid $\rho \in \{0.3, 0.5, 0.6, 0.7, 0.8\}$ peaks at $\rho = 0.5$ in Hybrid and at $\rho = 0.6$ in OL+BDC; at $\rho = 0.8$ it has fallen steeply in both, and at 0.9 their response block fails. The statement "AR(1) widens the feasible range of c_{bm}" therefore holds at $\rho = 0.7$ and fails at $\rho = 0.3$, where compound symmetry admits the higher ceiling in all three designs (Source: `docs/34`, Section 4.4, and `docs/44`, Step 6; verified). [`docs/34` evaluates $\rho = 0.3$, 0.5, 0.7, 0.8 and 0.9 only; on that grid the OL+BDC ceiling rises through 0.7, and the peak at 0.6 appears with the $\rho = 0.6$ row of `docs/44`.]

Table 11: No-carryover design ceilings at the end points of the ρ range, CO / Hybrid / OL+BDC; the rows $\rho = 0.5$, 0.7 and 0.8 are in Table 8 (Source: `docs/34`, Section 4.4; verified). "not PD": M itself is not positive definite.

ρ	CS, constant, graded	AR(1), decaying, graded (`covar`)
0.3	.520 / .504 / .504	.502 / .475 / .428
0.9	not PD	.449 / not PD / not PD

An evenly spaced schedule. On eight visits two weeks apart, on drug at the first four, at $\rho = 0.7$ the published specification has ceiling 0.346 without carryover and 0.841 at any positive half-life, against 0.670 for `covar` with graded coupling at $t_{1/2} = 1$ (at $\rho = 0.8$ the published values are 0.256 and 0.893). Holding the graded coupling at $t_{1/2} = 1$ fixed, the ceiling is 0.370 under compound symmetry with constant cross-factor entries, 0.646 when only the within-factor block is made AR(1), and 0.670 when the cross-factor entries also decay (Source: `docs/32`, Sections 3.3 and 4.3; verified). The like-for-like comparison attributes almost all of the gain to the within-factor form; the `orig`-against-`covar` comparison mixes two changes and reverses direction. [`docs/32`, Section 5.2, also quotes a fall from 0.34 to 0.16 when cross-factor decay is applied without AR(1), from `docs/54` under the 2024 coupling that leaves the first visit uncoupled; that figure was not recomputed. With every visit coupled the worst cell is 0.151 (Table 9).]

The cross-factor parameters. Under compound symmetry they enter almost entirely through $c_1 - c_\times$, as Equation (56) implies; under AR(1) the difference dominates, but the levels matter and a large difference invalidates M (Table 12).

`covar` against separable at $\rho = 0.7$. The design-level $\rho = 0.7$ values of the two structures are the F and C rows of Table 8 (binding paths A in all three designs). Table 13 adds a second Hybrid path and two further schedules: a multi-cycle design of 16 weekly visits in the pattern

Table 12: No-carryover ceiling on Hybrid path C by cross-factor parameters, $\rho = 0.7$, graded coupling (Source: `docs/34`, Section 5.1; verified).

$(c_1, c_\times)$	(0.1, 0)	(0.2, 0.1)	(0.3, 0.2)	(0.1, 0.1)	(0.2, 0.2)	(0.2, 0)	(0.3, 0.1)
$c_1 - c_\times$	0.1	0.1	0.1	0	0	0.2	0.2
Compound symmetry	0.346	0.346	0.345	0.378	0.377	0.261	0.261
AR(1) `covar`	0.485	0.465	0.435	0.504	0.491	not PD	not PD

on, on, off, off, and a dense schedule of eight visits half a week apart, half on drug (Source: `docs/35`, Section 6; verified).

Table 13: No-carryover ceilings under `covar` and the separable structure for paths and schedules not in Table 8, $\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$ (Source: `docs/35`, Section 6; verified).

Path	`covar`	Separable AR(1)
Hybrid, path C	0.465	0.491
OL+BDC, path A	0.452	0.474
Multi-cycle, 16 weekly visits	0.262	0.286
Dense, 0.5-week spacing	not PD	0.509

9.7 Ceilings at the published $\rho = 0.8$: the adjustment configurations

`docs/36` evaluates three proposed adjustments to the published construct (A: compound symmetry, constant cross-factor entries, step coupling): AR(1) within factors (B, step coupling, `covar` cross block), AR(1) with graded coupling (F, `covar` cross block; C, separable), and mean moderation (D on compound symmetry, E on the separable block), which has no ceiling. The OL design (eight visits 2.5 weeks apart, all on drug) is included (Source: `docs/36`, Section 6.5; verified, every entry; the OL schedule was identified by reproducing its ceilings).
[The OL rows at $t_{1/2} = 2$ repeat the constant value, since an all-on-drug path has no off-drug visit; the separable step rows at $t_{1/2} = 0.1$ to 2 repeat the `docs/37` "$t_{1/2} > 0$" value, which holds wherever the off-drug mean has not underflowed (Section 14.7).]

Three consequences follow (derived from the table and Equation (55)). First, AR(1) alone does not repair problem 1: it raises the ceiling where the coupling has an on/off contrast and *lowers* it where the pattern is constant, as in OL (from 0.893 to 0.576) and in every Hybrid and OL+BDC carryover cell. The quantity $c_{bm}^2 \tilde{g}^\top M^{-1} \tilde{g}$ is the squared multiple correlation of the biomarker on the whole trajectory; under compound symmetry the visits share a large person-level component that the biomarker can correlate with once, whereas under AR(1) little is shared, so the same correlation at every visit demands more of the biomarker. Second, graded coupling alone keeps the low ceiling (0.256 to 0.265 for $t_{1/2} \leq 0.2$), so applied alone it would make $c_{bm} = 0.3$ and 0.6 infeasible in every Hybrid and OL+BDC cell. Third, only the separable combination makes $c_{bm} = 0.3$ feasible everywhere; `covar` is marginal in Hybrid (0.291), and $c_{bm} = 0.6$ exceeds every covariance-moderation ceiling in Hybrid and OL+BDC at $\rho = 0.8$.

Compound symmetry against the separable form under graded coupling. Under the published step rule and any $t_{1/2} > 0$ the compound-symmetry ceiling exceeds the separable AR(1) one, because a constant coupling pattern is cheap under compound symmetry (Section 9.2, item 3). Under graded coupling the ordering does not reverse: at $\rho = 0.8$ the separable ceiling is the higher of the two in every design at every half-life examined (rows C and "graded alone" of Table 14), and the compound-symmetry ceiling is at most 0.469 (OL+BDC, $t_{1/2} = 2$) (Source: `docs/33`, Section 3.4, note of 2026-10-07; `docs/44`, Step 6; verified, `check-landmarks-hybrid-ceilings.R`, block 10). [The pre-restatement Paper 01 (Appendix B.7) reported that for any $t_{1/2} > 0$ the published compound-symmetry implementation admits

Table 14: Design-level ceilings by configuration and $t_{1/2}$ (weeks), $\rho = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$. This is the single table of $\rho = 0.8$ ceilings by structure and coupling rule; it is cited from Section 14.12. (Source: `docs/36`, Section 6.5; the $t_{1/2} = 2$ column and the separable step rows from `docs/37`, Section 5.8; verified, `check-closed-form-extensions.R`; the graded rows also agree with `ceilings-graded-by-rho.csv`.)

Configuration	Design	0	0.1	0.2	0.5	1	2
A published (CS, step)	OL	0.893	0.893	0.893	0.893	0.893	0.893
	OL+BDC	0.262	0.893	0.893	0.893	0.893	0.893
	CO	0.256	0.256	0.256	0.256	0.256	0.256
	Hybrid	0.256	0.893	0.893	0.893	0.893	0.893
B AR(1) `covar`, step	OL	0.576	0.576	0.576	0.576	0.576	0.576
	OL+BDC	0.303	0.594	0.594	0.594	0.594	0.594
	CO	0.627	0.576	0.576	0.576	0.576	0.576
	Hybrid	0.291	0.595	0.595	0.595	0.595	0.595
Separable, step	OL+BDC	0.468	0.588	0.588	0.588	0.588	0.588
	CO	0.637	0.567	0.567	0.567	0.567	0.567
	Hybrid	0.440	0.588	0.588	0.588	0.588	0.588
F AR(1) + graded, `covar`	OL	0.576	0.576	0.576	0.576	0.576	0.576
	OL+BDC	0.303	0.303	0.314	0.404	0.517	0.592
	CO	0.627	0.627	0.627	0.627	0.627	0.627
	Hybrid	0.291	0.291	0.302	0.386	0.486	0.559
C AR(1) + graded, separable	OL	0.567	0.567	0.567	0.567	0.567	0.567
	OL+BDC	0.468	0.469	0.476	0.526	0.578	0.609
	CO	0.637	0.637	0.637	0.637	0.637	0.637
	Hybrid	0.440	0.440	0.446	0.491	0.544	0.581
Graded alone (CS)	OL+BDC	0.262	0.262	0.265	0.289	0.348	0.469
	CO	0.256	0.256	0.256	0.256	0.256	0.256
	Hybrid	0.256	0.256	0.257	0.274	0.307	0.369

the higher ceiling; `docs/33`, Section 3.4, records that this is a property of the step rule, under which the interaction is zero at every positive half-life, so the higher ceiling carries no signal.]

The ceiling predicts every repair. A path's matrix fails the published check (`corpcor::is.positive.definite()`) if and only if $c_{bm} \geq c^*_{bm}$ for that path, since the response block is valid in every cell (Table 10). Counting one matrix per design path, c_{bm} and $t_{1/2}$ gives Table 15 (Source: `docs/36`, Section 6.5, where the count was checked against the `58b32a9` code's own repairs and against all 504 matrices; verified here from the ceilings).

Table 15: Matrices requiring repair, by configuration. Published grid: OL, OL+BDC, CO, Hybrid (nine paths), $c_{bm} \in \{0.3, 0.6\}$, $t_{1/2} \in \{0, 0.1, 0.2\}$. Common grid: the same paths, $c_{bm} \in \{0.25, 0.3\}$, $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$ (Source: `docs/36`, Section 6.5; verified).

Configuration	Published grid	Common grid
A published	20/54 (37%)	12/90 (13%)
B AR(1)	26/54 (48%)	3/90 (3%)
F AR(1) + graded, `covar`	26/54 (48%)	5/90 (6%)
C AR(1) + graded, separable	21/54 (39%)	0/90 (0%)
D, E mean moderation	0/54	0/90

The `covar` boundary at the published values. With $(c_1, c_\times) = (0.2, 0.1)$ the threshold of Equation (59) is $(c_1 - c_\times)/(1 - c_\times) = 1/9 = 0.111$. For equally spaced visits the smallest eigenvalue of A stays above $(1 - \rho^d)/(1 + \rho^d)$ (Section 9.10), so at $\rho = 0.8$ validity for any number of visits is guaranteed only for gaps $d \geq 1$ week, exactly the closest spacing of Hybrid and OL+BDC (derived; `docs/36`, Section 6.4). This is why the `covar` block in those designs

has smallest eigenvalue 0.010 (Table 10).

9.8 Schedule design under AR(1)

Under compound symmetry and binary coupling the ceiling depends only on n, n_{on} and the correlation parameters (Equation (57)); `docs/34` reports that it is 0.346 in all twenty 8-visit, 4-on schedules it examined, while the AR(1) ceiling ranges from 0.33 to 0.74 across the same schedules (quoted). Under AR(1) every schedule feature moves the ceiling (Table 16; `covar`, $\rho = 0.7$, no carryover).

Table 16: Effect of schedule features on the AR(1) (`covar`) ceiling, $\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$, no carryover (Source: `docs/34`, Section 5.2; verified except where marked quoted).

Feature	Effect
More visits (first half on, one switch)	Lowers it: 0.605, 0.564, 0.508, 0.434 for 4, 8, 16, 32 visits at 1-week spacing; 0.758, 0.619, 0.480, 0.358 at 2.5-week spacing. Compound symmetry: 0.480, 0.346, 0.247, 0.176
More on/off switches (16 visits, 8 on, equal blocks)	Lowers it sharply: 0.508, 0.358, 0.262, 0.149 for 1, 3, 7, 15 switches at 1-week spacing; 0.480, 0.428, 0.361, 0.278 at 2.5-week spacing
Shorter gap at the switch (8 visits, 4 on, study length 17.5 weeks)	Lowers it: 0.723, 0.619, 0.505, 0.331 at switch gaps of 8, 2.5, 1, 0.5 weeks
Uniform spacing (8 visits, 4 on)	Non-monotone: 0.564 at 1 week, 0.619 at 2.5, 0.494 at 10; at 0.5 weeks M is not positive definite
More on-drug visits (8 visits, 2.5-week spacing, one switch)	Lowers it: 0.867, 0.758, 0.678, 0.619, 0.573, 0.536, 0.506 for 1 to 7 on
Clustering on-drug visits	Raises it: 0.735 with 1-week gaps on drug and 4-week gaps off, against 0.555 for the reverse (quoted)
Order of on-drug visits	Matters: 0.526, 0.497, 0.447 for three orders with equal counts (quoted)

From the 4-to-32-visit rows, the compound-symmetry ceiling falls by 63%, the AR(1) ceiling by 28% at 1-week and 53% at 2.5-week spacing, so the two do not fall in similar proportion (`docs/34`, Section 8; verified). The number of switches is the most consequential decision, and its cost depends on spacing: going from one switch to three costs 0.150 at 1-week spacing and 0.052 at 2.5-week spacing. Crossing spacing and switch count for 16 visits, 8 on, reverses the spacing preference: with one switch closer spacing is better (0.524 at 1.5 weeks against 0.397 at 5), with fifteen it is worse (0.149 at 1 week against 0.331 at 5) (verified; `docs/34`, Section 5.3, reports that only 4 of its 24 grid designs can test 0.45, quoted). Replacing the paper 01 visit times by evenly spaced times over the same span, keeping the on/off order, raises the ceiling from 0.465 to 0.507 on Hybrid path C and from 0.452 to 0.545 on OL+BDC path A, which is 0.093 of the 0.167 gap between OL+BDC and CO (0.619) and 0.042 of the 0.154 gap for Hybrid path C (`docs/34`, Section 5.4; verified; Hybrid's binding path A was not equalized in the source).

A two-visit bound. For unit-variance B, BR_t, BR_s with $\text{Cor}(BR_t, BR_s) = A_{ts}$, the variable $BR_t - BR_s$ has variance $2(1 - A_{ts})$ and covariance $r_t - r_s$ with B, where $r_t = c_{bm} g_t$; Cauchy–Schwarz gives the necessary condition

$$|r_t - r_s| \le \sqrt{2(1 - A_{ts})} \tag{62}$$

(Source: `docs/34`, Section 5.5; derived). The exact condition from the 3×3 submatrix is $(r_t^2 + r_s^2 - 2 r_t r_s A_{ts})/(1 - A_{ts}^2) < 1$, which for $r_s = 0$ reads $|r_t| < \sqrt{1 - A_{ts}^2}$, tighter than Equation (62)

(0.714 against 0.775 at $A_{ts} = 0.7$; verified). Either form shows why covariance moderation, which needs a jump from c_{bm} to near zero between adjacent on and off visits, is constrained most where adjacent visits are most correlated; the exact accounting is Equation (61).

9.9 Shape of the coupling decay

If the off-drug coupling followed the Weibull family (Equation (19)), $g_t = \exp\{-(\lambda_w t_{sd,t})^k\}$ with $\lambda_w = (\ln 2)^{1/k}/t_{1/2}$, rather than the exponential, the shape would enter the ceiling only through the off-drug entries of g; it does not affect M and has no effect at $t_{1/2} = 0$, and changing the shape of the carryover *mean* alone leaves the ceiling unchanged because the means do not enter R (derived). Table 17 gives the `covar` ceilings (Source: `docs/34`, Section 5.6; verified). The package decays the correlation exponentially only, so this is a counterfactual.

Table 17: `covar` ceilings under Weibull-shaped coupling decay of shape k, $\rho = 0.7$. Multi-cycle: 16 weekly visits, on, on, off, off (Source: `docs/34`, Section 5.6; verified).

Design	$t_{1/2}$	$k = 0.25$	0.5	1 (exp.)	2	4	No carryover
Hybrid	0.5	0.543	0.531	0.506	0.469	0.456	0.456
Hybrid	1	0.551	0.551	0.544	0.527	0.518	
Hybrid	2	0.556	0.563	0.563	0.551	0.529	
OL+BDC	0.5	0.524	0.518	0.500	0.465	0.452	0.452
OL+BDC	1	0.530	0.533	0.533	0.523	0.516	
OL+BDC	2	0.534	0.540	0.546	0.542	0.526	
Multi-cycle	0.5	0.390	0.353	0.301	0.271	0.262	0.262
Multi-cycle	1	0.415	0.400	0.364	0.308	0.292	
Multi-cycle	2	0.440	0.448	0.451	0.430	0.395	
CO	any	0.619	0.619	0.619	0.619	0.619	0.619

At $t_{1/2} = 0.5$ and 1 the ceiling falls monotonically with k: faster washout acts like less carryover, and at $k = 4$, $t_{1/2} = 0.5$ the residual one week after stopping is 1.5×10^{-5}, so Hybrid, OL+BDC and the multi-cycle design return to their no-carryover ceilings. At $t_{1/2} = 2$ the ordering is not monotone, since the off-drug visits fall one to three weeks after discontinuation, where a small-k curve lies below the exponential; the residual at the visits actually sampled governs the ceiling. CO is unaffected because its binding path at positive half-lives is the placebo-first path. Because the no-carryover cell is the lowest across shapes and half-lives, a target below it is feasible whatever the shape (`docs/34`, Section 5.6; derived from the table).

The step rule under underflow. A Weibull curve never reaches zero analytically, so for $k \leq 2$ the step coupling is constant after exposure and the published-specification ceiling at $\rho = 0.7$ is 0.841, as under the exponential. At $k = 4$ the residual underflows to exactly zero in double precision (8.6×10^{-78} at two weeks and 0 at four weeks for $t_{1/2} = 0.5$), the step rule detects the zero, and the ceiling falls to 0.384 (Hybrid) and 0.467 (OL+BDC) at $t_{1/2} = 0.5$ and to 0.467 (Hybrid) at $t_{1/2} = 1$ (Source: `docs/34`, Section 5.6; verified). The ceiling of a step-coupled process under fast washout is then set by floating-point arithmetic, not by the model.

9.10 The `covar` spectrum, a spacing condition, and the published construct as two levels

Spectrum of `covar`. $K_b = (c_1 - c_\times)(J_3 - I_3)$ has eigenvalues $2(c_1 - c_\times)$ (once) and $-(c_1 - c_\times)$ (twice), and K_a and K_b share eigenvectors. Diagonalizing $A = Q\,\mathrm{diag}(\omega_1, \ldots, \omega_n)Q^\top$ and permuting to visit-major order turns M_{covar} into $\bigoplus_j (\omega_j K_a + K_b)$, so its eigenvalues are

$$\omega_j(1 + 2c_\times) + 2(c_1 - c_\times), \qquad \omega_j(1 - c_\times) - (c_1 - c_\times) \ \text{(twice)}, \qquad j = 1, \ldots, n, \tag{63}$$

from which Equation (59) follows (Source: `docs/35`, Section 4.4; derived; verified to 4×10^{-15} on Hybrid path 1). The white-noise term of Section 9.3 in the summed response is $s_t^\top K_b s_t = 2(c_1 - c_\times)(s_{TV}s_{PB} + s_{TV}s_{BR} + s_{PB}s_{BR}) > 0$ (derived).

A sufficient spacing condition. For equally spaced visits with step correlation ρ^d, the eigenvalues of the AR(1) Toeplitz matrix lie above the minimum of its spectral density, $(1 - \rho^d)/(1 + \rho^d)$, and approach it as n grows (Grenander and Szegő, 1958). Hence $M_{\text{covar}} \succ 0$ for every n if

$$\frac{1-\rho^d}{1+\rho^d} \geq \frac{c_1 - c_\times}{1 - c_\times} \iff d \geq \frac{\ln\{(1-k)/(1+k)\}}{\ln \rho}, \qquad k = \frac{c_1 - c_\times}{1 - c_\times} \tag{64}$$

(Source: `docs/35`, Section 4.4; derived). At $\rho = 0.7$ this requires gaps of at least 0.63 weeks for $(c_1, c_\times) = (0.2, 0.1)$, 0.70 weeks for $(0.3, 0.2)$ and 0.30 weeks for $(0.1, 0.05)$; at $\rho = 0.8$ and $(0.2, 0.1)$ exactly one week (verified). Consistently, on 16 visits at $\rho = 0.7$ the smallest eigenvalue of A is 0.090 at half-week spacing, below the thresholds 0.111 and 0.125 of the first two parameter sets, and 0.045 at quarter-week spacing, below the 0.053 of the third (verified). [`docs/35`, Section 4.4, gives 0.71 weeks for $(0.3, 0.2)$; the condition gives $\ln(7/9)/\ln 0.7 = 0.7046$, which rounds to 0.70.]

The published construct as person and occasion levels. Equation (56) is the covariance of $X_{c,t} = T_c + O_{c,t}$ with person-level effects $T \sim \mathcal{N}(0, K_p)$ and occasion-level effects $O_t \sim \mathcal{N}(0, K_o)$ independent across visits (Diggle, 1988). The implied cross-component correlations are

$$\text{Cor}(T_c, T_{c'}) = \frac{c_\times}{\rho_{cs}}, \qquad \text{Cor}(O_{c,t}, O_{c',t}) = \frac{c_1 - c_\times}{1 - \rho_{cs}}, \tag{65}$$

0.125 and 0.500 at the published $\rho_{cs} = 0.8$, 0.143 and 0.333 at $\rho_{cs} = 0.7$ (Source: `docs/35`, Section 4.5; derived; verified). The published construct is thus a coherent two-level model whose same-visit excess is carried by occasion-level variance; the `covar` construct keeps the excess but has no occasion-level variance to carry it, which is the indefinite K_b term.

One framework. All three response blocks are instances of the linear model of coregionalization, $M = \sum_j K_j \otimes A_j$ with correlation kernels A_j, which is positive semidefinite whenever every K_j is (each term is then a Kronecker product of positive semidefinite matrices). The published construct has two such terms, the separable construct one, and `covar` one valid term and one indefinite term (Source: `docs/35`, Section 4.7; derived). With only two components the separable factor term becomes $[K^{-1}]_{22} = 1/(1 - c_1^2)$, 1.042 at $c_1 = 0.2$ (`docs/35`, Section 7; derived); its consequence for the ceiling is Equation (68).

9.11 A four-visit worked example of the factored ceiling

Take four visits of Hybrid path A at weeks $(10, 11, 12, 16)$, on, off, off, on, with $\rho = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$ and $c_{bm} = 0.3$ (Source: `docs/36`, Appendices A.5 and A.7; verified). The gaps are $(1, 1, 4)$, so $\rho_t = (0.8, 0.8, 0.4096)$; A has eigenvalues 2.693, 0.825, 0.339, 0.143; and the smallest eigenvalue of the response block is 0.100 (published), 0.029 (`covar`) and 0.115 (separable). For `covar`, $\omega_{\min}(A) = 0.143$ exceeds the threshold 0.111, and $0.143 \times 0.9 - 0.1 = 0.029$ is the predicted smallest eigenvalue of Equation (63). With $[K^{-1}]_{33} = 1.0714$, Equation (61) decomposes the separable ceiling into per-step costs (Table 18).

Table 18: Per-step decomposition of $g^\top A^{-1} g$ and the separable ceiling, four-visit fragment (Source: `docs/36`, Appendix A.7.4; verified).

Coupling g	g_1^2	Step 1	Step 2	Step 3	$g^\top A^{-1} g$	c_{bm}^*
No carryover $(1, 0, 0, 1)$	1.000	1.778	0.000	1.202	3.979	0.4843
Graded, $t_{1/2} = 1$: $(1, 0.5, 0.25, 1)$	1.000	0.250	0.063	0.968	2.281	0.6397
Step, any carryover $(1, 1, 1, 1)$	1.000	0.111	0.111	0.419	1.641	0.7541

Without carryover the on-to-off step across one week costs $\rho_t^2/(1-\rho_t^2) = 1.778$ and the off-to-on step across four weeks $1/(1-\rho_t^2) = 1.202$; graded coupling softens both switches; the constant step-rule pattern pays almost nothing and carries no contrast. In the factored draw of Equation (71) for the graded pattern, the conditional-mean weights $c_{bm}[K^{-1}]_{c3}(A^{-1}g)_t$ on BR are 0.5357, −0.1607, −0.2759, 0.3467, those on TV and PB are $-1/6$ of them (−0.0893, 0.0268, 0.0460, −0.0578), removing the part of BR shared with the other components, and the conditional variance is $1 - c_{bm}^2\mathcal{Q} = 0.7801$, positive because $0.3 < 0.6397$ (verified). The ratio $-1/6$ is $[K^{-1}]_{13}/[K^{-1}]_{33} = -c_1/(1+c_1)$ (derived).

9.12 The number of response components and the ceiling

Papers 13 and 14 reduce the decomposition to two components (13 keeps $BR + TV$, 14 keeps $BR + PB$). With n_c components the joint correlation matrix is $(2+n_cn)\times(2+n_cn)$: 26 square at $n_c = 3$, $n = 8$, and 18 square at $n_c = 2$ (Source: `docs/31`, Sections 9 and 17; Paper 14, Appendix B.6). Both papers expected the reduction to relax the ceiling of Equation (55). It does not, and the reason is analytic.

Compound symmetry with n_c components. With common within-component correlation ρ_{cs}, same-visit cross-component correlation c_1 and different-visit cross-component correlation $c_\times$, the response block is $M = A_{n_c} \otimes I_n + B_{n_c} \otimes (J_n - I_n)$ in occasion-major order, with $A_{n_c} = I_{n_c} + c_1(J_{n_c} - I_{n_c})$ and $B_{n_c} = \rho_{cs}I_{n_c} + c_\times(J_{n_c} - I_{n_c})$; this is Equation (56) with n_c in place of 3 and the ordering of the Kronecker factors exchanged, which does not change the spectrum. Its eigenvalues are those of $A_{n_c} + (n-1)B_{n_c}$ (once each) and of $A_{n_c} - B_{n_c}$ (each with multiplicity $n-1$), and both are of the form $aI_{n_c} + b(J_{n_c} - I_{n_c})$, with eigenvalues $a + (n_c-1)b$ and $a - b$ (the latter $n_c - 1$ times). The four distinct eigenvalues are

$$\begin{aligned} &\big[1+(n-1)\rho_{cs}\big] + (n_c-1)\big[c_1 + (n-1)c_\times\big], \qquad &&\big[1+(n-1)\rho_{cs}\big] - \big[c_1 + (n-1)c_\times\big], \\ &(1-\rho_{cs}) + (n_c-1)(c_1 - c_\times), &&(1-\rho_{cs}) - (c_1 - c_\times), \end{aligned} \tag{66}$$

the second and fourth present only when $n_c \geq 2$ (derived; Paper 14, Appendix C.2; verified for $n_c = 1$ to 4 and $n = 4, 8, 12$ at $\rho_{cs} = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$, `check-landmarks-hybrid-ceilings.R`, block 7). At those values and $n_c = 3$, $n = 8$ they are 7.7, 5.0, 0.5 and 0.2. When $n_c \geq 2$, $c_\times \leq \rho_{cs}$ and $c_1 \geq c_\times$,

$$\omega_{\min}(M) = (1-\rho_{cs}) - (c_1 - c_\times), \tag{67}$$

which contains neither the number of visits nor the number of components: removing a component does not move the positive-definiteness boundary of the response block (Source: `docs/31`, Section 20; Paper 14, Appendix C.3; derived). As in §9.3, $\omega_{\min}$ denotes the smallest eigenvalue. [2026-10-09: notation. This subsection wrote the eigenvalue as $\lambda_{\min}$, which collides with the carryover decay rate λ, and used K both for the number of response components and for the cross-component correlation matrix; the count is now n_c and K is the matrix only.] Positive definiteness of the block requires $\rho_{cs} < 1 - (c_1 - c_\times)$, that is $\rho_{cs} < 0.9$ at the reference values. [`docs/31`, Section 20, states Equation (67) without conditions, and Paper 14 says only that it is "generally" the smallest. Both conditions are needed: with $c_\times > \rho_{cs}$ the second eigenvalue of Equation (66) is smaller (at $\rho_{cs} = 0.1$, $c_1 = 0.2$, $c_\times = 0.15$, $n_c = 3$, $n = 8$ the smallest eigenvalue is 0.45, not 0.85), and with $c_1 < c_\times$ the third is (at $\rho_{cs} = 0.7$, $c_1 = 0.05$, $c_\times = 0.2$ it is 0, against 0.45 from Equation (67)); verified, block 7. With $n_c = 1$ the smallest eigenvalue is $1 - \rho_{cs}$.]

Numerical ceilings: three against two components. The ceiling itself depends on the coupling pattern and the schedule, not only on $\omega_{\min}$. Under the package's non-separable AR(1) construct M_{covar} (Equation (59); within-component $\rho^{|w_t - w_s|}$, cross-component c_1 at the same visit and $c_\times\rho^{|w_t-w_s|}$ otherwise) and graded coupling, the design ceiling, minimized over paths, is given in Table 19; the three-component values are the `covar` row of Table 9 and are not

Table 19: Ceiling c^*_{bm} under the non-separable AR(1) construct with two components, and its gain over the three-component ceiling of Table 9 (`covar` row), graded coupling, $\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$, $n = 8$, minimum over paths (Source: Paper 14, Appendix C.4, quoted in `docs/31`, Section 20; all entries verified, `check-landmarks-hybrid-ceilings.R`, block 8).

Design	$t_{1/2}$	Two components	Gain
CO	0, 0.5, 1	0.627	0.008
Hybrid	0	0.469	0.013
Hybrid	0.5	0.516	0.011
Hybrid	1	0.552	0.008
OL+BDC	0	0.464	0.012
OL+BDC	0.5	0.509	0.009
OL+BDC	1	0.540	0.007

repeated here. The two two-component reductions give the same matrix, since all components share ρ and the biomarker couples to BR only.

The reduction buys between 0.007 and 0.013 of feasible c_{bm}. [`docs/31`, Section 20, and Paper 14, Appendix C.4, give the range as 0.008 to 0.013. The unrounded gains run from 0.0069 (OL+BDC, $t_{1/2} = 1$) to 0.0130 (Hybrid, $t_{1/2} = 0$), and Paper 14's own rounded table gives $0.540 - 0.533 = 0.007$ for the OL+BDC cell.] The ceilings rise with $t_{1/2}$ in Hybrid and OL+BDC and are constant in CO, where the placebo-first path binds (Section 14.11). Paper 14 also reports that, holding the design shape fixed, the three-component ceiling falls from about 0.81 at $n = 4$ to 0.58 at $n = 12$, and that the occasion-index exponent $\rho^{|j-j'|}$ in place of the week gap lowers the worst-cell ceiling from about 0.45 to 0.35 [quoted; not recomputed, because the "fixed design shape" at $n = 4$ and $n = 12$ is not specified in the source]. These ceilings belong to M_{covar} at $\rho = 0.7$ and are not comparable with the compound-symmetry and separable entries of Table 8 (`docs/31`, Section 20).

The separable form: a design-free ratio. Under the separable covariance of Equation (58) with n_c equicorrelated components, $K_{n_c} = (1 - c_1)I_{n_c} + c_1 J_{n_c}$ has $[K^{-1}_{n_c}]_{n_c n_c} = \frac{1+(n_c-2)c_1}{(1-c_1)(1+(n_c-1)c_1)}$, so Equation (60) holds with

$$R^2_{n_c} = \frac{(n_c - 1)\, c_1^2}{1 + (n_c - 2)\, c_1}, \qquad \frac{c^*_{bm}(n_c = 2)}{c^*_{bm}(n_c = 3)} = \sqrt{\frac{1 - R^2_2}{1 - R^2_3}} = \sqrt{\frac{(1 - c_1^2)(1 + c_1)}{1 + c_1 - 2c_1^2}}, \tag{68}$$

which reduces to Equation (60) at $n_c = 3$, and in which the schedule, the coupling pattern and ρ cancel (derived here, as a corollary of Equation (60); not stated in the sources). The ratio is 1.0042, 1.0142 and 1.0277 at $c_1 = 0.1$, 0.2 and 0.3; at $\rho = 0.7$, $c_1 = 0.2$ and $t_{1/2} = 0$ the separable design ceilings move from 0.616, 0.480 and 0.474 (CO, Hybrid, OL+BDC) to 0.625, 0.487 and 0.481 (verified, block 9). The reduction's gain is therefore at most a few percent under either construct, and `docs/31`'s conclusion that the case for a reduction is transparency and cost, not numerical headroom, holds under both.

9.13 The outcome covariance: a random intercept plus AR(1)

Let $s_t = (s_{TV,t}, s_{PB,t}, s_{BR,t})^\top$ be the component standard deviations at visit t and $S_t = \sum_c s_{c,t} X_{c,t}$ the summed response, so that $Y_{it} = \text{BL}_i - S_{it}$ with BL_i independent of the responses. Under Equation (58),

$$\text{Cov}(Y_{it}, Y_{is}) = \sigma^2_{BL} + (s_t^\top K s_s)\, A_{ts}, \tag{69}$$

which, when the component standard deviations are constant over visits, is a random intercept plus an AR(1) process: the residual structure of the random-intercept-plus-`corCAR1` analysis

model (Equation (136)), not of `corCAR1` alone. The placebo-belief standard deviation scales with the visit's expectancy weight η_t (1 open label, 0.5 blinded), so the match is close but not exact; `docs/35` reports a correlation factor of 0.976 between an open-label and a blinded visit, recomputed as 0.9755 in Equation (191) of §19.4. Under M_{covar} the same calculation adds $(s_t^\top K_b s_t)\,\mathbf{1}[t = s]$, which is positive, a white-noise term the analysis model omits. (Derived. Source: `docs/35`, Section 4.6; Paper 01, Appendix A.5.) [The 0.976 factor is confirmed in §19.4 (0.9755, configuration C, $\rho = 0.7$; verified, `check-stress-test-construct.R`, Section 5), which also shows that a random intercept with phase-specific residual variances reproduces every element of this covariance except the open-label by blinded pairs. The effect of the white-noise term on test calibration has not been measured in any source.]

9.14 The generalized ceiling with prognostic and baseline couplings

When the biomarker is also coupled to TV, PB or the baseline, write its correlations with the three components over the visits as vectors $\mathbf{r}_{TV}$, $\mathbf{r}_{PB}$, $\mathbf{r}_{BR}$ (with $\mathbf{r}_{BR} = c_{bm}\, g$) and its correlation with BL as $c_{bm,BL}$. Because BL is uncorrelated with the responses, the block of R below the biomarker row is $\operatorname{diag}(1, M)$, and the Schur complement with $M^{-1} = K^{-1} \otimes A^{-1}$ gives

$$R \succ 0 \iff M \succ 0 \;\text{ and }\; c_{bm,BL}^2 + \sum_{c,c'} [K^{-1}]_{cc'}\, \mathbf{r}_c^\top A^{-1} \mathbf{r}_{c'} < 1. \tag{70}$$

Equation (60) is the special case $c_{bm,BL} = 0$, $\mathbf{r}_{TV} = \mathbf{r}_{PB} = 0$. The off-diagonal elements of K^{-1} are negative, $-c_1/\big((1 - c_1)(1 + 2c_1)\big) = -0.179$ at $c_1 = 0.2$, so same-sign couplings to positively correlated components cost less than the sum of their separate costs. On Hybrid path A at $\rho = 0.7$, $c_1 = 0.2$, coupling 0.3 to BR alone costs 0.390, a constant prognostic coupling $\mathbf{r}_{TV} = 0.3 \cdot \mathbf{1}$ alone costs 0.342, and both together 0.663: the BR–TV cross term is -0.070. Adding $c_{bm,BL} = 0.3$ gives 0.7525, and the full matrix gives the same value, so these stress-test cells are feasible. (Derived; verified, `check-ceilings.R`. Source: `docs/35`, Section 4.8.) [`docs/35`, Section 4.8, does not state ρ for these figures; they reproduce at $\rho = 0.7$.] A biomarker that is predictive and prognostic in the same direction is therefore easier to accommodate than either coupling alone suggests.

9.15 Silent repair, and why the generator must stop

The published driver calls `generateData()` with `makePositiveDefinite = TRUE`, and a matrix that fails the check `corpcor::is.positive.definite()` is replaced, without notice, by `make.positive.definite(sigma, tol = 1e-3)`. At $\rho = 0.8$ and $c_{bm} = 0.6$ the repair lowers the on-drug biomarker–response correlation to 0.540–0.568, introduces 0.029–0.042 at off-drug visits, and moves other correlations by up to 0.097; at $c_{bm} = 0.3$ the largest change is 0.007 (`docs/36`, Section 4.3). The package keeps two repairs: `buildSigma()` repairs under the same flag and warns only when `verbose = TRUE`, and the Cholesky step falls back to a repaired matrix whether or not the flag is set (`R/generateData.R`, lines 377–388 and 391–393; inspected 2026-10-08). Just above the ceiling the perturbation is at most 0.004 (`docs/34`, Section 4.7, at $\rho = 0.7$), small enough to pass unnoticed.

Stopping is required, not merely preferable, for three reasons. Above c_{bm}^* no joint distribution with the stated correlations exists, so a repaired matrix is a different model, and the power reported for the requested c_{bm} belongs to a data-generating process that was never stated. The repair changes the effect in a direction and amount that depend on design, path and c_{bm}, and it can move coupling into off-drug visits, so the resulting bias cannot be undone after the fact. And because the step rule makes exactly the signal-free cells feasible (Section 9.2), repair counts and power together can describe a construct quite unlike the one in the Methods. Paper 01's refinement 1 (Appendix B.2) replaces the repair by a `stop()`. Under the separable structure

the condition is part of generation itself: drawing three AR(1) recursions, mixing them by L_K, and drawing the biomarker from

$$B_i \mid \mathbf{X} \sim \mathcal{N}\Big(c_{bm} \sum_c [K^{-1}]_{c3}\, X_c^\top A^{-1} g,\;\; 1 - c_{bm}^2\, \mathcal{Q}\Big), \qquad \mathcal{Q} = [K^{-1}]_{33}\, g^\top A^{-1} g, \tag{71}$$

gives a conditional variance that is positive exactly when $c_{bm} < \mathcal{Q}^{-1/2} = c_{bm}^*$, so an infeasible effect size cannot be simulated. (Derived; the generator was checked against its target distribution in `docs/36`, Section 6.7, where a request of $c_{bm} = 0.6$ against the ceiling of Hybrid path A under graded coupling at $t_{1/2} = 0.5$, separable, $\rho = 0.8$, is refused; that ceiling is 0.491, 0.490934, as in Table 14.) [2026-10-08: corrected from 0.494. `docs/36`, Section 6.7, and its check script, `07-factored-generator.R` in `analysis/scripts/quick-sim/cbm-ceiling/` (line 75), enter the coupling at visit 8 as 0.0625; that visit has $t_{sd} = 4$, so graded coupling at $t_{1/2} = 0.5$ gives $2^{-8} = 0.0039$. With the source's vector the ceiling is 0.494329, as reported; with the correct vector it is 0.490934 (verified, `check-closed-form-extensions.R`; the same value is in `design-power.csv`). The generator check itself (closed form against full matrix, sampled against target correlations) is unaffected, since it is internally consistent for whatever vector is supplied, and the refusal of $c_{bm} = 0.6$ stands.] [The factored generator and a pre-simulation ceiling check exist only in the simulation drivers. As of `docs/35`, Section 10, and the inspection above, the package's sampler still repairs silently; steps 1 to 3 of the `docs/35` implementation plan are pending.]

9.16 The 2024 revision of the published code: which change fixes the ceiling

The 2024 code couples the biomarker by the ratio-graded rule of Equation (39) and leaves the first visit uncoupled (§6.6). **The crossed decomposition.** The source builds one correlation matrix with four switches that turn the 2024 construct into the package's `covar` construct: `ar1` (AR(1) within components instead of compound symmetry), `cfd` (cross-component block $c_\times \rho^{|w_t - w_s|}$ instead of the constant $c_\times$), `bexp` (graded coupling instead of Equation (39)) and `occ1` (the first visit coupled). Table 20 gives the exact worst-cell ceiling of each of the 16 combinations over three designs and $t_{1/2} \in \{0, 0.5, 1\}$ at $\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$, and Table 21 the cells for four of them; its `covar` column is the first row of Table 9 to four decimals, repeated as the reference for the comparison. (Source: `docs/54`, Section 5, on a grid of step 0.01; recomputed exactly here, and every grid value of the source's Tables 5.2 and 5.3 is the floor of the exact value, `check-stress-test-construct.R`, Section 2.)

Table 20: Worst-cell ceiling over CO, Hybrid and OL+BDC and $t_{1/2} \in \{0, 0.5, 1\}$, by configuration, $\rho = 0.7$ (Source: `docs/54`, Section 5; exact values, verified). A switch marked 1 is on; all off is the 2024 construct, all on is `covar`.

ar1	cfd	bexp	occ1	Ceiling	ar1	cfd	bexp	occ1	Ceiling
1	0	0	0	0.4818	0	0	0	0	0.3460
1	0	1	0	0.4818	0	0	1	0	0.3460
1	1	0	0	0.4810	0	0	0	1	0.3460
1	1	1	0	0.4810	0	0	1	1	0.3460
1	1	0	1	0.4522	0	1	0	0	0.1645
1	1	1	1	0.4522	0	1	1	0	0.1645
1	0	0	1	0.4379	0	1	0	1	0.1512
1	0	1	1	0.4379	0	1	1	1	0.1512

The results, with how each was established:

(a) **AR(1) is the change that matters, but it is not better than `covar`.** Every configuration with `ar1` on reaches at least 0.438, and every one with it off is at most 0.346. The source's original claim that AR(1) alone (0.48) beats the full set (0.45) holds

Table 21: Ceiling by design and $t_{1/2}$ (weeks) for the 2024 construct, AR(1) alone with the first visit uncoupled (the source's "AR(1) only"), AR(1) alone with it coupled, and the full `covar` set, $\rho = 0.7$ (Source: `docs/54`, Section 5.3 and its 2026-10-07 note; exact values, verified).

Design	$t_{1/2}$	2024 construct	AR(1), $g_1 = 0$	AR(1), g_1 coupled	`covar`
CO	0, 0.5, 1	0.3460	0.5909	0.5909	0.6189
Hybrid	0	0.3460	0.4974	0.4521	0.4562
	0.5	0.3511	0.5119	0.4629	0.5059
	1	0.3682	0.5498	0.4899	0.5441
OL+BDC	0	0.3460	0.4818	0.4379	0.4522
	0.5	0.3511	0.4950	0.4477	0.4996
	1	0.3698	0.5286	0.4718	0.5332

only with the first visit uncoupled. With g_1 coupled, AR(1) alone reaches 0.4379 against `covar`'s 0.4522 (worst cell, both in OL+BDC at $t_{1/2} = 0$), and 0.4521 against 0.4562 in Hybrid (verified; this confirms the 2026-10-07 correction of the source and `docs/52`).

(b) **Why an uncoupled first visit raises the ceiling.** Setting an on-drug g_1 to zero removes a term from $\tilde{g}^\top M^{-1}\tilde{g}$ in Equation (55); under AR(1) it removes the g_1^2 term of Equation (61) and lowers the first step's cost, so the quadratic form falls and the ceiling rises (derived). The gain is feasibility for a different model, one with less coupling, not additional range for the stated one.

(c) **Under compound symmetry the first visit does not move the worst cell.** For a binary pattern the ceiling depends only on n and n_{on} (Equation (57)). The binding path is CO's placebo-first path, at 0.3460 for every $t_{1/2}$, whose first visit is before exposure and so uncoupled under every rule; the drug-first path is at 0.360 to 0.363 (verified; the source states the binding path as inferred).

(d) **The decaying cross-component block lowers the ceiling under compound symmetry** (0.346 to 0.165, or 0.151 with g_1 coupled) and changes it by -0.001 to $+0.014$ under AR(1). The response block is positive definite in every configuration (smallest eigenvalues 0.015 to 0.331 over designs), so the biomarker row is always the binding constraint (verified).

(e) **The 2024 construct's exact ceilings** at $\rho = 0.7$ are 0.346 without carryover and in CO, 0.351 at $t_{1/2} = 0.5$ in Hybrid and OL+BDC, and 0.368 (Hybrid) and 0.370 (OL+BDC) at $t_{1/2} = 1$ (verified; the first two are quoted in Section 4.2 of the source).

["AR(1) alone" names two different constructs in the sources: in `docs/36` and `docs/44` (configuration B) it keeps the decaying cross-component block, and in `docs/54` it keeps the constant block. At $\rho = 0.7$, no carryover and the first visit coupled, they give 0.452 and 0.438 in OL+BDC.]

10 A pharmacodynamic sensitivity model

The constructs of §6 specify carryover twice: once for the off-drug mean and once, through a separate rule, for the biomarker correlation (Equation (31)). The proposal of `docs/38` specifies it once, in a generative model of each participant's drug response, from which the mean, the moderation profile and the covariance all follow. Every result in this section is derived; none has been implemented in `R/generateData.R` or simulated in the compendium. (Source: `docs/38`, Sections 1–6; `docs/44`, Step 8; `docs/51`, Features 2–5 and 8.) The symbols ω_i, ζ, v_i and ξ_{it} are local; they replace the θ, κ, u and ε of the first draft of `docs/38`, which collide with canonical symbols. `docs/51` writes the serial noise as ε_{it} and its SD as σ_ε; the ξ_{it} of `docs/38` is used here.

10.1 The drug-response time course: a turnover model

For a weekly symptom outcome, pharmacokinetic carryover is negligible for prazosin (plasma half-life about 2.3 hours, Hobbs et al., 1978), so the half-life that matters is that of the response, written $t_{1/2}^{R}$ to distinguish it from the $t_{1/2}$ of the published construct. Let $h(t) \in [0, 1]$ be the population drug-response time course, scaled to 1 at full on-drug effect, and let $h^*(t)$ be the drug-driven target, 1 on drug and 0 off drug (written $E(t)$ in `docs/51`). The turnover, or indirect-response, model (Dayneka et al., 1993) is

$$\frac{dh}{dt} = k_{\text{out}}\big(h^*(t) - h(t)\big), \qquad k_{\text{out}} = \frac{\ln 2}{t_{1/2}^{R}}. \tag{72}$$

For a target constant within each on-drug or off-drug interval the solution is exact and piecewise. Off drug, from the level $h(t_{\text{stop}})$ reached at discontinuation,

$$h(t) = h(t_{\text{stop}})\, 2^{-t_{sd}/t_{1/2}^{R}}, \tag{73}$$

and on drug, from a starting level h_0 at time t_{in} into the interval,

$$h(t) = 1 - (1 - h_0)\, 2^{-t_{\text{in}}/t_{1/2}^{R}}. \tag{74}$$

[2026-10-09: notation. The drug-response time course was written $f(t)$, with f_0, f_{res} and f^*; it is $h(t)$, h_0, h_{res}, h^*, as in `docs/44` and `docs/51`, because f is the canonical absorbed share (§17). t_{in} (written τ_{od}) is the time into the current on-drug interval.] Both follow from the linear first-order equation with constant forcing (derived; the closed form agrees with an Euler solution of Equation (72) to 1.2×10^{-5} on a six-week on, four-week off schedule, `check-pd-features.R`). Three properties follow, each of which differs from the published process.

- **Anchored decay.** Equation (73) decays from the level at discontinuation. The published recursion (compare Equation (21)) multiplies the previous visit's already-decayed mean by $2^{-t_{sd}/t_{1/2}}$ with t_{sd} cumulative, so at off-drug visits one and two weeks after stopping the second sits at $2^{-(1+2)/t_{1/2}}$ rather than $2^{-2/t_{1/2}}$ (derived; `docs/37`, Appendix A).
- **Restart from the residual.** On re-exposure, Equation (74) applies with $h_0 = h_{\text{res}}$, the residual level. The package instead resets t_{od} to zero: under the modified Gompertz curve (Equation (2), $m = 10.99$, $d = 5$, $r = 0.42$) a participant near full response at week 10 of the Hybrid design is credited with $G(4) = 4.28$ of 10.99 (39%) at week 16 (`docs/51`, Feature 4; the value $G(4)$ verified in `check-pd-features.R`, the reset to $t_{od} = 4$ at week 16 taken from the source, not rechecked against the design). Anchored offset combined with a zero restart can step the mean down at re-exposure.
- **One rate for onset and offset.** The turnover onset is exponential, not sigmoid; a sigmoid onset arises only if the target h^* itself rises with titration. The Gompertz landmarks it replaces are an inflection at $\ln(d)/r = 3.8$ weeks, half the maximum at 4.7 weeks and 90% at 9.2 weeks (verified, `check-pd-features.R`; `docs/51`, Feature 2).

10.2 Sensitivity binding of the biomarker

A predictive biomarker acts on drug sensitivity. Write participant i's drug response as

$$BR_{it} = \omega_i\, h(t) + \xi_{it}, \qquad \omega_i = \bar{\omega}\,(1 + \zeta\, b_i) + v_i, \tag{75}$$

with b_i the standardized biomarker, ζ the strength of multiplicative moderation, $v_i \sim \mathcal{N}(0, \sigma_v^2)$ the sensitivity variation the biomarker does not explain, and ξ_{it} serial noise with SD σ_ξ and

correlation $r_\xi(t,s)$, independent of b_i and v_i. The mean response is $\bar{\omega}h(t)$ and the biomarker-dependent part, $\bar{\omega}\zeta b_i h(t)$, has the time course of the response. The *coupling profile* (the fraction of the full moderation present at visit t, g_t in `docs/37`) is therefore

$$g_t = h(t) : \tag{76}$$

small early in titration, rising with the response on drug, and decaying with it off drug. It is neither the published step rule (full moderation wherever any residual mean remains) nor the on-drug-only shift of the mean-moderation construct. The second moments follow (derived; `docs/38`, Section 5):

$$\mathrm{Var}(BR_t) = H\,h(t)^2 + \sigma_\xi^2, \qquad H = \bar{\omega}^2\zeta^2 + \sigma_v^2, \tag{77}$$

$$\mathrm{Cov}(BR_t, BR_s) = H\,h(t)h(s) + \sigma_\xi^2\, r_\xi(t,s), \tag{78}$$

$$\mathrm{Cor}(B, BR_t) = \frac{\bar{\omega}\zeta\, h(t)}{\sqrt{Hh(t)^2 + \sigma_\xi^2}}. \tag{79}$$

The covariance is a rank-one positive semidefinite term plus a positive definite noise covariance, so it is valid for every design, schedule and half-life; correlations are outputs, and there is no ceiling on the effect size. The drug-response variance shrinks to σ_ξ^2 as the effect washes out, whereas every current DGP keeps the full σ_{BR} (8 in Hendrickson et al.) at off-drug visits with zero mean (`docs/51`, Feature 8). The published c_{bm} becomes a derived quantity, most naturally the correlation at full effect,

$$c_{bm}^{\mathrm{peak}} = \frac{\bar{\omega}\zeta}{\sqrt{H + \sigma_\xi^2}}. \tag{80}$$

Equations (77)–(79) were confirmed by simulation of Equation (75) with 4×10^5 participants ($\bar{\omega} = 10$, $\zeta = 0.3$, $\sigma_v = 2$, $\sigma_\xi = 3$, $r_\xi = 0.5^{|t-s|}$; simulated $\mathrm{Cor}(B, BR_6) = 0.635$ against 0.635; `check-pd-features.R`).

10.3 Consequences for the interaction test

With the paired-difference statistic E9 (Equation (141)) written as $\Delta_i = \sum_t a_t Y_{it}$, contrast weights $a_t = 1/n^{\mathrm{on}}$ on drug and $-1/n^{\mathrm{off}}$ off drug, define the contrast-weighted exposure

$$F = \sum_t a_t h(t) = \bar{h}_{\mathrm{on}} - \bar{h}_{\mathrm{off}}. \tag{81}$$

Because $Y = \mathrm{BL} - (TV + PB + BR)$, the drug response enters Δ_i as $-\omega_i F - \sum_t a_t \xi_{it}$. Writing V_0 for the variance contributed by the serial noise and the other components, assumed independent of ω_i,

$$\beta_{bm:D} = -\frac{\bar{\omega}\zeta\, F}{\sigma_{bm}}, \qquad \mathrm{Var}(\Delta) = HF^2 + V_0, \qquad r_{\Delta B} = -\frac{\bar{\omega}\zeta\, F}{\sqrt{HF^2 + V_0}} \tag{82}$$

(derived; `docs/38`, Section 6, which writes $\rho_{\Delta B}$; $r_{\Delta B}$ is used here, as in §14, because ρ is the AR(1) parameter). In the simulation above ($F = 0.605$) the simulated slope, variance and correlation were -0.907, 18.19 and -0.426 against -0.908, 18.20 and -0.426 (verified, `check-pd-features.R`). [`docs/38`'s first draft gave $r_{\Delta B}$ without the minus sign; the sign follows from that of $\beta_{bm:D}$ and was corrected in the source on 2026-10-07. Power depends only on $|r_{\Delta B}|$.]

Power is monotone in $|F|$. $|r_{\Delta B}| = \bar{\omega}\zeta|F|/\sqrt{HF^2 + V_0}$ has derivative $\bar{\omega}\zeta V_0/(HF^2 + V_0)^{3/2} > 0$ in $|F|$ for $V_0 > 0$, and power is increasing in $|r_{\Delta B}|$ (`docs/37`, Section 4.4), so power is strictly increasing in $|F|$ (derived). Under anchored offset with the on-drug course held fixed, each off-drug $h(t)$ increases with $t_{1/2}^R$, so power falls monotonically with the response half-life. Under the full turnover model a longer $t_{1/2}^R$ also slows onset and raises the restart level, so F

need not be monotone in $t^R_{1/2}$ and must be checked per design. Unlike the published process, $\mathrm{Var}(\Delta)$ now depends on carryover through HF^2.

Reduction to graded coupling. Suppose $h(t) = 1$ at every on-drug visit and at discontinuation, and Equation (73) holds off drug with $t^R_{1/2} = t_{1/2}$. Then g_t of Equation (76) is the graded coupling profile of `docs/37` (1 on drug, $2^{-t_{sd}/t_{1/2}}$ off drug), $F = 1 - \bar{g}_{\text{off}}$ is its coupling gap, and Equation (82) becomes

$$\beta_{bm:D} = -\frac{\bar{\omega}\zeta}{\sigma_{bm}}\left(1 - \bar{g}_{\text{off}}\right) = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\left(1 - \bar{g}_{\text{off}}\right), \tag{83}$$

the coupling-gap result of `docs/37`, with the identification $c_{bm}\sigma_{BR} = \bar{\omega}\zeta$. This identification is exact when σ_{BR} is read as the full-effect SD $\sqrt{H + \sigma^2_\xi}$, by Equation (80) (derived). The two models still differ in the variance: the graded construct keeps $\mathrm{Var}(BR_t)$ constant, while Equation (77) lets it shrink off drug. [Under the pure turnover model, onset and offset share k_{out}, so $h = 1$ at every on-drug visit together with slow offset is not attainable; the reduction is exact for a construct with saturated onset and anchored turnover offset, and approximate otherwise. The mapping of `docs/37`'s $(1 - \bar{g}_{\text{off}})$ onto F is derived here, not stated in `docs/38` beyond its remark that F replaces the coupling gap.]

The power cap. For every F,

$$|r_{\Delta B}| < \frac{\bar{\omega}\zeta}{\sqrt{\bar{\omega}^2\zeta^2 + \sigma^2_v}}, \tag{84}$$

the square root of the share of sensitivity variation that the biomarker explains, approached as HF^2 dominates V_0. Since $f \in [0, 1]$ implies $|F| \leq 1$, the attainable bound is the lower value $\bar{\omega}\zeta/\sqrt{H + V_0}$ (in the simulation above, 0.832 and 0.583 respectively). No design and no exposure contrast can exceed Equation (84); it is a property of the biology encoded by σ_v, not of the simulation (derived; `docs/38`, Section 6).

10.4 Sensitivity binding against the other encodings

Sensitivity binding (Equation (75)) can be read in the terms of both other encodings. Write $G = \bar{\omega}\zeta$, $H = G^2 + \sigma^2_v$, Σ_ξ for the covariance of the serial noise ($\sigma^2_\xi r_\xi(t, s)$, positive definite), and h for the vector of $h(t)$ over the visits.

Conditional on the biomarker it is mean binding plus a random slope. Given b_i,

$$E(BR_{it} \mid b_i) = \bar{\omega}h(t) + G\, b_i\, h(t), \qquad \mathrm{Cov}(BR_{i\cdot} \mid b_i) = \sigma^2_v\, hh^\top + \Sigma_\xi \tag{85}$$

(derived; verified by simulation of 4×10^5 participants at $\bar{\omega} = 10$, $\zeta = 0.3$, $\sigma_v = 2$, $\sigma_\xi = 3$, $r_\xi = 0.5^{|t-s|}$ and an arbitrary profile h: conditional slope 3.00 against $Gf_4 = 3$, conditional covariance within 0.044 of Equation (85) on entries of order 10, within sampling error). This is mean binding (Equation (26)) with profile $h_t = h(t)$ and strength $\beta_{bm} = G/\sigma_{BR}$, with one addition: the unexplained sensitivity v_i contributes the rank-one term $\sigma^2_v hh^\top$, a random slope on the exposure profile. With $\sigma_v = 0$ it is exactly mean binding with profile h.

Unconditionally it is covariance binding with an implied coupling. The joint law of $(B_i, BR_{i\cdot})$ is normal with $\mathrm{Cov}(B, BR_t) = \sigma_{bm}Gh(t)$ and $\mathrm{Cov}(BR_{i\cdot}) = \Sigma_{BR} = Hhh^\top + \Sigma_\xi$ (Equations (77)–(79)), so it is a covariance construct whose coupling pattern, relative to the full-effect correlation of Equation (80), is

$$\frac{\mathrm{Cor}(B, BR_t)}{c^{\text{peak}}_{bm}} = h(t)\sqrt{\frac{H + \sigma^2_\xi}{Hh(t)^2 + \sigma^2_\xi}} \geq h(t), \tag{86}$$

equal to $h(t)$ only at $h = 0$ and $h = 1$ (derived). The implied correlation profile is therefore not proportional to the response, because the drug-response SD shrinks with it; in the simulation above, $h = 0.5$ gives a relative coupling of 0.670 (verified). The covariance encoding of Section 6.3 holds the SD at σ_{BR} and the coupling at g_t; the two coincide only in the reduction of Equation (83), where the comparison is made on the slope scale.

The Schur-complement ceiling is never reached. The squared multiple correlation of the biomarker on the drug-response trajectory is, by the Sherman–Morrison formula with $\mathcal{Q}_\xi = h^\top \Sigma_\xi^{-1} h$,

$$R^2_{B|BR} = G^2 h^\top \left(Hhh^\top + \Sigma_\xi\right)^{-1} h = \frac{G^2 \mathcal{Q}_\xi}{1 + H\mathcal{Q}_\xi} < \frac{G^2}{H} = \frac{\bar{\omega}^2 \zeta^2}{\bar{\omega}^2 \zeta^2 + \sigma_v^2} \leq 1 \tag{87}$$

(derived; 0.5280 both ways in the simulation's parameters, against the bound 0.6923, verified). The Schur-complement condition that defines the ceiling in Equation (55), $1 - R^2_{B|BR} > 0$, holds for every schedule, profile and strength, strictly even when $\sigma_v = 0$ because $\mathcal{Q}_\xi$ is finite. Under sensitivity binding the ceiling condition is a consequence of the model, not a constraint on its parameters. The bound G^2/H is the square of the power cap of Equation (84): the share of sensitivity variation that the biomarker explains bounds both how well the biomarker can be predicted from the response trajectory and how strongly it can correlate with any contrast of it. If the other response components are independent of ω_i and b_i, as assumed in Equation (82), adding them leaves $R^2_{B|BR}$ unchanged; correlated serial noise across components changes $\mathcal{Q}_\xi$ but not the bound (derived).

Feature comparison. Table 22 sets the proposal against the constructs in use (Source: `docs/38`, Section 7; argued, with the `covar` mean and restart entries inspected in the package source by the source authors).

Table 22: Constructs compared (Source: `docs/38`, Section 7). `orig`: published `58b32a9`; C: graded separable (`docs/36`); D: mean moderation. Updated 2026-10-09: the generator of C and D (`04-power-simulation.R`) now uses anchored carryover by default; the source's "recursive" describes the runs stored before that date.

Feature	`orig`	`covar`	C	D	Sensitivity binding
Off-drug mean	recursive	anchored	recursive	recursive	anchored, turnover
Restart after off-drug	from zero	from zero	from zero	from zero	from residual
Moderation on drug	full	full	full	full	proportional to h
Moderation off drug	full if any residual (step)	decays	decays	none	proportional to h
Drug-response variance off drug	full	full	full	full	shrinks with h
Valid for every schedule	no	no	block yes, ceiling remains	yes	yes
Ceiling on effect size	yes	yes	yes	none	none

11 Further features of a data-generating process

`docs/51`, Features 11–17, identifies seven properties of a trial that no current DGP of the compendium represents. This section carries the six with derived results (Features 11–16); Feature 17, the bounded outcome scale, is argued only in the source and is omitted. Each result is marked derived (analytic) or argued; the numerical checks cited are of the formulas themselves, by the script `check-pd-features.R` written for this section, and none of the features has been simulated in the compendium's DGPs. Two further options of the data-generating process close the section: the phase-specific measurement error of `docs/51`, Feature 9, and the prognostic pathways of the biomarker (`docs/49`). Here φ and Φ are the standard normal density and distribution function (ϕ is reserved for decay families, Equations (17)–(20)), and $C(z) = \varphi(z)/\{1 - \Phi(z)\}$ is the inverse Mills ratio. (Source: `docs/51`, Sections 13–18.)

11.1 Biomarker reliability

Let the observed biomarker be $\tilde{B}_i = B_i + e_{B,i}$ with independent error and reliability $R_B = \mathrm{Var}(B)/\mathrm{Var}(\tilde{B})$ (as in `docs/51`, which wrote the error ξ_i and the observed biomarker B_i^* before 2026-10-09; B_i^* is the reflected biomarker of §19.5).

Since $\mathrm{Cov}(\Delta, \tilde{B}) = \mathrm{Cov}(\Delta, B)$, the raw-scale interaction coefficient on $\tilde{B}$ is $R_B \beta_{bm:D}$ (regression dilution; Hutcheon et al., 2010), the coefficient per SD of the observed biomarker is attenuated by $\sqrt{R_B}$, and $r_{\Delta\tilde{B}} = \sqrt{R_B}\, r_{\Delta B}$. The noncentrality, proportional to $r_{\Delta B}/\sqrt{1 - r_{\Delta B}^2}$ (Equation (109)), therefore scales by

$$\sqrt{R_B}\sqrt{\frac{1 - r_{\Delta B}^2}{1 - R_B r_{\Delta B}^2}} \;\approx\; \sqrt{R_B}, \tag{88}$$

and the sample size for a given power rises by about $1/R_B$: about 25% at $R_B = 0.8$ (derived). The approximation holds when the interaction explains a small share of $\mathrm{Var}(\Delta)$; the exact factor is slightly below $\sqrt{R_B}$. Under covariance binding the visible correlation is $c_{bm}\sqrt{R_B}$, and because that encoding draws B_i inside the MVN, a binary or skewed biomarker can be represented only under mean or sensitivity binding. (Check: raw slope 0.800 at $R_B = 0.8$; simulated noncentrality ratio 0.887 against exact 0.889 and $\sqrt{0.8} = 0.894$.) [`docs/51` states the $\sqrt{R_B}$ scaling without the exact correction factor in Equation (88); the two agree to first order.]

11.2 Functional form of moderation

Write the moderation as a function $\mathcal{M}(b_i)$, so the on-drug response is scaled by $1 + \mathcal{M}(b_i)$; the linear case is $\mathcal{M}(b) = \beta_{bm} b$. A linear working model estimates the projection $\beta_{\mathrm{lin}} = \mathrm{Cov}\{\mathcal{M}(b), b\}/\mathrm{Var}(b)$. For a threshold $\mathcal{M}(b) = \Delta_h \mathbb{1}(b > b_{\mathrm{thr}})$ at the median of a standard normal biomarker ($b_{\mathrm{thr}} = 0$; as in `docs/51` since 2026-10-09, which wrote the effect Δ, the threshold b^* and the function $h(b)$; the effect is Δ_h to avoid the E9 contrast Δ_i and the function $\mathcal{M}$ to avoid the moderation profile h_t),

$$\beta_{\mathrm{lin}} = \Delta_h\, E\{b \mathbb{1}(b > 0)\} = \Delta_h\, \varphi(0) \approx 0.399\, \Delta_h, \qquad R^2_{\mathrm{lin}} = \frac{\varphi(0)^2}{1/4} = \frac{2}{\pi} \approx 0.637, \tag{89}$$

where R^2_{lin} is the share of the moderation variance $\mathrm{Var}\{\mathcal{M}(b)\} = \Delta_h^2/4$ captured by the linear term. When the moderation is small relative to the residual variance, the linear test has $\sqrt{2/\pi} \approx 0.80$ of the noncentrality of a test using the true form (derived; numerically confirmed: 0.3989, 0.6366, 0.7979; a 4000-replicate simulation with $N = 200$ gave a mean t-ratio of 0.79). A responder mixture, $\omega_i = \omega_R R_i$ with $\mathrm{logit}\,\Pr(R_i = 1 \mid b_i) = \psi_0 + \psi_g b_i$, is the DGP-side counterpart of §8; no projection result for it is stated.

11.3 Exposure versus assignment

Let exposure be $E_{it} = \mathcal{A}_{it}\,\mathrm{ramp}(t_{od})$, with adherence $\mathcal{A}_{it} \in [0,1]$ (`docs/51` writes a_{it}, which would collide with the contrast weights a_t) and $\mathrm{ramp}(t_{od}) = \min(1, t_{od}/T_{\mathrm{tit}})$, for example $T_{\mathrm{tit}} = 5$ weeks (Raskind et al., 2018). E_{it} can serve as the target h^* of Equation (72). If the response is linear in exposure and adherence is independent of the biomarker and of ω_i, then $E(\mathcal{A}_i\omega_i \mid b_i) = \bar{\mathcal{A}}\,E(\omega_i \mid b_i)$, so the assigned-drug contrast and the interaction are both scaled by the mean adherence $\bar{\mathcal{A}}$, the noncentrality falls by at least that factor, and the sample size rises by at least about $1/\bar{\mathcal{A}}^2$, variable adherence adding between-person variance (derived; check: slope 2.402 against $\bar{\mathcal{A}} \times 3 = 2.401$). If adherence depends on the biomarker, the interaction is biased in a direction set by that dependence (argued). [The linearity in exposure is exact for the turnover model, whose solution is linear in h^*, when adherence is constant within participant; with time-varying adherence F becomes a participant-specific weighted exposure and the scaling is by its mean.]

11.4 Integrity of blinding

In blinded phases let the expectancy be $\eta_{it} = \frac{1}{2} + \chi\,(D_{it} - \frac{1}{2})$, with $\chi = 0$ for intact blinding and $\chi = 1$ for full unblinding (`docs/51` uses χ; the canonical design weight is $\eta_t = 0.5$ in these phases, §12). With $PB_{it} = \eta_{it} PB_i^{\mathrm{full}}(t)$, the χ-dependent part of a blinded-phase contrast is $\chi \sum_t a_t (D_{it} - \frac{1}{2}) PB_i^{\mathrm{full}}(t) = \frac{\chi}{2}(\overline{PB}_{i,\mathrm{on}}^{\mathrm{full}} + \overline{PB}_{i,\mathrm{off}}^{\mathrm{full}})$, which is $\chi PB_i^{\mathrm{full}}$ when $PB_i^{\mathrm{full}}(t)$ is constant over the contrasted visits. In the outcome, which subtracts PB, the drug main effect is biased by $-\chi\overline{PB}^{\mathrm{full}}$. If the placebo response is prognostic, $PB_i^{\mathrm{full}} = \mu_{pb} + s_{pb}\,b_i$, the interaction acquires

$$-\chi\, s_{pb} \text{ per SD of } b, \qquad \text{that is,} \qquad \beta_{bm:D} \mapsto \beta_{bm:D} - \frac{\chi\, s_{pb}}{\sigma_{bm}}, \tag{90}$$

so unblinding converts a prognostic placebo pathway into a spurious predictive one (derived; check: slope -0.723 against $-\chi s_{pb} = -0.720$). A design-coded expectancy term is constant within the blinded phase and cannot absorb it. [On the compendium's convention (`NOTATION.md` Rule 5) the bias carries a minus sign, so a positive s_{pb} inflates the magnitude of a negative interaction. `docs/51` stated it without a sign until 2026-10-09 and now gives $-\chi s_{pb}$ per SD of the biomarker.]

11.5 Between-person variation in carryover

Let $t_{1/2,i} = t_{1/2}\exp(\sigma_{\log t_{1/2}} x_i)$, $x_i \sim \mathcal{N}(0,1)$, so $\lambda_i = \ln 2/t_{1/2,i}$, with off-drug coupling $g_{it} = 2^{-t_{sd}/t_{1/2,i}}$. The population-mean profile

$$\bar{g}(t_{sd}) = E\{e^{-\lambda_i t_{sd}}\} \tag{91}$$

is a mixture of exponentials. Its local decay rate, $-d\log\bar{g}/dt_{sd}$, is the mean of λ_i under weights proportional to $e^{-\lambda_i t_{sd}}$; its derivative is minus the weighted variance of λ_i, so the rate falls with t_{sd}. It starts at $E(\lambda_i) = \lambda e^{\sigma_h^2/2} > \lambda$, so $\bar{g}$ initially decays faster than the median-half-life profile $2^{-t_{sd}/t_{1/2}}$, and at long t_{sd} it is dominated by the longest half-lives and decays more slowly (derived). With $t_{1/2} = 1$ week and $\sigma_h = 0.5$: $\bar{g} = 0.689$ against 0.707 at 0.5 week, 0.266 against 0.250 at 2 weeks, 0.018 against 0.004 at 8 weeks; initial rate 0.785 against $\lambda = 0.693$. A common half-life in D_{bc} (Equation (13)) therefore misfits the mean profile at both ends. That power under a binary indicator follows the mean coupling gap to first order is argued, not derived. [`docs/51` states only the long-t_{sd} behavior; the faster initial decay and the crossing are added here.]

11.6 Selection on a screening score

Let the screening score be $Y_{i,\mathrm{scr}} = \mathrm{BL}_i + e_{i,\mathrm{scr}}$ with between- and within-person variances σ^2_{btw} and σ^2_{wth}, enrollment when $Y_{i,\mathrm{scr}} > Y_{\mathrm{elig}}$, and z_{elig} the threshold standardized by $\sqrt{\sigma^2_{\mathrm{btw}} + \sigma^2_{\mathrm{wth}}}$.

Regression to the mean. Later visits carry independent errors, so the expected fall after selection is $E(e_{i,\text{scr}} \mid \text{selected})$, which by normal truncation is (Barnett et al., 2005)

$$\frac{\sigma^2_{\text{wth}}}{\sqrt{\sigma^2_{\text{btw}} + \sigma^2_{\text{wth}}}}\, C(z_{\text{elig}}) \tag{92}$$

(derived; check: 0.555 simulated against 0.555 at $\sigma_{\text{btw}} = 1$, $\sigma_{\text{wth}} = 0.7$, $Y_{\text{elig}} = 1$). It can be mistaken for natural course or placebo response. Within- person blinded contrasts are largely protected, since it affects all post-screening visits alike; contrasts that lean on the baseline visit are not.

Range restriction. If the standardized biomarker has correlation $c_{bm,\text{scr}}$ with the standardized screening score (as in `docs/51`; written $r_{B,\text{scr}}$ here before 2026-10-09), then, since $\text{Var}(S \mid S > z) = 1 - C(z)\{C(z) - z\}$ for standard normal S,

$$\text{Var}(b \mid \text{selected}) = 1 - c^2_{bm,\text{scr}}\, C(z_{\text{elig}})\{C(z_{\text{elig}}) - z_{\text{elig}}\} \tag{93}$$

(derived; check at $z_{\text{elig}} = 1$: 0.930, 0.797, 0.487 simulated against 0.928, 0.800, 0.487 for $c_{bm,\text{scr}} = 0.3, 0.5, 0.8$, each within Monte Carlo error). Interaction precision is proportional to this variance, so power falls when the biomarker is prognostic for baseline severity (`docs/51`, Feature 6).

11.7 Phase-specific measurement error

Phase-specific measurement error adds a variance $\sigma^2_{\varepsilon,\text{phase}}$ to each visit's diagonal, which a homoscedastic working model omits (`docs/51`, Feature 9). The stress-test construct realizes it in one cell (Equation (187)).

11.8 Prognostic pathways and their implied slopes

A biomarker can also be coupled to the natural course, the placebo response or the baseline. Write s_0 and s_t for the true slopes of Y on the standardized biomarker b_i at baseline and at visit t. From $Y_{it} = \text{BL}_i - (TV_{it} + PB_{it} + BR_{it})$, with $\sigma_{TV} = 10$, placebo SD $10\eta_t$ and $\sigma_{BR} = 8$, the slopes per biomarker SD are (Source: `docs/49`, Section 2; re-derived; $\tilde{G}_{TV} = G_{TV} / \max G_{TV}$ is the normalized natural-course curve):

Table 23: Biomarker pathways and the slopes they imply.

Pathway	Kind	s_0	s_t, $t \geq 1$
Drug response, c_{bm}	predictive	0	$-c_{bm}\sigma_{BR}\, g_t$
Natural course, $c_{bm,TV}$	prognostic	0	$-c_{bm,TV}\sigma_{TV}$ (constant)
Natural course, β^{TV}_{bm}	prognostic	0	$-\beta^{TV}_{bm}\sigma_{TV}\tilde{G}_{TV}(t)$ (growing)
Placebo response, $c_{bm,PB}$	prognostic	0	$-10\, c_{bm,PB}\, \eta_t$
Baseline severity, $c_{bm,BL}$	prognostic	$c_{bm,BL}\sigma_{BL}$	$c_{bm,BL}\sigma_{BL}$

Only the first row is the estimand; in every other row a correct analysis reports no interaction. Write the prognostic slope as s_{prog} when it is constant after baseline. The *constancy assumption* is $s_0 = s_{\text{off}}$: it holds for a purely predictive biomarker ($s_0 = s_{\text{off}} = 0$) and for baseline severity, and fails for the natural-course and placebo pathways, whose slope is zero at baseline and nonzero afterwards.

12 Partitioning the treatment effect

Paper 06 asks what a single-indicator ("lumped") analysis of the treatment effect and its biomarker interaction actually estimates when the true generative process has three distinguishable components.

12.1 The outcome decomposition

$$Y_{it} = \mathrm{BL}_i - \left[BR_{it} + PB_{it} + TV_{it}\right] + \varepsilon_{it}, \tag{94}$$

with each component following the modified-Gompertz form (Equation (3) as Paper 06 writes it; Equation (2) as the code computes it), and PB_{it} additionally scaled by an *expectancy multiplier* η(phase), the phase-level form of the canonical design expectancy η_t, that operationalizes belief-driven placebo response (Kirsch, 1985):

$$PB(t \mid \text{phase}) = \eta(\text{phase}) \cdot m_{PB} \exp\!\big(-d_{PB} \exp(-r_{PB} t)\big) + (\text{offset}), \qquad \eta = \begin{cases} 1.0 & \text{open-label, on drug} \\ \approx 0.5 & \text{blinded discontinuation} \\ \in (0, 0.5) & \text{open-label placebo.} \end{cases} \tag{95}$$

Because η scales both the mean and SD of PB by the same factor, $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$, only the *product* $\eta \cdot m_{PB}$ is identified from a two-phase (open-label + blinded) design; recovering m_{PB} itself requires an open-label-placebo phase or a direct expectancy measurement.

A generalized, non-additive form allows a $BR \times PB$ interaction:

$$Y_{it} = \mathrm{BL}_i - \left[BR_{it} + PB_{it} + TV_{it} + \gamma \cdot BR_{it} \cdot PB_{it}\right] + \varepsilon_{it}, \tag{96}$$

with *subadditivity* of the two response mechanisms defined by the sign condition $\partial Y / \partial (BR \cdot PB) > 0$ on the reduction scale. Paper 06 explicitly distinguishes true $\gamma \neq 0$ subadditivity from an *apparent* subadditivity that a latent responder-class gating function $\pi(B)$ (§8) could generate through confounding, without any real BR–PB interaction being present.

12.2 What a one-component analysis actually estimates

Writing the within-participant symptom reduction as $R_{it} = \mathrm{BL}_i - Y_{it} = BR_{it} + PB_{it} + TV_{it} - \varepsilon_{it}$ and fitting the one-component model $R_{it} = \beta_0 + \beta_D D_{it} + s(t) + u_i + e_{it}$, with $s(t)$ a smooth time trend, Paper 06 derives:

$$\operatorname{plim} \hat{\beta}_D = m_{BR} + \Delta_{PB} + \Delta_{TV}, \tag{97}$$

where $\Delta_{PB} = E[PB \mid \text{on}] - E[PB \mid \text{off}]$ and Δ_{TV} is the natural-history contribution that fails to cancel under the contrast used (zero for a clean on/off within-participant contrast; $E[TV]$ itself for a baseline-anchored pre-post analysis). Both terms are generally non-negative for an improving outcome, so the one-component estimate systematically *overstates* the pharmacological effect m_{BR}. This is a property of the estimator's probability limit, unaffected as $N \to \infty$, not a finite-sample artifact.

[2026-10-08: notation. Paper 06 writes α, $g(t)$, b_{PB} and b_{TV}; these collide with the significance level α, the coupling profile g_t and the standardized biomarker b_i, and are written β_0, $s(t)$, Δ_{PB} and Δ_{TV} here.]

The same logic applied to the biomarker slope gives the compendium's key attenuation/inflation identity:

$$\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB}\, \beta_{bm}^{PB} + w_{TV}\, \beta_{bm}^{TV}, \qquad \beta_{bm}^{C} = \frac{\mathrm{Cov}(m_C, B)}{\mathrm{Var}(B)} \quad (C \in \{BR, PB, TV\}), \tag{98}$$

with $w_{PB}, w_{TV} \geq 0$ design-determined loadings. A biomarker correlated with placebo responsiveness or with the natural-history trajectory therefore registers a non-zero lumped slope even when its true pharmacological association β_{bm}^{BR} is exactly zero, and, because Equation (98) is an *identity*, no amount of additional sample size can separate a genuinely pharmacological biomarker signal from a confounded one whenever $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV} \neq 0$. Under $\gamma \neq 0$ subadditivity (Equation (96)), a further first-order correction proportional to $\mathrm{Cov}(m_{BR} \cdot m_{PB}, B)$ enters the slope; the identity above is the leading-order behavior in the additive case.

The operative confound is a biomarker's correlation with individual- level *variance* in placebo responsiveness, not with its *mean*:

$$\text{contamination driver} = c_{bm,PB} \cdot \sigma_{PB}, \qquad \text{not } c_{bm,PB} \cdot m_{PB}. \tag{99}$$

Paper 06's calibration target throughout its production studies is

$$\beta_{bm:D} = -\frac{c_{bm}\,\sigma_{BR}}{\sigma_{bm}} = -2.25, \tag{100}$$

against which Study A (uncontaminated biomarker, $c_{bm,PB} = 0$) shows bias within one Monte Carlo standard error of zero at every sample size tested ($N \in \{35, 70, 200, 300\}$, totals), consistent with Equation (98) giving zero bias when $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$. A contaminated-biomarker pilot (100 replicates per cell, $N \in \{70, 300\}$) at $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$ shows, at $N = 300$, bias moving from -0.06 and -0.12 to $+0.17$ and $+0.51$ (the last about 30 MCSEs from zero) as contamination increases. Paper 06 explains the positive sign as attenuation toward zero: high-$c_{bm,PB}$ participants' elevated placebo response is partially absorbed by the biomarker main effect, leaving the interaction coefficient pulled toward zero rather than pushed away from it.

Paper 06 emphasizes that the bias of Equation (97) lives in the *location* of the estimand, not in the test: the one-component coefficient has a valid Wald test of its own nullity, which rejects at the rate dictated by the displaced value, so a significance test cannot detect a displacement away from the null. Detecting it requires comparing the estimate with m_{BR} (or, for the slope, with β_{bm}^{BR}), which only a decomposition or a phase contrast makes available. For the contaminated interaction the displacement is *toward* zero, so it lowers power, but a test still cannot distinguish it from a smaller true effect.

[2026-10-08: corrected. This paragraph previously applied Paper 06's statement that "the bias does not move the coefficient toward zero" to $\hat{\beta}_{bm:D}$ under contamination, which contradicted the attenuation toward zero reported just above; in Paper 06 the statement concerns plim $\hat{\beta}_D$. A parenthetical stating that $c_{bm,PB}$ was "constrained below 0.45 by positive-definiteness" was removed: 0.45 is in the reported grid and the current Paper 06 does not state such a constraint.]

12.3 Recovery, and its dependence on trial design

A phase-augmented analysis model is the smallest step toward component-aware inference:

```
Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase
```
(101)

with random intercept and `corCAR1` residual correlation, where `phase` (open-label / blinded / post-crossover) absorbs systematic expectancy differences and `Dbc:phase`, `bm:Dbc:phase` test whether the apparent drug effect and its moderation shrink under blinding. Whether this, or full component-matched Gompertz-basis fitting, recovers the true pharmacological slope β_{bm}^{BR} is shown to be *design-dependent*, not a property of decomposition per se: under the compendium's standard Hybrid design, belief-covariate decomposition, blinded-stratum contrasts, and full Gompertz-basis decomposition all fail to recover β_{bm}^{BR}, performing no better (and sometimes worse) than the lumped analysis, because D_{bc} and the expectancy multiplier η are positively correlated ($\approx +0.6$) under that design. A belief-decoupling *balanced-placebo* design (three open-label, six covert on-drug with $\eta = 0$, and six open-placebo with $\eta = 1$ timepoints) reduces

that correlation to ≈ -0.66 and restores recovery: in the production run (1,000 replicates per cell, single-path design, $N \in \{70, 150\}$ totals, verified from `study-b-balanced-summary.rds`), decomposition-analysis bias at $N = 150$ is -0.022, $+0.008$, $+0.021$ and -0.026 across $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$ (MCSE about 0.02, so within 1.4 MCSE of zero), while the lumped estimator's bias grows monotonically, -0.014, $+0.161$, $+0.340$, $+0.476$ (verified from the stored summary).

[2026-10-08: Paper 06 states that every decomposition bias at the larger N is "within one MCSE" of zero; in the stored summary three of the four lie between 1.0 and 1.4 MCSE (1.05, 1.03 and 1.32). The substantive conclusion is unaffected.]

12.4 Rank of the phase-augmented model, and the drug-expectancy correlation by design

The phase-augmented formula is rank-deficient whenever a phase is entirely on drug. Write the phase factor of Equation (101) with levels ℓ and indicators I_ℓ, the open-label level as reference. If every visit of the open-label level is on drug, $D_{bc,it} = 1$ there, so

$$D_{bc,it} = I_{\mathrm{OL},it} + \sum_{\ell \neq \mathrm{OL}} D_{bc,it} I_{\ell,it} = 1 - \sum_{\ell \neq \mathrm{OL}} I_{\ell,it} + \sum_{\ell \neq \mathrm{OL}} D_{bc,it} I_{\ell,it}, \tag{102}$$

an exact linear combination of the intercept, the phase main effects and the `Dbc:phase` columns. The fixed-effects design matrix therefore loses one rank, whatever the coding of the factor (derived). The `bm:Dbc:phase` columns are not aliased, because the model has no `bm:phase` term. Consequences:

- With the three phases of `docs/24` (open label, blinded discontinuation, crossover) the extension adds six columns but only five identified parameters, not the six that `docs/24` counts. In the two-level coding of the Paper 06 driver (phase set by $\eta_t \geq 0.99$, in `phase_augmented_fit()` of `analysis/scripts/component-decomposition/00-common.R`), three columns add two identified parameters. (Verified by the rank of the model matrix for the 8-visit Hybrid paths of Table 4 and for the 16-visit two-path design of the Paper 06 recovery study, `check-gompertz-rmanova-cycles.R`.)
- `nlme::lme` cannot fit the full formula: on the 16-visit design it stops with "Singularity in backsolve" (verified). The driver catches the error and refits without `Dbc:phase`, so every phase-augmented result is from the reduced formula and is recorded with `formula_dropped = 'Dbc:phase'` (inspected). The rank deficiency encountered at $N = 35$ that `docs/28` (Section 3.4) attributes to near-collinear component trajectories is this exact aliasing of the formula, present at every N (derived); for $N < 70$ the driver omits the full formula without attempting it.
- In the reduced formula `Sx ~ ... + bm:Dbc + bm:Dbc:phase`, the biomarker slope is $\beta_B + \beta_{bm:D}$ at open-label visits and $\beta_B + (\beta_{bm:D} + \beta_{bm:D:\mathrm{BDC}}) D_{bc,it}$ at blinded visits. The reported coefficient $\beta_{bm:D}$ is therefore the open-label slope minus the blinded slope extrapolated to $D_{bc} = 0$: an open-label versus blinded contrast, in which drug state and expectancy change together, while the blinded-window interaction is the sum $\beta_{bm:D} + \beta_{bm:D:\mathrm{BDC}}$ (derived from the parameterization). [What this implies for the bias of the reported coefficient under a PB-coupled biomarker was not evaluated here; the sign of the coupling effect depends on the covariance construction, including any positive-definiteness repair (§9).]

(Source: `docs/24`, "A small extension that usually pays off"; `docs/28`, Sections 3.4 and 4.2.) [`docs/24` states that the extension "adds six fixed-effect parameters (two each for `phase`, `Dbc:phase` and `bm:Dbc:phase`)" and calls this "derived by counting". The count of columns is six; the count of identified parameters in any design whose open-label phase is entirely on drug is five.]

Drug-expectancy correlation by design. For a single-path design whose visits fall into an on-drug, full-belief block of n_1 visits, an on-drug, covert ($\eta = 0$) block of n_2, and an off-drug, full-belief block of n_3, with binary drug state, the pooled correlation of D and η over visits is

$$\mathrm{Cor}(D, \eta) = \frac{p_{11} - p_D p_\eta}{\sqrt{p_D(1-p_D)\, p_\eta(1-p_\eta)}}, \qquad p_D = \frac{n_1+n_2}{n},\ p_\eta = \frac{n_1+n_3}{n},\ p_{11} = \frac{n_1}{n}, \tag{103}$$

with $n = n_1 + n_2 + n_3$. For the balanced-placebo design of Paper 06 ($n_1 = 3$, $n_2 = n_3 = 6$) this is exactly $-2/3$; with the decayed D_{bc} at the driver's $t_{1/2} = 1$ week and 1.5-week spacing it is -0.656, which matches the "≈ -0.66" of Paper 06 and §12 (derived; verified, `check-gompertz-rmanova-cycles.R`). [The companion value "$\approx +0.6$" for the Hybrid design was not reproduced. For the 16-visit two-path design of the recovery study (`design_hybrid_full()`), pooled over visits with equal path weights, the correlation is 0.775 with binary D and 0.753 with D_{bc} at $t_{1/2} = 1$ (0.785 with a baseline row $D = \eta = 0$ added); for the 8-visit four-path Hybrid of Table 4 it is 0.509 and 0.485 (0.645 with the baseline row). How Paper 06 computed its figure, at which stage and over which rows, was not located.]

13 Informative dropout

Paper 09 evaluates a censoring mechanism whose hazard depends on the participant's own symptom trajectory, crossed with the four trial designs of §4.

13.1 The dropout hazard

Marginal per-timepoint censoring probability is the union of two independent components:

$$\Pr(\mathrm{drop}_{it}) = 1 - \left(1 - p_{it}^{\mathrm{flat}}\right)\left(1 - p_{it}^{\mathrm{bias}}\right), \tag{104}$$

with $p_{it}^{\mathrm{flat}} \propto t$ depending on elapsed time alone (calibrated to a marginal dropout fraction of $\beta_0^{\mathrm{dr}}(1 - \beta_1^{\mathrm{dr}})$) and $p_{it}^{\mathrm{bias}} \propto t\,(\Delta_{Y,i}(t) + 100)^{e_{b1}}$ depending on the symptom change $\Delta_{Y,i}(t) = Y_{it} - Y_{i0}$ since baseline (calibrated to $\beta_0^{\mathrm{dr}}\beta_1^{\mathrm{dr}}$), where β_0^{dr} is the overall dropout rate, β_1^{dr} the fraction attributable to the symptom-biased mechanism, and $e_{b1} = 2$ throughout (Paper 09, `R/censordata.R`).

[2026-10-08: notation. Paper 09 writes the dropout parameters as β_0, β_1 and the symptom change as $\Delta_{Sx}(t)$; the superscript separates the former from regression coefficients, and Y replaces the code name `Sx` (`NOTATION.md` Rule 3).] The additive shift of 100 inside the biased component keeps its convex argument positive across the realized range of symptom scores, so that large improvement lowers dropout probability while worsening (or a small change) raises it. Once a participant is censored at a timepoint, every later timepoint for that participant is also set missing (monotone/absorbing dropout).

13.2 Missingness classification

Because the flat component depends only on a fully observed design variable (elapsed time), it induces missingness that is ignorable for likelihood-based inference on its own. Because the biased component's censoring indicator is a function of $\Delta_{Y,i}(t)$, the quantity realized at, and thereby rendered missing by, that timepoint, Paper 09 argues the mechanism is MNAR for every dropout pattern with $\beta_1^{\mathrm{dr}} > 0$: missingness of Y_{it} depends on the value of Y_{it} itself, the defining condition for missing-not-at-random data. No manuscript in the compendium gives a formal probability statement for MCAR/MAR/MNAR beyond this argument; the classification rests on the hazard construction of Equation (104) alone.

13.3 Interaction with design

The estimand is the same interaction test as elsewhere, $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$, fit post-dropout. Paper 09 documents a "happy accident" selection effect, described qualitatively rather than by a closed-form path-conditional formula: participants with unambiguous open-label response are the most readily resolved and most prone to drop out early, which enriches the later, randomized (crossover/discontinuation) blocks for participants whose open-label response was ambiguous, the subgroup whose crossover contrasts are most informative for the interaction test. This selection effect is design-dependent: under the OL design, where D_{bc} is identically one throughout, there is no within- subject contrast for $bm{:}D_{bc}$ at all, so dropout there can only reduce power, never selectively concentrate information. A quantified, per-path decomposition of the happy-accident effect is explicitly scoped for a future production-grade rerun and is not delivered in the version inspected here; as with several other design comparisons in this compendium (§2), an earlier mismatched-N run of this same comparison had produced an apparent robustness finding (the Hybrid design appearing dropout-immune) that a matched-$N = 70$ re-run showed was a ceiling effect rather than a property of the design.

13.4 The MAR and MCAR mechanisms of the stress test

The stress test of `docs/46` (§19) uses two monotone mechanisms distinct from Paper 09's hazard (Equation (104)). With d_i the first missing post-baseline visit ($d_i = \infty$ if complete), t indexing the post-baseline visits $1, \ldots, 8$, and U_{it} independent uniforms,

$$\begin{aligned} \text{MCAR:} \quad & \Pr(d_i < \infty) = 0.2, \qquad d_i \mid d_i < \infty \sim \text{Uniform}\{3, \ldots, 8\}; \\ \text{MAR:} \quad & \operatorname{logit} p_{it}^{\text{MAR}} = \mu_h + \beta_h \, z_{i,t-1}, \quad \beta_h = 1, \quad t = 3, \ldots, 8, \\ & \frac{1}{n_p} \sum_{i \in p} \Big[1 - \prod_{t=3}^{8} \big(1 - p_{it}^{\text{MAR}}\big) \Big] = 0.2, \end{aligned} \tag{105}$$

where, under MAR, p_{it}^{MAR} is the hazard at visit t given presence at $t-1$, d_i is the first t with $U_{it} < p_{it}^{\text{MAR}}$, $z_{i,t-1}$ is the score at the previous visit standardized, visit by visit, across the n_p participants of randomization path p in the simulated trial, and μ_h is solved numerically for each path of each trial (Sources: `docs/46`, Section 5; `docs/51`, Feature 9; inspected, `11-strawman-stress-test.R`, `dropout_visit()`). Because the hazard depends only on the previously observed score, the mechanism is MAR and likelihood analysis of the observed data is ignorable for it, in contrast to the MNAR classification of Section 13 (derived). [`docs/51` writes the slope as β_h without a value; the driver fixes it at 1. Because μ_h is calibrated within each path of each trial to the realized scores, the 20% rate holds in expectation conditional on those scores rather than as a population parameter, μ_h itself varies across trials, and a participant's hazard also depends on the other participants' observed values: dropout depends only on observed data, but not through a fixed per-participant model.]

Part III
Closed-form theory of the interaction test

14 Biomarker coupling under carryover and the closed-form power of the interaction test

This section gives the closed-form account of how the carryover half-life $t_{1/2}$ acts on the power of the interaction test (Source: `docs/37`; Paper 01, Methods and Appendix A.1–

A.3). The estimand and the coupling rules are those of §6.4, and the conditional equivalence of the two encodings is Equation (30). Setting: the published process (commit `58b32a9`, configuration A of `docs/36`); the Hybrid design, visits at weeks 4, 8, 9, 10, 11, 12, 16, 20, expectancy $\eta_t = 1$ at the two open-label visits and 0.5 afterwards; $N = 70$ allocated $n_p = 18, 18, 17, 17$ to $P = 4$ paths (Table 4); compound symmetry (CS), $\rho_{cs} = 0.8$ within a component, cross-component correlation $c_1 = 0.2$ at the same visit and $c_\times = 0.1$ otherwise; $\sigma_{TV} = 10$, $\sigma_{PB,t} = 10\eta_t$, $\sigma_{BR} = 8$, $\sigma_{bm} = 15.36$; strength $c_{bm} = \beta_{bm} = 0.25$ (`NOTATION.md` Rule 2), below the CS ceiling of 0.256, so no matrix is repaired. Local symbols (a_t, g_t, h_t, ϕ_t, σ^2_Δ, ϖ_p, $\mathcal{G}$) collide with no canonical one. *Verified* below means recomputed by an independent check script written for this review (`check-coupling-closed-form.R`) and compared with `analysis/data/quick-sim/carryover-closed-form/` (`e9-per-path.csv`, `e9-power-vs-gap.csv`); *quoted* means taken from `docs/37` without recomputation.

Every Hybrid off-drug visit follows an exposure, so $t_{sd} \geq 1$ week there; several results below rest on this condition.

14.1 The paired-difference statistics E9 and E9b, and their moments

E9 (Equations (141)–(142), with β_1 written $\beta_{bm:D}$) forms $\Delta_i = \sum_t a_t Y_{it}$ with $a_t = 1/n_{\text{on}}$ on drug and $-1/n_{\text{off}}$ off drug, regresses Δ_i on B_i by least squares over all N participants, and tests the slope on $N - 2$ degrees of freedom. E9b counts baseline as one more off-drug visit (weight $-1/(n_{\text{off}} + 1)$ on each off-drug row), as the published `lmer` analysis codes it. All weights sum to zero, so BL_i cancels and $\Delta = -\sum_{t\geq 1} a_t S_t$, $S_t = TV_t + PB_t + BR_t$; the post-baseline weights sum to 0 (E9) or $1/(n_{\text{off}} + 1)$ (E9b) (derived). Within a path Δ and B are jointly normal, so $E(\Delta \mid B) = \mu + \beta_{bm:D}(B - \mu_{bm})$ exactly, with residual variance σ^2_Δ. [2026-10-09: notation. The residual variance of the contrast was written τ^2 (and τ^2_p per path) here and in `docs/37` to `docs/39` and Paper 01; it is σ^2_Δ, $\sigma^2_{\Delta,p}$, because τ^2 is the canonical between-participant variance of the drug effect (§20). The path share $q_p = n_p/N$ is now ϖ_p, since q is the canonical path index.]

Proposition 1 (variance). With $m_1 = \sum a_t$, $m_2 = \sum a_t^2$, $m_e = \sum a_t \eta_t$, $m_{ee} = \sum a_t^2 \eta_t^2$, $m_{2e} = \sum a_t^2 \eta_t$ and $\sigma_{PB} = 10$,

$$\begin{aligned}\text{Var}(\Delta) = {} & (\sigma^2_{TV} + \sigma^2_{BR})\big[\rho_{cs} m_1^2 + (1 - \rho_{cs}) m_2\big] + \sigma^2_{PB}\big[\rho_{cs} m_e^2 + (1 - \rho_{cs}) m_{ee}\big] \\ & + 2\sigma_{TV}\sigma_{BR}\big[c_\times m_1^2 + (c_1 - c_\times) m_2\big] + 2(\sigma_{TV} + \sigma_{BR})\sigma_{PB}\big[c_\times m_1 m_e + (c_1 - c_\times) m_{2e}\big],\end{aligned} \tag{106}$$

which is free of $t_{1/2}$: the half-life enters only through the means and the coupling, and neither appears (Source: `docs/37`, Section 4.1; Paper 01, Appendix A.1; derived). This holds for any construct whose response block is free of $t_{1/2}$, hence for unrepaired matrices only. Values: 44.13 (paths 1–2) and 45.03 (paths 3–4) for E9, 51.04 and 51.77 for E9b (verified: Equation (106) equals $a^\top \Sigma a$ from the full 24×24 response covariance).

Proposition 2 (slope). Only BR_t covaries with B, so $\beta_{bm:D} = -c_{bm}(\sigma_{BR}/\sigma_{bm}) \sum_t a_t g_t$ and, with $\bar{g}_{\text{off}}$ the mean coupling over the post-baseline off-drug visits,

$$\beta^{\text{E9}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\big(1 - \bar{g}_{\text{off}}\big), \qquad \beta^{\text{E9b}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\Big(1 - \tfrac{n_{\text{off}}}{n_{\text{off}}+1}\,\bar{g}_{\text{off}}\Big) \tag{107}$$

(Source: `docs/37`, Section 4.2; derived, and verified per path by direct covariance against `e9-per-path.csv`). Under the mean rules replace c_{bm} by β_{bm} and $\bar{g}_{\text{off}}$ by $\bar{h}_{\text{off}}$, which is 0 under plain mean moderation. The pooled slope is $\bar{\beta}_{bm:D} = \sum_p \varpi_p \beta_{bm:D,p}$, $\varpi_p = n_p/N$.

Residual variance. Covariance binding explains part of the existing variance; mean binding adds to it:

$$\sigma^2_{\Delta,\text{cov}} = \text{Var}(\Delta) - \beta^2_{bm:D}\sigma^2_{bm}, \qquad \sigma^2_{\Delta,\text{mean}} = \text{Var}(\Delta), \tag{108}$$

with $\text{Var}(\Delta)$ of Equation (106) in both (derived). Power depends on these through $r_{\Delta B} = \beta_{bm:D}\sigma_{bm}/\text{SD}(\Delta)$, about -0.30 without carryover (verified, -0.301 on path 1; `docs/37` writes $\rho_{\Delta B}$, renamed here to keep ρ for AR(1)).

Power. On a single path of n participants, $\mathrm{SS}_{bm} = \sum_i (B_i - \bar{B})^2 \sim \sigma^2_{bm}\chi^2_{n-1}$; given the biomarker values the t-statistic is exactly noncentral t on $n-2$ degrees of freedom with noncentrality $\beta_{bm:D}\sqrt{\mathrm{SS}_{bm}}/\sigma_\Delta$, power is its tail probability averaged over the χ^2 law, and $\mathrm{Var}(\hat{\beta}_{bm:D}) = \sigma^2_\Delta/\{\sigma^2_{bm}(n-3)\}$. At $\mathrm{SS}_{bm} = \sigma^2_{bm}(n-1)$,

$$\Lambda = \sqrt{n-1}\, r_{\Delta B}/\sqrt{1 - r^2_{\Delta B}} \tag{109}$$

(Paper 01, Appendix A.2; derived; Λ, the noncentrality, as in Paper 01 and `docs/37`; this section previously wrote ncp). Pooling paths with a common intercept gives, to first order in the path differences (argued), a noncentral t on $N-2$ degrees of freedom with noncentrality $\bar{\beta}_{bm:D}\sqrt{\mathrm{SS}_{bm}}/\sigma_{\mathrm{res}}$ and

$$\sigma^2_{\mathrm{res}} = \sum_p \varpi_p \left[\sigma^2_{\Delta,p} + (\mu_p - \bar{\mu})^2 + \sigma^2_{bm}(\beta_{bm:D,p} - \bar{\beta}_{bm:D})^2\right]. \tag{110}$$

The path-mean term $(\mu_p - \bar{\mu})^2$ is under 0.7% of σ^2_{res} in Hybrid but 54% in CO, where it is a period effect that a mixed model's time term absorbs and E9's common intercept does not (`docs/37`, Sections 4.4 and 5.7; the CO share is recomputed in §14.10).

14.2 Power through the coupling gap

Let $\mathcal{G} = \bar{g}_{\mathrm{on}} - \bar{g}_{\mathrm{off}} = 1 - \bar{g}_{\mathrm{off}}$, the *coupling gap*. Under covariance binding and E9, $\mathrm{Var}(\Delta)$ is fixed and $r_{\Delta B} = -c_{bm}\sigma_{BR}\,\mathcal{G}/\mathrm{SD}(\Delta)$ is exactly proportional to $\mathcal{G}$, so the coupling profile acts on power only through $\mathcal{G}$; which off-drug visits carry the coupling does not matter, since E9 weights them equally (derived). Fixing Λ at a target Λ^* in Equation (109) gives the exact single-path sample size

$$n - 1 = \Lambda^{*2}\left(\frac{1}{r^2_{\Delta B}} - 1\right) \approx \frac{\Lambda^{*2}\,\mathrm{Var}(\Delta)}{c^2_{bm}\sigma^2_{BR}\,\mathcal{G}^2} \qquad (r^2_{\Delta B} \ll 1), \tag{111}$$

so the required N scales roughly as $1/\mathcal{G}^2$ (derived). On Hybrid path 1, halving the gap from 1 to 0.5 raises $1/r^2_{\Delta B} - 1$ from 10.03 to 43.13, a factor of 4.3 (verified). Table 24 gives power along the gap.

[`docs/37`, Section 5.4, says the noncentrality is proportional to the gap. By Equation (109) only $r_{\Delta B}$ is; the noncentrality is approximately proportional when $r^2_{\Delta B}$ is small, as Paper 01 states, and the $1/\mathcal{G}^2$ rule is approximate to the same degree.]

Table 24: Closed-form power against the coupling gap (uniform off-drug coupling $1-\mathcal{G}$), Hybrid, CS, $N = 70$, $c_{bm} = 0.25$ (Source: `docs/37`, Section 5.4, `e9-power-vs-gap.csv`; reproduced to within 0.002).

$\mathcal{G}$	0	0.25	0.5	0.75	1
E9	0.050	0.094	0.236	0.468	0.721
E9b	0.081	0.161	0.294	0.468	0.654

14.3 Step coupling: the gap is zero at any positive half-life

With $t_{1/2} > 0$ the off-drug mean $\mu_{BR,t} = \mu_{BR,t-1}2^{-t_{sd,t}/t_{1/2}}$ is strictly positive at every off-drug visit after an exposure, so Equation (33) gives $g_t = 1$ there. In Hybrid (and OL+BDC) every off-drug visit follows an exposure, so $\bar{g}_{\mathrm{off}} = 1$, $\mathcal{G} = 0$, and by Equation (107)

$$\beta^{\mathrm{E9}}_{bm:D} = 0, \qquad \beta^{\mathrm{E9b}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}} \cdot \frac{1}{n_{\mathrm{off}} + 1} \qquad \text{for every } t_{1/2} > 0 \tag{112}$$

(Source: `docs/37`, Section 5.1; derived). Power is a step function of the half-life: 0.721 (E9) and 0.654 (E9b) at $t_{1/2} = 0$; 0.050 and 0.081 at every positive value (verified). The E9b remainder (-0.0325 on paths 1–2, -0.0260 on paths 3–4, pooled -0.029) is carried by the baseline row alone and matches the simulated published analysis (-0.026 to -0.032; quoted). The step rule is the $t_{1/2} \to \infty$ limit of graded coupling, applied at every positive half-life. The condition matters: in CO the placebo-first path's off-drug visits precede exposure, keep $g_t = 0$, and keep their full slope, so pooled E9 power falls only to 0.20 (`docs/37`, Section 5.7; recomputed as 0.199 in Table 32, §14.10).

14.4 Graded coupling: monotone decline

Under Equation (34) (Paper 01, Appendix A.3),

$$\bar{g}_{\text{off}}(t_{1/2}) = \frac{1}{n_{\text{off}}} \sum_{t\,\text{off}} 2^{-t_{sd,t}/t_{1/2}}, \qquad \frac{d\bar{g}_{\text{off}}}{dt_{1/2}} = \frac{\ln 2}{n_{\text{off}}\, t_{1/2}^2} \sum_{t\,\text{off}} t_{sd,t}\, 2^{-t_{sd,t}/t_{1/2}} > 0 \tag{113}$$

because every off-drug visit has $t_{sd,t} > 0$ (derived; derivative verified by finite difference). **Proposition 3.** On every path $|\beta_{bm:D,p}|$ is strictly decreasing in $t_{1/2}$ on $(0, \infty)$ and $\sigma^2_{\Delta,p} = \text{Var}(\Delta) - \beta^2_{bm:D,p}\sigma^2_{bm}$ strictly increasing, since $\text{Var}(\Delta)$ is fixed; so the noncentrality and the power of the path-stratified test are strictly decreasing (derived). For the pooled statistic the path-mean term also moves with $t_{1/2}$; power was found non-increasing on a 120-point grid from 0.03 to 4 weeks for E9 and E9b (`docs/37`, Section 5.2; quoted). Monotonicity is thus proved for the stratified statistic and checked numerically, not proved, for the pooled one. **Limits.** With $\min t_{sd} \geq 1$, $\bar{g}_{\text{off}} \leq 2^{-1/t_{1/2}}$, so the loss vanishes faster than any power of $t_{1/2}$ as $t_{1/2} \to 0^+$ (at most 2^{-10} at 0.1 week); and $\bar{g}_{\text{off}} \to 1$ as $t_{1/2} \to \infty$. The patient-weighted gap is 0.991, 0.906, 0.754 and 0.551 at $t_{1/2} = 0.2, 0.5, 1, 2$ weeks (verified).

Table 25: Closed-form pooled power by rule and $t_{1/2}$ (weeks), Hybrid, CS, $N = 70$, strength 0.25 (Source: `docs/37`, Sections 5 and 5.6; reproduced to within 0.003).

Rule	Variant	0	0.1	0.2	0.5	1	2
Step (published)	E9	0.721	0.050	0.050	0.050	0.050	0.050
	E9b	0.654	0.081	0.081	0.081	0.081	0.081
Graded	E9	0.721	0.721	0.712	0.631	0.473	0.278
	E9b	0.654	0.653	0.647	0.586	0.472	0.327
Mean moderation	E9	0.681	0.681	0.682	0.682	0.682	0.682
Decayed mean mod.	E9	0.681	0.681	0.674	0.599	0.454	0.272

14.5 Anchored versus compounding means in the closed form

For the closed form the choice between the anchored and the compounding means of Equation (23) is nearly immaterial: the means enter neither $\text{Var}(\Delta)$ nor $\beta_{bm:D}$, only μ_p and so only the path-mean term of Equation (110); the step rule is unchanged because both forms are positive after exposure; and graded coupling uses ϕ_t, not the recursive mean (derived). The closed-form and Monte Carlo results of this part now use the anchored mean; against the compounding mean they change by at most 0.002 in closed-form power in Hybrid and OL+BDC and up to 0.041 in CO, where the means enter through the period effect (verified, rerun of `01-e9-closed-form.R`, 2026-10-09). Its effect on the published mixed model has now been tested: in RH's pipeline the correction alone changes power by at most 0.006 (`docs/53`, Section 8.3).

14.6 Covariance binding as mean moderation

Conditioning the joint normal on the biomarker gives Equation (30) (§6). Hence, conditionally on the biomarker, *covariance binding with coupling g is mean moderation with profile $h_t = g_t$ and strength $\beta_{bm} = c_{bm}$, with the drug-response covariance reduced by the rank-one term $c_{bm}^2 \sigma_{BR}^2 g g^\top$.* Consequences (derived unless marked):

(i) Graded coupling and decayed mean moderation have identical per-path slopes at every half-life (Equation (107) with $h_t = g_t$).

(ii) Their residual variances differ by exactly $\beta_{bm:D}^2 \sigma_{bm}^2$ (Equation (108); 42.007 against 44.133 on path 1 at $t_{1/2} = 1$, verified). The covariance encoding thus has slightly more power at the same slope (0.721 against 0.681 without carryover), and decayed mean moderation loses power almost exactly as graded coupling does (Table 25). The carryover loss is set by the exposure profile of the interaction, not by the channel that carries it.

(iii) The ceiling is the condition that the conditional covariance of the whole response vector, $M - c_{bm}^2 u u^\top$ in Paper 01's correlation notation (M the response block, $c_{bm} u$ the biomarker correlations), be positive definite; given M positive definite this is equivalent to $c_{bm}^2 u^\top M^{-1} u < 1$, both being Schur-complement conditions on the full matrix. Without carryover it is 0.256 under the published compound-symmetry construct (Hybrid, $\rho = 0.8$), and 0.440 under separable AR(1) at $\rho = 0.8$ (0.474 at $\rho = 0.7$, 0.480 in the Hybrid design) (Paper 01, Results; `docs/34`; Table 8).

(iv) Under graded coupling g_t equals an exposure-decayed indicator at the true half-life, provided that indicator is 0 before first exposure, so the conditional mean of BR is linear in $b_i D_{bc,it}$ with the on-drug coefficient of Equation (32). This is the basis of Paper 01's statement that $D_{bc,it}$ at the true half-life recovers the on-drug moderation; recovery by a fitted `lme` also needs the remaining mean structure to be adequate (argued, not shown).

[This bears on §6, whose 2026-08 text attributed the divergence between the architectures under carryover to a structural asymmetry (exponential versus proportional decay of the biomarker signal); the 2026-10-08 revision of that section marks the comparison superseded. By Equation (30) the divergence is a difference of exposure profiles ($h_t = D_{it}$ against $g_t = \phi_t$ off drug), not of channels; a comparison reporting larger losses under covariance binding compares a fading interaction with a non-fading one (Paper 01, Discussion).]

14.7 Further moments of the paired-difference statistic

Pooled-slope variance. With the fixed allocation of the published code (path p receives exactly n_p participants), write $\beta_{bm:D,p} = \bar{\beta}_{bm:D} + \tilde{\beta}_p$, so that $\sum_p \varpi_p \tilde{\beta}_p = 0$ [2026-10-09: notation. The path deviation was written δ_p, which collides with the standardized bias δ; it is $\tilde{\beta}_p$ here.]. The pooled least-squares slope then has, to first order in $1/N$,

$$\mathrm{Var}(\hat{\beta}_{bm:D}) \approx \frac{\sum_p \varpi_p \left[\sigma_{\Delta,p}^2 + (\mu_p - \bar{\mu})^2\right]}{(N-3)\,\sigma_{bm}^2} + \frac{2}{N} \sum_p \varpi_p \, \tilde{\beta}_p^2 \tag{114}$$

(Source: `docs/37`, Section 4.4; derived). The first term uses $E(1/\mathrm{SS}_{bm}) = 1/\{\sigma_{bm}^2 (N-3)\}$ for $\mathrm{SS}_{bm} \sim \sigma_{bm}^2 \chi_{N-1}^2$. The second arises because, conditionally on the biomarker values, the pooled slope is the $(B_i - \bar{B})^2$-weighted mean of the path slopes; with the allocation fixed its mean term is exactly $\bar{\beta}_{bm:D}$ and its variance is $\sum_i \tilde{\beta}_{p(i)}^2 \mathrm{Var}\{(B_i - \mu_{bm})^2\}/(N\sigma_{bm}^2)^2 = 2\sum_p \varpi_p \tilde{\beta}_p^2 / N$, since a $\sigma_{bm}^2 \chi_1^2$ variable has variance $2\sigma_{bm}^4$. A Monte Carlo of the two-stage regression (4×10^4 trials, four paths of $18, 18, 17, 17$, slope heterogeneity far larger than in any design here) gave 0.002959 against 0.002992 from Equation (114), a relative difference of 1.1% (verified). [The

noncentrality of Equation (110) carries the slope heterogeneity as $\sigma^2_{bm}\tilde{\beta}^2_p$ inside σ^2_{res}, which corresponds to a factor $1/N$ rather than the $2/N$ of Equation (114). The two treatments agree to first order only in the sense that the term is negligible in every design evaluated: in Hybrid the path-mean term is at most 0.30 against residual variances of 41 to 49, and in OL+BDC at most 0.6% of the total (`docs/37`, Section 4.4; quoted).]

Power quadrature. Power is computed by averaging the noncentral-t tail probability over 400 quantiles of χ^2_{N-1}. `docs/37` (Section 4.4) records that a first version, which used N in place of $N-3$ and fixed SS_{bm} at its mean, overstated power by about 0.01 to 0.02, an error the direct simulation detected (quoted).

E9b variance. Counting baseline as an off-drug row raises $\text{Var}(\Delta)$ by about 15% (on paths 1–2, $51.04/44.13 = 1.157$), because the post-baseline weights no longer sum to zero ($m_1 = 1/(n_{\text{off}}+1)$) and the person-level ρ_{cs} terms of Equation (106) no longer cancel (derived; the exact split of the increase is Equation (158)). This is why E9 has more power than E9b under short carryover, while beyond about one week E9b has more, since its baseline row keeps zero coupling at every half-life (Table 25; `docs/37`, Section 5.2).

Mean slope. Table 26 gives the pooled slope $\bar{\beta}_{bm:D}$ that accompanies Table 25, together with the E9b rows of the two mean rules (verified; the decayed-mean E9b row is computed here and is not in the source).

Table 26: Closed-form pooled slope $\bar{\beta}_{bm:D}$ and the E9b power of the mean rules, Hybrid, compound symmetry, $N = 70$, strength 0.25, by $t_{1/2}$ (weeks) (Source: `docs/37`, Section 5; verified).

Quantity	Rule	Variant	0	0.1	0.2	0.5	1	2
Slope	Step	E9	−0.130	0	0	0	0	0
		E9b	−0.130	−0.029	−0.029	−0.029	−0.029	−0.029
	Graded	E9	−0.130	−0.130	−0.129	−0.118	−0.098	−0.072
		E9b	−0.130	−0.130	−0.129	−0.121	−0.105	−0.085
	Mean moderation	both	−0.130	−0.130	−0.130	−0.130	−0.130	−0.130
	Decayed mean mod.	E9, E9b	as graded coupling					
Power	Mean moderation	E9b	0.619	0.619	0.620	0.620	0.620	0.620
	Decayed mean mod.	E9b	0.619	0.619	0.613	0.557	0.453	0.318

[The graded E9b slope at $t_{1/2} = 1$ is -0.105475, so -0.105 to three places; `docs/37` printed -0.106, a double rounding, corrected there on 2026-10-08.]

The shape of the gap curve. Along Table 24 the curve is steepest near gap 1 and flattest near 0: the first 0.1 of gap lost costs 0.097 of E9 power (0.721 to 0.624), the last 0.1 only 0.007 (0.057 to 0.050) (Source: `docs/37`, Section 5.4; verified, 0.722, 0.625, 0.057, 0.050 with the path-mean term omitted). Short carryover costs almost nothing under graded coupling because the gap barely moves (0.991 at $t_{1/2} = 0.2$), not because the curve is flat there.

Floating-point underflow of the step rule. At very short half-lives the off-drug mean at the latest Hybrid off-drug visits underflows to exactly zero in double precision, so the step rule switches those visits' coupling off. Under the anchored mean used for the closed-form results this happens below about 0.0065 weeks (Hybrid, largest $t_{sd} = 7$) and 0.0093 weeks (CO); under the published compounding recursion, below about 0.012 weeks (about 2^{-1300} at the last off-drug visit of path 4 when $t_{1/2} = 0.01$) and 0.023 weeks (`docs/37`, Section 5.1, and its script; computed 2026-10-09). This is a property of the arithmetic, not of the model; the same mechanism under fast Weibull washout is treated in Section 9.9.

14.8 Power by interaction strength

The closed form extends to any strength below the ceiling. Table 27 gives it at 0.2 and 0.3, with the compound-symmetry ceiling of each covariance rule; cells at or above the ceiling have no joint distribution and are marked infeasible (Source: `docs/37`, Sections 5.5 and 5.6, Figure 3; verified). At zero strength every rule gives the test size, 0.05.

Table 27: Closed-form pooled power by strength (c_{bm} for the covariance rules, β_{bm} for the mean rules), rule and $t_{1/2}$ (weeks), Hybrid, compound symmetry, $N = 70$. "inf." marks a cell at or above the ceiling (Source: `docs/37`, Sections 5.5–5.6; verified).

Rule	Strength, variant	0	0.1	0.2	0.5	1	2
Ceiling, step		0.256	0.893	0.893	0.893	0.893	0.893
Ceiling, graded		0.256	0.256	0.257	0.274	0.307	0.369
Step	0.2, E9	0.521	0.050	0.050	0.050	0.050	0.050
	0.2, E9b	0.462	0.070	0.070	0.070	0.070	0.070
	0.3, E9	inf.	0.050	0.050	0.050	0.050	0.050
	0.3, E9b	inf.	0.095	0.095	0.095	0.095	0.095
Graded	0.2, E9	0.521	0.520	0.513	0.443	0.324	0.193
	0.2, E9b	0.462	0.462	0.456	0.407	0.323	0.225
	0.3, E9	inf.	inf.	inf.	inf.	0.630	0.379
	0.3, E9b	inf.	inf.	inf.	inf.	0.629	0.446
Mean moderation	0.2, E9	0.498	0.498	0.498	0.498	0.498	0.498
	0.3, E9	0.827	0.827	0.827	0.827	0.827	0.827
	0.3, E9b	0.771	0.771	0.772	0.772	0.772	0.772
Decayed mean mod.	0.2, E9	0.498	0.497	0.490	0.426	0.315	0.191
	0.3, E9	0.827	0.827	0.820	0.752	0.598	0.367

Three readings follow (`docs/37`, Section 5.5). Under the step rule the ceiling jumps to 0.893 at any $t_{1/2} > 0$ because a constant coupling pattern pays almost no switch cost (Section 9.4), so $c_{bm} = 0.3$ becomes feasible exactly where it carries no signal; its E9b power, 0.095, is close to the 0.12 and 0.10 published for Hybrid at $N = 70$, $c_{bm} = 0.3$, $t_{1/2} = 0.1$ and 0.2 (`docs/36`, Section 5.1; quoted), cells that were not repaired. Under graded coupling the ceiling rises with the half-life, so $c_{bm} = 0.3$ is feasible only from $t_{1/2} = 1$; that power there (0.630) exceeds power at $c_{bm} = 0.2$ without carryover (0.521) is a comparison across strengths, not a gain from carryover. Decayed mean moderation has no ceiling, so its decline can be traced at strengths the covariance rules cannot reach. For the path-stratified statistic its decline is strictly monotone by Proposition 3 with h_t in place of g_t, since $\sigma^2_{\Delta,p} = \mathrm{Var}(\Delta)$ is then fixed and only the slope shrinks (derived); for the pooled statistic it is non-increasing on every grid evaluated (`docs/37`, Section 5.6; quoted).

14.9 The published Hybrid construct: component means, outcome covariance and the expectancy regressor

This subsection collects the closed-form properties of the published construct (commit `58b32a9`) in the Hybrid design of Table 4 that `docs/40` uses to explain the design and its analysis model. Setting as in Section 14: visits at weeks $w_t = 4, 8, 9, 10, 11, 12, 16, 20$, $\eta_t = 1$ at the first two visits and 0.5 thereafter, compound symmetry with $\rho_{cs} = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$. The baseline has mean 83.07 and standard deviation $\sigma_{BL} = 18.48$ (18.48267 unrounded, $\sigma^2_{BL} = 341.6$), and the biomarker mean 124.33 and standard deviation $\sigma_{bm} = 15.36$ (15.36159) (quoted from `docs/40`, Section 3, which printed them from the `58b32a9` data; the unrounded values are those used in `check-stress-test-construct.R` and `check-baseline-design-search.R`; not re-read from the data here).

Component clocks and means. The three components run on different clocks:

$$E(BR_{it}) = D_{it}\, G(t_{od,it}; m_{BR}, 5, 0.42), \qquad E(PB_{it}) = \eta_t\, G(t_{pb,it}; m_{PB}, 5, 0.35),$$
$$E(TV_{it}) = G(w_t; m_{TV}, 5, 0.35), \tag{115}$$

without carryover, with standard deviations 8, $10\eta_t$ and 10. Time on drug restarts at zero on re-exposure, so in paths 1 to 4 the visit-by-visit t_{od} (weeks) is `4 8 9 10 0 0 4 0`, `4 8 9 10 0 0 0 4`, `4 8 9 0 0 0 4 0` and `4 8 9 0 0 0 0 4` (`docs/40`, Section 2, verified there against the published `buildtrialdesign()` output; recomputed here from the drug states). Expectancy stays positive after baseline in every path, so $t_{pb} = w_t$, and TV and PB have the same means in all four paths; only BR differs between paths. Table 28 gives the means.

Table 28: Expected component means by week, Hybrid, no carryover, at the exact shipped parameters (Source: `docs/40`, Section 3, and `docs/42`, Section 2, for the $TV + PB$ row; all entries verified, `check-landmarks-hybrid-ceilings.R`, block 3, and `check-baseline-design-search.R`).

Week	4	8	9	10	11	12	16	20
TV, all paths	1.86	4.79	5.24	5.59	5.85	6.03	6.39	6.48
PB, all paths	1.86	4.79	2.62	2.79	2.92	3.02	3.19	3.24
$TV + PB$, all paths	3.73	9.58	7.86	8.38	8.77	9.05	9.58	9.72
BR, path 1	4.28	9.22	9.79	10.19	0	0	4.28	0
BR, path 2	4.28	9.22	9.79	10.19	0	0	0	4.28
BR, path 3	4.28	9.22	9.79	0	0	0	4.28	0
BR, path 4	4.28	9.22	9.79	0	0	0	0	4.28

The total mean improvement peaks at 18.80 at week 8 and falls to 17.66 at week 9, although every path is still on drug, because the expectancy weight halves (verified). The carryover recursion and its anchored alternative for path 1 at $t_{1/2} = 1$ are Equation (23).

[The within-path contrast of the on-drug visits at weeks 8 and 9 is described in `docs/40`, Section 7, and `docs/31`, Sections 5 and 7.2, as identifying $0.5\,PB$. In mean it is $E(PB_8) - E(PB_9) = G(8) - 0.5\,G(9) = 2.17$, against $0.5\,G(9) = 2.62$, because the belief curve keeps rising between the visits (verified). The statement is approximate, as `docs/31` marks with $\approx$; the contrast identifies the product of the expectancy drop and the curve only once the curve's growth between the two visits is modeled.]

Per-path interaction slopes. Equation (107) evaluated per path gives Table 29. The estimand can be read on Hendrickson's trajectory template: participants one standard deviation above and below the biomarker mean differ in expected drug response by $2c_{bm}g_t\sigma_{BR}$ at visit t (4 points at full coupling and $c_{bm} = 0.25$), and the slope is minus the difference between the mean on-drug and mean off-drug band widths divided by $2\sigma_{bm}$ (`docs/40`, Section 5; derived from Equation (30)).

Table 29: E9 interaction slope by path, Hybrid, $c_{bm} = \beta_{bm} = 0.25$, $\sigma_{BR} = 8$, $\sigma_{bm} = 15.36$ (Source: `docs/40`, Section 5, `interaction-gap-by-path.csv`; all entries verified, `check-landmarks-hybrid-ceilings.R`, block 4).

Rule, $t_{1/2}$ (weeks)	Path 1	Path 2	Path 3	Path 4
Step or graded, 0	−0.130	−0.130	−0.130	−0.130
Step, 0.1 (any $t_{1/2} > 0$)	0	0	0	0
Graded, 1	−0.095	−0.097	−0.100	−0.101
Mean moderation, any	−0.130	−0.130	−0.130	−0.130

Under graded coupling at $t_{1/2} = 1$ the mean off-drug couplings are 0.271, 0.255, 0.234 and 0.221 (off-drug t_{sd} of $\{1, 2, 4\}$, $\{1, 2, 6\}$, $\{1, 2, 3, 4\}$ and $\{1, 2, 3, 7\}$ weeks), so path 4, whose off-drug visits lie furthest from a discontinuation, keeps the largest slope (derived).

The outcome covariance is not compound symmetric. With $Y_{it} = \mathrm{BL}_i - S_{it}$, $S_{it} = \sum_c s_{c,t} X_{c,t}$, $s_t = (s_{TV,t}, s_{PB,t}, s_{BR,t})^\top = (10, 10\eta_t, 8)^\top$ and $c_{bm} = 0$, the compound-symmetry construct of Equation (56) gives

$$\mathrm{Cov}(Y_{it}, Y_{is}) = \sigma^2_{BL} + s_t^\top K_{ts}\, s_s, \qquad K_{tt} = (1-c_1)I_3 + c_1 J_3, \quad K_{ts} = (\rho_{cs} - c_\times)I_3 + c_\times J_3 \ (t \neq s), \tag{116}$$

and $\mathrm{Cov}(Y_{i0}, Y_{it}) = \mathrm{Var}(Y_{i0}) = \sigma^2_{BL}$ for the baseline row, which carries no response component (derived; the compound-symmetry counterpart of Equation (69)). Table 30 gives the values.

Table 30: Outcome variances and covariances by phase under the compound-symmetry construct, Hybrid path 1, $c_{bm} = 0$, unconditional and conditional on baseline, at $\sigma_{BL} = 18.48267$ (Source: `docs/40`, Section 6; `docs/41`, Sections 3 and 6; all entries verified against the full 26×26 construct and Equation (116), `check-landmarks-hybrid-ceilings.R`, block 5, and `check-baseline-design-search.R`).

	Variance	Covariance within phase	Covariance with baseline	Conditional variance	Conditional covariance
Baseline	341.6				
Open label (weeks 4, 8)	709.6	604.8	341.6	368.0	263.2
Blinded (weeks 9 to 20)	598.6	526.8	341.6	257.0	185.2
Covariance between an open-label and a blinded visit: 555.8					

The variance falls by 111 at blinding because the placebo-belief standard deviation halves, so a random-intercept model, which implies one variance and one covariance for all post-baseline visits, misstates the outcome covariance even when the data-generating correlations are compound symmetric. Conditioning on baseline removes σ^2_{BL} from every entry but leaves the phase difference (368.0 against 257.0), so a random intercept on the post-baseline visits alone is still misspecified (`docs/41`, Section 6, item 5). [`check-landmarks-hybrid-ceilings.R` uses the rounded $\sigma_{BL} = 18.48$ and prints 341.5, 709.5, 604.7, 598.5, 526.7 and 555.7; the table uses the unrounded value. `docs/40` rounds the entries to 342, 710, 605, 599, 527 and 556. Its further statement that under AR(1) data the published analysis rejects 14% to 23% of null crossover trials is a simulation result of `docs/36`, Section 6.8, quoted there and not rechecked.]

The expectancy regressor and the drug indicator. In the Hybrid long data at the published allocation (18, 18, 17, 17), with the baseline row coded $D_{i0} = 0$, $\eta_0 = 0$, $w_0 = 0$, two different collinearity measures are in use, and both are correct for what they measure (verified, `check-landmarks-hybrid-ceilings.R`, block 6, and `check-reconciliations.R`):

- the allocation-weighted correlation of η_t with the binary D_{it} over the nine rows is $r = 0.588$, so $r^2 = 0.345$ and the variance inflation factor of either regressor in a model containing only the two is $1/(1 - r^2) = 1.53$;
- the squared multiple correlation of η_t on D_{it} *and* linear time w_t is $R^2 = 0.354$, so the variance inflation factor of η_t in a model that also contains linear time, as the published analysis model does, is $1/(1 - R^2) = 1.55$ (that of D_{it} there is 1.54).

Over the eight post-baseline rows the corresponding values are $r = 0.507$ ($1/(1 - r^2) = 1.35$) and $R^2 = 0.516$ ($1/(1 - R^2) = 2.07$): without the baseline row the time trend carries more of η_t. (Source: `docs/40`, Section 8, which gives 0.59, 0.35 and 1.5, the R^2 with time.) Every open-label visit is on drug and every row with $\eta_t = 0$ is baseline, so the collinearity that led Hendrickson et al. to drop the expectancy term is real but moderate. [The correlation $\approx +0.6$ of Section 12 is between D_{bc} and η in Paper 06's analysis data; the value here is for the binary indicator of the published analysis. With D_{bc} at $t_{1/2} = 1$ week in place of D_{it}, equal path weights and the baseline row included, the Hybrid value is 0.645 (0.485 without the baseline row; `check-gompertz-rmanova-cycles.R`), which is near Paper 06's figure, but which rows and half-life Paper 06 used was not located (§12.4).]

14.10 Other designs: OL+BDC and the crossover

Each path has its own weights a_t and coupling pattern, and pooling is as in Equation (110). The two further published designs, with $N = 70$ allocated 35 and 35, are in Table 31 (Source: `docs/37`, Section 5.7; verified, the t_{sd} values from the published `buildtrialdesign()` semantics as reproduced in the source's builder). The expectancy η_t is 1 at the four open-label OL+BDC visits and 0.5 thereafter, and 0.5 at every CO visit.

Table 31: OL+BDC and CO paths: visit weeks, drug state and t_{sd} (weeks) at visits 1–8 (Source: `docs/37`, Section 5.7).

Design	Weeks	Path	D_t	t_{sd}
OL+BDC	4, 8, 12, 16, 17, 18, 19, 20	1	1 1 1 1 1 1 0 0	0 0 0 0 0 0 1 2
		2	1 1 1 1 1 0 0 0	0 0 0 0 0 1 2 3
CO	2.5, 5, . . . , 20	1	1 1 1 1 0 0 0 0	0 0 0 0 2.5 5 7.5 10
		2	0 0 0 0 1 1 1 1	0 0 0 0 0 0 0 0

OL+BDC has the structure of Hybrid: every off-drug visit follows an exposure, one to three weeks after it, so Sections 14.3 and 14.4 apply unchanged. Under compound symmetry E9 power under the step rule falls from 0.598 to 0.050 at every positive half-life, and under graded coupling it is 0.598, 0.597, 0.586, 0.477, 0.305 and 0.157 at $t_{1/2} = 0$, 0.1, 0.2, 0.5, 1, 2 weeks, faster than in Hybrid because its off-drug visits come sooner after discontinuation (verified).

CO differs structurally: path 2 is placebo first, so its four off-drug visits precede any exposure, keep $g_t = h_t = 0$ under every rule, and keep the full slope -0.1302; the half-life acts on path 1 alone (derived). Under the step rule path 1 loses its E9 slope entirely and keeps $1/(n_{\text{off}} + 1) = 1/5$ of it under E9b, so the pooled slopes are

$$\bar{\beta}^{\text{E9}}_{bm:D} = \tfrac{1}{2}(0 - 0.1302) = -0.065, \qquad \bar{\beta}^{\text{E9b}}_{bm:D} = \tfrac{1}{2}\big(-\tfrac{1}{5}\,0.1302 - 0.1302\big) = -0.078 \tag{117}$$

at every $t_{1/2} > 0$ (derived; verified), against -0.075 to -0.082 in the simulated published analysis (quoted). Under graded coupling CO is robust to short carryover, because path 1's off-drug visits fall 2.5 to 10 weeks after discontinuation: E9 power is 0.622 at every half-life up to 0.2 weeks and 0.578 at one week (verified).

The period effect. E9's common intercept assumes the paths share one mean contrast. In CO they do not, since path 1 is on drug in the first half of the trial and path 2 in the second while the natural-course and expectancy components trend over time. Without carryover the mean contrasts are $\mu_1 = -2.29$ and $\mu_2 = -11.12$, so the path-mean term of Equation (110) is $\frac{1}{4}(\mu_1 - \mu_2)^2 = 19.5$ against $\sum_p \varpi_p \sigma^2_{\Delta,p} = 35.9$ (mean moderation), 54% of it; carryover narrows path 1's contrast ($\mu_1 = -0.50$ at $t_{1/2} = 2$), and the term grows to 28.2 (verified; the means μ_p from the source's builder). Setting the term to zero, which a path-specific intercept in the second stage would do, gives Table 32 (closed form, not simulated). In Hybrid and OL+BDC removing the term changes graded-coupling power by at most 0.001 (verified; `docs/37` states at most 0.003 over all rules).

With the period effect removed the three behaviors of the Hybrid design reappear exactly: a switch, a monotone decline and no dependence. The loss belongs to E9, not to the crossover: a simulation of the published `lmer` analysis (1000 replicates per cell) gave power 0.740 under mean moderation without carryover for the 35/35 crossover against 0.582 for E9 on the same replicates, and 0.736 to 0.770 when all 70 participants follow one sequence; adding a fixed path effect to the `lmer` model changed its power by at most 0.001 in any cell (`docs/37`, Sections 5.7 and 6; quoted). The mixed model's within-participant time term absorbs the trend that enters the two sequences' E9 contrasts with opposite signs.

Table 32: CO, compound symmetry, E9 power with and without the path-mean term, by rule and $t_{1/2}$ (weeks) (Source: `docs/37`, Section 5.7; verified).

Rule, path term	0	0.1	0.2	1	2
Step, as specified	0.622	0.199	0.199	0.192	0.178
Step, term removed	0.817	0.283	0.283	0.283	0.283
Graded, as specified	0.622	0.622	0.622	0.578	0.480
Graded, term removed	0.817	0.817	0.817	0.794	0.737
Mean moderation, as specified	0.590	0.590	0.590	0.572	0.531
Mean moderation, term removed	0.771	0.771	0.771	0.771	0.771

14.11 Measurement spacing and the coupling at the first off-drug visit

Under graded coupling (or with the exposure-decayed indicator of Equation (13)) the coupling at the first off-drug visit of a run is $2^{-d_1/t_{1/2}}$, with d_1 the gap between the last on-drug visit and that visit. In the crossover design ($d_1 = 2.5$ weeks) it is 0.177 at $t_{1/2} = 1$ and 0.031 at $t_{1/2} = 0.5$; in Hybrid and OL+BDC ($d_1 = 1$ week) it is 0.5 and 0.25 (Source: `docs/33`, Section 2.5; verified, `check-landmarks-hybrid-ceilings.R`, block 11). The crossover's off-drug visits are therefore nearly free of carryover at every half-life of the canonical grid, which is consistent with its insensitivity to carryover in the pre-restatement Paper 01 results (argued; `docs/33` labels the explanation inferred and untested, and proposes varying the crossover spacing at fixed duration as the test). [Paper 14, Appendix C.4, gives the same spacing argument for the crossover ceiling being constant in $t_{1/2}$ (Table 19). That attribution does not hold for the ceiling. The design ceiling is the minimum over paths, and it is set by the placebo-first path B, whose off-drug visits all precede exposure and so carry $g_t = 0$ at every half-life. The drug-first path A's own ceiling does rise with the half-life, from 0.619 at $t_{1/2} = 0$ to 0.623, 0.639 and 0.651 at 0.5, 1 and 2 weeks under the non-separable construct at $\rho = 0.7$, and from 0.637 to 0.644, 0.674 and 0.709 under the separable form at $\rho = 0.8$ (verified, `check-crossover-binding-path.R`).] [`docs/33` offers this in place of the pre-restatement manuscript's account (that OL+BDC sustains its off-drug window longest and that CO alternates quickly), which the design table contradicts: CO path A has the longest uninterrupted off-drug period of any path (four visits over ten weeks), and each CO participant switches once.]

14.12 AR(1) response structures: the factor-by-time decomposition

Proposition 2 does not involve the response correlation, and Proposition 1's independence of $t_{1/2}$ holds for any response block free of $t_{1/2}$, so the closed form applies to the AR(1) structures of Section 9.3 with only $\mathrm{Var}(\Delta)$ changed. Write each response correlation as a sum of factor-by-time Kronecker terms $M = \sum_\ell K^{(\ell)} \otimes T^{(\ell)}$ and $s_t = (\sigma_{TV}, \sigma_{PB,t}, \sigma_{BR})^\top$ for the component SDs at visit t. Then

$$\mathrm{Var}(\Delta) = \sum_\ell \sum_{t,s} a_t a_s \, T^{(\ell)}_{ts} \left(s_t^\top K^{(\ell)} s_s \right), \tag{118}$$

with the terms of Table 33 (Source: `docs/37`, Section 5.8; `docs/36`, Appendix A.7; derived, and the factored variance agrees with the full-matrix value to 10^{-13} in all 48 design-structure-statistic-path cases, quoted; verified here for the separable CO case). When η_t is constant over the contrast, as in CO, $s_t = s$ and the separable variance is a single product,

$$\mathrm{Var}(\Delta) = (s^\top K s)\,(a^\top A a) \tag{119}$$

(212.846 both ways on CO path 1; verified).

Why AR(1) costs power. Under compound symmetry the person-level term $K_p \otimes J$ is constant across visits, and E9's weights sum to zero, so $a^\top J a = (\sum_t a_t)^2 = 0$ and that term cancels, leaving the occasion-level term with factor variances $1 - \rho_{cs} = 0.2$ (derived). Under

Table 33: Factor-by-time terms of the three response structures (Source: `docs/37`, Section 5.8). $A_{ts} = \rho^{|w_t - w_s|}$; I the occasion kernel; J the all-ones (person-level) kernel.

Structure	Terms $K^{(\ell)} \otimes T^{(\ell)}$
Separable AR(1)	$K \otimes A$, $K = (1 - c_1)I_3 + c_1 J_3$
AR(1) `covar`	$K_a \otimes A + K_b \otimes I$, $K_a = (1 - c_\times)I_3 + c_\times J_3$, $K_b = (c_1 - c_\times)(J_3 - I_3)$
Compound symmetry (published)	$K_o \otimes I + K_p \otimes J$, $K_o = (1 - \rho_{cs})I_3 + (c_1 - c_\times)(J_3 - I_3)$, $K_p = \rho_{cs} I_3 + c_\times (J_3 - I_3)$

AR(1) there is no constant component to cancel: visits weeks apart are nearly uncorrelated ($0.8^{2.5} = 0.57$ at the CO spacing, $0.8^5 = 0.33$ at five weeks), so a contrast between them keeps most of each factor's variance. Table 34 gives the result; the slope is the same under every structure, so the comparison is at an identical interaction and only the noise differs.

Table 34: Mean E9 contrast variance Var(Δ) over paths, by design and response structure, $\rho = 0.8$ (Source: `docs/37`, Section 5.8; verified).

Design	Compound symmetry	AR(1) `covar`	Separable AR(1)
Hybrid	44.6	121.5	117.9
OL+BDC	58.4	186.0	188.3
CO	35.9	201.7	212.9

Table 35: Closed-form E9 power by design, structure and rule, strength 0.25, $N = 70$, $\rho = 0.8$ (Source: `docs/37`, Section 5.8; verified). Step: $t_{1/2} = 0$ / $t_{1/2} = 0.1$ (CO declines slowly beyond, to 0.178, 0.084 and 0.083 at $t_{1/2} = 2$). Mean moderation: range over $t_{1/2} \in [0, 2]$.

Design	Structure	Step	Graded, $t_{1/2} = 0$ / 0.5 / 1 / 2	Mean moderation
Hybrid	Compound symmetry	0.721 / 0.050	0.721 / 0.631 / 0.473 / 0.278	0.681–0.682
Hybrid	AR(1) `covar`	0.326 / 0.050	0.326 / 0.276 / 0.205 / 0.130	0.317
Hybrid	Separable	0.335 / 0.050	0.335 / 0.283 / 0.210 / 0.133	0.325
OL+BDC	Compound symmetry	0.598 / 0.050	0.598 / 0.477 / 0.305 / 0.157	0.568
OL+BDC	AR(1) `covar`	0.229 / 0.050	0.229 / 0.183 / 0.127 / 0.082	0.225
OL+BDC	Separable	0.226 / 0.050	0.226 / 0.181 / 0.126 / 0.082	0.222
CO	Compound symmetry	0.622 / 0.199	0.622 / 0.615 / 0.578 / 0.480	0.531–0.590
CO	AR(1) `covar`	0.199 / 0.086	0.199 / 0.198 / 0.189 / 0.168	0.191–0.196
CO	Separable	0.192 / 0.084	0.192 / 0.190 / 0.182 / 0.163	0.184–0.189

The contrast variance is 2.6 to 2.7 times larger in Hybrid and up to 5.9 times larger in CO, and power at strength 0.25 falls to between a third (CO) and a half (Hybrid) of its compound-symmetry value (Table 35). The shape of the dependence on the half-life is unchanged: a switch under the step rule, a monotone decline under graded coupling and decayed mean moderation, and no dependence under mean moderation except through CO's period effect. Separability changes power relative to `covar` by at most 0.009; its effect is on feasibility. The ceilings of every structure and rule at $\rho = 0.8$, including $t_{1/2} = 2$, are collected in Table 14 of Section 9.7. The separable ceiling factorizes as $c^*_{bm} = ([K^{-1}]_{33} \max_p g_p^\top A^{-1} g_p)^{-1/2}$ (Equation (60) minimized over paths), which agrees with the direct ceiling to 1.1×10^{-16} (verified).

In CO the graded ceiling does not move with the half-life, because the placebo-first path keeps its pre-exposure zeros and binds. Under AR(1) the CO step ceiling *falls* at positive half-lives (0.627 to 0.576): path 1's pattern becomes constant, and a constant pattern over eight weakly correlated visits demands more of the biomarker than an on/off pattern does (Section 9.7). The Hybrid compound-symmetry row of Table 35 is that of Table 25, repeated

for the comparison across structures.

14.13 Validation of the closed form

Against direct simulation. `docs/37` (Section 6) simulated the statistic from the full joint distribution, not from the formulas, in 283 cells (three designs, three structures, four rules, strengths 0.1 to 0.3), each with 5000 trials of 70 participants and both variants, 566 cell-variants in all; Monte Carlo standard errors are about 0.007 on power, 0.0007 on the mean slope and 0.0005 on its SD. Table 36 summarizes the agreement, with z the difference between simulated and closed-form power in binomial standard errors. Sixteen of 566 cell-variants (2.8%) lie beyond 2 standard errors, against about 5% expected by chance, with deviations of both signs; near the ceiling (graded coupling at $c_{bm} = 0.3$, $t_{1/2} = 1$, ceiling 0.307) the deviations are under 0.6 standard errors. The exact zero slope under the step rule, the floor at the test size, the CO floor and the monotone declines are reproduced without exception. (Quoted from `docs/37`; the block summaries were recomputed here from the stored `e9-closed-form-vs-mc.csv`, inspected, not re-simulated.)

Table 36: Closed form against direct Monte Carlo, by validation block (Source: `docs/37`, Section 6, `e9-closed-form-vs-mc.csv`).

Block	Cells	Max $\lvert z\rvert$	Mean z	$\lvert z\rvert > 2$	Max mean diff.
Hybrid, CS, strength 0.25, four rules	48	2.37	0.00	2	0.0014
Hybrid, CS, strengths 0.1, 0.2, 0.3	134	2.45	−0.21	2	0.0018
Hybrid, AR(1) structures	96	2.18	0.00	2	0.0027
OL+BDC, three structures	144	2.83	0.05	6	0.0038
CO, three structures	144	2.44	−0.09	4	0.0044

Table 37: E9 closed form (cf) against Monte Carlo (MC): mean slope, its SD, and power, Hybrid, compound symmetry, strength 0.25 (Source: `docs/37`, Section 6; stored output inspected). The MC columns are from the 2026-10-09 rerun with anchored carryover and per-cell seeds; the earlier values came from a legacy file with other seeds, so they change by up to about 0.02 mainly through reseeding (also at $t_{1/2} = 0$); paired on the same seeds the carryover-only change is at most 0.0022 in power.

Rule	$t_{1/2}$	Mean, cf	Mean, MC	SD, cf	SD, MC	Power, cf	Power, MC
Step	0	−0.1302	−0.1307	0.0507	0.0502	0.721	0.725
Step	0.1	0	0.0008	0.0531	0.0530	0.050	0.051
Step	0.5	0	0.0001	0.0531	0.0529	0.050	0.050
Step	2	0	0.0003	0.0531	0.0541	0.050	0.055
Graded	0	−0.1302	−0.1300	0.0507	0.0509	0.721	0.723
Graded	0.2	−0.1290	−0.1283	0.0508	0.0511	0.712	0.705
Graded	0.5	−0.1180	−0.1174	0.0511	0.0506	0.631	0.631
Graded	1	−0.0982	−0.0978	0.0517	0.0520	0.473	0.472
Graded	2	−0.0717	−0.0725	0.0524	0.0511	0.278	0.273
Mean moderation	0	−0.1302	−0.1296	0.0531	0.0525	0.681	0.685
Mean moderation	1	−0.1302	−0.1302	0.0531	0.0529	0.682	0.682
Mean moderation	2	−0.1302	−0.1301	0.0531	0.0533	0.682	0.673

Against the published mixed model. `docs/36` simulated the published process (configuration A, step rule) and mean moderation with the published `lmer` analysis, 500 replicates per cell, at strength 0.25, here repeated with anchored carryover on the same seeds (2026-10-09); E9b reproduces it (Table 38). The Monte Carlo standard error of the `lmer` power is about 0.02 and of its mean about 0.0026. Mean estimates agree to within 0.004 and SDs to within

0.003; power agrees to within 0.034 in nine of ten cells, the exception being mean moderation at $t_{1/2} = 0.1$ (0.672 against 0.619, about 2.6 standard errors). In OL+BDC and CO, E9b reproduces the `lmer` mean estimates in every cell, including the partial step floors (−0.078 against −0.075 to −0.082 in CO; −0.038 against −0.034 to −0.040 in OL+BDC), but its power is lower: in CO because the mixed model's time term absorbs the period effect (`lmer` SD 0.048 to 0.053 against 0.055 to 0.059), and in OL+BDC, where the SDs agree, because the `lmer` test appears anticonservative, consistent with the Type I error of 0.06 to 0.11 that `docs/36` (Section 6.8) reports there under compound symmetry (inferred in the source, not tested). The closed form is therefore a proxy for the published analysis's mean behavior in every design and for its power in Hybrid (quoted; the cell values match the stored `e9-vs-lmer.csv`, inspected).

Table 38: Published `lmer` analysis (500 replicates) against the E9b closed form, Hybrid, compound symmetry, strength 0.25 (Source: `docs/37`, Section 6; `e9-vs-lmer.csv`).

Rule	$t_{1/2}$	`lmer` mean	E9b mean	`lmer` SD	E9b SD	`lmer` power	E9b power
Step	0	−0.126	−0.130	0.055	0.055	0.632	0.654
Step	0.1	−0.028	−0.029	0.055	0.057	0.080	0.081
Step	0.2	−0.026	−0.029	0.058	0.057	0.096	0.081
Step	0.5	−0.029	−0.029	0.058	0.057	0.108	0.081
Step	1	−0.032	−0.029	0.060	0.057	0.106	0.081
Mean moderation	0	−0.131	−0.130	0.059	0.057	0.628	0.619
Mean moderation	0.1	−0.132	−0.130	0.058	0.057	0.672	0.619
Mean moderation	0.2	−0.127	−0.130	0.058	0.057	0.636	0.620
Mean moderation	0.5	−0.131	−0.130	0.059	0.057	0.646	0.620
Mean moderation	1	−0.129	−0.130	0.058	0.057	0.654	0.620

One path against the published allocation. The published code assigns participants to paths in fixed numbers rather than at random. `docs/37` (Section 6; script `06-single-path-lme.R`, 1000 replicates per cell, 72,000 fits, none failed) simulated every design-rule cell both with that allocation and with all 70 participants on one path, under `lmer`, `lmer` with a fixed path effect, and E9. Its findings (quoted): simulated E9 power lies within 2.61 standard errors of the closed form in all 72 cells, with 1 beyond 2; in Hybrid the mixed allocation lies within 0.038 of every single path under `lmer` and within 0.018 of their mean; in CO the mixed E9 power is 0.18 to 0.25 below the weaker sequence alone wherever neither sequence is at the step floor, which is the period effect, and mixed `lmer` power is not; and a fixed path effect changes `lmer` power by at most 0.002. The data-generating process studied associates the biomarker with the drug response only; path randomization protects against a biomarker-by-time confound that this process cannot produce, so the comparison does not test that role.

On the published code. Replacing only the coupling block of the `58b32a9` `generateData()` by graded coupling, with the published pipeline, design, $N = 70$, $c_{bm} = 0.25$ and 500 replicates, gave power 0.668, 0.638, 0.656 and 0.462 at $t_{1/2} = 0$, 0.1, 0.2 and 1 against 0.668, 0.126, 0.092 and 0.094 for the published code, with mean estimates −0.131, −0.129, −0.131, −0.107 against −0.131, −0.028, −0.029, −0.033 (`docs/37`, Appendix A; quoted, Monte Carlo standard error at most 0.022). The published-code power of 0.126 at $t_{1/2} = 0.1$ is about 2.4 combined standard errors above the independent reconstruction (0.080), which the source notes as larger than expected.

15 Design efficiency and time adjustment

Sources: `docs/39`, `docs/42`, Paper 08's short communication and Paper 12. Throughout, $a_t = 1/n_{\text{on}}$ on drug and $-1/n_{\text{off}}$ off drug, $\Delta_i = \sum_t a_t Y_{it}$, and C is the response covariance over visits.

15.1 The criterion

Under decayed mean moderation, the biomarker shifts the drug response by $\beta_{bm}\sigma_{BR}b_i h_t$, with $h_t = 1$ on drug and $e^{-\lambda t_{sd}}$ off drug. The slope of Δ_i on B_i is

$$\beta_{bm:D} = -\beta_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\big(1-\bar{h}_{\text{off}}\big), \qquad \sigma^2_\Delta = \mathbf{a}'C\mathbf{a}, \tag{120}$$

where σ^2_Δ is the residual variance of the contrast. Without carryover the slope is the same in every design, so a design acts only through σ^2_Δ and through $\bar{h}_{\text{off}}$. Power is governed by the signal-to-noise ratio $\beta^2_{bm:D}\sigma^2_{bm}/\sigma^2_\Delta$, which fixes the noncentrality $\sqrt{n-1}\,\beta_{bm:D}\sigma_{bm}/\sigma_\Delta$ (Source: `docs/39`, Section 2; Paper 01, Appendix A.2; derived). [`docs/39` calls this ratio the squared contrast-biomarker correlation, writing it $\rho^2_{\Delta B}$ ($r^2_{\Delta B}$ here, as in §14). Under mean moderation the shift adds variance, so the squared correlation is $r^2_{\Delta B} = \beta^2_{bm:D}\sigma^2_{bm}/(\sigma^2_\Delta + \beta^2_{bm:D}\sigma^2_{bm})$ and the source's quantity is $r^2_{\Delta B}/(1-r^2_{\Delta B})$. The two are monotone in each other, so no design ranking changes.]

15.2 The simplest hybrid design: an exact estimator

Paper 08's short communication takes p on-drug occasions followed by q off-drug occasions, equally spaced, with the additive model $Y_{it} = \mu + \beta_B b_i + \beta_D D_t + \beta_{bm:D}\, b_i D_t + \varepsilon_{it}$, errors with variance σ^2 and correlation matrix R. [The source writes β, γ, θ and φ for β_B, β_D, $\beta_{bm:D}$ and the AR(1) parameter; γ and θ are reserved, and ρ_{cs} is the compound-symmetry correlation, as everywhere in this document. The occasion counts p and q are the source's and are local to this subsection.] The contrast cancels μ, $\beta_B b_i$ and any participant offset, leaving $\Delta_i = \beta_D + \beta_{bm:D}b_i + \varepsilon_i$, so the least-squares slope of Δ_i on b_i is unbiased for every R, and, because every participant shares the design, OLS coincides with GLS:

$$\operatorname{Var}(\hat{\beta}_{bm:D}) = \frac{V_\varepsilon}{(N-1)s^2_b}, \qquad V_\varepsilon = \sigma^2\Big(\frac{S_{\text{on}}}{p^2} + \frac{S_{\text{off}}}{q^2} - \frac{2S_\times}{pq}\Big), \tag{121}$$

with S_{on}, S_{off} and $S_\times$ the sums of R over the on-drug block, the off-drug block and the cross block. Under compound symmetry and under AR(1) on unit spacing,

$$V_\varepsilon^{\text{CS}} = \sigma^2(1-\rho_{cs})\Big(\frac{1}{p}+\frac{1}{q}\Big), \tag{122}$$

$$V_\varepsilon^{\text{AR}} = \sigma^2\Big(\frac{S_p}{p^2} + \frac{S_q}{q^2} - \frac{2C_{pq}}{pq}\Big), \quad S_m = \frac{m(1+\rho)}{1-\rho} - \frac{2\rho(1-\rho^m)}{(1-\rho)^2}, \quad C_{pq} = \frac{\rho(1-\rho^p)(1-\rho^q)}{(1-\rho)^2}. \tag{123}$$

For a lag-only (Toeplitz) correlation ρ_k, $S_\times = \sum_{\ell=1}^{p+q-1}\min(\ell, p, q, p+q-\ell)\,\rho_\ell$, and a random time slope of variance σ^2_{slope} adds $\sigma^2_{\text{slope}}(\bar{t}_{\text{on}} - \bar{t}_{\text{off}})^2$ to V_ε. (Source: Paper 08, short communication; all four identities verified against the full quadratic form $\mathbf{a}'R\mathbf{a}$, `check-design.R`.) At $p = q = 4$ and $\rho_{cs} = \rho = 0.5$, $V_\varepsilon^{\text{CS}} = 0.25\,\sigma^2$ and $V_\varepsilon^{\text{AR}} = 0.81\,\sigma^2$, a ratio of 3.25; at $p = q = 8$ the ratio is 4.51 (verified). Under compound symmetry only the counts enter; under AR(1) the contiguous blocks share little noise and the contrast cancels little of it. [The source reads the ratio as the factor by which a compound-symmetry analysis understates the variance. That holds at a matched lag-one correlation; a fitted compound-symmetry model estimates an average correlation instead, and its understatement is smaller (compare `docs/45`, Section 4.3, $\kappa = 0.740$ for the crossover, computed in §15.5). Argued, not computed for this design.]

Three phases. With p_1 open-label on-drug, p_2 blinded on-drug and q blinded off-drug occasions, expectancy $\eta = 1$ and η_b ($= 0.5$) and the model extended by $\beta_\eta\eta_t + \beta_{bm:\eta}b_i\eta_t$, the class slopes are $s_1 = \beta_B + \beta_{bm:D} + \beta_{bm:\eta}$, $s_2 = \beta_B + \beta_{bm:D} + \eta_b\beta_{bm:\eta}$ and $s_3 = \beta_B + \eta_b\beta_{bm:\eta}$, so

$$\beta_{bm:D} = s_2 - s_3, \qquad \beta_{bm:\eta} = \frac{s_1 - s_2}{1-\eta_b}, \tag{124}$$

each the slope of a single within-participant contrast (derived; Paper 08 short communication). The drug interaction is identified within the blinded window, and its variance is (121) applied to the p_2 and q blinded occasions.

15.3 Contrast variance by design under CS and AR(1)

Compound symmetry. Expanding σ^2_Δ with Paper 01's Appendix A.1 formula (Equation (106)), the person-level component cancels and four terms remain. The balance term is proportional to $1/n_{\text{on}} + 1/n_{\text{off}}$; an expectancy-confounding term appears whenever $\sum_t a_t \eta_t \neq 0$, that is, whenever open-label visits enter the contrast. Arrangement does not enter at all (Source: `docs/39`, Section 3; derived). With $m_1 = \sum_t a_t = 0$ for E9, Equation (106) splits into four parts,

$$\mathrm{Var}(\Delta) = \mathcal{B} + \mathcal{P} + \mathcal{E} + \mathcal{X}, \qquad \mathcal{B} = (\sigma^2_{TV} + \sigma^2_{BR})(1-\rho_{cs})\, m_2, \qquad \mathcal{P} = \sigma^2_{PB}(1-\rho_{cs})\, m_{ee},$$
$$\mathcal{E} = \sigma^2_{PB}\, \rho_{cs}\, m_e^2, \qquad \mathcal{X} = 2(c_1 - c_\times)\big[\sigma_{TV}\sigma_{BR}\, m_2 + (\sigma_{TV} + \sigma_{BR})\sigma_{PB}\, m_{2e}\big] \tag{125}$$

(derived): the balance term, the occasion-level placebo term, the expectancy confound and the cross-component term. $\mathcal{E}$ is the person-level placebo variance that the contrast fails to cancel when $m_e = \sum_t a_t \eta_t \neq 0$, that is, when open-label visits enter the contrast; $\mathcal{P}$ carries the doubled placebo SD at those visits. $\mathcal{B} \propto 1/n_{\text{on}} + 1/n_{\text{off}}$, which is 0.500 at 4/4, 0.533 at 5/3 (6.7% more) and 0.667 at 6/2 (33% more). (Source: `docs/39`, Section 3, which tabulates $\mathcal{B}$, $\mathcal{E}$ and the total; verified.)

Table 39: Components of $\mathrm{Var}(\Delta)$ under the CS construct ($\rho_{cs} = 0.8$), and the time part and $\mathrm{Var}(\Delta)$ under separable AR(1) ($\rho = 0.8$ per week, no measurement error), per path (verified, `check-baseline-design-search.R`; agree with `design-contrast-structure.csv`). CO, ABAB and alternating designs at 2.5-week spacing.

Design, path	On/off	$\mathcal{B}$	$\mathcal{P}$	$\mathcal{E}$	$\mathcal{X}$	CS total	$\mathbf{a}'A\mathbf{a}$	AR(1) total
Hybrid, 1	5/3	17.49	3.87	3.20	19.57	44.13	0.384	110.9
Hybrid, 2	5/3	17.49	3.87	3.20	19.57	44.13	0.449	126.5
Hybrid, 3	4/4	16.40	4.38	5.00	19.25	45.03	0.361	107.3
Hybrid, 4	4/4	16.40	4.38	5.00	19.25	45.03	0.443	126.8
Hybrid schedule blinded, 1	5/3	17.49	2.67	0	18.13	38.29	0.384	98.6
Hybrid schedule blinded, 3	4/4	16.40	2.50	0	17.00	35.90	0.361	92.8
OL+BDC, 1	6/2	21.87	5.00	8.89	24.67	60.42	0.649	183.9
OL+BDC, 2	5/3	17.49	5.07	12.80	21.01	56.37	0.664	192.7
CO, either	4/4	16.40	2.50	0	17.00	35.90	0.828	212.8
CO, open-label first period	4/4	16.40	6.25	20.00	21.50	64.15	0.828	262.7
ABAB	4/4	16.40	2.50	0	17.00	35.90	0.352	90.6
ABAB at the Hybrid visit times	4/4	16.40	2.50	0	17.00	35.90	0.299	76.7
Alternating	4/4	16.40	2.50	0	17.00	35.90	0.165	42.3

Every blinded 4/4 design has the same CS variance, 35.90, whatever its arrangement. Under AR(1) with constant expectancy $\mathrm{Var}(\Delta) = (\mathbf{v}'K\mathbf{v})(\mathbf{a}'A\mathbf{a})$ with $\mathbf{v}'K\mathbf{v} = 257$ (for example $257 \times 0.384 = 98.6$ for the blinded Hybrid schedule); with open-label visits the expectancy weights break the product form, so the Hybrid and OL+BDC rows are not proportional to $\mathbf{a}'A\mathbf{a}$ (derived; verified). The alternating design has about a fifth of the crossover's AR(1) contrast variance (42.3 against 212.8).

Power consequences. The source reports that blinding the Hybrid schedule raises E9 power from 0.72 to 0.80 and that an open-label first period lowers the crossover's from 0.82 to 0.56. These are closed-form powers under graded covariance coupling at $c_{bm} = 0.25$, $t_{1/2} = 0$ (0.721 and 0.801 with the common intercept; 0.817 and 0.556 for CO with the path term removed; `02-design-characteristics.R`, `design-power.csv`; inspected). Under the decayed

mean moderation that `docs/39` adopts for its comparisons, with the stratified statistic of Section 15.6, the same contrasts are 0.663 against 0.739 and 0.765 against 0.524 (verified). [`docs/39`, Section 3, states these powers without naming the coupling rule, after Section 2 fixes decayed mean moderation for the paper; the direction and size of both effects are the same under either rule.]

AR(1). With constant expectancy and the separable covariance $K \otimes A$, $\sigma^2_\Delta = (\mathbf{v}'K\mathbf{v})(\mathbf{a}'A\mathbf{a})$, with $\mathbf{v}$ the factor standard deviations and $A_{ts} = \rho^{|w_t - w_s|}$, so designs differ only through the time part $\mathbf{a}'A\mathbf{a}$ (derived; values in Table 39, also verified by `check-design.R`). With $\mathbf{v}'K\mathbf{v} = 257$ for the blinded factors, the crossover's σ^2_Δ is 212.8, as the source states. Under AR(1) the contrast cancels noise only between nearby visits, so interleaving dominates; this, not the open-label phase, is the Hybrid design's advantage over the crossover.

15.4 Within-participant information of a binary exposure

For a binary exposure over T visits of which n_{on} are on drug, $\bar{D}_i = n_{\text{on}}/T$ and

$$\text{SS}_{\text{within}}(D_i) = \sum_t (D_{it} - \bar{D}_i)^2 = T\bar{D}_i(1 - \bar{D}_i) = \frac{n_{\text{on}} n_{\text{off}}}{T} = \left(\frac{1}{n_{\text{on}}} + \frac{1}{n_{\text{off}}}\right)^{-1} \le \frac{T}{4}, \qquad (126)$$

the reciprocal of the balance factor of Equation (122), whose values 0.500, 0.533 and 0.667 at 4/4, 5/3 and 6/2 on/off visits appear in §15.3 (derived). Under compound symmetry the interaction information of a random-intercept analysis is proportional to $\sum_i B_i^2 \,\text{SS}_{\text{within}}(D_i)$, and the GEE precision parameter $\bar{D}_i(1 - \bar{D}_i) = \text{SS}_{\text{within}}/T$ is maximized at equal on and off time (Source: `docs/23`, Section 13.1, items 3 and 4). The values quoted there, 1.5 (OL+BDC, 6 of 8 on), 2.0 (CO, 4 of 8) and 2.9 (N-of-1, 7 of 12), are correct for those visit counts; the current eight-visit Hybrid paths give 1.875 (5 on) and 2.0 (4 on), and no eight-visit design can exceed 2.0 (verified, block H). [Under AR(1) the information also depends on the arrangement of the on and off visits (Equation (123) and the AR(1) paragraph of §15), so $\text{SS}_{\text{within}}$ ranks designs only under compound symmetry. `docs/17`'s claim that short blocks suffer less from autocorrelation is the AR(1) interleaving result stated there.]

15.5 A crossover analyzed under compound symmetry when the correlation decays

The CO contrast compares two blocks of four visits, 2.5 weeks apart within a block. In units of the per-visit variance, with $\bar{r}_w$ and $\bar{r}_b$ the mean correlations within and between blocks, the variance of the difference of block means is exactly

$$\text{Var}(\bar{Y}_1 - \bar{Y}_2) = \frac{2(1 + 3\bar{r}_w)}{4} - 2\bar{r}_b, \qquad \text{and under a fitted CS model} \qquad \frac{2(1 - \bar{r})}{4}, \qquad (127)$$

with $\bar{r}$ the mean over all 28 pairs, which is what a single compound-symmetry correlation estimates (derived; Source: `docs/45`, Section 4.3). Under AR(1) at $\rho = 0.7$ per week, $\bar{r}_w = 0.272$, $\bar{r}_b = 0.069$ and $\bar{r} = 0.156$, so the true variance is 0.770 (equal to $\mathbf{a}'A\mathbf{a}$) and the CS variance 0.422, and the implied ratio of model-based SE to true SD is

$$\kappa \approx \sqrt{0.422/0.770} = 0.740 \qquad (128)$$

(verified), against 0.73 to 0.75 observed for `cs` and `csh` analyses in CO under the covariance study's C07 data (configuration C, $\rho = 0.7$; quoted). The approximation ignores the baseline row, the time term and the cross-component structure (argued). In Hybrid most off-drug visits are a week from an on-drug visit and the averaging does less harm (observed κ 0.92 to 0.97, quoted). The failure is a property of contrasting long, well-separated blocks under serial correlation, and Kenward–Roger does not repair it, because it assumes the working covariance is correct (argued; `docs/45`, Section 4.2). [This supplies the computation behind the comparison cited in the note after Equation (123).]

15.6 The stratified statistic and the exhaustive design search

Setting. Eight blinded post-baseline visits at spacing d, $N = 70$ in two counterbalanced sequences of 35 (a pattern and its complement), decayed mean moderation (Equation (36)) at $\beta_{bm} = 0.25$, $t_{1/2} \in \{0.5, 1, 2\}$ weeks, and three structures: CS, and separable AR(1) at $\rho = 0.8$ per week with a white measurement-error share $\omega \in \{0.2, 0.5\}$ in the time kernel (Source: `docs/39`, Sections 2 and 6).

The stratified statistic. E9 with sequence-specific intercepts regresses the within-sequence centered Δ_i on the within-sequence centered B_i. With $\tilde{\varpi}_p = (n_p - 1)/(N - P)$, $\bar{\beta} = \sum_p \tilde{\varpi}_p \beta_{bm:D,p}$ and

$$\sigma^2_{\text{res,s}} = \sum_p \tilde{\varpi}_p \left[\sigma^2_{\Delta,p} + \sigma^2_{bm}(\beta_{bm:D,p} - \bar{\beta})^2\right], \tag{129}$$

the t-statistic is, to first order in the slope differences, noncentral t on $N - P - 1$ degrees of freedom with noncentrality $\bar{\beta}\sqrt{\text{SS}_{bm}}/\sigma_{\text{res,s}}$, with $\text{SS}_{bm} \sim \sigma^2_{bm}\chi^2_{N-P}$, and power is the tail probability averaged over that law (argued, as for Equation (110); the path-mean term of the common-intercept form is absent; inspected in `04-design-search.R`). Under mean moderation $\sigma^2_{\Delta,p} = \mathbf{a}_p' C_p \mathbf{a}_p$ and the moderation shift adds $(\beta_{bm}\sigma_{BR}\sum_t a_t h_t)^2$ to $\text{Var}(\Delta_p)$ (Equation (108)). Designs are ranked by the information $\mathcal{I} = \sum_p (n_p/N)\, \beta^2_{bm:D,p}\sigma^2_{bm}/\text{Var}(\Delta_p)$. The source reports that direct simulation of 41 designs agrees with this power within two Monte Carlo standard errors (quoted).

Results. Of the 2^8 patterns, those with two to six on-drug visits and a first on-drug visit (a pattern and its complement being the same design) number 119; each was evaluated at $d \in \{0.5, 1, 1.5, 2, 2.5, 3, 4\}$ weeks in each of the nine conditions, 7,497 evaluations. Tables 40–42 were recomputed independently and agree with the stored search to 1.1×10^{-5} in power (verified, `check-baseline-design-search.R` against `design-search-best.csv`).

Table 40: Best design (pattern of the first sequence), spacing and power in each condition (Source: `docs/39`, Section 6; verified).

Structure	$t_{1/2}$	Pattern	Switches	d (weeks)	Power
CS	0.5	`11110000` (CO)	1	4.0	0.765
CS	1	`11110000`	1	4.0	0.759
CS	2	`11110000`	1	4.0	0.731
AR(1), $\omega = 0.2$	0.5	`10101010` (alternating)	7	2.0	0.547
AR(1), $\omega = 0.2$	1	`10101010`	7	3.0	0.435
AR(1), $\omega = 0.2$	2	`10101010`	7	4.0	0.314
AR(1), $\omega = 0.5$	0.5	`10101010`	7	2.5	0.400
AR(1), $\omega = 0.5$	1	`10101010`	7	4.0	0.352
AR(1), $\omega = 0.5$	2	`10011001`	4	4.0	0.265

Under CS the best design sits at the edge of the spacing grid and arrangement is irrelevant to the variance (Section 15.3), so the power bands are set by balance: at $t_{1/2} = 0.5$, 0.763 to 0.765 with four on-drug visits, 0.737 to 0.739 with three or five, and 0.643 to 0.645 with two or six (verified). More switches never help under CS and hurt once carryover is substantial, because each switch places an off-drug visit soon after a discontinuation, where moderation survives; under AR(1) they help, because each brings on and off visits closer, until long half-lives and large measurement error remove the gain (derived from Equations (120) and (125); the tables verify it). [`docs/39` rounds the robustness ratios to two decimals (0.78, 0.77, 0.77, 0.76).]

15.7 Open-label lead-in visits

With $n_{\text{OL}} = 0$ to 4 open-label on-drug visits for every participant ($\eta_t = 1$), then counterbalanced blinded sequences over the remaining $8 - n_{\text{OL}}$ visits, and the open-label visits either in the

Table 41: Best attainable power by the number of on/off switches in the first sequence, at the best spacing (Source: `docs/39`, Section 4; verified).

Structure	$t_{1/2}$	1	2	3	4	5	6	7
CS	0.5	0.765	0.765	0.764	0.764	0.763	0.763	0.763
CS	1	0.759	0.752	0.746	0.739	0.733	0.726	0.719
CS	2	0.731	0.703	0.675	0.645	0.616	0.584	0.559
AR(1), $\omega = 0.2$	0.5	0.268	0.386	0.453	0.512	0.510	0.522	0.547
AR(1), $\omega = 0.2$	1	0.221	0.282	0.339	0.389	0.397	0.404	0.435
AR(1), $\omega = 0.2$	2	0.210	0.239	0.269	0.291	0.297	0.301	0.314
AR(1), $\omega = 0.5$	0.5	0.257	0.315	0.347	0.378	0.381	0.385	0.400
AR(1), $\omega = 0.5$	1	0.243	0.275	0.305	0.326	0.332	0.340	0.352
AR(1), $\omega = 0.5$	2	0.230	0.247	0.259	0.265	0.261	0.258	0.257

Table 42: Design–spacing pairs with the highest minimum, over the nine conditions, of power relative to the condition's best (Source: `docs/39`, Section 6; verified).

Pattern (first sequence)	d (weeks)	Worst relative power	Mean relative power
`10101001` (tied with `10010101`)	4	0.781	0.918
`10100101`	4	0.772	0.912
`10101010` (alternating)	4	0.765	0.942
`10011001`	3	0.758	0.884

contrast (as E9 and the published analysis) or out of it, Table 43 gives the best stratified power over patterns and spacings (Source: `docs/39`, Section 7; verified, all entries within 0.001 of the source). Because the placebo factor is unrelated to the biomarker, the open-label phase changes only the noise, through $\mathcal{E}$ and $\mathcal{P}$ of Equation (125), and not the slope (derived).

Table 43: Best power by number of open-label visits, included in / excluded from the contrast (Source: `docs/39`, Section 7; verified).

Structure	$t_{1/2}$	0	1	2	3	4
CS	0.5	0.765	0.722 / 0.701	0.679 / 0.644	0.605 / 0.551	0.547 / 0.479
CS	1	0.759	0.707 / 0.686	0.661 / 0.627	0.583 / 0.530	0.521 / 0.456
CS	2	0.731	0.638 / 0.616	0.582 / 0.549	0.491 / 0.443	0.421 / 0.366
AR(1), $\omega = 0.2$	0.5	0.547	0.463 / 0.487	0.355 / 0.422	0.280 / 0.361	0.237 / 0.291
AR(1), $\omega = 0.2$	1	0.435	0.370 / 0.374	0.306 / 0.328	0.251 / 0.274	0.215 / 0.228
AR(1), $\omega = 0.2$	2	0.314	0.257 / 0.258	0.217 / 0.228	0.181 / 0.192	0.160 / 0.163
AR(1), $\omega = 0.5$	0.5	0.400	0.358 / 0.352	0.311 / 0.311	0.262 / 0.261	0.231 / 0.220
AR(1), $\omega = 0.5$	1	0.352	0.314 / 0.305	0.278 / 0.271	0.238 / 0.227	0.210 / 0.194
AR(1), $\omega = 0.5$	2	0.265	0.228 / 0.217	0.208 / 0.196	0.179 / 0.165	0.161 / 0.144

Each open-label visit that replaces a blinded one costs power in every condition; with the better analysis, two cost 0.09 to 0.15 under CS and 0.09 to 0.13 under AR(1) at $\omega = 0.2$ (verified). Inclusion is better under CS and exclusion under AR(1) at $\omega = 0.2$; the source's explanations of both are inferred. [One entry recomputes as 0.197 rather than 0.196 (AR(1), $\omega = 0.5$, $t_{1/2} = 2$, two visits, excluded), a quadrature difference in the last digit. The variant in which open-label titration is added before, not in place of, the blinded visits was not evaluated in the source.]

15.8 Blinded-discontinuation and crossover blocks

After the open-label visits at weeks 4 and 8, one block of four weekly visits (weeks 9–12) is run four ways on two paths of 35: BD 10/9 (`1100/1000`, the Hybrid's), BD 11/9 (`1110/1000`),

CO (`1100/0011`) and alternating (`1010/0101`). Separately, the Hybrid design is compared with three variants of its crossover block: equal visits (BD block at weeks 10, 12; CO block of four every two weeks, weeks 14–20), CO weekly (published BD block plus four weekly CO visits, weeks 13–16) and CO two-weekly (weeks 14–20) (Source: `docs/39`, Section 8). Tables 44 and 45 give the stratified power with the open-label visits in the contrast (closed form verified, `check-baseline-design-search.R`; Monte Carlo, 5000 trials, quoted).

Table 44: Power of four blinded block types after the open-label lead-in; closed form (simulated). AR(1) columns at $\omega = 0.2$.

Block	CS, 0	CS, 0.5	CS, 1	AR(1), 0	AR(1), 0.5	AR(1), 1
BD 10/9	0.528 (0.525)	0.423 (0.416)	0.274 (0.281)	0.231 (0.224)	0.186 (0.186)	0.129 (0.131)
BD 11/9	0.450 (0.444)	0.326 (0.322)	0.199 (0.208)	0.219 (0.215)	0.162 (0.171)	0.110 (0.107)
CO	0.532 (0.527)	0.407 (0.413)	0.249 (0.251)	0.277 (0.274)	0.211 (0.215)	0.137 (0.137)
Alternating	0.532 (0.530)	0.335 (0.339)	0.176 (0.170)	0.356 (0.349)	0.222 (0.231)	0.125 (0.118)

Table 45: Power of the published Hybrid design and three crossover-block variants; closed form (simulated). AR(1) columns at $\omega = 0.2$.

Design	CS, 0	CS, 0.5	CS, 1	AR(1), 0	AR(1), 0.5	AR(1), 1
Published	0.663 (0.666)	0.580 (0.571)	0.438 (0.435)	0.302 (0.297)	0.257 (0.267)	0.193 (0.194)
Equal visits	0.663 (0.673)	0.638 (0.640)	0.542 (0.551)	0.298 (0.300)	0.284 (0.285)	0.235 (0.236)
CO weekly	0.771 (0.772)	0.676 (0.674)	0.498 (0.502)	0.380 (0.385)	0.314 (0.314)	0.221 (0.228)
CO two-weekly	0.771 (0.776)	0.703 (0.698)	0.561 (0.580)	0.355 (0.351)	0.310 (0.303)	0.235 (0.234)

The pattern follows from Equations (120) and (125). Without carryover under CS only balance enters, so BD 10/9 and CO blocks of equal length tie (0.528, 0.532) and the unbalanced BD 11/9 loses 0.08. With carryover the slope factor $1 - \bar{h}_{\text{off}}$ decides: at $t_{1/2} = 1$ the mean off-drug moderation share is 0.333 (BD 10/9), 0.375 (CO), 0.396 (BD 11/9) and 0.500 (alternating), and 0.245 (published) against 0.134 (equal visits), 0.282 (CO weekly) and 0.223 (CO two-weekly) (verified). Under AR(1) the CO block is better because its off-drug visits lie between on-drug visits. The residual contrast variance σ^2_Δ, free of $t_{1/2}$, is 44.58 for the published and equal-visit designs and 34.27 for the ten-visit designs (verified). The BD 10/9 and CO blocks differ by 0.004 to 0.046, and the two added CO observations raise power by 0.028 to 0.123. Leaving the open-label visits out of the contrast under AR(1) gives 0.440 against 0.380 for CO weekly without carryover, but 0.263 against 0.298 for the equal-visit design (verified). [`docs/39`, Section 8, gives the CS contrast variance as falling "from 46.9 to 36.3–36.7". Those are path means of Var(Δ) including the mean-moderation shift at $t_{1/2} = 1$ (about $44.58 + (2 \times 0.755)^2 = 46.86$, with 0.755 the mean of $1 - \bar{h}_{\text{off}}$); the residual variance that determines precision falls from 44.58 to 34.27 at every half-life.]

15.9 Optimal spacing

With eight visits at spacing d, closer spacing shares more serially correlated noise, which the contrast cancels, and wider spacing lets more moderation wash out ($h = 2^{-d/t_{1/2}}$ at the first off-drug visit). For the alternating design with no measurement error, the time part is $\mathbf{a}'A\mathbf{a} \approx (4/n)(1-\rho^d)/(1+\rho^d)$, which is $\frac{1}{2}(1-\rho^d)/(1+\rho^d)$ at $n = 8$, and information is proportional to

$$f(d) = \left(1 - 2^{-d/t_{1/2}}\right)^2 \frac{1+\rho^d}{1-\rho^d} \tag{130}$$

(Source: `docs/39`, Section 5). At $\rho = 0.8$ its maxima are $d^* = 0.45$, 0.91, 1.88 and 4.34 weeks for $t_{1/2} = 0.25$, 0.5, 1 and 2, about twice the half-life (verified, `check-design.R`). Under compound symmetry the correlation does not depend on d, so f is monotone and there is no

interior optimum, only a plateau. With a white measurement-error share ω in the time kernel, $A'_{ts} = (1-\omega)\rho^{|w_t - w_s|} + \omega \mathbf{1}[t = s]$, close spacing buys less and the source's grid optima move to three to six half-lives (quoted). [The large-n approximation for the time part is coarse at $n = 8$: 0.136 against the exact 0.165 at $d = 2.5$, and 0.056 against 0.081 at $d = 1$ (verified). The source states that the maxima of f agree with the exact grid optimum; the agreement, to the 0.25-week grid resolution, is confirmed in §15.10.]

15.10 Optimal spacing on the exact grid

For the three counterbalanced arrangements at spacing $d \in \{0.25, 0.5, \ldots, 6\}$ weeks (common-intercept E9 with the path term removed, decayed mean moderation, $\beta_{bm} = 0.25$), the information maximum and the shortest spacing within 95% of the best power are given in Table 46 (Source: `docs/39`, Section 5; verified, `check-baseline-design-search.R`).

Table 46: Under CS: shortest spacing (weeks) within 95% of the best power. Under AR(1) ($\rho = 0.8$ per week): spacing of maximum information; "> 6" means at the edge of the grid. Verified.

Structure	Design	$t_{1/2} = 0.25$	0.5	1	2
CS, 95%	alternating	1.25	2.25	4.00	5.25
CS, 95%	ABAB	1.00	1.75	3.25	4.75
CS, 95%	CO	0.50	1.00	2.00	3.25
AR(1), $\omega = 0$, optimum	alternating	0.50	1.00	2.00	4.25
AR(1), $\omega = 0.2$, optimum	alternating	1.00	1.75	3.25	> 6
AR(1), $\omega = 0.5$, optimum	alternating	1.50	2.50	4.50	> 6
AR(1), $\omega = 0.2$, optimum	ABAB	0.50	1.00	2.25	> 6

Under CS there is no interior optimum (the information is monotone in d), and the 95% spacings are four to five half-lives for the alternating design at the shorter half-lives, about 2.4 to 4 for ABAB, and about two for CO, whose placebo-first sequence carries no carryover. Under AR(1) the optimum is interior and moves out with the measurement-error share: about two half-lives without it, three to four at $\omega = 0.2$, four and a half to six at $\omega = 0.5$. Without measurement error the exact grid optima, 0.50, 1.00, 2.00 and 4.25 weeks, agree with the maxima of Equation (130), 0.45, 0.91, 1.88 and 4.34, to the 0.25-week grid resolution (verified), which settles the point left open in the note after Equation (130). ABAB optima are shorter because the second visit of each off-drug block is already two spacings after discontinuation (argued).

15.11 A closed-form screen from visit schedules

Paper 12 approximates E1 under mean moderation by the same two-group contrast and writes, for a participant with k_{on} and k_{off} visits at fixed times,

$$\beta_{bm:D} = \beta_D \, \beta_{bm} \, A, \qquad A = 1 - \bar{D}^{\text{off}}_{bc}, \qquad \operatorname{Var}(\hat{\beta}_{bm:D}) \approx \frac{\sigma^2 v_1}{N}, \quad v_1 = \frac{\mathbf{1}' R_{\text{on}} \mathbf{1}}{k^2_{\text{on}}} + \frac{\mathbf{1}' R_{\text{off}} \mathbf{1}}{k^2_{\text{off}}}, \tag{131}$$

with power by the Wald-normal approximation (Paper 12, Section 3). [The source writes the drug effect as β_1 (here β_D) and the moderation as $\beta_1(1 + \beta_{bm} B_i)$, a multiplicative form on the drug effect, rather than the canonical additive shift $\beta_{bm} \sigma_{BR} b_i$, and uses γ_0 and η_i for the intercept and residual; η is the expectancy here.] [Correction. v_1 omits the cross term: the exact time part is $v_1 - 2\,\mathbf{1}' R_{\text{on,off}} \mathbf{1} / (k_{\text{on}} k_{\text{off}})$, as in (121). Under positive serial correlation the approximation always overstates the variance, and most for interleaved designs. At $\rho = 0.7$ it gives 0.909 against an exact 0.770 for the 2.5-week crossover, but 0.926 against 0.464 and 0.910 against 0.451 for Hybrid paths 1 and 3 (verified, `check-design.R`). This accounts for the source's residual pattern (crossover under-predicted, Hybrid and OL+BDC over-predicted after one common σ^2 is fitted), which it attributes to the same approximation

without quantifying it. Since the exact form costs no more to evaluate, the screen should use it.] The source also reports that $A(t_{1/2})$ ranks designs correctly but overstates the attenuation of the fitted coefficient, and recommends it only as a relative screen (quoted).

15.12 Separation of the biomarker block

Let the analysis include the biomarker only through the columns $B_i\,\mathbf{z}_t$, where $\mathbf{z}_t$ collects the regressors it multiplies (for the strawman analysis S of Table 55, $\mathbf{z}_t = (1, D_{bc,it})$). Because B_i is centered and independent of path, its cross-products with every column not multiplied by B_i (intercept, $f(t)$, D_{bc}, η_t) vanish in the probability limit, and the GLS normal equations separate. With Z_p the $9 \times q$ matrix of $\mathbf{z}_t$ on path p, V_p its working covariance, N_p its allocation and $\mathbf{s}_p$ the true slope vector,

$$\operatorname{plim} \hat{\boldsymbol{\beta}}_{\text{bm block}} = \Big(\sum_p N_p\, Z_p' V_p^{-1} Z_p \Big)^{-1} \sum_p N_p\, Z_p' V_p^{-1} \mathbf{s}_p \tag{132}$$

(derived here). Two consequences follow. The probability limit of $\beta_{bm:D}$ does not depend on the time adjustment $f(t)$, which enters only the other block (consistent with the agreement to 0.0009 across time adjustments in `docs/42`, Section 4). And if $\mathbf{s}_p$ lies in the column space of Z_p on every path, the fit is exact for *any* working covariance.

15.13 The most efficient contrast

Under mean moderation with profile $\mathbf{h} = (h_t)$ on a path whose response covariance is C, a contrast $\mathbf{a}$ has slope $-\beta_{bm}(\sigma_{BR}/\sigma_{bm})\,\mathbf{a}'\mathbf{h}$ and residual variance $\mathbf{a}'C\mathbf{a}$, so the squared noncentrality is proportional to $(\mathbf{a}'\mathbf{h})^2/\mathbf{a}'C\mathbf{a}$ (Equation (109) with $r^2_{\Delta B}/(1 - r^2_{\Delta B})$). Among contrasts with $\mathbf{1}'\mathbf{a} = 0$, which cancel the baseline level and any biomarker main effect, the Cauchy–Schwarz inequality in the C inner product gives the maximizer and the maximum

$$\mathbf{a}^* \propto C^{-1}\big(\mathbf{h} - \bar{h}_C \mathbf{1}\big), \quad \bar{h}_C = \frac{\mathbf{1}'C^{-1}\mathbf{h}}{\mathbf{1}'C^{-1}\mathbf{1}}, \qquad \max_{\mathbf{1}'\mathbf{a}=0} \frac{(\mathbf{a}'\mathbf{h})^2}{\mathbf{a}'C\mathbf{a}} = \mathbf{h}'C^{-1}\mathbf{h} - \frac{(\mathbf{1}'C^{-1}\mathbf{h})^2}{\mathbf{1}'C^{-1}\mathbf{1}}, \tag{133}$$

which is the reciprocal of the GLS variance of the coefficient of $B_i h_t$ in the biomarker block $(B_i, B_i h_t)$ with $V = C$ (Equation (132)): a correctly specified mixed model attains it (derived; Source: `docs/39`, Section 10, item 6). The E9 weights attain it only when $C^{-1}(\mathbf{h} - \bar{h}_C\mathbf{1})$ is proportional to the on/off weights. Relative efficiencies of E9 (Table 47) were computed with the GLS form, which matches Equation (133) to machine precision, and checked against 20,000 random sum-zero contrasts, none of which exceeds it (verified, `check-baseline-design-search.R`).

Table 47: Information of the E9 contrast relative to the optimum of Equation (133), one path (verified).

Design, path	Structure	$t_{1/2}$	E9 / optimum
Hybrid, 1	CS	0	0.950
Hybrid, 1	AR(1), $\omega = 0.2$	0	0.775
Alternating, 2.5-week spacing	AR(1), $\omega = 0.2$	1	0.958
CO, 2.5-week spacing	AR(1), $\omega = 0.2$	1	0.922
CO, 2.5-week spacing	CS	1	0.989

Equal weighting loses most where the design is irregular and the correlation decays (Hybrid under AR(1): 22.5% of the information). [`docs/39`, Section 10, item 6, states that the best contrast is proportional to C^{-1} times the moderation pattern; among contrasts invariant to a biomarker main effect it is $C^{-1}(\mathbf{h} - \bar{h}_C\mathbf{1})$, the projection that Equation (133) makes explicit.]

15.14 Time adjustment and the probability limit of the drug coefficient

The time term $f(t)$ must absorb the non-drug mean, which in the Hybrid design is common to all paths: a sigmoid natural course plus an expectancy curve that halves at week 9. Under working independence, the probability limit of the drug coefficient is the allocation-weighted least-squares fit to the population means, $\operatorname{plim}\hat{\beta}_D = \left[(X'WX)^{-1}X'W\boldsymbol{\mu}\right]_D$. With visit effects, Frisch-Waugh-Lovell gives

$$\operatorname{plim}\hat{\beta}_D = \frac{\sum_t \sum_p N_p\,(D_{pt} - \bar{D}_t)\,\mu_{pt}}{\sum_t \sum_p N_p\,(D_{pt} - \bar{D}_t)^2}, \qquad \bar{D}_t = \frac{\sum_p N_p D_{pt}}{\sum_p N_p}, \tag{134}$$

so only visits at which paths differ in drug state contribute, weeks 10, 16 and 20 in the Hybrid design (derived; `docs/42`, Section 3, states the identification in words).

Table 48: Probability limit of β_D under working independence, by time adjustment, Hybrid, graded coupling (Source: `docs/42`, Section 4; values inspected in `time-adjustment-plim.csv`).

Time adjustment	RMS misfit, $t_{1/2} = 0$	$t_{1/2} = 0$	$t_{1/2} = 1$
Linear t	3.18	−8.21	−6.38
Linear $t + \eta_t$	3.11	−7.28	−4.81
Natural spline, 3 df	1.93	−6.22	−3.67
Visit factor	0.80	−6.25	−4.49

Linear time credits about 2 points of open-label expectancy to the drug; visit effects remove the confounding at about 1.5 times the standard deviation (0.86 against 1.27, quoted). By the separation argument of (132), the probability limit of $\beta_{bm:D}$ is the same under every $f(t)$ when the biomarker is unrelated to the non-drug components. [These limits are for an independence working model; the random-intercept GLS of the fitted model converges elsewhere, which is why the simulated means (−8.16, −6.02, −5.99 at $t_{1/2} = 0$) differ from them, most for the flexible forms. The source script labels them illustrations.] The size excess of the flexible adjustments in the Hybrid carryover cell (0.075 spline and 0.080 visit factor against 0.056 linear under model-based inference; reproduced in `docs/45`, Section 4.4, and absent under CR2) is a property of inference, not of these limits; its mechanism is not established (quoted).

15.15 The non-drug mean of the Hybrid design

In the published process the natural course follows the modified Gompertz curve in calendar weeks, $E(TV_t) = G(w_t; 6.506, 5, 0.35)$ (Equation (2)), the placebo response the same curve scaled by expectancy, $E(PB_t) = \eta_t\, G(w_t; 6.506, 5, 0.35)$ (its clock t_{pb} equals w_t after baseline), and the drug response $E(BR_t) = G(t_{od}; 10.986, 5, 0.42)$ on drug. The non-drug mean is common to all paths; its values by week, with the drug-response means of every path, are in Table 28 (§14.9; Source: `docs/42`, Section 2; verified). The non-drug mean rises to week 8, falls by 1.71 at week 9 when expectancy halves, and then flattens, so no line follows it. At weeks 4, 8 and 9 every path is on drug and the drug response rises from 4.28 to 9.79 (10.19 at week 10), variation that a single D coefficient cannot represent and that a flexible time term absorbs; this is why the time adjustment moves $\operatorname{plim}\hat{\beta}_D$ so much (Table 48). [`docs/42` gives $TV + PB = 3.72$ at week 4, the sum of the rounded components; the exact value is 3.73.] The collinearity of the expectancy regressor with the drug indicator, which `docs/42`, Section 7, quotes from `docs/40` as 0.59 and 1.5, is reconciled in §14.9.

15.16 Cycle count and period length

`docs/27` plans a sweep (block S13 of Paper 05) over designs built by `build_cycle_design()`: an open-label run of T_{ol} weeks on drug ($\eta = 1$), then k blinded cycles of T_{on} weeks on drug and T_{off}

weeks off ($\eta = 0.5$), one observation per week, a single path in the `on_first` order (`analysis/scripts/nof1-design-sensitivity/00-common.R`, inspected). Optional run-in and run-out add T_{ri} and T_{ro} off-drug weeks, so the duration is $T_{\text{ri}} + T_{\text{ol}} + k(T_{\text{on}} + T_{\text{off}}) + T_{\text{ro}}$.

Information on the interaction. Take mean moderation without carryover ($D_{bc} = D$), independent within-participant errors of variance σ^2 with any random intercept, every participant on the same schedule, and the analysis model Equation (136). Centering the biomarker, the `bm:Dbc` column is orthogonal to every column that does not involve the biomarker, and of the biomarker columns the model contains only the main effect, so only the person-level mean of D is partialled out of it. Hence

$$\text{Var}(\hat{\beta}_{bm:D}) = \frac{\sigma^2}{(N-1)s_b^2}\,\frac{n_{\text{on}} + n_{\text{off}}}{n_{\text{on}}\, n_{\text{off}}}, \qquad n_{\text{on}} = T_{\text{ol}} + kT_{\text{on}}, \quad n_{\text{off}} = T_{\text{ri}} + kT_{\text{off}} + T_{\text{ro}}, \tag{135}$$

the special case of Equation (122) with independent within-participant errors (derived). Adding a linear time term or an expectancy main effect η_t leaves it unchanged; adding a biomarker-by-expectancy term $b_i\eta_t$ (as in Equation (124)) removes the open-label visits, so $n_{\text{on}} = kT_{\text{on}}$ (derived; all three verified to machine precision for every Tier 1 cycle structure from the GLS information of the full design matrix with a random intercept, `check-gompertz-rmanova-cycles.R`). Three statements follow.

1. **Leading order.** Without open-label, run-in or run-out visits the information is

$$I(\beta_{bm:D}) = \frac{(N-1)s_b^2}{\sigma^2}\; k\,\frac{T_{\text{on}}T_{\text{off}}}{T_{\text{on}} + T_{\text{off}}},$$

   exactly linear in k, which is `docs/27`'s Section 4 formula (Source: `docs/27`, Section 4; derived). The factor $T_{\text{on}}T_{\text{off}}/(T_{\text{on}} + T_{\text{off}})$ is half the harmonic mean of the two period lengths.

2. **The open-label phase counts.** Under the analysis model the open-label weeks enter n_{on} directly. With $T_{\text{ol}} = 8$ and $k = 2$, $T_{\text{on}} = T_{\text{off}} = 4$, the information per unit $(N-1)s_b^2/\sigma^2$ is 5.33 against the leading-order 4.00 (verified). Linearity in k then fails: at $T_{\text{on}} = T_{\text{off}} = 2$ it is 1.67, 3.00, 4.20 and 5.33 for $k = 1$ to 4.

3. **The symmetric optimum moves.** At a fixed cycle length $S = T_{\text{on}} + T_{\text{off}}$ the information is maximized at $T_{\text{on}}^* = S/2 - (T_{\text{ol}} - T_{\text{ri}} - T_{\text{ro}})/(2k)$, truncated to $[0, S]$ (derived, maximizing $n_{\text{on}}n_{\text{off}}$ at fixed total). For the Tier 2a sweep ($k = 2$, $S = 8$, $T_{\text{ol}} = 8$) this is $T_{\text{on}}^* = 2$, not 4: the information is 6.00, 5.83, 5.33, 4.50 and 3.33 at $T_{\text{on}} = 2, \ldots, 6$. With a $b_i\eta_t$ term in the model it is 3.00, 3.75, 4.00, 3.75 and 3.00, symmetric about 4 (verified).

[`docs/27` (Section 2a) argues that open-label observations contribute to the interaction only indirectly, "their $D_{bc,it} = 1$ value is identical to the within-cycle on-drug observations, so they do not enlarge the on/off contrast", and (Sections 3b and 4) predicts a symmetric optimum at (4, 4) without carryover. Both hold only for an analysis that includes a biomarker-by-expectancy (or biomarker-by-phase) term. Under the planned E2 analysis, which has none, the open-label weeks are on-drug observations of the contrast, and the no-carryover optimum of the Tier 2a sweep lies at the shortest on-drug period. The open-label visits also carry $\eta_t = 1$, so under a PB-coupled biomarker they are the visits through which expectancy confounding enters (§15, Table 39). Because the S13 designs have a single path, drug state is partly confounded with time, which costs information on β_D when a time term is fitted; it does not affect $\beta_{bm:D}$, by the orthogonality above (derived). AR(1) errors replace Equation (135) by Equation (123).]

Tier 1 grid size. With $T_{\text{ri}} = T_{\text{ro}} = 0$, $T_{\text{ol}} = 8$ and a 32-week cap, the constraint is $k(T_{\text{on}} + T_{\text{off}}) \le 24$ over $k \in \{1, 2, 3, 4\}$ and $T_{\text{on}}, T_{\text{off}} \in \{2, 4, 6\}$. It keeps all nine period pairs at $k = 1$ and 2, six at $k = 3$ and three at $k = 4$: 27 cycle structures, hence $27 \times 12 = 324$ planned cells over two architectures, three half-lives and two values of c_{bm}, and $27 \times 30 = 810$ cells in the driver, which

uses five values of c_{bm} (derived; verified by enumeration, `check-gompertz-rmanova-cycles.R`; Source: `docs/27`, Section 3a and status note).

Part IV
Analysis

16 The analysis model and its carryover specifications

This section states the analysis model common to the compendium, its open-label branch, and the carryover specifications of Paper 02, followed by two historical analyses of carryover.

16.1 The common analysis model

Across the compendium both architectures are fit identically:

$$\texttt{Sx ~ bm + t + Dbc + bm:Dbc} \tag{136}$$

via `nlme::lme` with `corCAR1(form = ~t|ptID)` residual correlation and a participant random intercept. The inferential machinery of §18 does not know which architecture generated the data it is fitting.

The restated Paper 01 departs from this convention in two places. The published analysis it reproduces is the `lmer` model $Y_{it} = \beta_0 + \beta_B B_i + \beta_D D_{it} + \beta_T t + \beta_{bm:D} B_i D_{it} + u_i + \varepsilon_{it}$, with the *binary* D_{it} and baseline as a row with $D_{i0} = 0$; under carryover a binary indicator estimates an attenuated on-minus-off contrast, so part of the power loss reflects the analysis, not the data-generating process (Paper 01, Estimand). Its stress test compares fourteen analyses under configuration C at $\rho = 0.7$, the covariance construct of the refined process, with the published recursive carryover (§19.1), and selects a random-intercept model with continuous-time AR(1) residuals, expectancy terms and CR2 standard errors (Paper 01, Results; `docs/46`, `docs/47`). That selection comes from an exploratory comparison.

[2026-10-08: superseded. A note here previously stated that Paper 01's Methods give $\theta_{\text{true}} = -c_{bm}\sigma_{BR}$ and that this is consistent with mean estimates of -0.23 to -0.28. The restated Paper 01 gives the estimand as $-c_{bm}\sigma_{BR}/\sigma_{bm}$ (covariance) and $-\beta_{bm}\sigma_{BR}/\sigma_{bm}$ (mean), which reduces to $-c_{bm}\sigma_{BR}$ only when $\sigma_{bm} = 1$, as in the Paper 02 driver (Equation (179)). The consistency claim was not supported: with $c_{bm} = 0.45$ it would require $\sigma_{BR} \approx 0.5$ to 0.6, and it is withdrawn.]

16.2 The open-label branch of the analysis model and the biomarker threshold

Open-label branch. When no participant's drug state varies after baseline, `lme_analysis()` replaces the exposure terms of Equation (136) by a biomarker-by-time interaction,

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_T t + \beta_{bm:t}\, B_i t + u_i + \varepsilon_{it}, \tag{137}$$

with a random intercept, `corCAR1` residuals and the raw biomarker (code `Sx ~ bm + t + bm:t`; `R/lme_analysis.R`, inspected; Source: `docs/23`, Section 4.1). The coefficient $\beta_{bm:t}$ moderates the whole trajectory, whose slope contains BR, PB and TV, so it is not the drug-specific estimand $\beta_{bm:D}$; and its within-participant information is $\sum_t (t - \bar{t}_i)^2$ rather than the exposure information of §15.4 (`docs/23`, Section 13.3; argued). [The source writes β_1, β_2, β_3 for β_B, β_T, $\beta_{bm:t}$. Under both architectures as implemented the biomarker-dependent part of the mean is constant over the post-baseline open-label visits (constant shift, or $g_t = 1$), and zero at the baseline row, so it reaches $\beta_{bm:t}$ only through the projection of that step onto linear time (argued). This is consistent with the near-nominal open-label power quoted in `docs/23`, Section 6.3 (0.06 to 0.10), but was not derived.]

Biomarker threshold. With the raw biomarker in the model, the fitted drug effect at biomarker value B is $\hat{\beta}_D + \hat{\beta}_{bm:D}B$, so the biomarker at which it equals a clinical threshold y^* is

$$B_{\text{thr}} = \frac{y^* - \hat{\beta}_D}{\hat{\beta}_{bm:D}}, \qquad \text{or} \qquad B_{\text{thr}} = \bar{B} + \frac{y^* - \hat{\beta}^c_D}{\hat{\beta}_{bm:D}} \text{ if the biomarker is centered at } \bar{B} \text{ before the fit,} \tag{138}$$

with $\hat{\beta}^c_D$ the drug coefficient of the centered fit (derived). [docs/23, Section 9.1, flags that `analyze_trial_extended()` combined the uncentered coefficient with the $\bar{B}$ offset. The working-tree `R/carryover_analysis.R` now computes the first form and its `predicted_effect` as $\hat{\beta}_D + \hat{\beta}_{bm:D}B$ (inspected 2026-10-08; the change is uncommitted).]

16.3 Analysis specifications for carryover: E1–E9

Paper 02 fixes the analysis model's carryover treatment across nine specifications, all fit as variants of

```
nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~t | ptID))
```
(139)

differing only in the construction of the exposure predictor X and, for the CR-variant rows, the standard-error estimator (§18.6). Table 49 reproduces the full specification table under the stored codes.

Labels. `NOTATION.md` (Part 3) defines E1, E2 and E3 and, since 2026-10-09, lists the extended E series (E4 to E7, E9, E9b and the CR2 variants) so that no label is reused; it requires stored and reported labels to be identical. The further codes E7, E9 and the CR2 variants are the values stored in Paper 02's data files; Paper 02 now displays all nine as G1–G9 (or by name) through a render-time mapping in `spec-labels.R`, which departs from that rule. Paper 01 uses E9 for the paired-difference statistic and E9b for the same statistic with baseline counted as an off-drug visit. This document uses the stored E codes.

Table 49: The nine carryover analysis specifications (Paper 02).

Stored (Paper 02 display)	Name	X / model	Target
E1 (G1)	Unadjusted	$X = D_{it}$ (binary), no carryover term	$bm{:}D_{it}$
E3 (G2)	Lag-adjusted	$X = D_{it}$, plus nuisance $L_{it} = D_{i,t-1}(1 - D_{it})$	$bm{:}D_{it}$
E2 (G3)	Exposure-weighted	$X = D_{bc,it}$	$bm{:}D_{bc,it}$
E7 (G4)	AIC-selected $\hat{t}_{1/2}$	E2 form, $\hat{t}_{1/2}$ chosen by AIC over $\{0.25, 0.5, 1.0, 2.0\}$ wk	$bm{:}D_{bc,it}$
E9 (G5)	Paired-difference	two-stage OLS on per-patient on/off means	biomarker slope
E1cr2, E3cr2, E2cr2, E7cr2 (G6–G9)	CR2 variants	E1/E3/E2/E7 formula, CR2 cluster-robust SE	(as above)

E3, the lag-adjusted (Jones–Kenward) specification. The nuisance indicator $L_{it} = D_{i,t-1}(1 - D_{it})$ marks off-drug timepoints that immediately follow an on-drug one; it enters as a main effect only, so the interaction actually tested remains $bm{:}D_{it}$, not $bm{:}L_{it}$.

E7, the AIC-selected half-life. With $M(t_{1/2})$ the E2 fit at a candidate half-life,

$$\hat{t}_{1/2} = \underset{t_{1/2} \in \mathcal{T}}{\arg\min}\, \text{AIC}\big(M(t_{1/2})\big), \qquad \text{AIC}(M) = -2\log L(M) + 2p, \qquad \mathcal{T} = \{0.25, 0.5, 1.0, 2.0\}, \tag{140}$$

with the reported coefficient, SE, and p-value read from $M(\hat{t}_{1/2})$ alone. Paper 02 notes the post-selection-inference risk explicitly (Berk et al., 2013): treating $\hat{t}_{1/2}$ as fixed in advance understates the sampling variability of the reported interaction estimate.

E9, the paired-difference specification. A correlation- structure-agnostic two-stage estimator. For patient i with $n_i^{\text{on}}, n_i^{\text{off}} > 0$,

$$\bar{Y}_i^{\text{on}} = \frac{1}{n_i^{\text{on}}} \sum_{t:\, D_{it}=1} Y_{it}, \qquad \bar{Y}_i^{\text{off}} = \frac{1}{n_i^{\text{off}}} \sum_{t:\, D_{it}=0} Y_{it}, \qquad \Delta_i = \bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}}, \tag{141}$$

followed by a second-stage OLS regression on the biomarker,

$$\Delta_i = \beta_0 + \beta_{bm:D}^{\text{E9}} B_i + e_i, \qquad i = 1, \ldots, N', \tag{142}$$

with the interaction test on $\beta_{bm:D}^{\text{E9}}$ and $N' \leq N$ patients contributing (those without both on- and off-drug observations are dropped). Paper 01 (as restated) derives the moments and power of this statistic in closed form: its variance does not depend on $t_{1/2}$, and its population slope is $-c_{bm}(\sigma_{BR}/\sigma_{bm})(1-\bar{g}_{\text{off}})$, with $\bar{g}_{\text{off}}$ the mean coupling over a path's post-baseline off-drug visits (Paper 01, Propositions 1 and 2, Appendix A.1).

[2026-10-08: notation. $S_{x,it}$ and bm_i were replaced by the canonical Y_{it} and B_i (`NOTATION.md` Rule 3), and the second-stage slope and residual, formerly β_1 and ε_i, were renamed so they do not collide with the treatment effect (Paper 07's β_1, now β_D) of §6 and the observation-level residual ε_{it}.]

An earlier, retired pair of specifications, E5 (adding $bm{:}L_{it}$) and E6 (adding $bm{:}t_{sd}$ as a direct carryover-by-biomarker interaction), underperformed the Unadjusted specification at every cell tested and were replaced by E7/E9; they are recorded here as historical context, not part of the active nine-specification set.

16.4 Analysis strategies outside the E1–E9 set

The pre-restatement Paper 01 (Section 6, as summarized in `docs/23`, Section 10) listed analysis strategies for recovering power under covariance moderation that are not among the nine specifications of Table 49. Two carry a model statement.

Weighted analysis. Each observation receives weight 1 on drug and $e^{-\lambda t_{sd,it}}$ off drug, that is, the weight $D_{bc,it}$ of Equation (13), on the argument that off-drug visits carry less coupling (argued).

Two-stage random slopes. The first stage fits a random drug slope without the biomarker,

$$Y_{it} = \beta_0 + \beta_T t + \beta_D D_{it} + u_{0i} + u_{1i} D_{it} + \varepsilon_{it}, \tag{143}$$

and the second regresses the predicted drug slopes on the biomarker,

$$\hat{u}_{1i} = \xi_0 + \xi_1 B_i + e_i, \tag{144}$$

testing $\xi_1 = 0$. (Source: `docs/23`, Section 10, item 6.4.) [Not evaluated in any source. Because the biomarker is omitted from the first stage, $\hat{u}_{1i}$ is a best linear unbiased predictor shrunk toward zero by a participant-specific reliability factor, so $\hat{\xi}_1$ estimates the interaction attenuated by that factor; the second-stage test also ignores the prediction error of $\hat{u}_{1i}$ (argued; standard properties of predicted random effects, not checked for these designs). `docs/23` renamed the source's δ_0, δ_1, η_i to ξ_0, ξ_1, e_i; the first-stage time slope, written β_1 in the source, is β_T here as in §16.]

The within-subject contrast of the same list is E9 (Equation (142)). A third proposal, a linear t_{sd} main-effect covariate added to the binary specification (`Sx ~ bm + t + Db + bm:Db + tsd`, "Approach B" of `docs/23`, Section 12.2), is the package option `op$simplecarryover` (`R/lme_analysis.R`, inspected); it is neither E3 nor any member of Table 49, and no stored result for it was located.

16.5 Carryover bias of the classical two-period crossover estimator

For an AB/BA crossover with period effects $\mu_{(1)}$, $\mu_{(2)}$, period-specific treatment effects $\beta_D^{(1)}$, $\beta_D^{(2)}$ (active minus control), and additive carryover χ_1 into the second period after active treatment and χ_0 after control, the expected cell means are $\mu_{(1)} + \upsilon_{A,1}$, $\mu_{(2)} + \upsilon_{B,2} + \chi_1$ (sequence AB) and $\mu_{(1)} + \upsilon_{B,1}$, $\mu_{(2)} + \upsilon_{A,2} + \chi_0$ (sequence BA). The basic estimator $\hat{\beta}_D = \frac{1}{2}[(\bar{Y}_{AB,1} - \bar{Y}_{AB,2}) - (\bar{Y}_{BA,1} - \bar{Y}_{BA,2})]$ then has expectation

$$E(\hat{\beta}_D) = \tfrac{1}{2}\big(\beta_D^{(1)} + \beta_D^{(2)}\big) - \tfrac{1}{2}\big(\chi_1 - \chi_0\big) \tag{145}$$

(derived; checked numerically, `check-historical-results.R`, block B). The period effects cancel, and the estimator is biased by half the difference of the two carryover effects; with no carryover from control ($\chi_0 = 0$) the bias is $-\chi_1/2$, toward zero when the carryover has the sign of the treatment effect. (Source: `docs/09a`, Section 2.3, attributing the result to Shi and Ye, 2024.) [`docs/09a` writes $E(\hat{\tau}) = \frac{1}{2}(\tau_1 + \tilde{\tau}_2 - \omega_0 - \omega_1)$, with the carryover terms summed. Under the additive model above the two carryovers enter with opposite signs; the source's form agrees only if ω_0 is defined with the opposite sign (for example, as an effect on the control contrast), and both forms agree when $\chi_0 = 0$. The Shi and Ye (2024) full text was not available in PubMed Central (only the abstract, which states the direction of the bias under a sign condition), so their convention was not checked. Symbols renamed: τ_1, $\tilde{\tau}_2$, ω_0, ω_1 of the source are $\beta_D^{(1)}$, $\beta_D^{(2)}$, χ_0, χ_1 here; the cell-level $\upsilon_{A,1}, \ldots, \upsilon_{B,2}$ (treatment, period) are local and are not the contrast variance σ^2_Δ of §14.1. 2026-10-09: notation. The cell effects were written $\tau_{A,1}, \ldots$; τ^2 is the canonical heterogeneity variance.]

16.6 A constant-exposure analysis under carryover: the historical OL+BDC study

`docs/09a` (a pmsimstats2025-era study, reviewed 2026-10-07) simulates a single-path design: on drug at weeks 1–8, off drug from week 9, visits at weeks 1–12, 16 and 20, $N = 70$. Its data-generating process is mean moderation with linear growth,

$$BR_{it} = \begin{cases} a_{BR}\, t_{od,it}\,(1 + \beta_{bm} b_i) & \text{on drug,} \\ a_{BR}\, t_{od}^{\text{disc}}\,(1 + \beta_{bm} b_i)\, 2^{-t_{sd,it}/t_{1/2}} & \text{off drug, } t_{1/2} > 0, \\ 0 & \text{off drug, } t_{1/2} = 0, \end{cases} \qquad Y_{it} = 50 - BR_{it} + \varepsilon_{it}, \tag{146}$$

with $a_{BR} = 0.5$ per week, $\beta_{bm} = 0.3$, $t_{od}^{\text{disc}} = 8$, $\varepsilon_{it} \sim \mathcal{N}(0, 2^2)$ independent, no random intercept and no serial correlation. The analysis regresses Y on b_i, t_{od} and $b_i t_{od}$ with a random intercept, where t_{od} stays at 8 after discontinuation. On drug, $E(Y \mid b_i) = 50 - a_{BR} t_{od} - a_{BR}\beta_{bm}\, b_i t_{od}$, so the target is

$$\beta_{bm:D} = -a_{BR}\,\beta_{bm} = -0.15 \tag{147}$$

per week on drug and per standard deviation of the biomarker (derived; Source: `docs/09a`, Section 3.3). [The source writes a for a_{BR}; a_t is used in §14.1 for contrast weights. t_{sd} is counted from the first off-drug visit (week 9, $t_{sd} = 0$), unlike the package, where the first off-drug visit has $t_{sd} \geq 1$. Under this convention the week-9 response is full at every $t_{1/2} > 0$ and zero at $t_{1/2} = 0$, a discontinuity at $t_{1/2} = 0$ of the same kind as the step rule's (§14.3); whether it accounts for part of the simulated change between $t_{1/2} = 0$ and 0.05 was not tested.]

Retention. The off-drug retention $2^{-t_{sd}/t_{1/2}}$ at the measured off-drug weeks is given in Table 50 (verified, `check-historical-results.R`, block A).

The misspecification integral. Off drug the true response is $Q_i\, 2^{-s/t_{1/2}}$, $Q_i = a_{BR} t_{od}^{\text{disc}}(1 + \beta_{bm} b_i)$ the response at discontinuation and s the time since discontinuation, while a predictor constant off drug predicts a constant μ^*. Over an off-drug span of length T_{off}, with $k = \ln 2 / t_{1/2}$,

$$\mathcal{V}(t_{1/2}) = \int_0^{T_{\text{off}}} \big[Q_i\, 2^{-s/t_{1/2}} - \mu^*\big]^2\, ds. \tag{148}$$

Table 50: Off-drug retention $2^{-t_{sd}/t_{1/2}}$ in the `docs/09a` design (Source: `docs/09a`, Section 5.4; verified).

Week	t_{sd}	$t_{1/2} = 0.1$	0.2	0.5	1.0	2.0
9	0	1.000	1.000	1.000	1.000	1.000
10	1	0.001	0.031	0.250	0.500	0.707
12	3	0.000	0.000	0.016	0.125	0.354
16	7	0.000	0.000	0.000	0.008	0.088
20	11	0.000	0.000	0.000	0.000	0.022

Two choices of μ^* give opposite conclusions (derived; closed forms checked against numerical quadrature, `check-historical-results.R`, block `A2`):

- $\mu^* = Q_i$, the on-drug level implied by a constant t_{od} with a common slope:

$$\mathcal{V} = Q_i^2 \Big[T_{\text{off}} - \frac{2(1 - e^{-kT_{\text{off}}})}{k} + \frac{1 - e^{-2kT_{\text{off}}}}{2k} \Big], \tag{149}$$

  which falls strictly and monotonically from $Q_i^2 T_{\text{off}}$ as $t_{1/2} \to 0$ to 0 as $t_{1/2} \to \infty$, since the integrand falls pointwise in $t_{1/2}$. There is no interior maximum, and $t_{1/2} = 0$ is the case of largest mismatch.

- μ^* the least-squares constant over the span:

$$\mathcal{V} = Q_i^2 \Big[\frac{1 - e^{-2kT_{\text{off}}}}{2k} - \frac{(1 - e^{-kT_{\text{off}}})^2}{k^2 T_{\text{off}}} \Big] = T_{\text{off}} \operatorname{Var}_s \big(Q_i 2^{-s/t_{1/2}} \big), \tag{150}$$

  which is 0 in both limits and positive between them. Since $\mathcal{V}/(Q_i^2 T_{\text{off}})$ depends on $t_{1/2}$ only through $t_{1/2}/T_{\text{off}}$, its maximum is at $t_{1/2} = 0.214\, T_{\text{off}}$, where it equals 0.066 (verified, numerical optimization).

(Source: `docs/09a`, Section 3.5, which states the two cases and the monotone limits; the closed forms and the location of the maximum are derived here.) The fitted model is neither case exactly, since one interaction slope serves both phases; the simulated attenuation of `docs/09a`, Section 5.1 (largest at $t_{1/2} = 0$, where the mean estimate changes sign, and shrinking with $t_{1/2}$) behaves like the first case, so the source's bias-variance account of a power minimum at an intermediate half-life is not supported (`docs/09a`, review of 2026-10-07; argued). [`docs/09a` originally claimed an interior maximum without qualification and gave the $t_{1/2} \to 0$ limit as μ^{*2}; both were corrected in the source. Its simulated power (0.06 at $t_{1/2} = 0$, 0.00–0.02 at 0.05–0.5 weeks, 0.36 at 2 weeks; 100 replicates, MCSE at most 0.05) is quoted, not recomputed. The published Hendrickson collapse is explained by the step coupling rule (§14.3), which a mean-moderation model of this kind cannot exhibit.]

17 Baseline, a prognostic biomarker, and the shared biomarker effect

The published Hybrid analysis treats baseline as a ninth response row, coded off drug at $t = 0$. This section collects what that implies for the interaction (Sources: `docs/41`, `docs/45`, `docs/48`, `docs/49`; Paper 01, Appendix A.4). Local symbols: β_B, the analysis model's biomarker main effect (not the canonical multiplier β_{bm}); $\beta_{BL,t}$, the slope of visit t on baseline, and $\boldsymbol{\beta}_{BL}$ the vector of these slopes [2026-10-09: notation. Written ϕ_t, $\boldsymbol{\phi}$ and ϕ_{BL} before; ϕ is reserved for the decay families]; s_0, s_t, the true slopes of Y on the standardized biomarker b_i at baseline and at visit t; η_t, the expectancy (0 at baseline, 1 open label, 0.5 blinded); n_{on}, n_{off}, a path's post-baseline on- and off-drug visits. The prognostic pathways and the slopes they imply are listed in §11.8, and the separation of the biomarker block, on which the leak computations rest, is §15.12.

17.1 Two codings of baseline and five propositions

Write $\mathbf{Y}_i = (Y_{i0}, \mathbf{Y}'_{ip})'$, with $E(Y_{i0}) = \mu_0 + \beta_B B_i$, $E(\mathbf{Y}_{ip}) = X_{ip}\boldsymbol{\beta}$, and partitioned covariance with blocks σ_{00}, $\boldsymbol{\sigma}_{p0}$ and Σ_{pp}. Baseline is off drug, so neither β_D nor $\beta_{bm:D}$ enters its mean. The *covariate coding* fits only the post-baseline visits,

$$Y_{it} = \beta_0 + \beta_{BL}\,\mathrm{BL}_i + \beta_B B_i + \beta_D D_{it} + f(t) + \beta_{bm:D}\,B_i D_{it} + \varepsilon_{it}, \qquad t = 1, \ldots, 8. \tag{151}$$

Assume multivariate normal outcomes, a correct mean model and complete data. (Source: `docs/41`, Sections 2 and 4.)

Proposition 1 (the covariate model is the conditional model).

$$\mathbf{Y}_{ip} \mid Y_{i0} \sim \mathcal{N}\Big(X_{ip}\boldsymbol{\beta} + \boldsymbol{\beta}_{BL}\,(Y_{i0} - \mu_0 - \beta_B B_i),\ \Sigma_{pp} - \boldsymbol{\sigma}_{p0}\boldsymbol{\sigma}'_{p0}/\sigma_{00}\Big), \qquad \boldsymbol{\beta}_{BL} = \boldsymbol{\sigma}_{p0}/\sigma_{00}. \tag{152}$$

Conditioning changes the intercept and the biomarker main effect but not β_D or $\beta_{bm:D}$, so the interaction estimand is the same under both codings (derived). In the published process $Y_{it} = \mathrm{BL}_i - S_{it}$ with BL_i independent of every response component and of the biomarker, so $\sigma_{t0} = \sigma_{00}$ and $\boldsymbol{\beta}_{BL} = \mathbf{1}$: the covariate analysis is a change-from-baseline analysis (verified in the source to six decimals, `02-baseline-propositions.R`).

Proposition 2 (no gain for a within-participant contrast with equal slopes). For weights $\mathbf{a}$ on the post-baseline visits with $\sum_t a_t = 0$,

$$\mathrm{Var}(\mathbf{a}'\mathbf{Y}_{ip} \mid Y_{i0}) = \mathbf{a}'\Sigma_{pp}\mathbf{a} - \sigma_{00}(\mathbf{a}'\boldsymbol{\beta}_{BL})^2, \tag{153}$$

and the reduction vanishes whenever $\beta_{BL,t} \equiv \bar{\beta}_{BL}$ (derived). Measurement error in baseline attenuates every slope by the same reliability factor and so does not create a gain; a gain requires baseline to predict on-drug and off-drug visits differently, for example severity-moderated response or a time-varying natural course confounded with drug state. In the construct the E9 contrast variance is 44.13 (paths 1, 2) and 45.03 (paths 3, 4) with or without conditioning. [Recomputed here from the compound-symmetry moment formula of Paper 01, Appendix A.1, as 44.13 and 45.02 (`check-design.R`); the last digit differs by rounding.]

Proposition 3 (between participants). $\mathrm{Var}(Y_{it} - wY_{i0})$ is minimized at $w = \beta_{BL,t}$, with minimum $\sigma_{tt} - \sigma_{t0}^2/\sigma_{00}$ (derived): for a blinded visit, 599 against 257, a gain for β_B and path means, not for the interaction.

Proposition 4 (the baseline row as an off-drug visit). The response coding's paired-difference analogue, E9b, includes baseline in the off-drug mean. Substituting $Y_{i0} = \mathrm{BL}_i$,

$$\Delta_i^{\mathrm{E9}} = -\bar{S}_{i,\mathrm{on}} + \bar{S}_{i,\mathrm{off}}, \qquad \Delta_i^{\mathrm{E9b}} = -\bar{S}_{i,\mathrm{on}} + \frac{n_{\mathrm{off}}}{n_{\mathrm{off}}+1}\,\bar{S}_{i,\mathrm{off}}. \tag{154}$$

With g_t the coupling weight (1 on drug, $e^{-\lambda t_{sd}}$ off drug after exposure), the E9b slope is $-c_{bm}(\sigma_{BR}/\sigma_{bm})\big(\bar{g}_{\mathrm{on}} - \frac{n_{\mathrm{off}}}{n_{\mathrm{off}}+1}\bar{g}_{\mathrm{off}}\big)$. Under the published step rule with any carryover, $\bar{g}_{\mathrm{on}} = \bar{g}_{\mathrm{off}} = 1$, so the E9 slope is zero and the E9b slope is $1/(n_{\mathrm{off}}+1)$ of the no-carryover value: the power floor of the published analysis. Writing $S_{it} = s_i + d_{it}$ with shared level variance σ_s^2, E9b retains $-s_i/(n_{\mathrm{off}}+1)$ and so adds $\sigma_s^2/(n_{\mathrm{off}}+1)^2$ to the contrast variance before the partial offset from the shrunken visit-specific terms (derived). The source reports 44.13 to 51.04 and 45.03 to 51.77 (verified there). [The two algebraic identities were re-derived; the exact split of the increase, and the totals, are recomputed in Equation (158) of §17.2.]

Proposition 5 (working covariance). The GLS estimator under working V has variance $A^{-1}MA^{-1}$, $A = \sum_i X_i'V^{-1}X_i$, $M = \sum_i X_i'V^{-1}\Sigma V^{-1}X_i$, and the model-based A^{-1} is correct when $V \propto \Sigma$. The baseline row has no noise of its own ($\sigma_{00} = \sigma_{t0}$), so no random-intercept V over nine rows is proportional to Σ (derived).

cLDA and ANCOVA. With complete data and an unstructured covariance the likelihood factorizes as $f(Y_{i0})f(\mathbf{Y}_{ip} \mid Y_{i0})$, and the row and covariate models coincide *provided* β_B is free at baseline (derived). The published row coding shares β_B between baseline and the post-baseline off-drug visits and does not coincide. [`docs/41`, Section 7, applied the equivalence to the shared row coding; `docs/45`, Section 6, corrected this, verified by its `us-KR-sepbm` arm.]

17.2 The construct's outcome covariance and the exact cost of the baseline row

The covariance. In the published process $Y_{it} = \mathrm{BL}_i - S_{it}$ for $t \geq 1$, with BL_i independent of the biomarker and of every response component, so

$$\sigma_{t0} = \sigma_{00} \text{ for every } t \geq 1, \qquad \Sigma_{pp} = \sigma_{00}\, J + \mathrm{Cov}(\mathbf{S}_{ip}), \tag{155}$$

with J the matrix of ones (derived; Source: `docs/41`, Section 3). With $\sigma_{BL} = 18.48$, $\sigma_{TV} = 10$, $\sigma_{PB,t} = 10\eta_t$, $\sigma_{BR} = 8$, $\rho = 0.8$, $c_1 = 0.2$ and $c_\times = 0.1$, $\mathrm{Var}(S_t) = \sigma_{TV}^2 + \sigma_{PB,t}^2 + \sigma_{BR}^2 + 2c_1(\sigma_{TV}\sigma_{PB,t} + \sigma_{TV}\sigma_{BR} + \sigma_{PB,t}\sigma_{BR})$ is 368.0 at an open-label visit and 257.0 at a blinded one; with $\sigma_{00} = \sigma_{BL}^2 = 341.6$ these give the unconditional and conditional entries of Table 30 (path 1, no interaction; verified, `check-baseline-design-search.R`). Baseline has no noise of its own (its covariance with every later visit equals its variance), so its correlation with later visits, 0.694 (open label) and 0.755 (blinded), is lower than theirs with each other, 0.852 and 0.880 (derived).

Proposition 3 in full. For a single visit,

$$\mathrm{Var}(Y_{it}) = \sigma_{tt}, \qquad \mathrm{Var}(Y_{it} - Y_{i0}) = \sigma_{tt} + \sigma_{00} - 2\sigma_{t0}, \qquad \mathrm{Var}(Y_{it} \mid Y_{i0}) = \sigma_{tt} - \sigma_{t0}^2/\sigma_{00}, \tag{156}$$

the cases $w = 0$, $w = 1$ and $w = \beta_{BL,t}$ of $\mathrm{Var}(Y_{it} - wY_{i0})$. The conditional variance is never the largest, and it equals the change-score variance exactly when $\beta_{BL,t} = 1$ (derived). At a blinded visit the three are 598.6, 257.0 and 257.0 (verified).

The residual factor for the interaction. Adding baseline to the E9 regression of $\Delta_i = \mathbf{a}'\mathbf{Y}_{ip}$ on B_i multiplies the residual variance by $1 - r^2_{\Delta,\mathrm{BL}}$, with

$$r^2_{\Delta,\mathrm{BL}} = \frac{(\mathbf{a}'\boldsymbol{\sigma}_{p0})^2}{\sigma_{00}\,\mathbf{a}'\Sigma_{pp}\mathbf{a}} \tag{157}$$

and costs one degree of freedom (derived; Source: `docs/41`, Proposition 2; the source renamed ρ^2 to $r^2_{\Delta,\mathrm{BL}}$). By Equation (155), $\mathbf{a}'\boldsymbol{\sigma}_{p0} = \sigma_{00}\sum_t a_t = 0$ for every contrast, so the factor is exactly 1 in the construct.

Proposition 4, exactly. Since $\Delta_i^{\mathrm{E9b}} = \Delta_i^{\mathrm{E9}} - \bar{S}_{i,\mathrm{off}}/(n_{\mathrm{off}} + 1)$ (Equation (154)),

$$\mathrm{Var}(\Delta^{\mathrm{E9b}}) = \mathrm{Var}(\Delta^{\mathrm{E9}}) + \frac{\mathrm{Var}(\bar{S}_{\mathrm{off}})}{(n_{\mathrm{off}}+1)^2} - \frac{2\,\mathrm{Cov}(\Delta^{\mathrm{E9}}, \bar{S}_{\mathrm{off}})}{n_{\mathrm{off}}+1} \tag{158}$$

(derived). The second term is the shared response level that E9b retains; the third is the offset from the shrunken off-drug mean. On paths 1 and 2 ($n_{\mathrm{off}} = 3$), $\mathrm{Var}(\bar{S}_{\mathrm{off}}) = 209.13$ and Cov = 12.33, so the addition is 13.07 and the offset 6.17 (47%), giving 44.13 → 51.04; on paths 3 and 4 ($n_{\mathrm{off}} = 4$), 203.15 and 3.45, an addition of 8.13 and an offset of 1.38 (17%), giving 45.03 → 51.77 (verified). [`docs/41`, Proposition 4(b), approximates the addition by $\sigma_s^2/(n_{\mathrm{off}}+1)^2$ with $\sigma_s^2 \approx 200$, that is 12.5 and 8.0, and the offsets by 45% and 16% of those; the exact split of Equation (158) is 13.07 and 8.13, 47% and 17%. The totals agree.]

The closed-form E9 and E9b powers that follow are those of Table 25 and, for mean moderation under E9b, of Table 26. The standard deviations of the pooled slopes, $\mathrm{SD} = \sigma_{\mathrm{res}}/(\sigma_{bm}\sqrt{N-3})$ with σ_{res} of Equation (110), are 0.0507 (E9) and 0.0549 (E9b) under the step rule at $t_{1/2} = 0$, 0.0531 and 0.0571 at every positive half-life, and 0.0531 and 0.0572 (0.0571 at $t_{1/2} = 1$) under mean moderation (Hybrid, CS, $N = 70$, strength 0.25; Source: `docs/41`, Section 6, from `docs/37`; verified, `check-baseline-design-search.R`; the E9 values agree with Table 37). Under the step rule the SD rises from 0.0507 to 0.0531 because covariance binding stops explaining variance once the slope vanishes (Equation (108)); under mean moderation it is 0.0531 throughout. [The source gives one mean-moderation row for $t_{1/2} = 0$ to 1 (E9b power 0.619); the E9b power moves to 0.620 at $t_{1/2} = 1$ through the path-mean term of Equation (110).]

17.3 The leak through a shared biomarker effect

The probability limit of the biomarker block is Equation (132) (§15.12).

The leak, and which factor is correct. For S under a natural-course pathway with no interaction, $\mathbf{s} = s_{\text{prog}}(0, 1, \ldots, 1)'$. Under working independence with binary D, the fit reproduces the on-drug rows exactly and gives $\hat{\beta}_B$ the weighted mean of the baseline and post-baseline off-drug slopes. With n_0 and n_3 the row counts of the baseline rows and of the post-baseline off-drug rows ([2026-10-09: notation. Written w_0, w_3 before; w_t is the calendar week, and these weights are row counts.]),

$$\hat{\beta}_B \to f\, s_{\text{prog}}, \qquad \hat{\beta}_{bm:D} \to (1-f)\, s_{\text{prog}}, \qquad f = \frac{n_3}{n_0 + n_3}, \quad 1 - f = \frac{n_0}{n_0 + n_3} \tag{159}$$

(derived). So with f defined as the share of s_{prog} that the shared β_B absorbs, as in `docs/48`, `docs/49` and Paper 01, Appendix A.4, the spurious interaction is $(1-f)\, s_{\text{prog}}$, and $1-f$ is the effective share of the baseline row. Per path with equal visit weights, $1 - f = 1/(n_{\text{off}} + 1)$, the same factor as Proposition 4. Pooled over the Hybrid paths at the published allocation (18, 18, 17, 17), $n_0 = 70$ and $n_3 = 244$, so $1 - f = 0.223$ (verified, `check-prognostic.R`). The leak is a fixed bias, so its rejection rate grows with N. Under a correlated working covariance the bias is still proportional to s_{prog}, but its factor is the second element of (132), not a simple weight share, and it can be larger than under independence.

Computed factors plim $\hat{\beta}_{bm:D}/s_{\text{prog}}$ in the `04-power-simulation.R` construct, the published construct in the CS row and the `04-power-simulation.R` construct with AR(1) components in the AR(1) rows (Hybrid, $N = 70$; D_{bc} at an assumed $t^a_{1/2} = 0.5$ week, pre-exposure coded 0; verified, `check-prognostic.R` and `check-prognostic-car1.R`):

Table 51: Spurious interaction per unit prognostic slope. "True" uses the construct's own outcome covariance as V (an approximation to an unstructured fit); RI+CAR1 uses the covariance to which a large ML fit converged ($\sigma^2_u = 356$, $\sigma^2_\varepsilon = 248$, CAR(1) parameter $\hat{\rho}_a = 0.64$ per week).

Construct, working V	Pathway	S	F1	ES
any, independence	natural course, constant	0.271	−0.020	0
any, independence	placebo, $\propto \eta_t$	0.369	0	0.239
CS, true	natural course, constant	0.091	−0.013	0
AR(1) $\rho = 0.8$, true	natural course, constant	0.430	0.171	0
AR(1) $\rho = 0.7$, RI+CAR1	natural course, constant	0.276	0.089	0
AR(1) $\rho = 0.7$, RI+CAR1	placebo, $\propto \eta_t$	0.311	0	0.139

The ordering of S across working covariances matches the stress-test rejection rates of `docs/48`, Section 2 (S 0.099, unstructured 0.416), which supports, but does not prove, the weighting account that the source labels inferred. [These factors are computed in the `04-power-simulation.R` construct (with AR(1) components in the AR(1) rows; 2026-10-09: previously called the published process, which under `NOTATION.md` is compound symmetry only), not in the stress-test construct (configuration C, $\rho = 0.7$; Table 58); they illustrate the mechanism and are not a reproduction of the stress-test rates.]

17.4 Freeing the baseline slope, and the expectancy terms

ES adds $B_i \cdot \mathbf{1}\{t = 0\}$, giving baseline its own slope $\beta_B + \beta_{B,0}$. For any pathway constant after baseline, $\mathbf{s} = s_{\text{prog}}(\mathbf{1} - \mathbf{1}\{t = 0\})$ lies in the span of $(1, D_{bc}, \mathbf{1}\{t = 0\})$, so by (132) the leak is exactly zero under every working covariance (derived; Table 51). ES is equivalent to the covariate coding (`docs/45`, Section 6).

F1 adds $\eta_t + B_i\eta_t$, with coefficient $\beta_{bm:\eta}$ on $B_i\eta_t$. Because $\eta_0 = 0$, the baseline slope is freed implicitly, and an expectancy-scaled pathway $\mathbf{s} \propto (\eta_0, \ldots, \eta_8)'$ lies in the span, so F1 is exact for

it under every working covariance (derived). For a pathway constant after baseline it is not in the span. With binary D and working independence, group the rows into baseline, open label on drug, blinded on drug and blinded off drug, with row counts n_0, n_1, n_2, n_3. The fitted slopes $(\beta_B,\ \beta_B+\beta_{bm:\eta}+\beta_{bm:D},\ \beta_B+\frac{1}{2}\beta_{bm:\eta}+\beta_{bm:D},\ \beta_B+\frac{1}{2}\beta_{bm:\eta})$ satisfy the single linear restriction $\hat{s}_0+\hat{s}_1=\hat{s}_2+\hat{s}_3$, and the weighted projection of $s_{\text{prog}}(0,1,1,1)$ onto it gives

$$\hat{\beta}_{bm:D} \to s_{\text{prog}}\,\frac{1/n_3-1/n_2}{1/n_0+1/n_1+1/n_2+1/n_3} \tag{160}$$

(derived here). The misfit therefore cancels exactly if and only if the blinded on-drug and blinded off-drug visits carry equal total weight, $n_2=n_3$, whatever the weights of baseline and open label; this is the condition stated in `docs/49`, Section 3. In the Hybrid design at the published allocation $n_2=176$ and $n_3=244$, so the condition fails and the factor is -0.051 (binary D; verified, `check-prognostic.R`). [`docs/49` says a residual bias "can remain" with unequal weights or serial correlation; the computation shows it does remain in the Hybrid design, small under independence (-0.020 with D_{bc}) but 0.089 under the RI+CAR1 working covariance and 0.171 under the true AR(1) covariance. The simulated F1 rejection of 0.047 in the constant-pathway cell is consistent with a bias of this size, but F1's robustness there is approximate, not exact.] Conversely, ES is not exact for the expectancy-scaled pathway (factor 0.139 to 0.239), although its simulated rejection rate was nominal (0.049). [`docs/48`, Section 3, states that under ES "nothing that happens at baseline can leak"; that is correct for baseline, but an expectancy-scaled slope still leaks into ES through the open-label visits.] In F1 the interaction rests mainly on the blinded on-drug versus blinded off-drug comparison, at equal expectancy.

17.5 The power cost of freeing the baseline slope

Size-adjusted power, each analysis rejecting at its own null 5% quantile (Source: `docs/48`, Sections 1 and 3; `docs/49`, Section 4; verified there from the stress-test outputs):

Table 52: Size-adjusted power of the shared (S) and freed (ES, F1) analyses.

Cell	S	ES	F1
AR(1), $c_{bm}=0.30$	0.514	0.364	0.364
AR(1), $c_{bm}=0.45$	0.808	0.656	0.672
CS, $c_{bm}=0.25$	0.554	0.662	0.522

Under AR(1) data the baseline row is an informative off-drug observation of the biomarker slope that the participant effect does not replace, and freeing it costs about 30% of the power; an analytic account of this gain has not been derived (`docs/45`, Section 12, question 4). [The sources summarize the CS row as "the cost largely disappears"; the table shows ES 0.108 *above* S there, which is a gain, while F1 is 0.032 below. The mechanism of the ES gain under CS is not established in the sources.]

17.6 The assumed half-life of D_{bc}

The probability limit under an assumed half-life. Let the data follow graded covariance binding (Equation (34)) with half-life $t_{1/2}$, so that on path p the true slope of Y on B is $\mathbf{s}_p=-c_{bm}(\sigma_{BR}/\sigma_{bm})(0,g_1,\ldots,g_8)'$, the leading 0 for the baseline row. Let the analysis use $D_{bc,it}$ built at an assumed $t^a_{1/2}$: $x_t=1$ on drug, $2^{-t_{sd}/t^a_{1/2}}$ off drug after exposure, 0 at baseline and before exposure, with the biomarker block $(B_i, B_i x_t)$. By Equation (132), if $t^a_{1/2}=t_{1/2}$ then $\mathbf{s}_p=-c_{bm}(\sigma_{BR}/\sigma_{bm})\,\mathbf{x}_p$ lies in the span of $Z_p=(\mathbf{1},\mathbf{x}_p)$ on every path, so

$$\operatorname{plim}\hat{\beta}_{bm:D}=-c_{bm}\,\sigma_{BR}/\sigma_{bm}\qquad(t^a_{1/2}=t_{1/2}), \tag{161}$$

the full on-drug interaction, under every working covariance (derived). Otherwise the limit is the second element of Equation (132) and depends on V; $\beta_{bm:D}$ is then the interaction per unit of modeled exposure, so its scale depends on $t^a_{1/2}$, and an assumed half-life longer than the true one overshoots because the analysis attributes exposure the off-drug visits do not carry. Table 53 evaluates the limit with V the true outcome covariance conditional on B (the target of an unstructured fit) and under independence.

Table 53: plim $\hat{\beta}_{bm:D}$, Hybrid, CS construct, graded coupling, $c_{bm} = 0.25$, allocation 18, 18, 17, 17; the estimand is -0.1302 (derived and computed, `check-baseline-design-search.R`). Last column: simulated mean of `us-KR`, 250 replicates (Source: `docs/45`, Section 7; quoted).

Exposure regressor	$t_{1/2}$	True V	Independence	Simulated
D_{it}	0	-0.1302	-0.1302	-0.135
D_{it}	1	-0.1010	-0.1056	-0.100
D_{bc}, $t^a_{1/2} = 0.5$	0	-0.1393	-0.1372	
D_{bc}, $t^a_{1/2} = 0.5$	1	-0.1135	-0.1166	
D_{bc}, $t^a_{1/2} = 1$	0	-0.1511	-0.1447	
D_{bc}, $t^a_{1/2} = 1$	1	-0.1302	-0.1302	-0.129
D_{bc}, $t^a_{1/2} = 2$	0	-0.1672	-0.1511	-0.168
D_{bc}, $t^a_{1/2} = 2$	1	-0.1529	-0.1445	-0.151

The limits under the true covariance reproduce every simulated mean the source reports to within Monte Carlo error, which supports reading the `us-KR` estimates as these limits (verified for the limits; the agreement is with quoted simulation means). [`docs/45`, Section 7, states the recovery at $t^a_{1/2} = t_{1/2}$ and the overshoot from simulation only; Equation (161) and the table are derived here.]

18 Inference: from a fitted coefficient to a calibrated test

18.1 A closed-form comparator: repeated-measures ANOVA

Paper 08's principal analytic contribution, stated for comparison against the compendium's simulation-based test procedures, is a full closed-form treatment of the classical split-plot RM-ANOVA under a dichotomized biomarker. In the symbols of `docs/26` (renamed on 2026-10-07 so that none collides with a canonical symbol), the model is

$$Y_{gjit} = \mu + \xi_g + \phi_j + (\xi\phi)_{gj} + u_{i(g)} + \upsilon_t + (\xi\upsilon)_{gt} + (\phi\upsilon)_{jt} + (\xi\phi\upsilon)_{gjt} + \varepsilon_{gjit}, \tag{162}$$

for biomarker stratum $g \in \{1, \ldots, a\}$, treatment phase $j \in \{1, \ldots, b\}$, participant $i \in \{1, \ldots, n\}$ within stratum, and within-phase timepoint $t \in \{1, \ldots, T\}$, with $u_{i(g)} \sim \mathcal{N}(0, \sigma^2_u)$ the random participant effect and $\varepsilon_{gjit} \sim \mathcal{N}(0, \sigma^2_\varepsilon)$ independent within-participant error. The effect symbols ξ, ϕ, υ are local to this subsection (ϕ_j is not the decay function ϕ of §4). [2026-10-09: notation. The time effect was written τ_t, as in `docs/26`; τ^2 is the canonical heterogeneity variance, so it is υ_t here.] Here n is the number of participants *per stratum*, so the total is $N = an$. Model (162) implies compound symmetry; the F-tests need only the weaker condition of sphericity of each within-participant factor, and a factor with two levels is always spherical. The biomarker-treatment interaction $(\xi\phi)_{gj}$ is the target; its test statistic is

$$F_{AB} = \frac{MS_{AB}}{MS_{B\times S(A)}} \sim F(df_{AB},\, df_{B\times S(A)}) \text{ under } H_0, \tag{163}$$

which specializes, for $a = b = 2$, to a closed form. Writing the interaction contrast as $\hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2$, with $\bar{\Delta}_g$ the stratum-g mean of each participant's phase-1-minus-phase-2 difference

of within-phase means Δ_{gi}, for which $\mathrm{Var}(\Delta_{gi}) = 2\sigma_\varepsilon^2/T$ and $\mathrm{Var}(\hat{L}) = 4\sigma_\varepsilon^2/(nT)$,

$$F_{AB} = \frac{nT\,\hat{L}^2}{4\,\hat{\sigma}_\varepsilon^2}, \qquad t_{AB} = \sqrt{\tfrac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_\varepsilon}, \qquad t_{AB}^2 = F_{AB}, \qquad df_1 = 1,\ df_2 = 2(n-1). \tag{164}$$

The 2×2 interaction F-test is therefore the ordinary pooled two-sample t-test on the within-participant differences Δ_{gi}. Because that t-test needs only that the Δ_{gi} be iid normal with a common variance in the two strata, the identity and the exactness of the test hold for *any* within-participant covariance, AR(1) included, once each phase is reduced to its mean (`docs/26`, Section 5b; derived). Under AR(1) errors the variance of Δ_{gi} is no longer $2\sigma_\varepsilon^2/T$, so the power formula below must use the actual variance of the phase difference. Power follows the noncentral F distribution with noncentrality

$$\Lambda = \frac{nT\,L_1^2}{4\sigma_\varepsilon^2}, \qquad \pi = \Pr\big[F_{AB} > F_{1,\,2(n-1),\,1-\alpha} \mid \Lambda\big], \tag{165}$$

where L_1 is the value of the contrast under H_1, an effect size on the raw outcome scale (`docs/26`, eqs. 17–18). The variance identities and the nominal size of the pooled t at $df_2 = 2(n-1)$ were also checked by a short simulation (`check-existing-sections.R`: $n = 30$, $T = 3$; simulated $\mathrm{Var}(\hat{L}) = 0.0433$ against 0.0444, rejection rate 0.050 over 2,000 replicates).

[2026-10-08: corrected and renamed. This subsection previously used Paper 08's classical symbols α_i, β_j, $\pi_{l(i)}$, e_{ijlk}, λ and $1-\beta$, which collide with the canonical α, π and λ; Paper 08 itself still uses them, with Δ_0 for the effect size, written L_1 here as in `docs/26`. It also described the error as satisfying "compound-symmetric sphericity", conflating the two conditions; Paper 08 still states that sphericity means a compound-symmetric within-subject covariance. Finally, it repeated Paper 08's claim that this is the compendium's only fully analytic power result. That no longer holds: the restated Paper 01 (Appendix A.1–A.2) and `docs/37` give the moments and noncentral-t power of the paired-difference statistic E9 in closed form under both coupling rules. GEE, Mancl–DeRouen and the LME/`corCAR1` model are still evaluated by simulation, with the sandwich/CR2 machinery of §18.6 providing the small-sample correction.]

18.2 RM-ANOVA: the sum-of-squares decomposition, sphericity, and sample size

This subsection completes §18.1 in the same local symbols: stratum $g \le a$, phase $j \le b$, participant $i \le n$ within stratum, within-phase time $t \le T$, and a dot for averaging over a subscript. (Source: `docs/26`, Sections 4 to 7.)

Sums of squares. The between-participant and the phase-related within-participant terms of the balanced split-plot decomposition, with their degrees of freedom, are

$$\begin{aligned}
SS_A &= bnT \sum_g (\bar{Y}_{g\cdots} - \bar{Y}_{\cdots})^2, & df_A &= a-1,\\
SS_{S(A)} &= bT \sum_{g,i} (\bar{Y}_{g\cdot i\cdot} - \bar{Y}_{g\cdots})^2, & df_{S(A)} &= a(n-1),\\
SS_B &= anT \sum_j (\bar{Y}_{\cdot j\cdot\cdot} - \bar{Y}_{\cdots})^2, & df_B &= b-1,\\
SS_{AB} &= nT \sum_{g,j} (\bar{Y}_{gj\cdot\cdot} - \bar{Y}_{g\cdots} - \bar{Y}_{\cdot j\cdot\cdot} + \bar{Y}_{\cdots})^2, & df_{AB} &= (a-1)(b-1),\\
SS_{B\times S(A)} &= T \sum_{g,j,i} (\bar{Y}_{gji\cdot} - \bar{Y}_{g\cdot i\cdot} - \bar{Y}_{gj\cdot\cdot} + \bar{Y}_{g\cdots})^2, & df_{B\times S(A)} &= a(n-1)(b-1),
\end{aligned} \tag{166}$$

and the interaction test of Equation (163) is

$$F_{AB} = \frac{SS_{AB}/[(a-1)(b-1)]}{SS_{B\times S(A)}/[a(n-1)(b-1)]} \sim F\big((a-1)(b-1),\, a(n-1)(b-1)\big) \text{ under } H_0 \tag{167}$$

and sphericity of the phase factor, which holds automatically when $b = 2$. Under Model (162) and balance, $E[MS_{B\times S(A)}] = \sigma_\varepsilon^2$, because each participant's phase means have residual variance σ_ε^2/T and the factor T rescales it (derived). For $a = b = 2$, the sum-to-zero interaction estimates are $\widehat{(\xi\phi)}_{gj} = \pm\hat{L}/4$, so

$$SS_{AB} = nT \cdot 4\,(\hat{L}/4)^2 = \frac{nT\,\hat{L}^2}{4}, \tag{168}$$

which with $df_{AB} = 1$ and $df_{B\times S(A)} = 2(n-1)$ gives Equation (164). (Derived. Every sum of squares, the degrees of freedom and the F ratio agree with `aov()` fitted with error strata `subj/(phase*time)`, for (a, b, n, T) equal to $(2, 2, 10, 3)$ and to $(3, 3, 8, 4)$; the estimate $\widehat{(\xi\phi)}_{11}$ equals $\hat{L}/4$ to machine precision; `check-gompertz-rmanova-cycles.R`.)

The time-resolved test. Whether the biomarker modifies the treatment effect differently over within-phase time is the three-way term $(\xi\phi\upsilon)_{gjt}$, tested by

$$F_{AB\upsilon} = \frac{MS_{AB\upsilon}}{MS_{B\upsilon\times S(A)}}, \qquad df_1 = (a-1)(b-1)(T-1), \quad df_2 = a(n-1)(b-1)(T-1) \tag{169}$$

(classical split-plot; the degrees of freedom agree with the `subj:phase:time` stratum of `aov()`, verified). Its within-participant factor has T levels, so for $T > 2$ the exactness of §18.1 does not carry over and sphericity is required.

Sphericity and Box's measure. For a within-participant factor with m levels and covariance Σ, let C be an $m \times (m-1)$ matrix of orthonormal contrasts and $S = C^\top \Sigma C$. Box's measure is

$$\epsilon_{\text{Box}} = \frac{(\operatorname{tr} S)^2}{(m-1)\operatorname{tr}(S^2)}, \qquad \frac{1}{m-1} \le \epsilon_{\text{Box}} \le 1, \tag{170}$$

with $\epsilon_{\text{Box}} = 1$ if and only if $S \propto I$ (sphericity) and the lower bound attained when S has rank one (derived, Cauchy–Schwarz on the eigenvalues of S). Sphericity is equivalent to all pairwise differences of levels having equal variance. Under AR(1), $\operatorname{Var}(y_j - y_k) = 2\sigma^2(1-\rho^{|j-k|})$ varies with $|j-k|$ once $m \ge 3$, so $\epsilon_{\text{Box}} < 1$ for every $0 < \rho < 1$, while $m = 2$ is always spherical (derived). Table 54 gives values at unit spacing. The Greenhouse–Geisser correction multiplies both degrees of freedom by $\hat{\epsilon}_{\text{Box}}$; the Huynh–Feldt variant is less conservative and is the conventional choice when $\hat{\epsilon}_{\text{Box}} > 0.75$ (Source: `docs/26`, Section 6; the 0.75 threshold is a rule of thumb).

Table 54: Box's ϵ_{Box} under AR(1) at unit spacing, and its lower bound (computed here, `check-gompertz-rmanova-cycles.R`; compound symmetry gives 1 at every m).

m	$\rho = 0.5$	$\rho = 0.8$	$1/(m-1)$
2	1.000	1.000	1.000
3	0.925	0.849	0.500
4	0.859	0.725	0.333
8	0.722	0.469	0.143

[`docs/26` states the range of ϵ_{Box} and that it is strictly below 1 under AR(1) for more than two levels, but gives neither the formula nor values; Equation (170) and Table 54 are supplied here.]

Sample size. Inverting the noncentrality of Equation (165),

$$n = \frac{4\sigma_\varepsilon^2\,\Lambda}{TL_1^2}, \tag{171}$$

per stratum, where Λ must deliver the target power at $df_2 = 2(n-1)$. Because the critical value depends on n, the equation is solved by iteration (derived; Source: `docs/26`, Section 7, eq. 19). For 80% power at $\alpha = 0.05$: $L_1/\sigma_\varepsilon = 1$ needs $n = 33$ per stratum at $T = 1$ ($\Lambda = 8.25$) and

$n = 9$ at $T = 4$; $L_1/\sigma_\varepsilon = 0.5$ needs $n = 127$ at $T = 1$ (computed here by direct search over n with the noncentral F, `check-gompertz-rmanova-cycles.R`). Under AR(1) errors σ_ε^2/T must be replaced by the actual variance of a within-phase mean, as in §18.1.

Relation to the mixed-model test. `docs/26` (Section 9) argues, without derivation, that the strict RM-ANOVA F and the Wald t on `bm:Dbc` are asymptotically equivalent when the biomarker enters as the stratum indicator, the within-participant covariance is exactly compound-symmetric, and the drug indicator is binary (E1), and that the mixed model otherwise uses more information (argued).

18.3 The efficiency cost of dichotomizing the biomarker

The 2×2 RM-ANOVA of Equation (164) is a two-sample t-test of the within-participant contrast Δ_i across a median split of the biomarker, whereas E9 (Equation (142)) regresses Δ_i on the continuous biomarker. For a normal biomarker and $\Delta_i = \beta b_i + e_i$, the split indicator $x_i = \mathbf{1}(b_i > 0)$ has $\mathrm{Cov}(b_i, x_i) = \varphi(0)$ and $\mathrm{Var}(x_i) = 1/4$, so

$$\frac{\mathrm{Corr}(\Delta_i, x_i)}{\mathrm{Corr}(\Delta_i, b_i)} = 2\varphi(0) = \sqrt{2/\pi} \approx 0.798, \qquad \text{relative efficiency} = \frac{2}{\pi} \approx 0.637, \tag{172}$$

when the moderation is small relative to the residual variance: the dichotomized test has about 0.80 of the noncentrality and needs about $\pi/2 \approx 1.57$ times the sample size (derived; a 4,000-replicate simulation at $N = 200$ gave a mean t-ratio of 0.798, block I). (Source: `docs/23`, Section 13.1, items 2 and 6, from `docs/17`, which state the 36 percent loss and the 0.64 relative efficiency without derivation.) The constant is the one of Equation (89) for the reverse situation, a threshold moderation analyzed with a linear term. [`docs/17` adds that sphericity corrections reduce RM-ANOVA power further; for $a = b = 2$ the within-participant factor has two levels and is always spherical (§18.1), so no correction applies in that case.]

18.4 A general decomposition of realized Type I error

Paper 10, written in deliberately general Wald-test notation (applicable to any fixed-effect interaction coefficient, not only $\beta_{bm:D}$), decomposes an estimator into a standardized component:

$$\hat\theta = \mu + \sigma Z^*, \qquad \mu = E(\hat\theta), \qquad \sigma = \mathrm{SD}(\hat\theta), \qquad Z^* = \frac{\hat\theta - \mu}{\sigma}, \tag{173}$$

and rewrites the Wald statistic $T = \hat\theta/\widehat{\mathrm{SE}}$ in terms of two dimensionless channels,

$$T \approx \frac{\mu + \sigma Z^*}{\kappa\sigma} = \frac{1}{\kappa}\big(Z^* + \delta\big), \qquad \kappa = \frac{\overline{\widehat{\mathrm{SE}}}}{\sigma}, \qquad \delta = \frac{\mu}{\sigma}. \tag{174}$$

Realized Type I error follows directly:

$$\alpha_{\mathrm{real}} = \Pr\big(|T| > t_{\nu,1-\alpha/2}\big) \approx \Pr\big(|Z^* + \delta| > \kappa\, t_{\nu,1-\alpha/2}\big), \tag{175}$$

and exactly *four* channels can move it away from nominal: a standardized bias δ, a scale-distortion κ (model-based SE too large or small relative to the true sampling SD), a degrees-of-freedom channel ν, and the shape of Z^*'s distribution (departures from normality). Calibration is recovered exactly when $\delta = 0$, $\kappa = 1$, $\nu = \infty$, and $Z^* \sim \mathcal{N}(0,1)$. With the bias and reference-distribution channels inert, this reduces to a closed form in κ alone,

$$\alpha_{\mathrm{real}} \approx 2\,\Phi\big(-\kappa\, t_{\nu,1-\alpha/2}\big) \;\approx\; 0.05 - 0.23\,(\kappa - 1) \text{ near } \kappa = 1, \tag{176}$$

with elasticity $d\log\alpha_{\mathrm{real}}/d\log\kappa \approx -4.6$ at $\kappa = 1$ (both constants verified numerically, `check-existing-sections.R`: slope -0.229, elasticity -4.58): a small overstatement of the

model-based standard error produces a disproportionate fall in realized Type I error. Paper 10 uses this to explain why ordinary least squares, which ignores the participant random effect, is strongly *conservative* for this within- participant interaction (standard-error ratio near 1.3), not anti-conservative as the familiar warning about ignored clustering suggests. OLS charges a within-participant contrast the full residual variance, and in the idealized case of a pure within-participant contrast with exchangeable errors its scale-distortion factor is

$$\kappa^2_{\mathrm{OLS}} \approx \frac{\sigma_u^2 + \sigma_\varepsilon^2}{\sigma_\varepsilon^2}, \tag{177}$$

which exceeds one by the ratio $\sigma_u^2/\sigma_\varepsilon^2$ of random-intercept to residual variance. The anti-conservative case (the Moulton pitfall) concerns between-cluster effects.

[2026-10-08: corrected. This paragraph previously stated that the OLS analysis is anti-conservative, which contradicts both Equation (177) ($\kappa > 1$ is conservative) and Paper 10, Section on the working-covariance ladder; it also said the excess over one is "exactly the random-intercept variance", where it is that variance relative to σ_ε^2. Paper 10 writes the random-intercept variance as σ_b^2; the canonical symbol is σ_u^2, since b_i is the standardized biomarker.]

Paper 02 gives the compendium's operational estimator of κ directly on the interaction coefficient, using the notation $\hat{\beta}_{bm:X}$ for the fitted coefficient under carryover specification X:

$$\kappa = \frac{\overline{\widehat{SE}}}{\mathrm{SD}(\hat{\beta}_{bm:X})} = \frac{\frac{1}{n_{\mathrm{sim}}}\sum_{r=1}^{n_{\mathrm{sim}}} \widehat{SE}_r}{\sqrt{\frac{1}{n_{\mathrm{sim}}-1}\sum_{r=1}^{n_{\mathrm{sim}}} (\hat{\beta}_{bm:X,r} - \bar{\hat{\beta}}_{bm:X})^2}}, \tag{178}$$

and the calibrated true value against which bias is computed throughout the carryover-specification manuscript,

$$\theta_{\mathrm{true}} = -c_{bm}\,\sigma_{BR} = -3.6 \quad (c_{bm} = 0.45,\ \sigma_{BR} = 8). \tag{179}$$

This is the general estimand $-c_{bm}\sigma_{BR}/\sigma_{bm}$ of Paper 01 with $\sigma_{bm} = 1$, the biomarker SD in Paper 02's driver (`default_baseline_param()` in `analysis/scripts/carryover-sensitivity/simulation-core.R`, inspected). It is the target of the exposure-weighted specification only; E1 and E3 estimate the coefficient on the binary D_{it}, so their bias against θ_{true} reflects an estimand mismatch (Paper 02, headline table).

18.5 What a random intercept absorbs under AR(1)

Let the within-participant deviations have variance σ_T^2 and correlation ρ^h at lag h (weeks). A working model with a random intercept of variance $\sigma_u^2 < \sigma_T^2$ and independent residuals assigns the remainder $\Sigma - \sigma_u^2 J$ to the residual, with variance $\sigma_\varepsilon^2 = \sigma_T^2 - \sigma_u^2$ and correlation

$$\mathrm{Corr}(\varepsilon_{it}, \varepsilon_{is}) = \frac{\sigma_T^2\rho^h - \sigma_u^2}{\sigma_\varepsilon^2} = \rho^h - \left(1 - \rho^h\right)\frac{\sigma_u^2}{\sigma_\varepsilon^2}, \qquad h = |w_t - w_s|, \tag{180}$$

positive when $\rho^h > \sigma_u^2/\sigma_T^2$ (nearby visits) and negative beyond (derived; verified, `check-historical-results.R`, block E; Source: `docs/23`, Section 4.3, which corrected an earlier form without the rescaling by σ_ε^2). A random intercept therefore absorbs only the long-lag part of AR(1) correlation, and a model that treats the remainder as independent misstates the variance of any contrast between visits. [`docs/23` concludes that such a model "underestimates standard errors". The direction depends on the contrast: for contiguous on and off blocks the AR(1) contrast variance exceeds the compound-symmetry one (ratio 3.25 at $p = q = 4$, Equation (123)), so the standard error is understated and the test anti-conservative, as in the quoted `lmer` crossover sizes of 0.13 to 0.17 (200 replicates; central 95% binomial range $[0.020, 0.080]$ at size

0.05, verified, block F); for interleaved designs the ratio is smaller (Section 15). This is argued from Equation (123), not computed for each design.]

[A related claim in `docs/23`, Section 13.1, item 5 (from `docs/17`), that ordinary least squares inflates Type I error to about 15 to 25 percent at an intraclass correlation of 0.3 with $T = 8$, holds for a *between*-participant effect: the design effect is $1 + (T-1)\,0.3 = 3.1$, giving a realized size of $2\Phi(-1.96/\sqrt{3.1}) = 0.266$ (verified, block K). For the within-participant interaction $\beta_{bm:D}$ OLS is conservative, Equation (177).]

18.6 Cluster-robust sandwich variance and the CR2 correction

The naive (Liang–Zeger) sandwich estimator, clustering on participant,

$$\widehat{\mathrm{Var}}_{\mathrm{sandwich}}(\hat\beta) = \Big(\textstyle\sum_i X_i'\hat V_i^{-1}X_i\Big)^{-1}\Big(\sum_i X_i'\hat V_i^{-1}\hat e_i\hat e_i'\hat V_i^{-1}X_i\Big)\Big(\sum_i X_i'\hat V_i^{-1}X_i\Big)^{-1}, \tag{181}$$

is downward-biased at moderate cluster counts. Paper 10 gives the general CR family in the GLS-transformed metric,

$$\hat V_{\mathrm{CR}} = (\tilde X'\tilde X)^{-1}\Big(\textstyle\sum_i \tilde X_i' A_i\tilde e_i\tilde e_i' A_i\tilde X_i\Big)(\tilde X'\tilde X)^{-1}, \tag{182}$$

indexed by the adjustment matrix A_i: $A_i = I$ (CR0, unadjusted); $A_i = (I - H_{ii})^{-1}$ (the jackknife form, equal to the Mancl–DeRouen correction in the marginal/GEE setting); and

$$A_i = (I - H_{ii})^{-1/2} \qquad \text{(CR2)}, \tag{183}$$

the symmetric square root that renders $\hat V_{\mathrm{CR}}$ exactly unbiased when the working covariance is correctly specified, the cluster generalization of the HC2 estimator of MacKinnon and White. Its Satterthwaite degrees of freedom are obtained by matching the first two moments of a weighted sum of χ^2_1 variables with weights ℓ_g, the eigenvalues of the relevant quadratic form,

$$\nu = \frac{\big(\sum_g \ell_g\big)^2}{\sum_g \ell_g^2}. \tag{184}$$

[2026-10-08: the eigenvalues were written λ_g; λ is the canonical carryover decay rate, so they are written ℓ_g.]

Paper 02's null-cell diagnostics find κ from 1.05 to 1.10 across specifications (empirical Type I error 0.029 to 0.039 against nominal 0.05) for the model-based standard errors, and κ from 1.07 to 1.26 in the information-rich cells of its block S6, restored to about one by CR2. These values are consistent, via Equation (176) (which gives 0.040 at $\kappa = 1.05$ and 0.031 at $\kappa = 1.10$; verified numerically), with the mild conservatism of the compendium's model-based Wald tests. Paper 02 attributes it to the mismatch between the random-intercept plus `corCAR1` working covariance and the covariance the data-generating process induces. The conservatism is not general: in the restated Paper 01's stress test (configuration C at $\rho = 0.7$, the covariance construct of the refined process), the published random-intercept analysis with model-based inference rejected at 0.069 pooled over 11 non-prognostic null cells (0.081 at $N = 35$), and analyses sharing the biomarker effect with baseline were anti-conservative when the biomarker was also prognostic (Paper 01, Results).

19 The stress-test construct and its fourteen analyses

This section states as model formulas the data-generating process and the fourteen analyses of the exploratory stress test of the Hybrid design (Sources: `docs/46`, Sections 4 and 5 and Amendments 1 and 2; `docs/47`; `docs/43`, Section 3.9), read against the driver `11-strawman-`

`stress-test.R` in `analysis/scripts/quick-sim/carryover-closed-form/` (inspected 2026-10-08). The selection criteria are in §21.2. Local symbols: $B_i^c = B_i - \bar{B}$, the biomarker centered at the trial's sample mean (code `bmc`); X_{it}, the analysis exposure regressor; $\iota_t = \mathbf{1}\{t = 0\}$ (code `bl0`); $k(t) \in \{0, \mathrm{OL}, \mathrm{bl}\}$, the phase of visit t (baseline, open label, blinded); ρ_a, the working CAR(1) parameter; σ_ε, $\sigma_{\varepsilon,k}$, working residual standard deviations; $\mathcal{T}_a$, the grid of analysis half-lives.

19.1 The data-generating process

Visits are at weeks $w \in \{0, 4, 8, 9, 10, 11, 12, 16, 20\}$, $t = 0$ being baseline, on the four Hybrid paths of Table 4, with $N_p = \lfloor N/4 \rfloor + \mathbf{1}\{p \le N \bmod 4\}$ (18, 18, 17, 17 at $N = 70$). For participant i on path p,

$$(B_i, \mathrm{BL}_i, TV_{i\cdot}, PB_{i\cdot}, BR_{i\cdot}) \sim \mathcal{N}\big(\mu_p,\ S_p R_p S_p\big), \tag{185}$$

with $\mu_{bm} = 124.33$, $\sigma_{bm} = 15.36$, $\mu_{BL} = 83.07$, $\sigma_{BL} = 18.48$; $\sigma_{TV} = 10$, $\sigma_{PB,t} = 10\eta_t$ and $\sigma_{BR} = 8$ at every visit; $E(TV_{it}) = G_{TV}(w_t)$ and $E(PB_{it}) = \eta_t G_{TV}(t_{pb,t})$, with G_{TV} the modified Gompertz curve of Equation (2) at $(m, d, r) = (6.506, 5, 0.35)$; and $E(BR_{it})$ from $G_{BR}(t_{od,t})$ at $(10.986, 5, 0.42)$ with the recursive, compounding carryover of Equation (23), $\mu_{BR,t} = G_{BR}(t_{od,t}) + \mu_{BR,t-1}\,\phi(t_{sd,t})$ off drug after exposure. The correlation matrix R_p is Equation (54) with the response block $M = K \otimes A$ (configuration C, $\rho = 0.7$, $c_1 = 0.2$) or M_{CS} (Equation (56), $\rho_{cs} = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$), and the biomarker row

$$\begin{aligned} \mathrm{Cor}(B_i, BR_{it}) &= c_{bm}\, g_t, & \mathrm{Cor}(B_i, TV_{it}) &= c_{bm,TV}, \\ \mathrm{Cor}(B_i, PB_{it}) &= c_{bm,PB}, & \mathrm{Cor}(B_i, \mathrm{BL}_i) &= c_{bm,BL}, \end{aligned} \tag{186}$$

where g_t is graded coupling (Equation (31)) with ϕ exponential or Weibull of shape $k \in \{0.5, 2\}$ (Equation (19)), the same ϕ as in the mean. A cell whose R_p is not positive definite stops; none did. The outcome is

$$Y_{i0} = \mathrm{BL}_i, \qquad Y_{it} = \mathrm{BL}_i - \big(TV_{it} + \beta_{bm}^{TV}\,\sigma_{TV}\, b_i\, \tilde{G}_{TV}(w_t) + PB_{it} + BR_{it}\big) + \varepsilon_{it}^{\mathrm{me}}, \quad t \ge 1, \tag{187}$$

with $b_i = (B_i - \mu_{bm})/\sigma_{bm}$, $\tilde{G}_{TV} = G_{TV}/\max_t G_{TV}(w_t)$, and $\varepsilon_{it}^{\mathrm{me}} \sim \mathcal{N}(0, s_{k(t)}^2)$ independent, $s_{\mathrm{OL}} = 3$ and $s_{\mathrm{bl}} = 6$ in the measurement-error cell and zero otherwise. Dropout is monotone, MCAR or MAR as in Equation (105) of §13.4 (inspected). The 41 cells vary one factor at a time from the base cell (configuration C, $N = 70$, $t_{1/2} = 0.5$ week, exponential decay, no dropout, no prognostic coupling): c_{bm} along the two power curves, measurement error, $N \in \{35, 150\}$, $t_{1/2} \in \{0, 1\}$, Weibull shape, MCAR and MAR dropout, and one of $c_{bm,TV}$, β_{bm}^{TV}, $c_{bm,PB}$, $c_{bm,BL}$ set to 0.3 (`docs/46`, Section 5). The slopes these pathways imply are those of Table 23, and the generalized ceiling is Equation (70). At the base half-life of 0.5 week the Hybrid ceiling is 0.518 under configuration C and 0.274 under compound symmetry (verified, `check-stress-test-construct.R`, Section 1), so the curve maxima 0.45 and 0.25 are feasible and 0.30 under compound symmetry is not.

[`docs/46`, Section 3, bounds the compound-symmetry curve by "that construct's ceiling of 0.256"; that is the value at $t_{1/2} = 0$. At the curve's $t_{1/2} = 0.5$ week the graded ceiling is 0.274; the grid and the exclusion of 0.30 are unaffected.] [The base construct (configuration C, $\rho = 0.7$) keeps the published compounding recursion on the drug-response mean, not the anchored form of Equation (22) (`docs/44`, Step 3; inspected), and allocates participants to paths in fixed numbers. Configuration C is the covariance construct of the refined process of Paper 01, but the stress test did not run the refined process in full (inspected, `11-strawman-stress-test.R`).]

19.2 The fourteen analyses

Every analysis fits a sub-model of

$$Y_{it} = \beta_0 + \beta_B B_i^c + \beta_T t + \beta_D X_{it} + \beta_{bm:D}\, B_i^c X_{it} + \underbrace{\beta_\eta \eta_t + \beta_{bm:\eta}\, B_i^c \eta_t}_{\text{F1, F1A, G1, G2}} + \underbrace{\beta_{bm:T}\, B_i^c t}_{\text{F2}} + \underbrace{\beta_{B,0}\, B_i^c \iota_t}_{\text{ES, EA}} + e_{it}, \qquad t \in \{0, 4, \ldots, 20\}, \tag{188}$$

over all nine rows, baseline included, and tests $H_0\colon \beta_{bm:D} = 0$ two-sided at $\alpha = 0.05$. The regressor X_{it} is $D_{bc,it}$ of Equation (13) at an assumed half-life $t^a_{1/2}$, with $\lambda = \ln 2 / t^a_{1/2}$, zero at baseline and before exposure, and $X_{it} = D_{it}$ when $t^a_{1/2} = 0$. The working covariances of $\mathbf{Y}_i$, with u_i the random intercept where present, are

$$\begin{aligned}
\text{RC:}\quad & \operatorname{Cov}(Y_{it}, Y_{is}) = \sigma_u^2 + \sigma_\varepsilon^2\, \rho_a^{|t-s|}, \\
\text{RCH:}\quad & \operatorname{Cov}(Y_{it}, Y_{is}) = \sigma_u^2 + \sigma_{\varepsilon,k(t)}\, \sigma_{\varepsilon,k(s)}\, \rho_a^{|t-s|}, \qquad k \in \{0, \mathrm{OL}, \mathrm{bl}\}, \\
\text{US:}\quad & \operatorname{Cov}(\mathbf{Y}_i) = \Sigma, \ \text{unrestricted over the nine visits}, \\
\text{CS:}\quad & \operatorname{Cov}(Y_{it}, Y_{is}) = \sigma^2\big[(1-r)\mathbf{1}\{t = s\} + r\big],
\end{aligned} \tag{189}$$

RC being `nlme::lme` with `random = ~1 | ptID` and `corCAR1(form = ~t | ptID)` on weeks, RCH the same with `varIdent(form = ~1 | phase)`, and US and CS the `mmrm` structures `us(visit | ptID)` and `cs(visit | ptID)` without a random intercept. Table 55 lists the analyses.

Table 55: The fourteen stress-test analyses (Sources: `docs/46`, Section 4 and Amendment 2; `docs/47`, notation table; inspected in `11-strawman-stress-test.R`). Terms: those of Equation (188) beyond the core $\beta_0, \beta_B, \beta_T, \beta_D, \beta_{bm:D}$. p, q: numbers of fixed-effect and covariance parameters. AIC: $t^a_{1/2}$ chosen over $\mathcal{T}_a = \{0, 0.25, 0.5, 1, 2\}$ weeks (§19.3).

ID	Terms	$t^a_{1/2}$	Covariance	Inference	Status	p	q
S	core only	0.5	RC	CR2	protocol (strawman)	5	3
DS	core only	AIC	RC	CR2	protocol; reference, Amend. 1	5	3
C1	core only	0.5	RCH	CR2	protocol	5	5
C2	core only	0.5	RCH	model-based	protocol	5	5
F1	η_t, $B_i^c \eta_t$	0.5	RC	CR2	protocol; selected	7	3
F2	$B_i^c t$	0.5	RC	CR2	protocol	6	3
ES	$B_i^c \iota_t$	0.5	RC	CR2	protocol	6	3
A	core only	0.5	US	Kenward–Roger	protocol	5	45
DA	core only	AIC	US	Kenward–Roger	protocol	5	45
EA	$B_i^c \iota_t$	0.5	US	Kenward–Roger	protocol	6	45
R	core only	0 ($X = D_{it}$)	CS	Satterthwaite	protocol (published ref.)	5	2
G1	η_t, $B_i^c \eta_t$	0.5	RCH	CR2	post hoc	7	5
G2	η_t, $B_i^c \eta_t$	0.5	RCH	model-based	post hoc	7	5
F1A	η_t, $B_i^c \eta_t$	AIC	RC	CR2	post hoc	7	3

Inference. CR2 is Equation (183) with the Satterthwaite degrees of freedom of Equation (184) (`clubSandwich::coef_test`, clustered on participant). Model-based is the `nlme` Wald t on its between-within (containment) degrees of freedom; `nlme` offers no Satterthwaite option for these models. Kenward–Roger is the small-sample covariance adjustment and degrees of freedom of Kenward and Roger (1997) as implemented in `mmrm` in its linear variant (`Kenward-Roger-Linear`), which omits the second-derivative terms of the covariance parameterization. Satterthwaite is the `mmrm` model-based standard error with Satterthwaite degrees of freedom. The covariance-parameter counts q are those behind the tie rule of `docs/46`, rule 5: F1 has 3 and G2 has 5 (inspected).

[`docs/47` describes `bmc` as the standardized biomarker; it is centered but not scaled (inspected). Test statistics and p-values are invariant to rescaling B_i^c, so no reported rate depends on this, but $\beta_{bm:D}$ in

these fits is per unit of the biomarker, the raw-scale estimand of Equation (32). `docs/46` describes R as a random intercept with model-based inference; the driver fits the marginal compound-symmetry structure of `mmrm` with Satterthwaite degrees of freedom, which equals a random intercept with independent errors when $r \geq 0$ (inspected).]

The expectancy terms and the drug indicator. The collinearity of η_t with D_{it} in the Hybrid design ($r = 0.588$ over the nine rows, 0.507 over the post-baseline rows; verified, `check-stress-test-construct.R`, Section 6) is reconciled with the variance inflation factors quoted in the sources in §14.9; `docs/40`, quoted in `docs/43`, Section 3.1, gives it as the reason Hendrickson et al. dropped `De`. The biomarker block of F1, $B_i^c(1, X_{it}, \eta_t)$, is nonetheless identified, since the rows fall into four groups (baseline; open label, on drug; blinded, on drug; blinded, off drug) with distinct (X, η) (derived). Its leak properties are Equations (132) and (160).

Table 56: The published analysis and its options, and two further proposals (Source: `docs/43`, Sections 2.2, 2.11, 3.1 and 3.6; the published options inspected there in commit `58b32a9`). All use the uncentered B_i and `lmer`, baseline as a row.

ID	Model	Status
O1	$Y_{it} = \beta_0 + \beta_B B_i + \beta_D D_{it} + \beta_T t + \beta_{bm:D} B_i D_{it} + u_i + \varepsilon_{it}$	published default; §16
O2	O1 + $\beta_\eta \eta_t$ (`useDE = TRUE`)	proposed; not simulated as specified
O3	O1 with $u_{0i} + u_{1i} t$ (`t_random_slope = TRUE`)	proposed
O4	OL design: $Y_{it} = \beta_0 + \beta_B B_i + \beta_T t + \beta_{bm:T} B_i t + u_i + \varepsilon_{it}$	evaluated; tests a biomarker-by-time slope, confounded with natural course and expectancy
E4	$X = D_{it}$ plus a main effect of $t_{sd,it}$ (`simplecarryover`)	retired
D6	random intercept, random post-baseline shift and random expectancy coefficient, residual variance by phase	proposed; G1, G2 realize it with fixed expectancy terms and no random shift

Table 57: Working covariances named in `docs/43`, Section 2.7, for visits t, s (`nlme` and `mmrm` definitions, the latter from the package's covariance vignette).

Name	$\mathrm{Cov}(Y_{it}, Y_{is})$
independent	$\sigma^2 \mathbf{1}\{t = s\}$, with or without u_i
`corCAR1`	$\sigma_\varepsilon^2 \rho_a^{\lvert t-s \rvert}$ on the time scale (weeks), with or without u_i (RC with u_i)
`cs`	$\sigma^2[(1-r)\mathbf{1}\{t = s\} + r]$
`csh`	$\sigma_t \sigma_s[(1-r)\mathbf{1}\{t = s\} + r]$, a standard deviation per visit
`us`	Σ_{ts} unrestricted: 45 parameters over nine visits
`sp_exp`	$\sigma^2 \rho_a^{d_{ts}}$, d_{ts} the distance between visit times; homogeneous
`varIdent` by phase	multiplies the residual standard deviation by a factor per phase, one fixed at 1 (RCH)
GEE working	independence, exchangeable or AR(1) working correlation with a sandwich variance (§18.6)

19.3 AIC selection of the analysis half-life

DS, DA and F1A choose

$$\hat{t}_{1/2}^{a} = \arg\min_{h \in \mathcal{T}_a} \mathrm{AIC}_{\mathrm{ML}}(h), \qquad \mathcal{T}_a = \{0, 0.25, 0.5, 1, 2\} \text{ weeks}, \tag{190}$$

with the candidate models fitted by maximum likelihood, and refit the selected model by REML for inference; $h = 0$ is the binary D_{it} (`docs/46`, Section 4; inspected). The candidates differ only in the values of the one regressor X_{it}, so they have the same number of parameters and $\mathrm{AIC}_{\mathrm{ML}}(h) - \mathrm{AIC}_{\mathrm{ML}}(h') = -2\{\ell(h) - \ell(h')\}$: the AIC rule is the maximum-likelihood estimate of $t^a_{1/2}$ over the grid (derived). The reported test then treats an estimated nonlinear parameter as known, the post-selection problem noted for E7 at Equation (140). [The grid differs from E7's $\{0.25, 0.5, 1, 2\}$ of Table 49 by including 0. The size excess the selection produces (pooled 0.053 for DS, 0.057 for F1A, 0.065 for DA, against 0.048, 0.051 and 0.057 at the fixed 0.5 week; `docs/47`, Section 3) is simulated, not derived.]

19.4 The implied outcome covariance and phase-specific variances

Under configuration C ($\rho = 0.7$), with constant component standard deviations within a phase, Equation (69) gives three variance levels and one cross-phase factor. With $s_t = (10, 10\eta_t, 8)$ and $q_{ts} = s_t^\top K s_s$,

$$q_{\mathrm{OL}} = 368, \qquad q_{\mathrm{bl}} = 257, \qquad q_{\mathrm{OL,bl}} = 300, \qquad \frac{q_{\mathrm{OL,bl}}}{\sqrt{q_{\mathrm{OL}}\, q_{\mathrm{bl}}}} = 0.9755, \tag{191}$$

so $\mathrm{Var}(Y_{it})$ takes the three values of Table 30 (341.6, 709.6 and 598.6; variances do not depend on the correlation structure), plus 9 and 36 in the measurement-error cell (derived; verified, `check-stress-test-construct.R`, Section 5). The factor 0.9755 confirms the correlation factor of 0.976 quoted from `docs/35` in §9 (Equation (69)). The same q_{OL} and q_{bl} are the conditional variances of Table 30, since conditioning on baseline removes exactly σ^2_{BL}. Hence the RCH structure of Equation (189) with $\sigma^2_u = \sigma^2_{BL}$, $\rho_a = \rho$, $\sigma^2_{\varepsilon,\mathrm{OL}} = 368$, $\sigma^2_{\varepsilon,\mathrm{bl}} = 257$ and $\sigma_{\varepsilon,0} \to 0$ reproduces every element of the construct's 9×9 outcome covariance except the open-label by blinded pairs, which it overstates by the factor $1/0.9755$ (largest discrepancy 5.3; verified). The baseline row has no noise of its own (Proposition 5 of §17), which RCH can represent by $\sigma_{\varepsilon,0} \to 0$ and RC cannot (derived). With measurement error the construct adds a white-noise term that neither structure contains.

The leak factors of Equation (132) under this covariance are in Table 58. Under working independence the computation reproduces the "any, independence" rows of Table 51 (S, ES, F1: 0.271, 0, −0.020 for the constant natural course and 0.369, 0.239, 0 for the placebo pathway), which checks it; those rows are not repeated.

Table 58: Spurious interaction per unit prognostic slope, plim $\hat\beta_{bm:D}/s_{\mathrm{prog}}$, in configuration C ($\rho = 0.7$, Hybrid, $N = 70$, X at $t^a_{1/2} = 0.5$ week), by the mean model and the working covariance V (derived from Equation (132); verified, `check-stress-test-construct.R`, Section 5). "True" is the construct's outcome covariance; "RCH limit" the structure described above.

	Natural course, constant			Placebo, $\propto \eta_t$		
Working V	S	ES	F1	S	ES	F1
true	0.629	0	0.162	0.423	0.070	0
RCH limit	0.628	0	0.172	0.420	0.069	0

A working covariance that can approach the truth (RCH in C1, C2, G1, G2; US in A, DA, EA) makes the leak of the shared baseline slope larger, not smaller: 0.63 for S's mean model under the constant natural-course pathway, against 0.27 under independence. The stress-test null rejection rates in that cell order the same way (S 0.099, C1 0.382, A 0.416; G1 0.055 and G2 0.062 against F1 0.047; `docs/47`, Section 3), which supports, but does not prove, this account of why the phase-variance models fail the prognostic cells (argued). It also gives content to the statement of `docs/50` that the phase-variance models "weight the shared baseline slope heavily": a near-zero baseline residual variance gives the baseline rows large weight in $\hat\beta_B$. [The

factor under the RC fit itself was not computed for configuration C, because the converged RC parameters for this construct are not stored; Table 51 gives it for the published process (0.276). Rejection rates also depend on the standard errors, which this calculation does not address.]

19.5 Sign symmetry of power

Suppose the biomarker enters the data-generating process only through its correlation with BR, the cells without prognostic coupling, and is normal (more generally, symmetric about its mean). Let $B_i^* = 2\mu_{bm} - B_i$. It has the law of B_i, and its covariance with every other variable changes sign: with $D_{\mathrm{r}} = \mathrm{diag}(-1, 1, \ldots, 1)$ acting on the biomarker coordinate of the joint covariance Σ of Equation (185),

$$\Sigma(-c_{bm}) = D_{\mathrm{r}}\, \Sigma(c_{bm})\, D_{\mathrm{r}}, \tag{192}$$

so by joint normality the law of $(B_i^*, \text{the rest})$ under c_{bm} is that of $(B_i, \text{the rest})$ under $-c_{bm}$, with the outcomes and the dropout indicators unchanged. Every analysis of Table 55 uses the biomarker only through columns linear in B_i^c, and $B_i^{*c} = -B_i^c$ whatever the centering; negating those columns negates their coefficients and leaves fitted values, likelihood, working covariance, every standard error (model-based, CR2 or Kenward–Roger), the degrees of freedom and the two-sided p-value unchanged. Hence

$$\hat{\beta}_{bm:D}(-c_{bm}) \overset{d}{=} -\hat{\beta}_{bm:D}(c_{bm}), \qquad \pi(-c_{bm}) = \pi(c_{bm}), \tag{193}$$

for every analysis, unadjusted and size-adjusted, since both signs share the null cell (derived; Source: `docs/48`, Section 4, cited by `docs/46`, Amendment 1, and `docs/47`, Section 5). Checks: Equation (192) holds to machine precision for the Hybrid compound-symmetry construct (`check-baseline-design-search.R`); on one simulated trial, refitting S and F1 with the reflected biomarker reproduced the CR2 and model-based p-values to six decimals and negated the estimate (`check-stress-test-construct.R`, Section 8); and an `lme` fit with random intercept and `corCAR1` on simulated Hybrid data gave identical model-based (0.0293568) and CR2 (0.0210449) p-values with the reflected biomarker and a log-likelihood difference of 5×10^{-13} (`check-baseline-design-search.R`) (all verified).

With a prognostic coupling the reflection negates $c_{bm,TV}$, $c_{bm,PB}$, $c_{bm,BL}$ and β_{bm}^{TV} as well, so the symmetry is the joint one, $\pi(-c_{bm}, -c_{\mathrm{prog}}) = \pi(c_{bm}, c_{\mathrm{prog}})$, not symmetry in c_{bm} alone (derived). It fails if only some couplings are reflected (for example a predictive coupling reversed while a prognostic one is not), and, for the distributional step, if the biomarker is not symmetric about its mean. [`docs/48` states the identity for cells in which the biomarker is correlated only with the drug response; the joint condition is its general form.] The symmetry screen of `docs/46`, Section 3, tested this identity with independent replicates, and its failures were Monte Carlo error. Generating the $-c_{bm}$ trials from the $+c_{bm}$ draws with the biomarker reflected would make sampling asymmetry impossible (argued; `docs/48`, Section 4).

Part V
The treatment main-effect track and shared performance measures

20 The treatment main-effect track

Paper 04 targets a different estimand under a different, simpler DGP: the treatment *main* effect β_D, with **no biomarker moderation channel at all**, and a linear (not Gompertz) data- generating model.

20.1 Data-generating model

For the N-of-1 arm,

$$Y_{ij} = \beta_0 + \beta_D\, D_{bc,ij} + \beta_T\, t_{ij} + u_i + \varepsilon_{ij}, \qquad u_i \sim \mathcal{N}(0, \sigma_u^2), \tag{194}$$

with $\boldsymbol{\varepsilon}_i$ following a continuous-time AR(1) covariance $\mathrm{Cov}(\varepsilon_j, \varepsilon_k) = \sigma_w^2 \rho^{|t_j - t_k|}$, $\sigma_w = 2.3$, $\rho = 0.3$, with t_j the period index; carryover is applied as an additive exponential-decay term to the period immediately following a drug period, $\beta_D^{\mathrm{co}} = \beta_D \exp\{-\ln(2)\, L/t_{1/2}\}$, with $L = 7$ days the period length and $t_{1/2} = 3$ days at the primary cell. For the comparator parallel-group RCT arm, drawing patient baselines from a shared pool,

$$Y_i = \mu_0 + u_i + \beta_D Z_i + \varepsilon_i, \qquad u_i \sim \mathcal{N}(0, \sigma_u^2), \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_w^2), \qquad Z_i \in \{0, 1\}, \tag{195}$$

with $\mu_0 = 8.0$ nightmares per week and $\sigma_u = 1.5$ ([2026-10-09: notation. Paper 04 writes this SD σ_τ; τ^2 is the canonical heterogeneity variance of Equation (196), and u_i is the random intercept, so its SD is σ_u.]). The two arms share the same $N = 35$ patient pool (participant-count-matched, not observation-count-matched, a limitation the paper itself notes), fit respectively by `nlme::lme(Sx ~ Dbc + t_wk, ..., correlation = corCAR1(...))` and by ANCOVA, `lm(endpoint ~ trt + baseline)`.

[2026-10-08: three corrections. (i) The carryover term was written δ_c, which collides with the standardized bias δ; it is β_D^{co} here. (ii) Paper 04 draws the baselines from $\mathcal{N}(\mu_0, \sigma_u^2)$ and calls u_i "the patient's baseline draw" while also adding μ_0 in the model, which counts μ_0 twice; the consistent reading, used above, is $u_i \sim \mathcal{N}(0, \sigma_u^2)$ with baseline $\mu_0 + u_i$. (iii) Paper 04 describes its test as using Satterthwaite degrees of freedom "from `nlme`"; `nlme::lme` reports containment (between/within) degrees of freedom, not Satterthwaite. Which the driver uses was not checked.]

[Internal inconsistency, resolved.] Paper 04 previously stated $\sigma_w = 1.0$ in the AR(1) covariance definition of its DGP (§2.1) while separately citing $\sigma_w = 2.3$ as "moderate within-patient measurement noise" in its optimal-design-conditions discussion (§6), with the two values never reconciled in the source text. Inspection of the production driver (`analysis/scripts/treatment-main-effect/production-run.R`, constant `SIGMA_WITHIN <- 2.3`) confirms $\sigma_w = 2.3$ is the value the simulation actually uses; §2.1's $\sigma_w = 1.0$ was the error, and has been corrected in the manuscript to match both §6 and the code.

20.2 Classical closed-form power theory (background, not the paper's own simulation)

Paper 04's literature-review section reproduces the closed-form power apparatus that most of this compendium replaces with simulation, due to Senn (2018) and Zucker et al.:

$$d_{ij} = \Delta_i + \varepsilon_{ij}, \qquad \Delta_i \sim \mathcal{N}(\Delta, \tau^2), \qquad \varepsilon_{ij} \sim \mathcal{N}(0, 2\sigma^2) \tag{196}$$

$$\mathrm{Var}(\hat{\Delta}) = \frac{\tau^2 + 2\sigma^2/k}{n} \qquad \text{(random-effects, patient-mean analysis)} \tag{197}$$

$$\Lambda_{\mathrm{RE}} = \frac{\Delta}{\sqrt{(\tau^2 + 2\sigma^2/k)/n}}, \quad \nu = n - 1 \tag{198}$$

$$\Lambda_{\mathrm{FE}} = \frac{\Delta}{\sqrt{2\sigma^2/(nk)}}, \quad \nu = n(k-1) \qquad \text{(fixed-effects, pools within-patient SS)} \tag{199}$$

[2026-10-08: the patient-level effect and the noncentrality parameters were written δ_i and λ, which collide with the canonical δ and λ; they are Δ_i and Λ here. Λ_{RE} and Λ_{FE} are noncentral-t parameters, not the noncentral-F Λ of Equation (165).] for n patients and k paired treatment cycles each, with τ^2 the between-patient (treatment-by-patient interaction) variance and $2\sigma^2$ the within-patient, within-cycle variance of paired differences. The paper notes standard software assuming $n - 1$

denominator df underestimates power here whenever $k > 2$, because the fixed-effects analysis correctly pools within-patient degrees of freedom that the random-effects analysis discards. Zucker et al.'s equivalent precision formula,

$$\mathrm{Var}(\hat{\Delta}) = \frac{\tau^2 + \sigma_w^2/k}{n}, \tag{200}$$

which is Equation (197) with $\sigma_w^2 = 2\sigma^2$ the within-patient variance of a paired difference (here σ_w is not Paper 04's AR(1) residual SD of Equation (194)), makes explicit that the marginal benefit of additional cycles k diminishes once $\sigma_w^2/k \ll \tau^2$: past that point, adding cycles to existing patients buys far less precision than adding new patients, because the between-patient variance floor dominates.

Paper 04's own simulation (2,000 replicates per cell, effect-size grid $\beta_D \in \{0, -0.5, -1.0, -2.0\}$ nightmares/week, carryover half-life sweep $t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days) does not invoke Equations (197)–(200) directly; it estimates power and Type I error empirically, at the MCSE rates given in §21 below.

21 Shared performance measures

The manuscripts of the compendium report the same Monte Carlo performance quantities, following the ADEMP framework of Morris, White, and Crowther (2019):

$$\hat{\pi} = \frac{1}{n_{\text{reps}}} \sum_{r=1}^{n_{\text{reps}}} \mathbf{1}[p_r < \alpha] \qquad \text{(empirical power / Type I error)} \tag{201}$$

$$\mathrm{MCSE}(\hat{\pi}) = \sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}} \tag{202}$$

$$\mathrm{Bias} = \frac{1}{n_{\text{reps}}} \sum_r \hat{\theta}_r - \theta_{\text{true}}, \qquad \mathrm{MCSE(Bias)} = \frac{s_{\hat{\theta}}}{\sqrt{n_{\text{reps}}}} \tag{203}$$

$$\mathrm{MCSE(EmpSE)} = \frac{s_{\hat{\theta}}}{\sqrt{2(n_{\text{reps}} - 1)}} \tag{204}$$

with $s_{\hat{\theta}}$ the empirical standard deviation of the replicate-level point estimates, EmpSE $= s_{\hat{\theta}}$. Equation (204) is the Monte Carlo standard error of the *empirical* standard error (Morris et al., Table 6), not of the mean model-based standard error $\overline{\widehat{SE}}$. Here $\hat{\theta}_r$ denotes a generic replicate estimate; in Paper 10's sense it is the estimator of §18.4. A replicate count of $n_{\text{reps}} = 625$ is the recurring reference, the smallest for which $\mathrm{MCSE}(\hat{\pi}) \leq 0.02$ at the worst case $\hat{\pi} = 0.5$ ($0.25/0.02^2 = 625$). Production runs use various counts, typically 500 to 1,000 replicates for power and bias cells and up to 5,000 for null cells. In null cells the MCSE at $\hat{\pi} \approx 0.05$ is smaller for a given n_{reps} (0.0069 at 1,000), but a size check must resolve departures that are small relative to 0.05, which is why null cells receive more replicates.

[2026-10-08: corrected. Equation (204) was labeled MCSE($\widehat{SE}$); the paragraph previously said that a smaller rejection rate near α "tightens the MCSE requirement", where the binomial MCSE in fact falls as $\hat{\pi}$ moves from 0.5 toward 0.05. "Every manuscript reports the same three quantities" was softened: not every paper reports bias (Paper 07 reports it only at the null).]

Some sources also refer to the mean squared error (one of the whitepapers inventoried for this revision states MSE = variance + squared bias in prose),

$$\begin{aligned} \mathrm{MSE} &= \frac{1}{n_{\text{reps}}} \sum_r \left(\hat{\theta}_r - \theta_{\text{true}}\right)^2 = \mathrm{Bias}^2 + \frac{n_{\text{reps}} - 1}{n_{\text{reps}}} s_{\hat{\theta}}^2, \\ \mathrm{MCSE(MSE)} &= \sqrt{\frac{\sum_r \left[(\hat{\theta}_r - \theta_{\text{true}})^2 - \mathrm{MSE}\right]^2}{n_{\text{reps}}(n_{\text{reps}} - 1)}} \end{aligned} \tag{205}$$

(definition and MCSE as in Morris et al., Table 6; the decomposition is derived). [2026-10-08: added, because the term is used without definition in the sources; which papers report it in their tables was not surveyed.]

For a difference-of-differences comparison across two conditions, the pre-restatement version of Paper 01 gives

$$\mathrm{SE}(\Delta_B - \Delta_A) = \sqrt{\frac{\hat{\pi}_{B,0}(1-\hat{\pi}_{B,0})}{n} + \frac{\hat{\pi}_{B,1}(1-\hat{\pi}_{B,1})}{n} + \frac{\hat{\pi}_{A,0}(1-\hat{\pi}_{A,0})}{n} + \frac{\hat{\pi}_{A,1}(1-\hat{\pi}_{A,1})}{n}}, \qquad \Delta_X = \hat{\pi}_{X,0} - \hat{\pi}_{X,1}, \tag{206}$$

the standard error of the difference between two within-condition carryover losses, each estimated from n independent replicates per cell. The form assumes the four power estimates are independent; with shared seeds across cells (paired designs, as in the restated Paper 01's study 2) it is conservative or must be replaced by a paired estimate (argued). It is used to test whether two designs', architectures', or specifications' power differences are distinguishable from Monte Carlo noise.

[2026-10-08: the restated Paper 01 no longer uses Equation (206); the formula is retained from the pre-restatement version (`archive/report-2026-10-07-pre-restatement.Rmd`).]

21.1 Size-adjusted power

When a test's realized size departs from α, power at the nominal critical value mixes sensitivity with miscalibration. Size-adjusted power replaces the nominal critical value by an empirical one from the matching null cell c_0, the cell that shares design, architecture and $t_{1/2}$ (or the same stress, or the zero point of the same curve). Two forms are in use:

$$\tilde{\pi} = \frac{1}{n_{\text{reps}}} \sum_{r=1}^{n_{\text{reps}}} \mathbf{1}\big[\,|z_r| > q_{1-\alpha}\big] \quad \text{or} \quad \tilde{\pi} = \frac{1}{n_{\text{reps}}} \sum_{r=1}^{n_{\text{reps}}} \mathbf{1}\big[\,p_r < \hat{q}_\alpha\big], \tag{207}$$

with $z = \hat{\beta}_{bm:D}/\widehat{SE}$, $q_{1-\alpha}$ the $(1-\alpha)$ quantile of $|z|$ over the null-cell replicates (`docs/33`, glossary and Section 2.4, and its Appendix 1 script), and $\hat{q}_\alpha$ the empirical α quantile of the analysis's two-sided p-values in the null cell (`docs/46`, Section 6; `docs/47`, glossary; the decision script `12-strawman-stress-decision.R`, inspected). The two forms coincide when every replicate's test uses the same reference distribution; with replicate-specific degrees of freedom (CR2 with Satterthwaite degrees of freedom, Kenward–Roger) p is not a fixed function of $|z|$ and they can differ slightly (derived). Both inherit the Monte Carlo error of the estimated quantile, which with 1,000 null replicates rests on about 50 exceedances, so `docs/33` treats its size-adjusted figures as indicating magnitude only; no MCSE for $\tilde{\pi}$ is given there. [An MCSE that accounts for the estimated quantile would need, for example, a bootstrap over the null replicates; none has been computed. The binomial MCSE that the stress test uses (§21.2) ignores this component.] Applied to the pre-restatement Paper 01 output, size adjustment roughly halved the covariance-minus-mean gap in carryover loss (0.071 to about 0.029 in Hybrid, 0.217 to about 0.097 in OL+BDC), because the ratio κ of model-based to empirical standard error rose with carryover from 1.12 to 1.26 in the OL+BDC covariance cells (quoted from `docs/33`, Section 2.4; reanalysis of stored replicates, not rechecked here).

21.2 Size, power screens and the selection criteria of the stress test

The exploratory stress test (§19) judges each analysis a by the following quantities (Sources: `docs/46`, Sections 3, 6 and 7; `docs/47`, glossary; inspected in the decision script `12-strawman-stress-decision.R`).

Nominal size. With $\hat{\pi}_{a,0}$ the rejection rate pooled over the R replicates of the non-prognostic null cells,

$$\big|\hat{\pi}_{a,0} - 0.05\big| \le 2\sqrt{0.05 \times 0.95/R} \quad \text{and} \quad \max_{\text{null cells}} \hat{\pi}_{a,0,\text{cell}} \le 0.075, \tag{208}$$

the binomial MCSE of Equation (202) evaluated at the nominal 0.05. For the 11 non-prognostic null cells of 1,000 replicates, $R = 11{,}000$ and the band is 0.0458 to 0.0542; for the eight null cells of 500 replicates in `docs/45`, the band is 0.0431 to 0.0569, against which `docs/46`, Section 8, judges the mean of 0.058 (verified, `check-stress-test-construct.R`, Section 9). An analysis is flagged if any prognostic null cell exceeds 0.075.

Size-adjusted power. $\tilde{\pi}_a(c)$ is the p-value form of Equation (207) for analysis a in alternative cell c, with $\hat{q}_{a,c_0}$ the empirical 5% quantile of a's p-values in the matching null cell c_0 (the same stress, or the zero point of the same curve) and R_c replicates. Its MCSE is computed as binomial, at most $\sqrt{0.25/500} = 0.022$ for 500 replicates, which ignores the sampling error of $\hat{q}_{a,c_0}$; the difference between two independent cells of 500 then has an MCSE of at most 0.032 (`docs/47`, Section 5, "about 0.03"; verified).

Shortfall and its minimax. Over the set $\mathcal{S}$ of analyses passing the size and sensitivity screens, flagged ones included,

$$\Delta_a(c) = \max_{a' \in \mathcal{S}} \tilde{\pi}_{a'}(c) - \tilde{\pi}_a(c), \qquad m_a = \max_{c \in \mathcal{C}} \Delta_a(c), \tag{209}$$

with $\mathcal{C}$ the non-prognostic alternative cells (or, in the second reading, all 26), and the rule selects $\arg\min_a m_a$ among the unflagged survivors, with ties within 0.02 broken by the detectable coupling and then by the smaller q of Table 55. [The sources call m_a the "maximin shortfall"; selecting the analysis with the smallest largest shortfall is a minimax-regret rule. The best analysis in each cell is taken over $\mathcal{S}$, which includes the flagged C1 and C2 (inspected), so the shortfalls of the robust analyses are measured against the phase-variance models, which have the highest power in the cells `docs/47` tabulates (inferred from `docs/47`, Section 6).]

Sensitivity. On the size-adjusted power curve over the grid $0 = c_0 < c_1 < \cdots$ of positive couplings, with $\tilde{\pi}(0) = 0.05$: monotonicity requires $\tilde{\pi}(c_{j+1}) - \tilde{\pi}(c_j) \geq -2\,\text{MCSE}$ of the difference for every j; the steepness is $\{\tilde{\pi}(0.30) - \tilde{\pi}(0.15)\}/0.15$; and the detectable coupling is the linear interpolation

$$c_{0.8} = c_{j-1} + \frac{(0.8 - \tilde{\pi}(c_{j-1}))(c_j - c_{j-1})}{\tilde{\pi}(c_j) - \tilde{\pi}(c_{j-1})}, \qquad j = \min\{j : \tilde{\pi}(c_j) \geq 0.8\}, \tag{210}$$

undefined when no grid point reaches 0.8, as for every robust analysis (`docs/47`, Section 5). [`docs/46` states monotonicity "on either side of zero"; the decision script checks the positive side only, the negative side having the single point -0.30 (or -0.20). The symmetry component was dropped in Amendment 1; by Equation (193) it tested an identity.]

Paired comparison. Two analyses fitted to the same trials are compared in an alternative cell by McNemar's test on the paired size-adjusted rejections, in its exact form: with n_{10} and n_{01} the trials rejected by one analysis only,

$$p = \Pr\left(|K - n/2| \geq |n_{10} - n/2|\right), \qquad K \sim \text{Binomial}(n, 1/2), \quad n = n_{10} + n_{01} \tag{211}$$

(two-sided exact binomial test; inspected). The strawman rule of `docs/46` replaces the reference analysis only if a survivor beats it on m_a by more than 0.05 with $p < 0.05$ in at least half the alternative cells.

Summary: which manuscript supplies which piece

Paper	Contribution to this compendium	Key equations
01	Mean- and covariance-moderation architectures; step versus graded coupling; anchored carryover; conditional-mean identity linking the two encodings; closed-form moments and power of the paired-difference statistic E9 (restated October 2026); §§5, 6, 14	(22), (31), (26)–(30), (142)
02	Carryover decay families; nine analysis specifications (stored E1, E2, E3, E7, E9 and four CR2 variants; displayed G1–G9); κ calibration; sandwich/CR2; §§5, 16, 18	(17)–(20), (31), (139)–(142), (178)–(179)
03	Latent-class reading of covariance moderation; MVN moment-matching approximation; §8	(50)–(53)
04	Treatment main effect; simpler linear DGP; classical Senn/Zucker power theory; §20	(194)–(200)
05	Simple vs. complex designs; effective-N shrinkage (exchangeable form); Gompertz and carryover-recursion restatements; §§3, 4, 5	(12), (2), (21)
06	Partitioning the treatment effect into $BR/PB/TV$; lumped- slope bias identity; expectancy scaling; §12	(94)–(101)
07	Modified-Gompertz curve and alternative trajectory families; architecture-dependent shape sensitivity; §§3, 6	(1)–(6), (37), (38)
08	Closed-form RM-ANOVA derivation; the 2×2 interaction test as an exact pooled two-sample t (corrected in `docs/26`); §18	(162)–(165)
09	Informative-dropout hazard; MNAR argument; design-dependent "happy accident" selection effect; §13	(104)
10	General four-channel (δ, κ, ν, shape) decomposition of realized Type I error; CR family; §18	(173)–(184)
11	Dual-channel architecture; additive covariance, super-additive power, non-identifiability of the two channel weights; §7	(43)–(47)
12	Screening formula for design efficiency (on/off cross-covariance omitted; corrected in §15)	(131)
Whitepapers and Paper 01 appendices (added 2026-10-08)		
`docs/37`, 01 App. A.1–A.3	Closed-form E9 moments and power; coupling gap; step-rule collapse; conditional equivalence; anchored versus compounding carryover; §§5.2, 6.4, 14	(32)–(36), (30), (106)–(113), (23)
`docs/32`, `34`–`36`, 01 App. A.4–A.5	Coupling ceiling under compound symmetry and separable AR(1); factorization of the ceiling; repair of non-positive-definite matrices; §9	(54)–(71)
`docs/41`, `45`, `48`, `49`	Baseline as response or covariate; prognostic pathways; spurious interaction $(1 - f)\, s_{\text{prog}}$; separation of the biomarker block; residual bias of F1 and ES; §§11.8, 15.12, 17	(151)–(160), (132)
`docs/39`, `42`, Paper 08	Design criterion and contrast variances by correlation structure; optimal spacing; limits of the drug coefficient by time adjustment; §15	(120)–(134)
`docs/38`	Turnover model of drug effect; coupling profile; reduction to graded coupling; power cap; §10	(72)–(84)

Paper	Contribution to this compendium	Key equations
`docs/51`	Reliability, threshold response, adherence, unblinding, person-level half-lives, regression to the mean, range restriction; §11	(88)–(93)
`docs/43`, `46`, `47`, `48`	Stress-test construct and its fourteen analyses; AIC choice of the analysis half-life; phase variances; sign symmetry of power; size and selection criteria; §§19, 21.2	(185)–(193), (208)–(211)
Inventory of `docs/09a`–`docs/52` (added 2026-10-08, second pass)		
`docs/09a`, `19`, `21`–`23`	Two-period crossover bias; historical OL+BDC study; pharmacokinetic argument for graded coupling; historical mean-moderation forms and their calibration; open-label branch; random intercept under AR(1); cost of dichotomizing; binary-exposure information	(145), (150), (40), (42), (180), (172), (126)
`docs/24`–`28`	Stable Gompertz evaluation; Hill family; component clocks; rank deficiency of the phase-augmented model; RM-ANOVA sums of squares, sphericity and sample size; cycle count	(9)–(11), (14), (102), (166)–(171), (135)
`docs/29`–`31`, `33`, `40`; Papers 13, 14	Inflection share and maximum slope of the Gompertz curve; the published Hybrid construct; components and the ceiling; size-adjusted power	(7), (8), (116), (67), (68), (207)
`docs/32`, `34`–`38`	Pooled-slope variance; other designs; factor-by-time variance; validation of the closed form; crossed structures, configurations and schedules for the ceiling; `covar` spectrum and spacing condition; sensitivity binding against the other encodings	(114)–(119), (62)–(65), (85)–(87)
`docs/39`, `41`, `42`, `45`, `49`, `51`	Exact cost of the baseline row; assumed half-life of D_{bc}; components of the contrast variance; crossover under compound symmetry; stratified statistic and design search; open-label lead-in; optimal contrast; stress-test dropout	(155)–(158), (161), (13), (125), (127), (129), (133), (105)
Covar whitepaper, Architecture C plan	2024 revision of the published code; dual channel conditional on the biomarker	(24), (39), (48), (49)